Essays in the Industrial Organization of Grocery Retail Markets

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Introduction

> *In general, if any branch of trade, or any division of labour, be advantageous to the public, the freer and more general the competition, it will always be the more so.*
>
> — Adam Smith
> *The Wealth of Nations* (1776)

Advanced economies place a remarkable degree of trust in the market mechanism to ensure the supply of essential goods. Few sectors illustrate this reliance more clearly than grocery retail: households depend on decentralized markets to deliver food at affordable prices, with adequate variety and quality, while accommodating diverse consumer tastes. This arrangement is widely accepted because markets are expected to coordinate the allocation of resources more effectively than centralized alternatives.

However, the legitimacy of market allocation is fundamentally conditional. It performs its intended social function only if competition effectively constrains the exercise of market power. Competitive pressure—the threat of losing demand to rivals—disciplines firms' pricing, improves product offerings, and incentivizes investment. When markets become highly concentrated, when entry is blocked, or when consumers face frictions in comparing offers, this mechanism falters and may fail to pass efficiency gains on to consumers.

These concerns are particularly salient in the Norwegian grocery retail market, which, like many in Western economies, has undergone substantial consolidation. The sector has transformed from a landscape of independent retailers in the 1980s to one organized under four major umbrella groups by the mid-1990s, and ultimately to just three chains following further consolidation in 2015. This structural shift presents a classic trade-off: while scale economies in procurement, logistics, and distribution have likely improved productive efficiency, the resulting concentration raises concerns about weakened competitive discipline. If market power remains unchecked, the cost savings associated with consolidation may accrue to firms rather than benefit households.

Consequently, ownership concentration and market power in grocery retail have been the subject of intense scrutiny and sustained public debate. With food and non-alcoholic beverages accounting for a substantial share of household consumption

(approximately 12%; Statistics Norway, 2023) and prices exceeding the European average by roughly 50% (Eurostat, 2022*a*), three lines of argument have been particularly central to these discussions. First, competition policy proposals advocate tighter merger control or structural remedies, such as the forced divestiture of dominant chains.[1] Second, critics of the trade regime argue that high import barriers shield incumbents from international entrants, thereby dampening rivalry and inflating prices relative to neighbors such as Sweden.[2] Finally, a third perspective highlights demand-side frictions, suggesting that consumers struggle to monitor prices effectively, prompting calls for greater transparency and improved access to information.[3]

This thesis consists of three chapters, each examining a distinct aspect of competitive pressure in grocery retail and its implications for market and consumer outcomes. The first chapter studies the effects of store-format conversions following Coop's 2015 acquisition of ICA, contributing to the merger evaluation literature by showing how merger-enabled changes in store identity and positioning can reshape local rivalry even when divestment remedies limit changes in local ownership concentration. The second chapter examines cross-border shopping as an external competitive constraint, exploiting Norway's COVID-19 border closures to quantify how access to lower-priced retail options abroad affects domestic demand and commodity tax revenue. The third chapter turns to consumer-side frictions, combining survey-based measures of price knowledge with linked transaction histories to study whether and how informed consumers pay lower prices, and whether improved consumer information can strengthen competitive pressure in markets where households face a complex basket of prices rather than a single posted price, as in, for example, gasoline retail markets.

The first chapter, titled **The Competitive Effects of Store-Format Repositioning: Evidence from a Norwegian Grocery Acquisition**, provides an ex post evaluation of a major horizontal merger in the Norwegian grocery retail market: Coop's 2015 acquisition of ICA. While the transaction increased national ownership concentration, its local competitive impact operated primarily through extensive post-merger store-format conversions, particularly toward the fast-growing soft-discount segment. The chapter's central question is how nearby rival rebrandings and closures affected the prices, product assortments, and sales of NorgesGruppen's soft-discount incumbent, Kiwi, and what these responses reveal about the competitive effects of merger-enabled repositioning.

The empirical analysis exploits staggered variation in exposure to rival store conversions across local markets between 2015 and 2016. Because ICA was not present in all markets prior to the acquisition, some locations experienced no post-merger

[1] See Næringskomiteen (2025).
[2] See Aftenposten (2025).
[3] See Regjeringen (2025).

changes and serve as natural controls. Using store-level difference-in-differences designs that allow for heterogeneous treatment effects under staggered adoption, the chapter combines nationwide data on store locations and characteristics with high-frequency transaction data from one of Norway's three major discount chains to trace local responses in prices, product variety, and sales.

The results show that the competitive effects of store-format conversions depend sharply on the post-merger format. Rival rebrandings to Coop's soft-discount concept Extra lead to substantial demand diversion from Kiwi, reducing Kiwi's sales by 4.4–12.9% across specifications. In contrast, rebrandings to convenience-discount formats (Prix or Bunnpris) have effects that are generally small and statistically indistinguishable from zero, while rival store closures increase Kiwi's sales by 2.4–11.1%. Consistent with uniform national pricing in Norwegian grocery retail, there is no evidence of local adjustments in product-level prices and only modest changes in average store-level prices and assortment variety following rival conversions.

To shed light on aggregate effects, the chapter also examines national food prices using harmonized consumer price indices and cross-country comparisons. Real food prices in Norway decline by 3.8–5.2% relative to neighboring countries following the acquisition. Taken together, the findings suggest that the merger reallocated complementary, firm-specific assets—ICA's store locations and Coop's soft-discount format—in a way that intensified within-segment competition and likely benefited consumers despite higher national concentration. The chapter highlights that competitive pressure depends not only on ownership concentration but also on firms' identities and capabilities, and that merger remedies preserving the number of owners may fail to preserve competition if they disrupt such complementarities.

The second chapter, titled **The Effect of Cross-Border Shopping on Commodity Tax Revenue: Results from Norway's COVID-19 Border Closings**, studies cross-border shopping as an external competitive constraint and quantifies its fiscal consequences for commodity taxation. Using Norway's COVID-19 border restrictions as a natural experiment, the chapter estimates how access to lower-priced grocery shopping in Sweden affects sales in Norwegian stores and, in turn, value-added tax (VAT) and excise tax revenue. The setting is particularly well suited for identification. Norway's long border with Sweden and its dispersed geography generate a wide range of travel distances, producing a natural control group in areas far from the border. Moreover, Norwegian grocery chains largely follow uniform national pricing, limiting concerns about endogenous local price responses near the border and allowing the analysis to focus on quantity adjustments.

The empirical analysis combines weekly store-category-level data from a large Norwegian grocery umbrella chain with detailed information on the locations of Norwegian and Swedish stores and driving times between them. Treatment is defined

by the travel duration from each Norwegian store to the closest accessible Swedish store. The central variation arises from the abrupt closure of cross-border shopping in mid-March 2020 and the subsequent, limited and temporary reopening for selected Swedish counties during late summer and early autumn. The chapter estimates difference-in-differences models with store and week fixed effects, allowing the effect of cross-border shopping to vary flexibly across 30-minute driving-time bins.

The results show that cross-border shopping has large and economically meaningful effects on Norwegian retail activity close to the border, with impacts that decline gradually with distance. When the Swedish option is available, sales in Norwegian stores within 30 minutes of the border are about one quarter lower, with sizable effects also for stores 30–60 minutes away and detectable effects extending to around two hours' drive. Decompositions into store visits and basket size indicate that reduced store traffic dominates close to the border, while basket-size effects remain relevant further inland, consistent with consumers making less frequent but more targeted cross-border trips at longer distances.

Effects are especially pronounced for categories that are easy to transport and stockpile and for which cross-country price gaps are large. For prominent cross-border shopping categories—including beer, cigarettes, snus, carbonated soft drinks, sweets, meat, and cheese—the estimated reductions are substantial and extend further inland than for perishable or convenience-oriented categories. For tobacco products in particular, the implied volume losses near the border are very large, consistent with strong incentives created by excise tax and price differences.

The chapter then aggregates the estimated local treatment effects to the national level by combining the store-level estimates with the full national distribution of grocery stores by distance to Sweden. Using 2019 as a pre-pandemic benchmark, it computes counterfactual tax revenue under a closed-border regime. The estimated fiscal consequences are sizable: cross-border shopping reduces VAT revenue from grocery-store sales by 3.6% nationally. For excise-taxed goods, the implied revenue losses are larger, with national excise tax revenue reduced by 8.1% for carbonated soft drinks and 11.9% for cigarettes. Summing VAT and excise taxes across grocery-store sales, the estimated tax leakage amounts to roughly 2.3 billion NOK per year. The chapter concludes that cross-border shopping materially constrains independent commodity tax policy—particularly for excise taxes motivated by revenue and health objectives—and that the relevant competitive constraint extends well beyond immediate border regions.

The third chapter, titled **Do Informed Consumers Pay Less? Evidence from a Survey with Linked Grocery Purchase Data**, examines whether consumers who are better informed about grocery prices actually pay lower prices, and through which mechanisms this occurs. The chapter combines an online survey of Trumf loyalty

program members with detailed transaction records from Norway's largest grocery group. In the survey, more than 2000 households are asked to recall current shelf prices (including sales, excluding individualized discounts) for a set of representative products in two chains (a leading discount chain and a leading supermarket). These recall data—over 70,000 store–product price guesses—are linked to 18 months of individual-level purchase histories covering the surveyed products and a broader set of close substitutes (over 700 products). This linkage makes it possible to connect measured price knowledge to realized shopping outcomes.

Price knowledge is measured as the negative percentage absolute deviation between recalled and true shelf prices, aggregated to an individual-level score. The core outcome is "discount utilization": for each purchase, the chapter compares the price the consumer paid to the average shelf price over a quarterly window and then averages this measure across product–chain combinations. This construction is designed to isolate the timing of purchases relative to temporary price reductions, rather than differences in store choice or the inherent cheapness of the products consumers buy.

The main finding is that better-informed consumers systematically pay lower prices, primarily by timing purchases to coincide with sales. Quantitatively, a 10-percentage-point improvement in price knowledge (roughly an interquartile-range shift) is associated with about a 1.3 percent reduction in prices paid through higher discount utilization. Moving from very poor to quite good price knowledge (e.g., from a 40% to a 10% recall error) corresponds to roughly a 4-percentage-point lower average price paid. These relationships are robust to controls for demographics, shopping history (such as store visit frequency, basket composition, and loyalty program membership), and self-reported measures of price sensitivity and learning effort. Importantly, the chapter provides direct evidence for a central mechanism in Varian-style models of sales: informed consumers pay lower prices by exploiting temporary discounts.

The chapter also quantifies the direct consumer savings implied by these effects. Holding quantities and the price environment fixed, improvements in price knowledge translate into meaningful (though not enormous) annual savings for a representative household, and the gains are larger when the consumer's basket contains products with greater price variation. Heterogeneity analyses show stronger effects for consumers facing higher basket price volatility and, as expected, somewhat larger effects for non-perishable products, though the relationship is present for perishable goods as well.

Finally, the chapter studies what predicts price knowledge. Demographics explain little on their own. Instead, behavioral and product-level factors matter: price learning effort (actively checking prices, flyers, and comparisons) is strongly associated with better knowledge, while self-reported price sensitivity is not once learning effort is

controlled for. Price knowledge is higher with frequent chain visits, recent purchase experience, and higher spending on the product, and lower for large categories with many close substitutes and for products with greater price variability. Overall, the chapter links consumer information frictions to concrete household outcomes and suggests that policies or tools that reduce the costs of acquiring price information can strengthen competitive pressure by enabling more consumers to exploit sales.

Taken together, the three chapters underscore that competitive pressure in grocery retail is shaped along several margins: firm characteristics and positioning (Chapter 1), the availability of consumers' outside options through cross-border shopping (Chapter 2), and consumers' ability to observe and act on price variation (Chapter 3). One lesson from the first chapter is that interventions aimed at one margin—for example, structural divestment remedies—may mitigate market power while also disrupting complementarities that matter for rivalry and consumer welfare. The second chapter highlights that policies motivated by objectives beyond competition, including elements of trade and agricultural policy, can limit competitive pressure from foreign retailers and suppliers; characterizing these consumer-relevant trade-offs, and assessing the scope for attenuating their competitive side effects, is a natural avenue for future research. The third chapter, in turn, points to levers that may strengthen competitive pressure within existing institutional constraints, including initiatives that lower consumers' costs of acquiring and using price information.

Chapter 1

The Competitive Effects of Store-Format Repositioning: Evidence from a Norwegian Grocery Acquisition

Abstract: This paper evaluates Coop's 2015 acquisition of ICA, which raised national concentration in Norwegian grocery retail and reshaped local competition through extensive store-format repositioning toward soft-discount. Store-level difference-in-differences estimates for a leading soft-discount incumbent show that rival conversions to the acquirer's soft-discount format reduced sales by 4.4–12.9%, while closures increased them by 2.4–11.1%. Convenience-discount conversions had no sales effect. Local product prices were unchanged, consistent with uniform national pricing, and assortments adjusted little. Nationally, food prices fell 3.8–5.2% relative to neighboring countries. These findings suggest the merger reallocated complementary firm-specific assets—ICA's locations and Coop's soft-discount concept—strengthening competition despite greater concentration.

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1.1 Introduction

How can mergers between competing firms benefit consumers? Horizontal mergers are conventionally viewed as harming consumers by increasing market power and worsening the terms they face, unless efficiency gains are sufficiently large to offset these effects. Since Williamson (1968), such efficiencies have typically been understood as reductions in marginal production costs that lower prices (Kwoka, 2015).[1] However, mergers may also generate *repositioning efficiencies*: by combining complementary assets, firms may be able to reduce the cost of changing product characteristics.

In this chapter, I study the competitive effects of merger-enabled repositioning in the context of a major acquisition in the Norwegian grocery retail market, where the merger's local impact was driven primarily by store-format conversions rather than changes in local ownership concentration. Merger-enabled repositioning can benefit consumers through both direct and indirect channels. Directly, repositioning may improve quality or introduce attributes that better match consumer preferences. Indirectly, repositioning can reshape competitive interactions: as products move closer in characteristic space, substitutability increases, intensifying competitive pressure on prices and non-price attributes (though repositioning can also weaken competition if it instead leads to greater differentiation or product exit; Asker and Nocke, 2021). A central example—and the focus of this chapter—is the Norwegian grocery retailer Coop's development of its soft-discount format Extra. The concept was designed to compete with the fast-growing soft discounters Kiwi and Rema 1000, but scaling it required access to suitable store locations. Coop's 2015 acquisition of ICA provided precisely such locations, including stores from the convenience-discount chain Rimi and the supermarket chain ICA Supermarked, making large-scale store-format conversion feasible.

Norway's grocery retail sector is dominated by a small number of vertically integrated umbrella chains, a structure that has long raised concerns about weak competition, high food prices, and limited quality and variety. These concerns sharpened in 2015 when Coop acquired ICA, increasing national ownership concentration as the market consolidated from four to three umbrella chains that together accounted for 96% of grocery sales. The Norwegian Competition Authority (NCA) cleared the acquisition subject to a divestiture remedy: Coop was required to sell 93 of ICA's roughly 550 stores while retaining the remaining locations. As a result, local ownership concentration often changed little. Instead, the most salient post-merger changes in local market structure arose from store-format conversions—particularly into Coop's soft-discount concept Extra. Consequently,

[1] A growing empirical literature seeks to measure such productive efficiency gains directly using granular data on physical inputs and outputs (e.g., Braguinsky et al., 2015; Demirer and Karaduman, 2025).

the acquisition provides a setting in which the merger's local competitive impact is more likely to operate through changes in stores' identities and characteristics than through changes in ownership concentration. The aim of this chapter is to estimate the competitive effects of these merger-enabled store conversions.

The main research question addressed in this chapter is: how do rival store rebrandings and closures following Coop's acquisition of ICA affect the prices, product assortments, and sales of NorgesGruppen's soft-discount chain Kiwi? Prices and assortments capture firms' competitive responses to rival repositioning, while sales reflect consumers' purchasing behavior. Together, these outcomes provide evidence on how merger-enabled repositioning alters competitive pressure and consumer choice in local grocery markets. To complement the local analysis, I also examine whether the acquisition coincided with changes in aggregate food prices in Norway relative to neighboring countries. This comparison is informative about national price effects arising from chain-level efficiencies, changes in bargaining power or competitive intensity, or shifts in the composition of store formats following the merger.

The empirical strategy exploits variation in exposure to rival store rebrandings and closures across local grocery markets following Coop's acquisition of ICA. Because the acquired umbrella chain was not present in all local markets prior to the acquisition, some markets experienced no post-acquisition store conversions and thus serve as natural controls. Comparing treated Kiwi stores—whose local markets experienced a rival rebranding or closure—with control stores allows me to estimate the local effects of rival repositioning on prices, product assortments, and sales, while differencing out national- and chain-level effects.

Because store-format conversions occurred at different points in time between April 2015 and July 2016, I use difference-in-differences estimators that allow for heterogeneous treatment effects across cohorts and over time.[2] The empirical strategy combines nationwide data on the locations, characteristics, and local demographics of most grocery stores in Norway with detailed transaction data from the incumbent soft-discount chain Kiwi, enabling me to track prices, assortments, and sales at the store level across the country.

The national price analysis draws on harmonized consumer price indices from Eurostat and combines a structural break analysis of Norwegian food prices with difference-in-differences and synthetic control methods that compare price developments in Norway with those in neighboring countries around the time of the acquisition.

I find that the competitive effects of store-format conversions depend sharply on the post-merger format. When a nearby rival store rebrands to Coop's soft-discount

[2]See de Chaisemartin and D'Haultfœuille (2022) for a survey of difference-in-differences methods accounting for heterogeneous treatment effects.

concept Extra, sales at affected Kiwi stores fall by 4.4–12.9%, indicating substantial demand diversion and that Extra is a materially tougher competitor than the predecessor formats.[3] By contrast, rebrandings to convenience-discount formats (Coop's Prix or the independent chain Bunnpris) generally yield smaller sales responses that are statistically indistinguishable from zero. Rival store closures increase Kiwi sales by 2.4–11.1%, consistent with the removed store having exerted meaningful competitive pressure.

Despite these sizable demand responses, I find no evidence that Kiwi adjusts product-level prices locally in response to either rebrandings or closures, consistent with uniform national pricing in the Norwegian grocery market (Ozhegova, 2025; Friberg, Steen, and Ulsaker, 2022; Meile, 2020). However, as shown by Ozhegova (2025), firms can still respond by adjusting the product assortment at the store level toward higher- or lower-priced products. Evidence on *effective store prices*—store-level averages of product prices weighted by expenditure shares—is less clear and varies across specifications, but effective prices tend to increase modestly following rival closures (0.5–2.0%) and rebrandings to Bunnpris (1.2–1.8%), consistent with shifts in basket composition rather than changes in product prices.

On the quality margin, local assortment responses are limited. Kiwi's product variety declines modestly—by 0.6–2.0%—following a rival rebranding to Extra, while there is no clear evidence of systematic variety changes following rebrandings to Prix or Bunnpris. Following rival closures, variety is slightly higher in some specifications. The small magnitude of these local adjustments, together with evidence of average assortment expansion at the national level, is consistent with assortment decisions being negotiated and implemented largely at the chain level, with limited scope for local discretion.[4]

Finally, in the national analysis using harmonized consumer price indices, I find that real food prices in Norway declined by 3.8–5.2% after the acquisition relative to comparable neighboring countries. This decline is consistent with local market conversions that, in the aggregate, shifted the national format mix toward soft discount and intensified within-segment rivalry, but it may also reflect national-level channels such as increased buyer power in negotiations with suppliers or scale efficiencies.

Related Literature. This chapter contributes to the empirical literature evaluating the competitive effects of horizontal mergers using ex post data (see Asker and Nocke, 2021 for a recent summary of merger retrospectives). Most quasi-experimental merger evaluations operationalize treatment through changes in local ownership concentration,

[3]The reported intervals reflect variation across specifications using different local market definitions, as discussed in Section 1.4.1.

[4]Assortment discretion varies across retail formats. Soft-discount chains such as Kiwi typically operate with more standardized national assortments, whereas supermarkets and general stores allow greater store-level flexibility, making local assortment responses more pronounced in those formats (see Ozhegova (2025)).

because post-merger incentives to raise prices arise when some of the sales lost to a rival are recaptured by an entity the firm now owns (e.g., Allain et al., 2017; Pires and Trindade, 2018; Rickert, Schain, and Stiebale, 2021). In my setting, however, divestiture remedies limited changes in local ownership concentration in many markets, while the salient post-merger change was extensive store-format rebranding. This creates a setting in which the merger's local competitive impact may operate primarily through changes in stores' identities and product characteristics—i.e., merger-enabled repositioning—rather than through ownership concentration.

A first contribution is therefore to isolate the local competitive impact of store-format rebrandings as a form of product repositioning.[5] A close conceptual antecedent is Hastings (2004), who studies sharp conversions of independent gasoline stations into vertically integrated formats and finds that such changes increased prices by about 5%, illustrating that outlet identity and characteristics can affect competitive pressure even absent changes in the number of competitors. My setting differs in a key institutional dimension: Norwegian grocery retail features chain-level uniform pricing (Friberg, Steen, and Ulsaker, 2022), which sharply limits unilateral local price responses. Consistent with this, I find no evidence of systematic local adjustments in product-level prices following rival rebrandings, and only limited evidence of shifts in effective store prices through changes in basket composition. Precisely because local prices are largely rigid, sales responses can be interpreted as demand diversion and as a measure of changes in competitive pressure. They therefore provide suggestive revealed-preference evidence on the direction of consumer welfare effects for marginal shoppers who gain or lose access to preferred alternatives. For the analysis of merger remedies, in particular, this setting allows me to isolate the effect of buyer heterogeneity—showing that preserving the number of competing owners need not preserve competitive pressure when alternative buyers possess materially different assets (see Brown, Eckert, and Shaffer, 2023 on the importance of allocation across heterogeneous firms in electricity markets).

A second contribution is to provide suggestive evidence on national price effects in a uniform pricing environment. Store-level difference-in-differences designs are well suited to identifying local competitive effects, but they difference out nationwide changes by construction—including changes in buyer power, chain-level efficiencies, and centralized pricing decisions. Related studies illustrate this limitation in different

[5] A related literature studies how mergers affect product positioning itself (e.g., Gandhi et al., 2008; Sweeting, 2010; Fan, 2013; Wollmann, 2018; Mazzeo, Seim, and Varela, 2018; Herrera-Araujo and Piechucka, 2019; Argentesi et al., 2021; Li et al., 2022; Atalay et al., 2023). Whereas that work treats repositioning as an endogenous outcome of the merger, I treat store-format conversions as quasi-experimental variation in product characteristics and study their downstream effects on a rival. In line with Sweeting (2010)'s finding that merged firms may reposition closer to competitors, Coop frequently replaced ICA's formats with the soft-discount format Extra, increasing head-to-head competition with the soft discounters Kiwi and Rema 1000.

ways: Allain et al. (2017) identify local price responses among rivals with local pricing, but not national price changes for centrally priced grocery chains, while Aguzzoni et al. (2016) infer national price effects for books using within-country benchmarks—competitors' prices for the same titles and prices for the merging parties' top-selling titles—that may themselves respond to the merger, complicating magnitude interpretation. To shed light on aggregate effects of the ICA acquisition, I therefore compare real food prices in Norway with those in comparable neighboring countries using harmonized consumer price indices. This approach partially mitigates the absence of a domestic comparison group by netting out shocks common across countries, though causal interpretation necessarily remains cautious.

The next section provides an overview of the structure and segmentation of the Norwegian grocery retail market and describes Coop's acquisition of ICA. Section 1.3 presents the data and descriptive statistics. Section 1.4 outlines the research design used to analyze the local competitive effects of retail-format rebrandings and closures, while Section 1.5 reports results for prices, product variety, and sales and presents the aggregate national price analysis based on harmonized consumer price indices. Section 1.6 discusses identification and presents robustness checks. Finally, Section 1.7 concludes by summarizing the key findings and discussing their implications for the competitive effects of mergers and divestiture remedies.

1.2 The Norwegian Grocery Retail Market

1.2.1 Market Structure

Since 1994, the Norwegian grocery retail market has been dominated by four large umbrella chains accounting for approximately 96% of total sales: NorgesGruppen, Coop, Reitangruppen (Rema 1000), and ICA, until Coop acquired ICA in 2015. Each umbrella chain is vertically integrated with its own wholesaler and distribution system, and all except Reitangruppen operate several chain formats across different segments (see Table A4 in Appendix 1.B for an overview).[6] The evolution of market shares and ownership concentration over the past two decades is shown in Figure 1.1.[7]

Before 2014, ICA steadily lost market share to the other chains (from 25% in 2001 to 11% in 2014), especially to the market leader NorgesGruppen and Rema 1000, which grew by 7 and 8 percentage points, respectively, over the same period. Coop saw a slight decline before increasing its share by 7 percentage points following the acquisition. Bunnpris has maintained a small but growing presence (from 1.6% in

[6]Bunnpris is the largest retail chain outside the four umbrella chains and shares distribution with NorgesGruppen.

[7]Ownership concentration is measured using the Herfindahl–Hirschman index: $HHI = \sum_{i=1}^{N}(S_i)^2 \in [0, 1]$, where S_i is the market share of firm i in decimals.

2001 to 3.4% in 2020), while the hard discounter Lidl exited the market in 2009 after failing to gain a foothold over five years (peaking at 1.7% in 2007). From the early 2000s to 2020, the market transitioned from four to three large umbrella chains, with the remaining firms largely maintaining their relative standings.

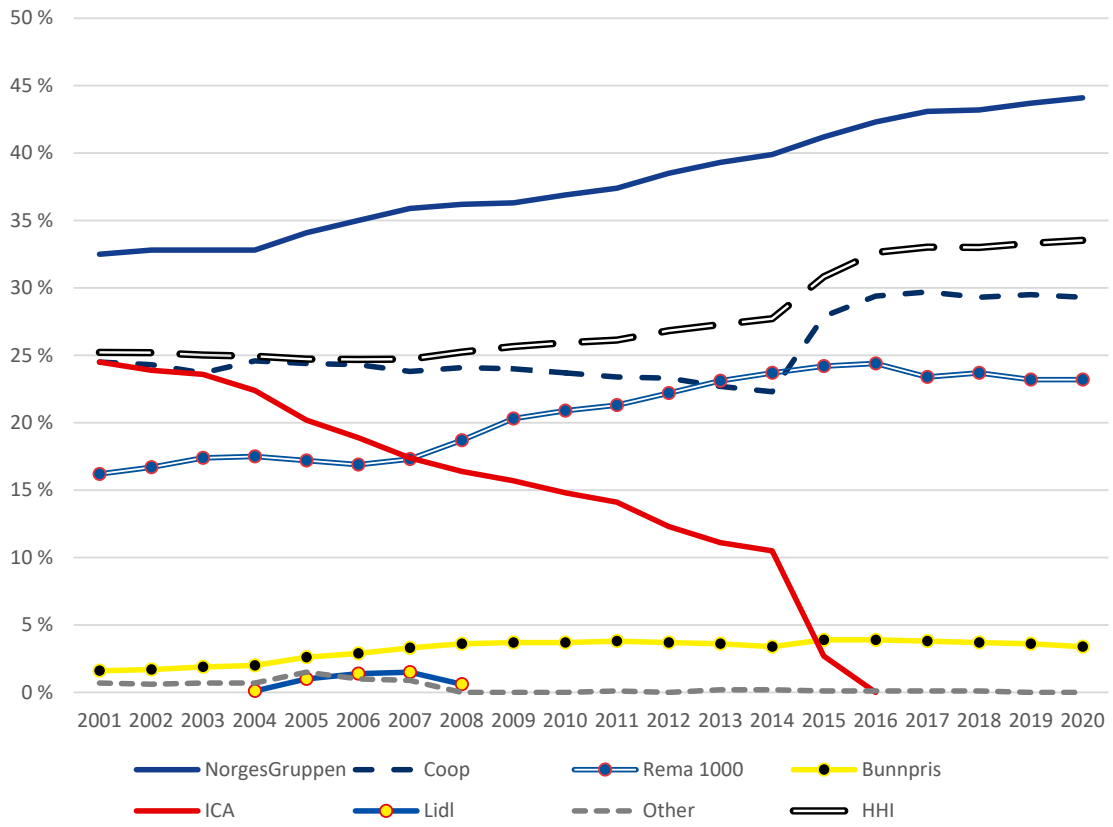


Figure 1.1: Evolution of market shares and the Herfindahl–Hirschman Index (HHI) by umbrella chain, 2001–2020. *Source:* AC Nielsen.

In 2002, the Herfindahl–Hirschman Index (HHI) in Norway (0.25; $N^e = 4$) and other Nordic countries was already high. For comparison, Germany (0.16; $N^e = 6.25$) and the United Kingdom (0.15; $N^e = 6.67$) exhibited substantially lower concentration levels (Nordic Competition Authorities, 2005).[8] However, it increased by only 0.03 over the 14 years leading up to the acquisition, before rising by another 0.05 by the end of 2016 ($HHI = 0.33$; $N^e = 3$). The acquisition reignited concerns about limited competition and its implications for product variety and food prices (Dagens Næringsliv, 2015; e24, 2014). As of 2026, the NCA continues to list competition in the grocery retail market as one of its priority issues (Ministry of Trade, Industry and Fisheries, 2026).[9]

[8]The effective number of firms is defined as $N^e = \frac{1}{HHI} \leq N$, with equality when firms are of equal size. It can be interpreted as the hypothetical number of symmetric firms that would yield the same HHI as the actual firms of unequal size.

[9]According to the U.S. Department of Justice and FTC's *Horizontal Merger Guidelines*, an HHI above 0.25 indicates high market concentration. In such markets, mergers that increase the HHI by more than

Although increased ownership concentration at the national level may be cause for concern, transportation costs ensure that competition remains local, as consumers are unlikely to include distant stores in their consideration set. Strategic decisions such as pricing and base assortment must ultimately reflect local market conditions, even if they are determined at the national level.

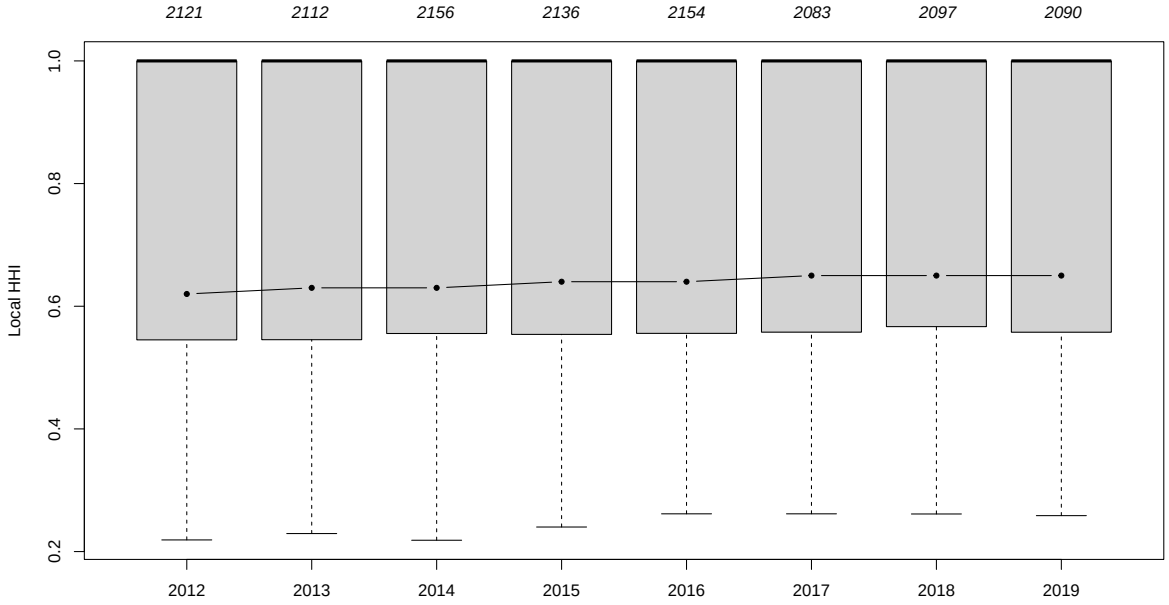


Figure 1.2: Box plots of the Herfindahl–Hirschman Index (HHI) at the *postcode* level, 2012–2019, calculated using store revenues aggregated at the umbrella chain level. The data cover the vast majority of grocery stores in Norway. The weighted mean, based on postcode sales shares, is shown as black dots connected by lines. The number of local markets in the sample is shown above each box plot. *Source:* Geodata.

To examine concentration at the local level, Figure 1.2 displays annual box plots of postcode-level HHI, summarizing the distribution across postcode areas from 2012 to 2019. The size of the postcode area a store resides in tends to vary inversely with the surrounding population and store density and is therefore correlated with consumers' willingness to travel. As a result, postcode areas serve as approximations to true markets (I return to this point in Section 1.4.1). The distribution of local market concentration clearly displays bunching at the upper endpoint ($HHI = 1$); the median local market is a monopoly. However, several of these markets are rural and served by a single small general store, with sales that likely account for only a minor share of the umbrella chains' national totals. A better measure of the competitive pressure imposed on the umbrella chains is the weighted mean of HHI, using each

0.02 are presumed likely to increase market power. Similarly, the European Commission cannot rule out competition concerns when the post-merger HHI exceeds 0.2 and the change in HHI is greater than 0.015. For a discussion of how HHI relates to competition intensity and social welfare, see Spiegel (2021).

local market's share of nationwide sales. This is shown as the black dots connected by lines in Figure 1.2. Since national chains are not equally represented in all markets, local ownership concentration tends to be substantially higher than at the national level. The sales-weighted mean HHI across postcode areas increased only marginally, from 0.63 ($N^e = 1.59$) in 2014 to 0.64 ($N^e = 1.56$) in 2016—a change of 0.01 points ($\Delta N^e = -0.03$). This small change reflects that the merging parties did not compete in all local markets, and that the NCA imposed divestment remedies in most of the markets where they did, in order to limit competition concerns (Konkurransetilsynet, 2015, Clause 1744).

Similar plots of local HHI for the four alternative geographical units—Basic Statistical Units (BSUs), postal areas (larger than postcode areas), municipalities, and counties—are shown in Figures A5–A8 in Appendix 1.B. There are approximately 14,000 BSUs in Norway, but they are too small to capture the relevant market, as the vast majority contain one or no stores. As we consider increasingly large market delineations—namely postal areas, municipalities, and counties—the sales-weighted HHI decreases mechanically toward the national level depicted in Figure 1.1. However, the national increase in HHI following the acquisition is only clearly reflected at the county level. Hence, there is limited scope for the acquisition to have affected competition through changes in local ownership concentration.

1.2.2 Market Segmentation

The store formats in the Norwegian grocery industry are traditionally categorized into general stores, discount stores, supermarkets, and hypermarkets (AC Nielsen).[10] General stores are typically small and offer a limited selection of goods and services to meet the basic needs of local communities.[11] Supermarkets are large, full-service grocery stores offering a wide variety of products, including fresh produce. Discount chains fall somewhere in between in terms of assortment size and focus on cost efficiency, low prices, and private labels. Hypermarkets are even larger than supermarkets and tend to also stock non-food items such as clothing and electronics. They usually offer lower prices due to economies of scale but are typically located on the outskirts of cities or in large shopping centers. Since the mid-2000s, the market share of discount chains has steadily grown at the expense of the other segments—from 47% in 2005 to 68% in 2020.

[10] In Norwegian: *nærbutikk*, *lavpris*, *supermarked*, and *hypermarked*.

[11] *Nærbutikk* is sometimes translated as "convenience store", but should not be confused with small outlets such as kiosks and petrol stations, which are often located in high-traffic areas. These smaller stores typically offer a narrow range of on-the-go products, such as prepared food, drinks, and tobacco, and have long opening hours.

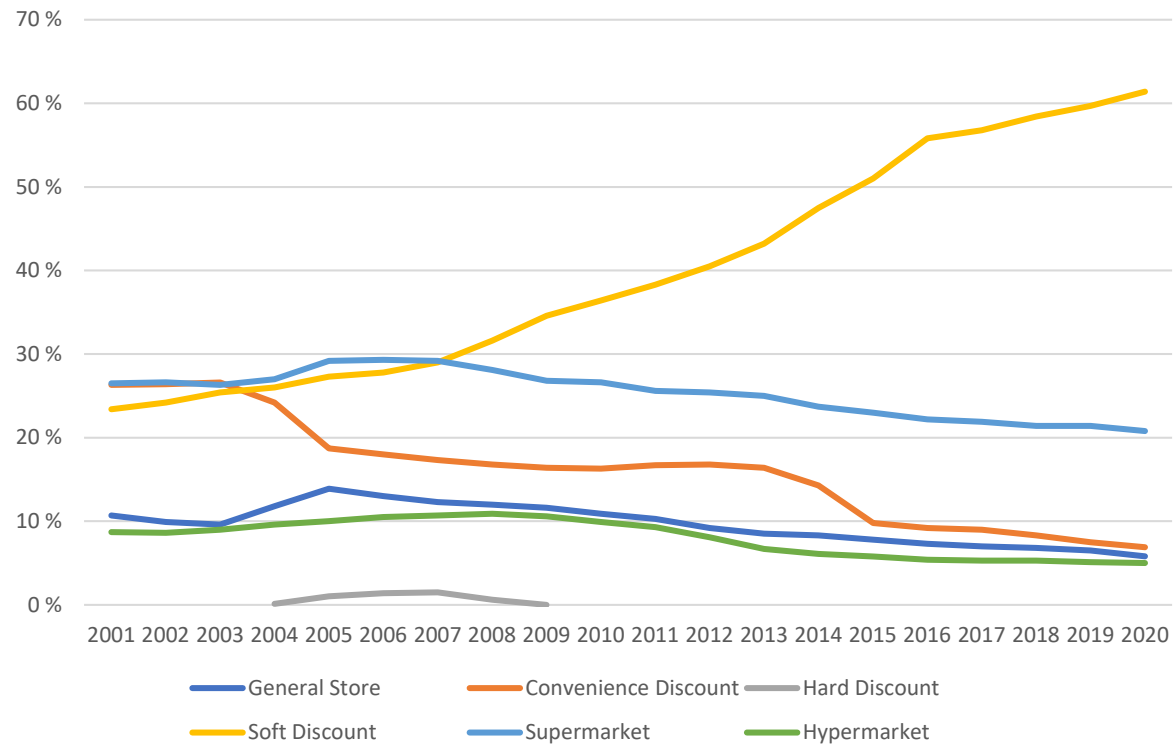


Figure 1.3: The evolution of market shares by retail segment, 2001–2020. *Source:* AC Nielsen.

In the Norwegian context, it is useful to further divide the discount segment into hard, soft, and convenience discount. Hard discount stores are no-frills outlets that carry a narrow selection of essential items, predominantly private labels. This enables high volumes of basic products, efficient operations, and consequently, very low prices.[12] In comparison, soft discounters are characterized by larger assortments and a greater presence of national brand names. Convenience-discount stores are typically smaller and emphasize quick and easy shopping, offering ready-to-eat meals, convenience foods, and on-the-go products such as salad bars, while still providing basic groceries at low cost.

Figure 1.3 shows the evolution of market shares across retail segments. The soft-discount segment has grown substantially over the past two decades, increasing from 23% in 2001 to 61% in 2020, largely at the expense of all other formats. The convenience-discount segment shrank the most, falling from 26% in 2001 to 7% in 2020, with a sharp drop in 2015 when ICA's convenience discounter Rimi was discontinued following the acquisition.

Engelund and Eimind (2018) find that the share of Norwegian consumers reporting discount chains as their preferred store format increased steadily from 2008 to 2016.

[12]Examples of quintessential hard discount chains include Aldi and Lidl, the latter of which had a brief spell in Norway from 2004 to 2008. Import restrictions, low population density, and the absence of national brands are commonly cited as likely reasons for Lidl's exit (Selmani and Førre, 2023).

They also report that accessibility, assortment size, and low prices are the most important store attributes, with the latter growing in importance over time. The success of the soft discounters aligns with their favorable positioning on these characteristics, especially price.[13] Given their growing market share, effective competition in the Norwegian grocery retail market depends heavily on the soft-discount segment.

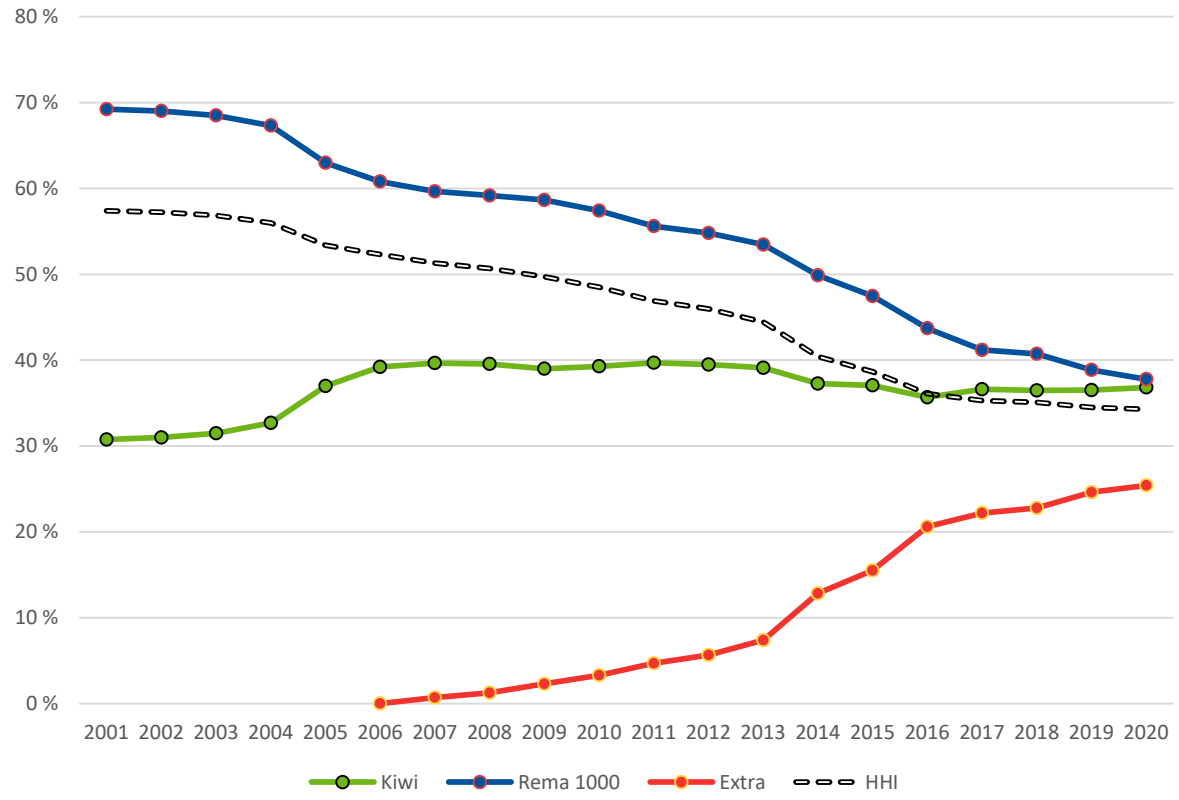


Figure 1.4: The evolution of market shares and the Herfindahl–Hirschman Index (HHI) in the soft-discount segment, 2001–2020. *Source:* AC Nielsen.

Rema 1000, inspired by the German hard discounter Aldi, spearheaded the development of the soft-discount segment in Norway. As shown in Figure 1.4, Rema 1000 dominated the segment throughout the 2000s and early 2010s, but concentration within the soft-discount segment has declined over time. Extra grew steadily after entering the market in 2006—with a noticeable acceleration following the ICA acquisition—and Kiwi has also slightly improved its position. In 2001, Rema 1000 and Kiwi held market shares of 70% and 30%, respectively, yielding a soft-discount HHI of 0.57 ($N^e = 1.75$). By 2020, the segment had evolved into a triopoly of Reitangruppen's Rema 1000, NorgesGruppen's Kiwi, and Coop's Extra, with market shares of 38%, 37%, and 25%, respectively, and an HHI of 0.34 ($N^e = 2.94$).

[13]The three soft discounters consistently dominate VG's matbørs, a recurring blind test of grocery prices conducted by one of Norway's leading newspapers (see https://www.vg.no/spesial/matborsen/).

The increase in soft-discount presence and within-segment rivalry—driven in part by merger-enabled format conversions and expansion—is a central focus of this chapter.

1.2.3 Coop Acquires ICA

The ICA group, with its chain concepts Rimi (convenience discount), ICA Supermarked, and Matkroken (general store), struggled with low profits and steadily lost market share throughout the 2000s and early 2010s. In 2014, the NCA blocked a purchasing and distribution agreement with NorgesGruppen that was intended to improve ICA's profitability through lower wholesale prices and more efficient distribution. Consequently, ICA instead sought to sell its 554 Norwegian stores to Coop, a transaction the NCA approved conditional on divestment remedies—43 stores were sold to the independent chain Bunnpris and 50 to NorgesGruppen.

The NCA employed a sequential screening process to select these stores. First, it filtered out 227 ICA stores that, if acquired by Coop, were unlikely to harm competition based solely on the number of competing stores nearby and their market shares. Second, 93 of the remaining stores were identified as problematic based on a high degree of competition between the merging parties (as measured by diversion ratios) and post-acquisition incentives to raise quality-adjusted prices.

The NCA considered acquisition-specific efficiency gains but concluded that they were insufficient to offset the costs of adjustment and the loss of competition (Konkurransetilsynet, 2015, Clause 1733).[14] It also deemed the likelihood of disciplining future entry low, citing significant barriers such as import restrictions, irreversible investments, economies of scale, and vertical integration (Konkurransetilsynet, 2015, Clause 156). Ultimately, the NCA approved the acquisition conditional on divestment remedies intended to mitigate its anti-competitive effects (Konkurransetilsynet, 2015, Clause 1753), but emphasized that the remedies would not go beyond what was necessary, in accordance with the principle of proportionality (Konkurransetilsynet, 2015, Clause 1754). The public documents did not consider whether the acquisition could result in a net pro-competitive outcome or examine the potential competitive effects of store-format rebrandings on non-merging stores, motivating an investigation into how such rebrandings may have affected competition.

[14]The details of the proposed efficiency gains are withheld from the public.

1.3 Data and Descriptive Statistics

1.3.1 Data Sources and Construction

The analysis draws on several data sources. The first is an annual panel compiled by Geodata, which tracks characteristics such as geographical location, size, retail format, ownership, and opening hours of Norwegian grocery stores.[15] Store-level annual sales figures in the panel are based on data from Nielsen and Iper. The dataset also includes demographic variables—including population, average earnings, and education—at various levels of geographical aggregation, such as basic statistical units (BSUs), municipalities, and counties.

The second source is a schedule of the staggered store rebrandings and closures that followed the acquisition, provided by the acquirer. It details which format replaced each store and at what time. Combined with store location data, this information allows me to determine whether a store experienced a rival's rebranding or closure in its vicinity. In a few cases where the exact reopening date was not provided, the reopening month was obtained from local newspapers.

The third data source consists of high-frequency transaction records provided by NorgesGruppen, covering product-level sales, prices, and assortment sizes for its soft discounter, Kiwi. The data are aggregated to the monthly level, and average monthly product prices are calculated by dividing total sales in NOK by the number of units sold, where units are defined as items, weight, or volume, depending on the product type. The product categories included in the sample are listed in Table A6 in Appendix 1.B, along with their expenditure shares. Most major categories are represented, but fruits and vegetables are notable exceptions—primarily due to substantial seasonality and variation in their sales and assortment, as well as less reliable recording. As a result, the main product category that typically exhibits non-uniform pricing within chains is excluded from the analysis. This exclusion also limits the occurrence of prices measured by weight or volume (e.g., kilograms or liters), as most remaining products are sold in discrete units or packages.

Combining these sources yields a panel of stores observed at monthly intervals from 2014 to 2016, with a rich set of store and product characteristics. The analysis focuses on the soft-discount chain Kiwi, which accounted for an average of 19% of the Norwegian grocery market during the sample period. These data also allow me to characterize local market structure and to trace store-level format rebrandings before and after the acquisition in detail.

[15] See https://geodata.no/.

1.3.2 Descriptive Statistics

The following descriptive statistics and visualizations provide an overview of the scope, timing, and geographic distribution of store rebrandings and closures. Table 1.1 reports the frequency of the different format transformations for ICA Supermarked and Rimi stores following the acquisition.[16] The analysis focuses on rebrandings to Coop Extra (129), Coop Prix (83), Bunnpris (32), and store closures (46). Although rebrandings to Bunnpris were less frequent, they are included because they resulted from divestment remedies and belong to the same segment as Prix. Rebrandings to NorgesGruppen are excluded because the analysis focuses on the impact of rival rebrandings on Kiwi stores; a rebranding to NorgesGruppen always involves changes in ownership concentration, not just format. The remaining transformation types are rare or difficult to interpret and are therefore not included in the main analysis. Double change refers to cases where a store underwent two successive changes, such as being rebranded twice, rebranded and then closed, or experiencing a prolonged closure before reopening during 2015 or 2016.

Figure 1.5 illustrates how these store rebrandings and closures accumulated over time. The transformation process took place between April 2015 and July 2016, with most activity concentrated in late 2015 and early 2016; roughly half of the 385 total transformations occurred in September 2015. Except for 27 transformations in the five months leading up to September 2015, the remainder took place gradually until July 2016.

Figure 1.6 provides a geographical overview of the stores included in the analysis. The left panel displays all Kiwi, ICA Supermarked, and Rimi stores operating in Norway in 2014, prior to the onset of the rebranding process. The right panel depicts the outcomes for each ICA store following the rebrandings and closures in 2015–2016, together with the Kiwi stores operating in 2016. Both chains were present in all counties, with Kiwi stores more concentrated in the south and ICA stores relatively more prevalent in the north. The maps also highlight areas of direct competition—where Kiwi and ICA stores are co-located—as well as regions where only one chain operated. In many cases, the spatial separation is such that a broad range of plausible market definitions would yield the same classification of whether stores are exposed to the rebranding.

[16] The general store *Matkroken* is not included here or in the analysis. Coop acquired most Matkroken stores without rebranding, though a few were divested and rebranded. This, together with a few double-change cases, accounts for the discrepancy between the 93 divested stores reported by the competition authority and the 72 Bunnpris and NorgesGruppen stores in Table 1.1.

Table 1.1: Frequency of rebranded ICA stores before and after the acquisition

	ICA Supermarked	Rimi	Sum ICA
Coop Extra	25	104	129
Coop Prix	1	82	83
Closed	6	40	46
NorgesGruppen	8	32	40
Bunnpris	4	28	32
Coop Mega	26	1	27
Double Change	3	18	21
Rema 1000	2	5	7
Sum Rebrandings	75	310	385

Notes: The table summarizes how Rimi and ICA Supermarked stores were rebranded or converted following the acquisition. Rows indicate the new format or status, while columns show the original ICA format. Cell values are the number of stores in each category, with row and column sums reported at the margins.

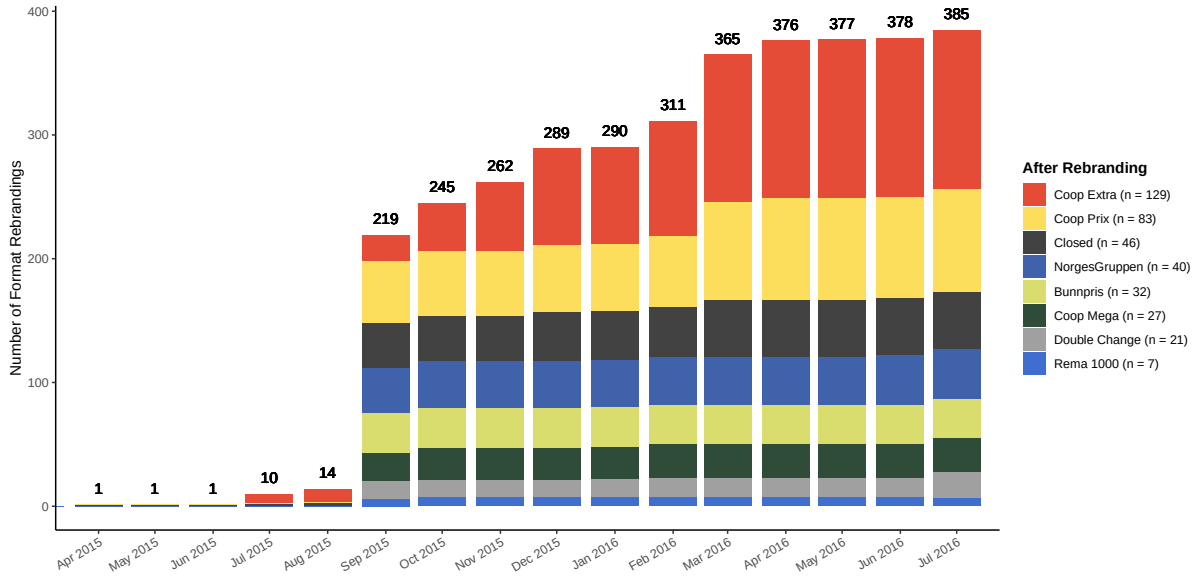


Figure 1.5: The figure displays the cumulative number of store rebrandings from April 2015, when the first rebranding occurred, to July 2016, by which time all stores had been transformed. The colors indicate the share of each type of store transformation.

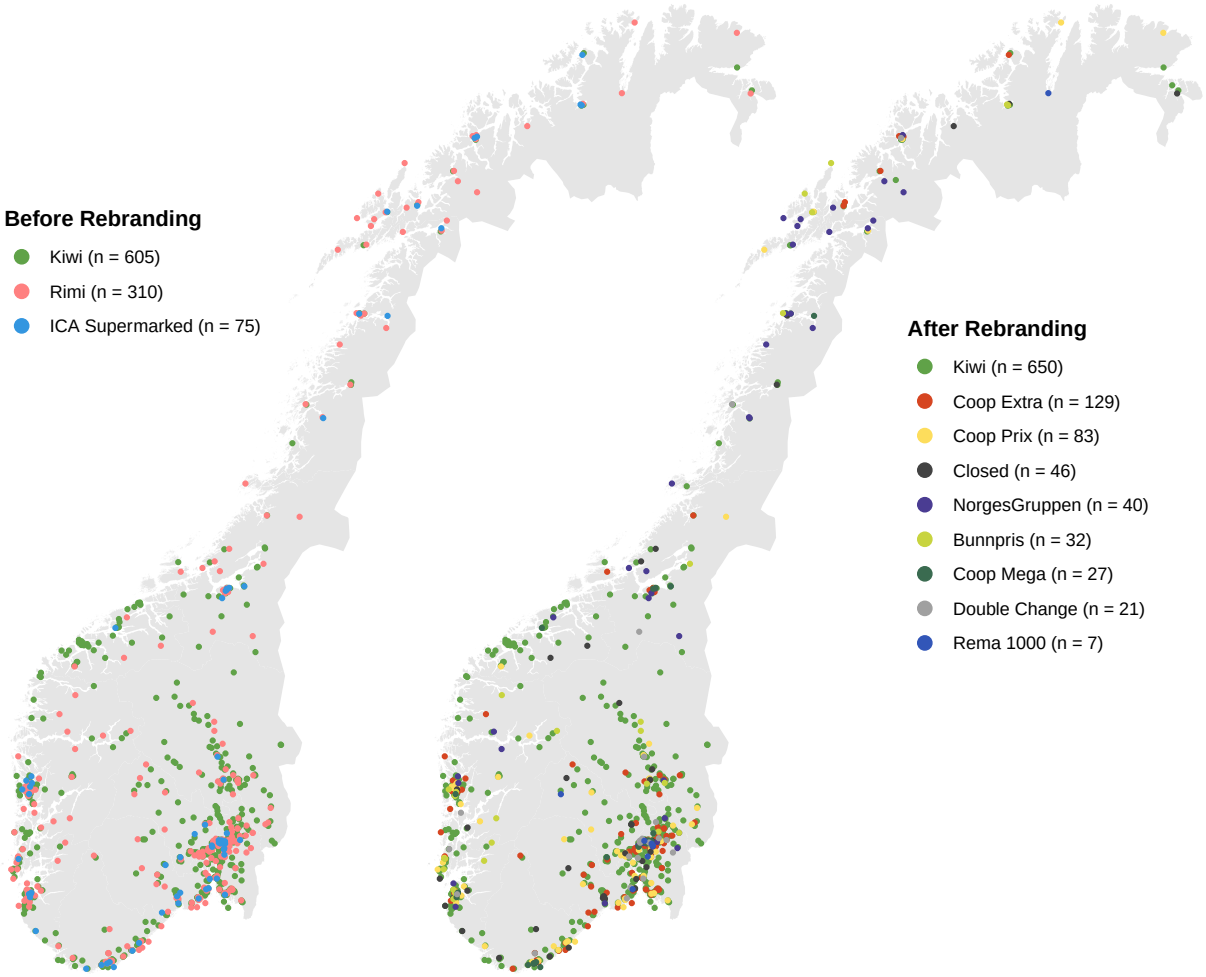


Figure 1.6: The left map shows the locations of Kiwi, Rimi, and ICA Supermarked stores operating in Norway in 2014. The right map shows ICA store outcomes in 2015–2016 and Kiwi stores operating in 2016. During 2014–2016, Kiwi opened 72 stores and closed 27, for a net increase of 45 stores. Dots are layered in decreasing order of category size so that less common formats are more visible.

1.4 Empirical Strategy

The absence of ICA stores in some local markets enables a difference-in-differences identification strategy: I compare outcomes for stores in markets where a rival rebranded or closed (treated) with those in control markets, before and after the conversions.[17]

1.4.1 Local Market Definition

I assume that stores compete if they are located within a given radius of one another, with this radius defining each store's local market. This discretized definition of competition is standard in ex post merger evaluations in retail settings (Hastings, 2004; Allain et al., 2017; Rickert, Schain, and Stiebale, 2021). Nevertheless, because competitive pressure plausibly declines smoothly with distance, such a binary classification may introduce discretization error. This concern is mitigated, however, for stores that are spatially clustered into hubs: in such cases, a range of reasonable market radiuses yields identical treatment assignments. As illustrated in Figure 1.6, many store pairings fall into this category, so moderate variation in the assumed market size does not alter the classification of treated and control stores, limiting sensitivity to the particular discretization rule.

To minimize discretization error, the chosen market size should capture most relevant competitors while excluding as many marginal or non-competing stores as possible. The optimal market size likely varies across settings and geographic contexts. For instance, Hastings (2004) considers gasoline retailers within a one-mile radius (approximately 1.6 km), Choné and Linnemer (2012) analyzes competition among parking lots within 650–1000 meters, and Allain et al. (2017) defines grocery markets as extending 10 km from the city center (20 km for hypermarkets), finding similar results with a 5 km (10 km) definition.

The appropriate market size in Norway is likely smaller than that used by Allain et al. (2017). They base their kilometric market size on the French Competition Authority's assumption that consumers are, on average, willing to drive 10 to 15 minutes to reach a supermarket or discount store. The NCA applies similar driving times in Norway, except in the largest cities and most remote areas, where it uses thresholds of 5 and 20 minutes, respectively (Konkurransetilsynet, 2015, Clause 90). However, it notes that survey evidence suggests Norwegians may be even less willing to spend time traveling, particularly in urban areas (Konkurransetilsynet, 2015, Clause 93), which points to smaller kilometric market sizes in the Norwegian context.[18]

[17] A store is defined as *treated* if a rival converted within its local market. Although national-level strategic adjustments may indirectly affect all markets, such effects are absorbed by time fixed effects in the difference-in-differences design.

[18] Lower willingness to travel is consistent with Norway's high store density, which reduces the need

Moreover, a given driving time likely corresponds to a shorter distance in Norway due to its topography. I therefore consider smaller market sizes in the range of 0.5–10 km.

Market sizes are also unlikely to be uniform across Norway, particularly due to variation in store density and accessibility. In areas with low store density, consumers must often travel farther to reach a store, leading geographically distant stores to compete for the same customers. In contrast, when store density is high, consumers located between two stores are more likely to be intercepted by intermediate stores, weakening or severing the competitive link between the original pair. A simple way to account for this heterogeneity is to use BSU population density as a proxy for store accessibility, as densely populated BSUs are more likely to have better access to stores in and around them.[19] This approach, for instance, ensures that market sizes are smaller on average in city centers—where many stores are within walking distance—than in rural areas, where only a few stores may be reachable by car. Table 1.2 presents five specifications with varying market sizes across six quantiles of BSU population density, assigning larger market sizes to stores located in sparsely populated BSUs.

Table 1.2: Market sizes across BSU population density quantiles

	Quantiles					
Specifications	1	2	3	4	5	6
10–1.5 km	10 km	7 km	5 km	4 km	2 km	1.5 km
10–1 km	10 km	7 km	5 km	3 km	1.5 km	1 km
10–0.5 km	10 km	7 km	5 km	2 km	1 km	0.5 km
6–1.5 km	6 km	5 km	4 km	3 km	2 km	1.5 km
5–0.5 km	5 km	4 km	3 km	2 km	1 km	0.5 km

Notes: The table reports market radiuses (in kilometers) assigned to stores across six quantiles of BSU population density, under five different specifications. In each specification, stores located in more densely populated BSUs are assigned smaller market radiuses, reflecting higher store accessibility and density.

Some studies (e.g., Aguzzoni et al., 2016; Pires and Trindade, 2018; Argentesi et al., 2021) define local markets using pre-determined geographical units such as cities. In a similar spirit, I include a specification that defines markets based on postcode areas. Postcodes reflect the geographic layout of cities, districts, and townships, with large cities typically divided into several distinct postcode areas. A key advantage of this approach is that the number and size of postcode areas in a given region tend

for long trips to reach a grocery store (Friberg et al., 2020, p. 37). This is especially true in the capital, Oslo, where the average consumer has access to 137 stores within 10 minutes of travel (Friberg et al., 2020, p. 50).

[19]Store density at the BSU level is less informative, as most BSUs contain zero or only one store, even in dense urban areas where consumers routinely shop across BSU boundaries. The next aggregation level at which store density is observed is the municipality, which is too coarse and larger than most local grocery markets.

to scale with population and store density. For example, a fixed area in Oslo will typically contain more—and smaller—postcode areas than an equally sized area in rural Norway. As a result, the size of the postcode area in which a store is located is likely correlated with local store density and consumers' willingness to travel, making the postcode specification a flexible complement to the radius-based definitions.

1.4.2 Treated and Control Stores

Stores are labeled as *treated* if their local market contained an ICA store prior to the acquisition, and *control* otherwise. Within the treated group, I define separate treatment arms based on the outcome of the nearby ICA store: format rebrandings into Extra, Prix, or Bunnpris, or permanent store closure.[20] To avoid confounding the effects of format rebrandings with changes in local ownership concentration, I restrict attention to local markets where the umbrella group of the rebranded format was not already present. Markets with multiple format transformations are also excluded.

Using the postcode market definition, Table A5 in Appendix 1.B reports mean values of the outcome and control variables before and after the acquisition, by treatment group, along with their percentage changes. The number of stores in each group is shown in parentheses next to the group label. Sales exhibit an upward trend across all groups, except for the Extra group, which remained flat on average. As expected, the average number of competitors changed only for stores that experienced a rival's exit. Treated stores initially had higher sales and faced more competitors, the latter consistent with the likelihood of experiencing a rival rebranding increasing with the number of competitors in the market. Assortment sizes and price levels rose over the sample period, but by similar magnitudes across groups.[21] Notably, initial price levels and assortment sizes were very similar across groups, despite differences in sales, competitor numbers, and market concentration, suggesting limited scope for stores to adjust these variables locally.

1.4.3 Staggered Adoption Design

Since the rebrandings occurred at different points in time, the empirical setting follows a *staggered adoption design* (Athey and Imbens, 2021). Under the assumptions of common trends and no anticipation, store outcomes can be modeled using the following two-way fixed effects specification (Borusyak, Jaravel, and Spiess, 2024):

[20] A few irregular cases—such as rebranded stores that later closed or rebranded a second time during the 2014–2016 sample period—are excluded.

[21] The price indices are explained in Section 1.5.1.

$$Y_{it} = \gamma_i + \lambda_t + \beta_{it} T_{it} + \delta^{'} \boldsymbol{Z_{it}} + \epsilon_{it} \tag{1.1}$$

In this specification, Y_{it} denotes the outcome variable for store i at time t, T_{it} is an indicator for whether store i is treated at time t, and $\boldsymbol{Z_{it}}$ is a vector of covariates. The error term ϵ_{it} captures unobserved shocks, which are assumed to be uncorrelated across municipalities but may exhibit serial correlation within stores or contemporaneous correlation within municipalities. The terms γ_i and λ_t are store and time fixed effects, respectively, while β_{it} and $\delta^{'}$ are the parameters to be estimated. Treatment effects β_{it} are allowed to vary arbitrarily across stores and time, and the Average Treatment Effect on the Treated (ATT) is defined as the expectation $\beta = E[\beta_{it} \mid T_{it} = 1]$. The common-trends and no-anticipation assumptions are evaluated in Section 1.6, and the main conclusions remain robust to specifications that relax these restrictions.

1.4.4 Accounting for Treatment Effect Heterogeneity

As recently shown in methodological contributions such as Goodman-Bacon (2021), Sun and Abraham (2021), Borusyak, Jaravel, and Spiess (2024), Callaway and Sant'Anna (2021), and De Chaisemartin and d'Haultfoeuille (2020), estimating Equation 1.1 using ordinary least squares does not yield consistent estimates of the ATT when treatment effects are heterogeneous across units or over time and treatment is staggered.[22] Instead, I employ the estimator proposed independently by Sun and Abraham (2021) (S&A) and Callaway and Sant'Anna (2021).[23] Imputation-based estimators, such as those proposed by Gardner et al. (2024) and Borusyak, Jaravel, and Spiess (2024), often yield similar results but are more sensitive to violations of the parallel-trends assumption (see Roth et al., 2023). By contrast, the S&A estimator is more sensitive to anticipation effects; however, these can be diagnosed and mitigated by shifting the event timing and excluding affected pre-event periods (see Section 1.6). A further advantage of the S&A estimator is that it allows for easily interpretable event-study plots, as its coefficients measure the mean differences between ever-treated and never-treated units relative to a specified reference period—here, the last period before treatment.

[22] For example, treatment effects may vary due to differences in market concentration or seasonal variation coinciding with the timing of rebranding.

[23] Implementation is facilitated by the `fixest` package in R (Bergé, 2018).

1.5 The Competitive Effects of Retail Format Rebrandings

In this section, I quantify the local effects of rival ICA stores rebranding or closing on Kiwi's prices, product assortment, and sales. Consistent with centralized or chain-level pricing, Kiwi stores do not appear to adjust prices unilaterally in response to local competitive changes. I therefore also examine the evolution of national food prices in Norway relative to neighboring countries.

1.5.1 Food Prices

Using product-level sales data derived from receipt records, I calculate the average unit price of product j in store i and month t as:

$$P_{ijt} = \frac{\text{Sales}_{ijt}}{\text{Quantity}_{ijt}}$$

where Sales_{ijt} denotes the total monthly turnover in NOK and Quantity_{ijt} the total quantity sold—measured, depending on the product, in number of items, grams, or liters.[24] These unit prices are then aggregated into a store-level price index using weights:

$$w_{ijt} = \frac{\text{Sales}_{ijt}}{\sum_j \text{Sales}_{ijt}}$$

which represent the share of product j in total sales at store i during month t.[25]

I consider two indices to capture both product- and store-level price variation. The first is the sales-weighted average of individual product prices:

$$\hat{P}^1_{it} = \sum_j w_{ijt} P_{ijt}$$

The second is the sales-weighted average of the log-deviations of product prices from their cross-sectional average across stores:

$$\hat{P}^2_{it} = \sum_j w_{ijt} \left[\ln\left(\frac{P_{ijt}}{\overline{P}_{jt}} \right) \right] = \sum_j w_{ijt} \left[\ln(P_{ijt}) - \ln(\overline{P}_{jt}) \right]$$

Each product contributes its approximate percentage deviation from the average price across stores.[26]

[24] Since fruits and vegetables are excluded from the analysis, most product prices refer to items.

[25] The analysis could alternatively be conducted at the store-product level, but this is computationally demanding due to the large number of products.

[26] Because the index is an average of percentage deviations, it is invariant to the units in which

The price indices can change either due to changes in individual product prices or in quantities. However, $\hat{P}^2_{it}$ remains fixed at zero and is unaffected by changes in weights (i.e., average basket composition) if product prices are strictly uniform across stores within a chain. In contrast, the effective price $\hat{P}^1_{it}$ can change even in the absence of local price deviations if consumers substitute toward higher- or lower-priced products in response to local market conditions. In fact, stores can influence such shifts by adjusting their product assortment, as noted by Ozhegova (2025). Thus, $\hat{P}^1_{it}$ captures both individual product prices and basket composition, whereas $\hat{P}^2_{it}$ isolates deviations in product-level prices across stores.

Table 1.3 reports sample means and standard deviations for the price indices under the postcode market definition. The price index $\hat{P}^2_{it}$ is close to zero on average as expected. The very small standard deviation is consistent with previous findings in the literature (Ozhegova, 2025; Friberg, Steen, and Ulsaker, 2022; Meile, 2020; DellaVigna and Gentzkow, 2019): product prices are largely uniform across stores, leaving little scope for local product-level price adjustments. In contrast, the effective price index $\hat{P}^1_{it}$ varies across Kiwi stores and may respond to local market conditions—for example, through assortment adjustments.

Table 1.3: Mean and standard deviation of price indices

	$\hat{P}^1_{it}$	$\hat{P}^2_{it}$
Mean	49.98 kr	0.00
Standard deviation	10.88 kr	0.01

Notes: $\hat{P}^1_{it}$ is in NOK and $\hat{P}^2_{it}$ is a percentage deviation from the average product price across stores.

Theoretically, prices can either rise or fall following a change in a competitor's identity, depending on consumer preferences and substitution patterns (Hastings, 2004). Suppose, however, that grocery stores within the same store-format segment are closer substitutes. Formally, this can be modeled using a nested logit framework with nests defined by store-format segments, as in Anderson and de Palma (1992). In such a model, stores in the same segment are closer substitutes than others if there is less taste heterogeneity within segments than across segments (i.e., $\mu_1 \geq \mu_2$ in Anderson and de Palma, 1992). In this case, price competition should be more intense in markets with stores in the same segment, all else equal.[27]

I report the estimated effects for Equation 1.1 using the two price indices as outcome variables in Tables 1.4 and 1.5, across six different local market definitions. After taking the natural logarithm, changes in $\hat{P}^1_{it}$ can be interpreted as percentage changes in the price level. I include the natural logarithms of municipality population and median

products are measured—e.g., grams, liters, or package sizes.

[27]Pricing could also be affected by differences in costs related to the new format or ownership.

Table 1.4: Effect of rival rebrandings on effective store prices

	$\ln(\hat{P}^1_{it})$					
	(1)	(2)	(3)	(4)	(5)	(6)
T_{Extra}	-0.002	0.005	0.007*	-0.001	0.006	0.002
	(0.003)	(0.004)	(0.003)	(0.003)	(0.004)	(0.003)
T_{Prix}	0.016	0.006	0.001	0.016	-0.002	0.004
	(0.012)	(0.011)	(0.007)	(0.010)	(0.007)	(0.008)
T_{Bunnpris}	0.012	0.015	0.018	0.015	0.018	0.013
	(0.016)	(0.013)	(0.010)	(0.017)	(0.010)	(0.009)
T_{Closure}	0.016	0.015	0.019*	0.016	0.020*	0.005
	(0.011)	(0.010)	(0.010)	(0.010)	(0.010)	(0.008)
Market Definition	10–1.5 km	10–1 km	10–0.5 km	6–1.5 km	5–0.5 km	Postcode

Notes: Each column reports event-specific ATT estimates for a given local market definition, estimated following Sun and Abraham (2021). The data span all months of 2014–2016, and the standard errors shown in parentheses are clustered by municipality. Each ATT is retrieved from separate regressions reported fully in Tables A7–A10 in Appendix 1.B. Controls for municipality-level population and income are included in each regression. *** $p < 0.001$; ** $p < 0.01$; * $p < 0.05$.

Table 1.5: Effect of rival rebrandings on local product price deviations

	$\hat{P}^2_{it}$					
	(1)	(2)	(3)	(4)	(5)	(6)
T_{Extra}	0.000	-0.001	-0.001	0.000	-0.001	0.000
	(0.000)	(0.000)	(0.000)	(0.000)	(0.000)	(0.000)
T_{Prix}	0.001	0.000	0.001	0.001	0.001	0.000
	(0.001)	(0.001)	(0.001)	(0.001)	(0.001)	(0.001)
T_{Bunnpris}	0.002	0.002	0.001	0.002	0.002	0.000
	(0.001)	(0.001)	(0.001)	(0.001)	(0.001)	(0.001)
T_{Closure}	0.001	0.001	0.000	0.001	0.000	0.000
	(0.001)	(0.001)	(0.001)	(0.001)	(0.001)	(0.001)
Market Definition	10–1.5 km	10–1 km	10–0.5 km	6–1.5 km	5–0.5 km	Postcode

Notes: Each column reports event-specific ATT estimates for a given local market definition, estimated following Sun and Abraham (2021). The data span all months of 2014–2016, and the standard errors shown in parentheses are clustered by municipality. Each ATT is retrieved from separate regressions reported fully in Tables A11–A14 in Appendix 1.B. Controls for municipality-level population and income are included in each regression. *** $p < 0.001$; ** $p < 0.01$; * $p < 0.05$.

income as controls to account for potentially different demographic trends across treated and control stores.[28]

The results in Table 1.5 are consistent with the limited variation in $\hat{P}_{it}^2$: I do not detect any economically significant local adjustments in product prices in response to rival rebrandings. More interestingly, there is also no clear evidence of substantial changes in effective store prices. One specification shows a 0.7% price increase associated with a rival rebranding to the soft discounter Extra, statistically different from zero at the 5% level, but the other specifications yield smaller and even negative estimates, none of which are statistically significant. The absence of effects for Coop-owned Prix or the independent chain Bunnpris is not surprising, given that both formats are similar to ICA's Rimi chain. However, Bunnpris rebrandings are generally associated with higher effective prices in the range of 1.2–1.8%, perhaps because Bunnpris exerts weaker pressure on Kiwi to offer low-priced alternatives. A potential price increase following a Bunnpris rebranding is particularly notable since these rebrandings resulted from divestment remedies. When local market concentration is held constant, there is no evidence that the remedied rebrandings are more pro-competitive than alternatives, which suggests potential limitations of divestment remedies. Still, confidence in the Bunnpris estimates is limited by the small number of ICA stores that rebranded to Bunnpris, as reflected in the large standard errors. Rival closures are associated with effective price increases for Kiwi in the range of 0.5–2%, with two specifications producing estimates statistically different from zero at the 5% level. This finding is consistent with reduced competition following a rival ICA store closure, allowing Kiwi to profitably offer slightly fewer and pricier products, thereby increasing effective store prices, as in Ozhegova (2025).

The fact that rival rebrandings to Extra—a soft discounter like Kiwi—do not induce Kiwi to shift its assortment toward cheaper products is somewhat surprising in light of Ozhegova (2025), who finds that effective store prices are lower in more competitive markets.[29] This, along with the moderate effect of rival closures, could be explained by soft discounters like Kiwi allowing relatively little discretion for individual stores to adjust their assortments. In contrast, convenience stores and supermarkets included in the sample of Ozhegova (2025) may be permitted to cater more to their local customer bases.

[28]The estimators of Sun and Abraham (2021) and Callaway and Sant'Anna (2021) currently do not support multiple treatment variables in a single estimation procedure in either Stata or R. Hence, I run separate regressions for each treatment variable and exclude stores exposed to other treatments.

[29]Note that effective prices could also change without stores adjusting product prices or assortments if rival rebrandings lead consumers to change their basket composition, for example, by buying some products at the new store while still purchasing others at the old store. However, the scope for such asymmetric substitution is limited if consumers tend to do most of their shopping at the same store.

National Food Prices After the Acquisition

Overall, I find no evidence of local adjustments in product-level prices in response to the acquisition or subsequent rebrandings. The evidence on effective store prices is less clear, owing to both imprecise estimates and variation across model specifications, although effective prices consistently increase slightly following rival closures and Bunnpris rebrandings. Since product prices are essentially fixed in the cross-section, these changes must reflect shifts in basket composition.

While these findings suggest limited local price effects, an important remaining margin is national pricing: the acquisition may have affected prices at the national level, which would not be captured in the store-level analysis due to the inclusion of time fixed effects. Identifying national price effects is inherently difficult in quasi-experimental settings, as domestic comparison groups are also exposed to national shocks (see Allain et al., 2017; Aguzzoni et al., 2016). To examine this possibility, I use the Harmonized Index of Consumer Prices (HICP), drawing on monthly data for Norway and neighboring countries from 2005 to 2021 provided by Eurostat.[30] The HICP is based on a national basket of representative consumer products selected by national statistical institutes.

The HICP for Food and Non-Alcoholic Beverages (F&B; hereafter food prices) increased by approximately 5.5% from 2014 to 2017, whereas the All-Items HICP rose by about 8% over the same period. To measure the evolution of real food prices, I deflate food prices by overall prices and take logarithms: $\ln(P_t^{Food}) - \ln(P_t^{All})$. This yields a measure of real food prices that can be interpreted as the approximate percentage deviation of food prices from overall prices. Figure 1.7 plots the F&B and All-Items HICP, as well as real food prices in Norway from 2005 to 2021. There appears to be a negative shift in the trend of real food prices following ICA's exit from the market in 2016 (see Appendix 1.A.1 for a formal structural break analysis).

[30] https://ec.europa.eu/eurostat/web/hicp/data/database

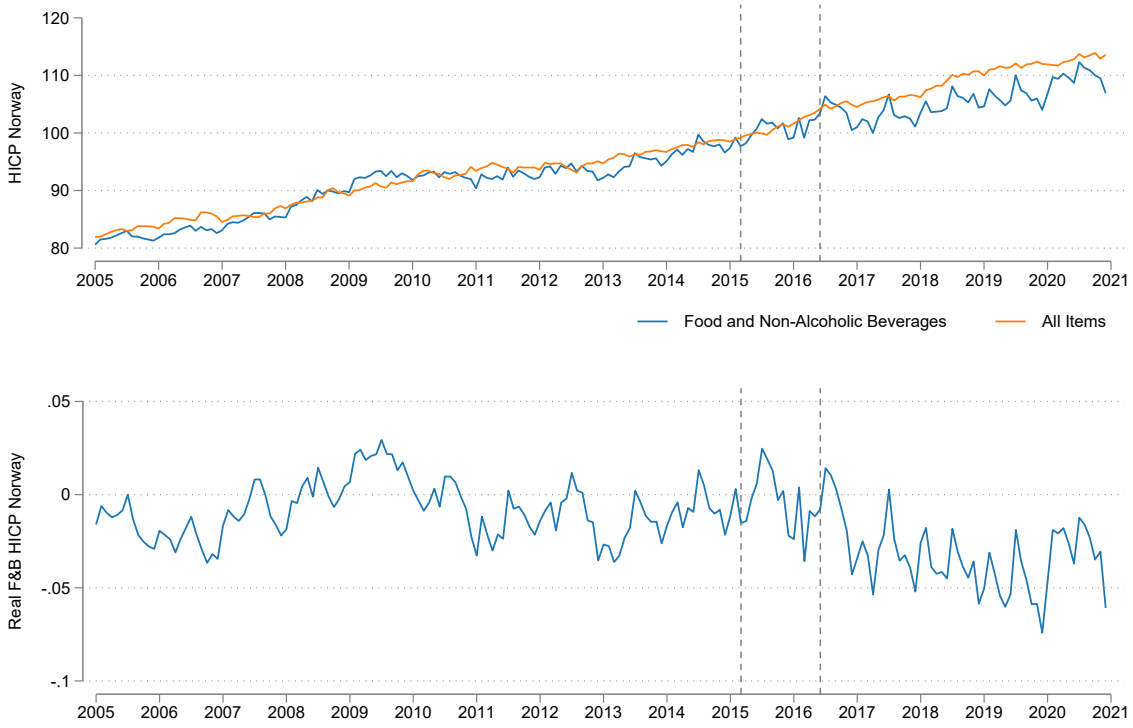


Figure 1.7: The top panel shows the F&B and All-Items HICP in Norway, with mean index values normalized to 2015 = 100. The bottom panel shows the real F&B HICP, calculated as $\ln(P_t^{Food}) - \ln(P_t^{All})$. The two vertical lines indicate the transition period, from just before the first store rebranding (March 2015) to just before the final rebranding (June 2016). *Source:* Eurostat.

Figure 1.8 illustrates the best linear fits of real food prices in Norway before and after the ICA acquisition, covering the period from 2011 to 2020. The transition period, during which only some ICA stores were rebranded, is excluded. The annualized trend growth rate of real food prices in Norway was 0.24% prior to the acquisition, changing to -1.19% thereafter. By the end of 2019, the trend line for real food prices was 5.85% lower than the trajectory predicted by the pre-acquisition trend.

The shift in real food prices following the acquisition could, of course, be driven fully or partially by changes in confounding variables. However, some of these underlying factors—such as weather, trade and transportation conditions, pandemics, wars, or other supply-chain disruptions—would also affect neighboring countries through global commodity prices (e.g., wheat or coffee beans). These explanations become less plausible if neighboring countries did not experience structural breaks comparable to those observed in Norway.

Figures A1 and A2 in Appendix 1.A.2 plot the F&B and All-Items HICP, as well as real food prices, for a set of candidate comparison countries. By inspection, Sweden, Denmark, Germany, and Iceland appear to follow common linear trends with Norway from the Great Recession until the acquisition. Figure 1.9 shows that real food prices in these countries, including Norway, track similar paths prior to the acquisition, and

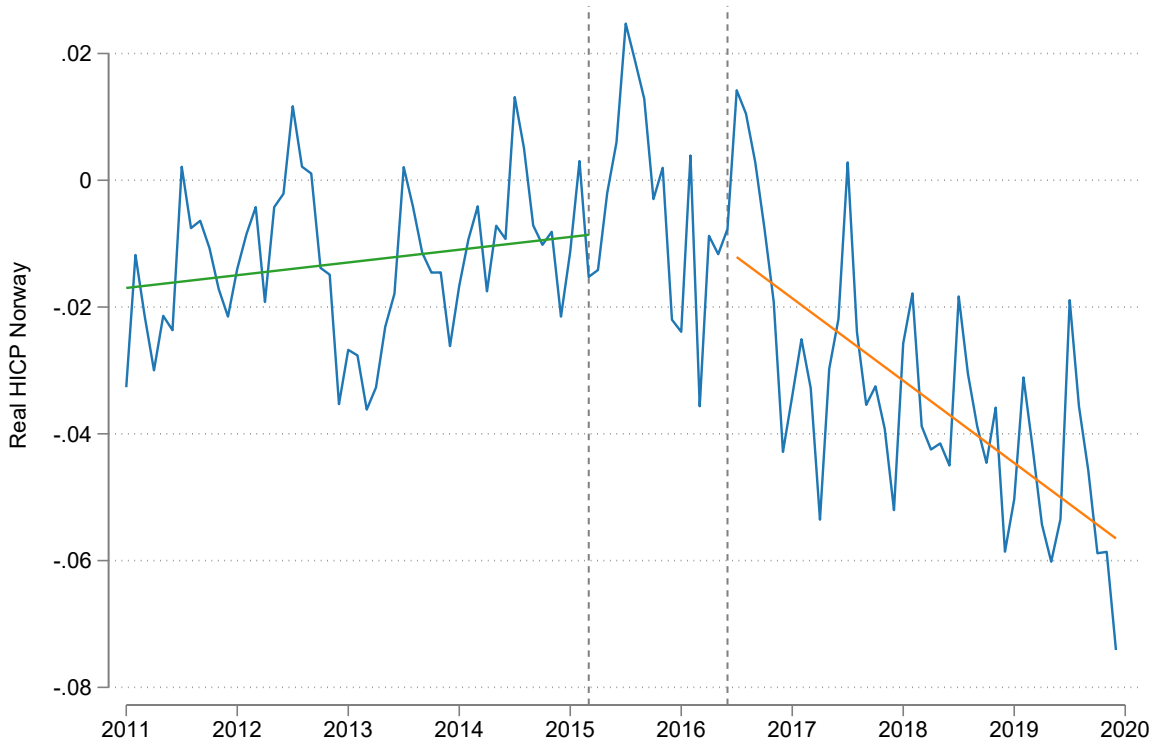


Figure 1.8: Best linear fit of real F&B HICP, calculated as $\ln(P_t^{Food}) - \ln(P_t^{All})$, for 2011–2019, before and after the acquisition. The transition period (April 2015–June 2016) is excluded. *Source:* Eurostat.

that Norway experiences a negative trend shift while the others remain on trend. A formal difference-in-differences analysis (Appendix 1.A.2), using Sweden, Denmark, Germany, and Iceland as comparison countries, indicates that real food prices fell by an average of 5.1% in Norway after the acquisition relative to those countries. Given the absence of comparable breaks in the comparison countries, purely common-shock explanations appear less plausible; accounting for Norway's break would require either a Norway-specific shock coinciding with the acquisition or heterogeneous pass-through of a common shock.

To relax the assumptions of common or linear trends, I employ the synthetic control method proposed by Abadie and Gardeazabal (2003). I extend the time series back to 2005 and construct a synthetic comparison country as a weighted average of the Nordic countries—Denmark, Finland, Iceland, and Sweden. The deseasonalized real food prices for Norway, the synthetic control, and their difference are shown in Figure A3 in Appendix 1.A.3. The synthetic comparison closely tracks Norway until the acquisition, after which a gap emerges as real food prices begin to decline in Norway. Consistent with the difference-in-differences analysis, Norwegian food prices are estimated to fall by an average of 5.2% relative to the synthetic control, suggesting a downward trend shift in Norwegian real food prices not observed in neighboring countries. For additional details and alternative specifications, see Appendix 1.A.3.

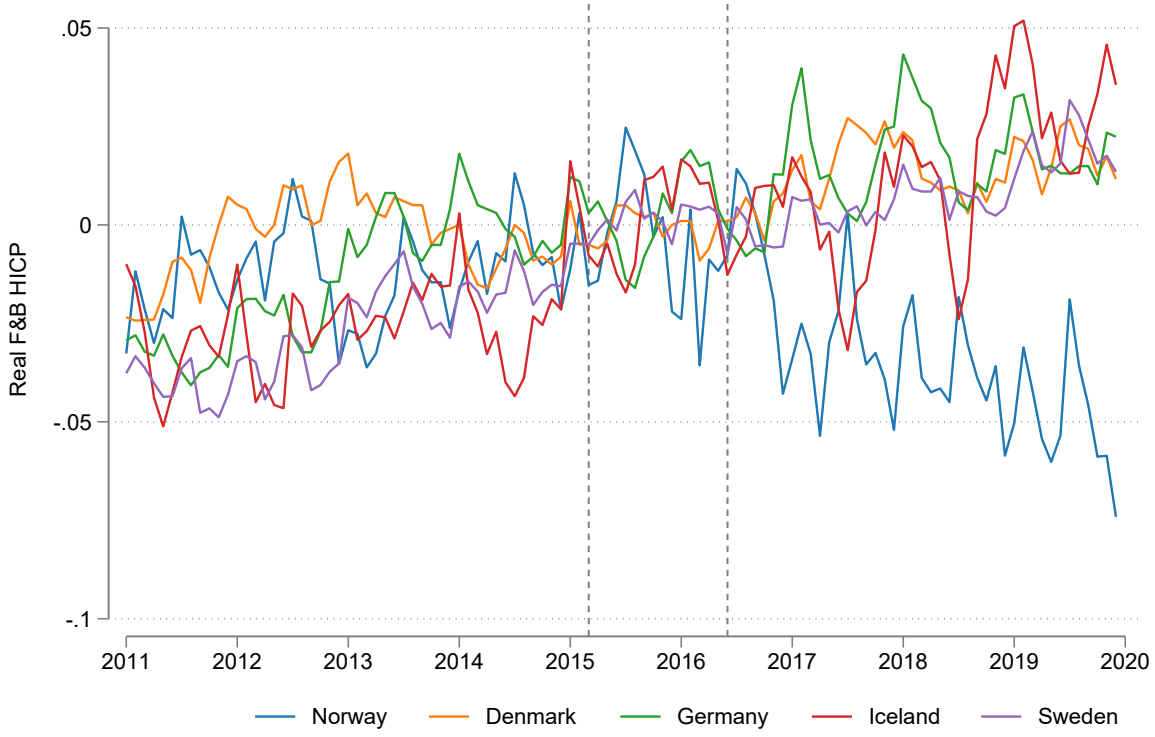


Figure 1.9: Real F&B HICP for Norway and comparison countries, calculated as $\ln(P_t^{Food}) - \ln(P_t^{All})$. The transition period (April 2015–June 2016) is indicated by the vertical dashed lines. *Source:* Eurostat.

Even if the acquisition caused the observed price decline, the underlying mechanism remains unclear. Mergers and acquisitions can alter factors such as bargaining power, ownership concentration, economies of scale, and product positioning. Yet the evidence indicates that the average market experienced a shift in store formats from convenience discount to soft discount without changes in local ownership, as shown in Figures 1.3 and 1.2 in Section 1.2.2. With the disappearance of ICA's convenience discounter Rimi and the large-scale rebranding of stores to Coop Extra, the market basket tilted toward the soft-discount segment. This compositional shift plausibly reduced average prices, since soft discounters like Extra were consistently cheaper than convenience discounters like Rimi.[31] At the same time, the growing presence of soft discounters increased the frequency of head-to-head competition within the segment (see Figure 1.4), presumably among closer substitutes. With uniform prices determined by competitive pressure across markets, the combination of a lower-price format mix and intensified rivalry aligns with the observed decline in national food prices.

Lower prices could also reflect increased bargaining power or economies of scale. ICA's exit removed one outside option for suppliers, while higher chain-level

[31] Rimi was about 5% more expensive than Extra in 2014 (see https://www.vg.no/spesial/matborsen/matborser/22/).

volumes may have generated efficiencies in distribution, administration, and marketing. However, the NCA did not deem the proposed efficiency gains large enough to offset adjustment costs or the loss of competition in the event of a complete takeover by Coop (Konkurransetilsynet, 2015, Clause 1733).[32]

1.5.2 Product Variety

Even if stores are unable to change prices in response to shifts in their competitive environment, they may adjust quality. In the grocery retail market, a key component of perceived quality is product variety and accessibility. For example, previous research on shocks to market structure—such as the entry of Walmart—finds significant effects on inventory shortfalls. Matsa (2011) argues that the increased risk of losing customers incentivized supermarkets to reduce stock shortfalls by about one-third. However, when considering assortment size more broadly, greater competitive pressure can also disincentivize the introduction of new products if fixed costs are involved. The business-stealing effect of a fiercer competitor may render a large product assortment unprofitable. These opposing mechanisms suggest that the effect of rival rebrandings on quality and assortment is uncertain and must be determined empirically.

To examine how stores adjust assortment sizes in response to rival rebrandings, we must address the measurement error that arises from not directly observing the full assortment, but rather the number of unique products purchased. In a given month, some offered products may not be purchased and thus remain unobserved. To mitigate this issue, I only count products that sold at least 100 units nationally during every period of the sample. This yields a measure that should change only if the store alters its assortment offering. The trade-off is that we do not capture changes in the assortment of low-volume products.

The percentage effects of a rival rebranding to Extra on assortment size are reported in Table 1.6. Across specifications, Kiwi appears to respond to rivals rebranding to Extra by reducing its product assortment by roughly 1%, although two of the estimates are not statistically significant at the 5% level. There is no clear evidence that rival rebrandings to Prix or Bunnpris led Kiwi to adjust its assortment size at the local store level. A rival closure, however, is associated with a slightly larger assortment size across specifications, but in only half of the cases is the effect statistically significant at the 5% level.

Counting the number of unique products offered provides an intuitive measure of product variety, but it does not account for the value to consumers of each individual product. Consider a store offering a large set of marginal products along with a few very popular ones and compare it to a store with a large offering of products with

[32]The specific details of those anticipated gains remain confidential.

Table 1.6: Effect of rival rebrandings on assortment size

	ln(Variety)					
	(1)	(2)	(3)	(4)	(5)	(6)
T_{Extra}	-0.009	-0.009*	-0.013***	-0.007	-0.011**	-0.007*
	(0.005)	(0.003)	(0.004)	(0.004)	(0.004)	(0.003)
T_{Prix}	-0.001	-0.007	-0.005	-0.002	-0.004	0.000
	(0.004)	(0.004)	(0.004)	(0.004)	(0.003)	(0.004)
T_{Bunnpris}	0.002	-0.003	-0.003	-0.002	-0.001	0.001
	(0.005)	(0.003)	(0.005)	(0.005)	(0.005)	(0.005)
T_{Closure}	0.006*	0.006	0.007	0.007*	0.004	0.004*
	(0.003)	(0.004)	(0.005)	(0.003)	(0.004)	(0.002)
Market Definition	10–1.5 km	10–1 km	10–0.5 km	6–1.5 km	5–0.5 km	Postcode

Notes: Each column reports event-specific ATT estimates for a given local market definition, estimated following Sun and Abraham (2021). The data span all months of 2014–2016, and the standard errors shown in parentheses are clustered by municipality. Each ATT is retrieved from separate regressions reported fully in Tables A15–A18 in Appendix 1.B. Controls for municipality-level population and income are included in each regression. *** $p < 0.001$; ** $p < 0.01$; * $p < 0.05$.

similar sales shares. The latter kind of variety is more valuable to consumers, as revealed by the expenditure shares. Alexander (1997) proposes measuring product variety as entropy (Shannon, 1948):

$$E_{it} = -\sum_{j=1}^{J_{it}} w_{ijt} \ln(w_{ijt})$$

where w_{ijt} is the sales share of product j, and J_{it} is the assortment size of store i at time t. This measure increases with J_{it} and reaches its maximum for a given J_{it}, $E_{it} = \ln(J_{it})$, when $w_{ijt} = 1/J_{it}$. If a single product accounts for all sales, i.e., $w_{1it} = 1$ and $w_{ijt} = 0$ for $2 \leq j \leq J_{it}$, then $E_{it} = 0$. For asymmetric but strictly positive sales shares across multiple products, $0 \leq E_{it} \leq \ln(J_{it})$. Unlike the simple count of unique products, the entropy measure is unaffected by unobserved products that are offered but not sold, since they contribute nothing to the sum.

The estimated effects of rival rebrandings on variety entropy are shown in Table 1.7. All specifications find a decrease in the entropy measure between 0.6% and 2% following a rival rebranding to Extra, but only three specifications are statistically significant at the 5% level. For rivals rebranding to Prix, there is no clear evidence of an effect across specifications, but it can be noted that all estimates are slightly positive. For Bunnpris, five estimates are small and negative, while one is positive and statistically significant at the 0.1% level. There is no statistically significant effect of rivals closing down.

The most robust finding in Tables 1.6 and 1.7 is that product variety at Kiwi declines modestly following rival rebrandings to Extra. One explanation is that

Table 1.7: Effect of rival rebrandings on variety entropy

	E_{it}					
	(1)	(2)	(3)	(4)	(5)	(6)
T_{Extra}	-0.007	-0.006	-0.016***	-0.007	-0.020***	-0.010**
	(0.005)	(0.005)	(0.005)	(0.005)	(0.005)	(0.004)
T_{Prix}	0.003	0.004	0.005	0.003	0.008	0.005
	(0.007)	(0.006)	(0.004)	(0.007)	(0.005)	(0.007)
T_{Bunnpris}	0.020***	-0.002	-0.007	-0.002	-0.009	-0.005
	(0.006)	(0.009)	(0.009)	(0.011)	(0.010)	(0.008)
T_{Closure}	0.002	0.003	0.001	0.002	-0.002	-0.007
	(0.006)	(0.007)	(0.007)	(0.007)	(0.005)	(0.005)
Market Definition	10–1.5 km	10–1 km	10–0.5 km	6–1.5 km	5–0.5 km	Postcode

Notes: Each column reports event-specific ATT estimates for a given local market definition, estimated following Sun and Abraham (2021). The data span all months of 2014–2016, and the standard errors shown in parentheses are clustered by municipality. Each ATT is retrieved from separate regressions reported fully in Tables A19–A22 in Appendix 1.B. Controls for municipality-level population and income are included in each regression. *** $p < 0.001$; ** $p < 0.01$; * $p < 0.05$.

increased competitive pressure renders some products locally unprofitable. An alternative, demand-driven explanation is that consumers who switch from Kiwi to Extra previously purchased niche items, reducing sales of those products or leaving them unsold. Both explanations are consistent with the finding below in Section 1.5.3 that Extra captures more revenue from Kiwi than other formats. At the same time, Table A5 in Appendix 1.B shows that both assortment size and entropy-based variety increase over time at the national level. Taken together, these patterns suggest that while competitive pressure increased and modestly reduced variety locally, the primary response occurred at the chain level, where overall assortment variety expanded.

1.5.3 Sales

Having established that (i) product-level prices are uniform across stores within chains and do not adjust locally, and (ii) local assortment responses are limited, I interpret changes in store-level sales following nearby rebrandings or closures primarily as shifts in demand. In this setting, lower sales at Kiwi after a rival's transformation reflect stronger demand diversion to that rival—that is, greater competitive pressure—rather than price adjustments.

I estimate Equation 1.1 using the natural logarithm of monthly sales as the outcome, and report results across multiple local market definitions in Table 1.8. As in the price and assortment analyses, I include demographic controls to absorb differential local trends.

Rebrandings of rivals to the soft discounter Extra are consistently associated with 7.0–12.9% declines in Kiwi's sales across market definitions, indicating that

Table 1.8: Effect of rival rebrandings on sales

	ln(Sales)					
	(1)	(2)	(3)	(4)	(5)	(6)
T_{Extra}	-0.093***	-0.101***	-0.126***	-0.070***	-0.129***	-0.117***
	(0.016)	(0.018)	(0.014)	(0.015)	(0.015)	(0.021)
T_{Prix}	-0.010	-0.008	-0.015	-0.025	-0.025	-0.033*
	(0.017)	(0.018)	(0.022)	(0.018)	(0.021)	(0.018)
T_{Bunnpris}	0.081***	0.023	0.025	0.037	-0.001	-0.045
	(0.030)	(0.035)	(0.044)	(0.032)	(0.053)	(0.039)
T_{Closure}	0.078***	0.083***	0.076***	0.082***	0.073***	0.024
	(0.019)	(0.023)	(0.021)	(0.022)	(0.025)	(0.019)
Market Definition	10–1.5 km	10–1 km	10–0.5 km	6–1.5 km	5–0.5 km	Postcode

Notes: Each column reports event-specific ATT estimates for a given local market definition, estimated following Sun and Abraham (2021). The data span all months of 2014–2016, and the standard errors shown in parentheses are clustered by municipality. Each ATT is retrieved from separate regressions reported fully in Tables A23–A26 in Appendix 1.B. Controls for municipality-level population and income are included in each regression. *** $p < 0.001$; ** $p < 0.01$; * $p < 0.05$.

Extra imposed stronger competitive pressure on Kiwi than the predecessor formats. Rebrandings to convenience-discount formats (Coop's Prix or Bunnpris) yield smaller effects that are generally statistically indistinguishable from zero. By contrast, rival store closures increase Kiwi's sales by 2.4–8.3%, consistent with reduced competitive pressure. Overall, these results suggest that convenience-discount stores such as Rimi and Prix were weaker competitors to Kiwi than Extra, though they still exerted some pressure. Given that 129 stores rebranded to Extra while only 46 closed, the net effect of rebrandings was an increase in competitive pressure on Kiwi.

The sales results have implications for consumer welfare. Based on revealed preference, consumers who switched away from Kiwi following rival rebrandings were plausibly better off, while those who switched to Kiwi following a rival rebranding or closure plausibly lost access to a preferred alternative. This interpretation is supported by uniform prices and limited assortment changes at Kiwi, as demand responses to national price or assortment changes are differenced out by the responses at stores that did not experience a rival rebranding. The estimated sales response thus approximates the average demand response of a consumer whose choice set still includes the unchanged alternative of Kiwi.[33] That said, the minor reduction in assortment variety at Kiwi following Extra rebrandings could theoretically contribute

[33]The ability to infer welfare changes for both groups of consumers (those switching away from, and those switching to, Kiwi) relies on the characteristics of Kiwi being fixed in the cross-section of stores. However, if Kiwi had become more attractive, say by lowering prices locally, we could still infer that the switchers-away benefited, but the welfare impact on switchers-to would be ambiguous, as the price decrease might compensate for the loss of their preferred alternative. Similarly, increased prices would mean that switchers-to surely suffered, while switchers-away might have chosen to remain absent the price increase.

to the demand diversion, although this adjustment may itself be a consequence of the reduced demand rendering niche items unprofitable or unsold.

The observed demand responses at Kiwi capture only part of the overall welfare implications of the rebrandings, which themselves represent just one component of the broader effects of the acquisition. I make no claim about the total welfare effects of either the acquisition or the rebrandings based solely on the sales results. Nevertheless, the findings highlight a potential synergistic effect of the acquisition that could benefit consumers—namely, the combination of a preferred store-format concept on the acquirer's side and scarce store locations on the acquiree's side. The acquisition may thus generate consumer gains when the acquiree cannot readily replicate the preferred format and the acquirer cannot readily obtain comparable locations elsewhere.

1.6 Identification and Robustness Checks

1.6.1 Dynamic Event Study Design

The causal interpretation of the store-level regression results rests on two identifying assumptions: parallel trends and no anticipation. To evaluate their validity, I estimate a variant of Equation 1.1 in which treatment effects vary nonparametrically over time—a dynamic event-study design. This approach serves two purposes. First, it tests for systematic differences in pre-treatment trends between treated and control stores. Second, it detects potential anticipatory effects in the periods preceding rival rebrandings. The dynamic specification is:

$$Y_{it} = \gamma_i + \lambda_t + \sum_{l \neq \{-1,-\infty\}} \mu_l \mathbb{1}\{t - E_i = l\} + \delta' \boldsymbol{Z_{it}} + \epsilon_{it} \tag{1.2}$$

where $\mu_l = E[\mu_{itl}] = ATT_l$ and $\mathbb{1}\{\cdot\}$ denotes an indicator function. E_i is the rebranding month in store i's market, and $t - E_i = l$ measures the number of periods since the rebranding. The coefficients of interest are the average treatment effects μ_l at relative time l after treatment. To avoid perfect collinearity among the relative-time indicators, one period must be omitted as the reference; following convention, I exclude $l = -1$. Markets in the untreated group correspond to $l = -\infty$. Under the parallel-trends and no-anticipation assumptions, trajectories should be indistinguishable across treated and untreated stores for all $l < -1$, implying $\mu_l = 0$.

This chapter reports a large number of regressions. Event-study estimates for each store-level regression are presented in Figures A9–A28 in Appendix 1.B. For consistency and readability, all figures display a common event window of $l \in [-18, +10]$ months around the event. Longer horizons are omitted to ensure that

most cohorts contribute to each plotted coefficient, since extreme leads and lags are supported by few observations, producing wide confidence intervals and potentially unrepresentative point estimates.

Overall, the estimates support the validity of the parallel-trends and no-anticipation assumptions. In some cases, the linear pre-trends differ slightly or the reference period $l = -1$ appears as an outlier among the pre-periods, but these deviations are small and cannot plausibly rationalize the qualitative implications of the post-event coefficients. Because the data are monthly and somewhat noisy, the event-study plots may appear erratic; in this setting, the overall pattern of the pre-event coefficients is more informative than whether individual leads happen to differ from zero. In the few cases where visible deviations occur, the main conclusions are likely robust to controlling for linear pre-trends or shifting the reference period. This is particularly true for price and variety outcomes, where the estimated effects are small.[34]

Event-study plots: ln(Sales), 10–1 km market definition

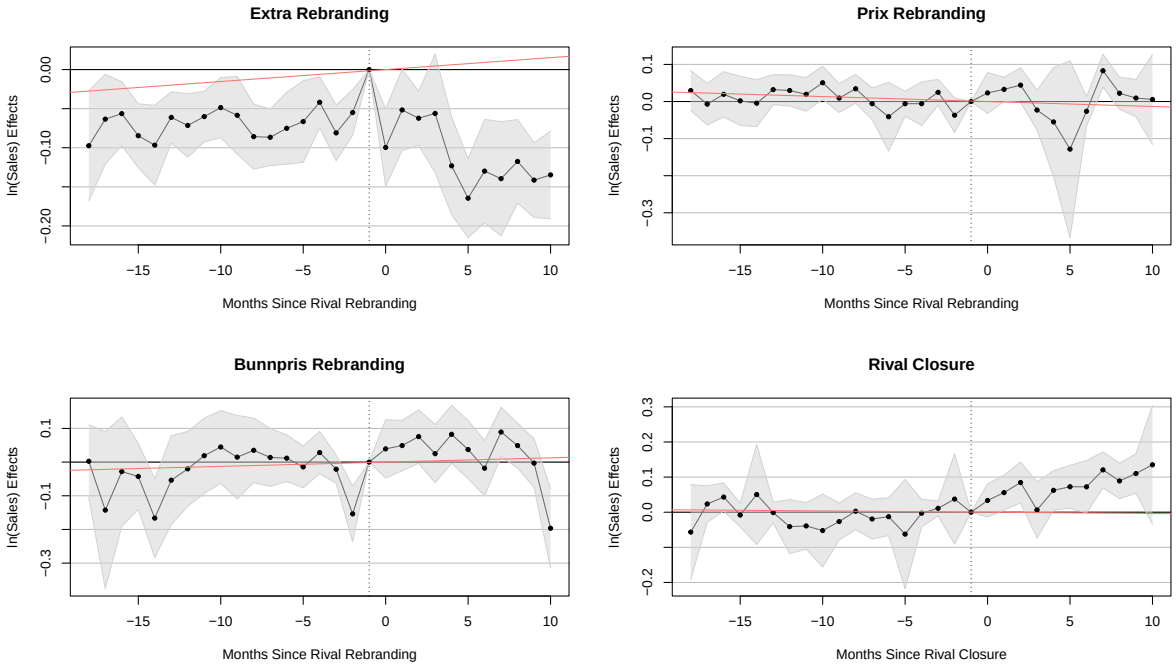


Figure 1.10: Estimated percentage sales effects of rival rebrandings and closures over time. The horizontal axis displays months relative to the event, $l \in [-18, +10]$, with the vertical line marking $l = -1$—the last month before the event. Point estimates are shown with 95% confidence intervals, based on standard errors clustered at the municipality level. A fitted linear pre-trend is displayed and extrapolated into the post-event period for comparison.

The estimated effects on sales are substantially larger than those for price and assortment outcomes, so I examine the corresponding event-study plots in greater

[34]Potential exceptions include effective store prices after closures (likely attenuating toward zero), variety after Extra rebrandings (somewhat reduced), variety after Bunnpris rebrandings (possibly turning slightly negative), and variety entropy after Bunnpris rebrandings (where the positive effect likely vanishes).

detail. Figure 1.10 reports event-study estimates from the sales specification under the 10–1 km market definition, which is broadly representative of the other definitions, separately by rival rebranding and closure type. Sales decline noticeably following a rival rebranding to Extra relative to the control group. However, sales are unusually high in the reference period and exhibit a mild positive pre-trend, which could bias the estimated effect. Sales following Prix and Bunnpris rebrandings appear broadly similar before and after the rebranding, although there are some outliers—particularly for Bunnpris, which had relatively few rebrandings in the sample. Finally, sales are similar across groups prior to rival closures, after which treated Kiwi stores begin to experience higher sales.

1.6.2 Adjusting for Anticipation and Linear Pre-trends

In this section, I address potential violations of the no-anticipation and parallel-trends assumptions. The former may arise in this context if stores experience an adjustment period during rebranding and thus do not operate normally. To mitigate this concern, I exclude the final month before the event and use the penultimate month as the reference period when estimating Equations 1.1 and 1.2, such that neither the pre- nor post-event periods are measured relative to a potential transition month immediately preceding treatment. The ATT estimates are reported in Table 1.9, and Figure 1.11 presents the corresponding event-study plots for the 10–1 km specification. The complete set of results is shown in Figures A29–A32 in Appendix 1.B.

Table 1.9: Effect of rival rebrandings on sales: last period before event excluded

	ln(Sales)					
	(1)	(2)	(3)	(4)	(5)	(6)
T_{Extra}	-0.072***	-0.051***	-0.096***	-0.053***	-0.091***	-0.094***
	(0.017)	(0.018)	(0.015)	(0.018)	(0.013)	(0.022)
T_{Prix}	0.034	0.019	-0.030	0.007	-0.045	-0.065**
	(0.031)	(0.026)	(0.026)	(0.033)	(0.036)	(0.030)
T_{Bunnpris}	0.255***	0.176***	0.132***	0.223***	0.136***	0.019
	(0.061)	(0.040)	(0.032)	(0.063)	(0.041)	(0.056)
T_{Closure}	0.044	0.046	0.056	0.044	0.051	-0.061
	(0.064)	(0.079)	(0.061)	(0.071)	(0.078)	(0.068)
Market Definition	10–1.5 km	10–1 km	10–0.5 km	6–1.5 km	5–0.5 km	Postcode

Notes: Each column reports event-specific ATT estimates for a given local market definition, estimated following Sun and Abraham (2021). The data span all months of 2014–2016 except the last period before the event, and the standard errors shown in parentheses are clustered by municipality. Each ATT is retrieved from separate regressions reported fully in Tables A27–A30 in Appendix 1.B. Controls for municipality-level population and income are included in each regression. *** $p < 0.001$; ** $p < 0.01$; * $p < 0.05$.

Event-study plots: ln(Sales), 10–1 km market definition, last period before event excluded

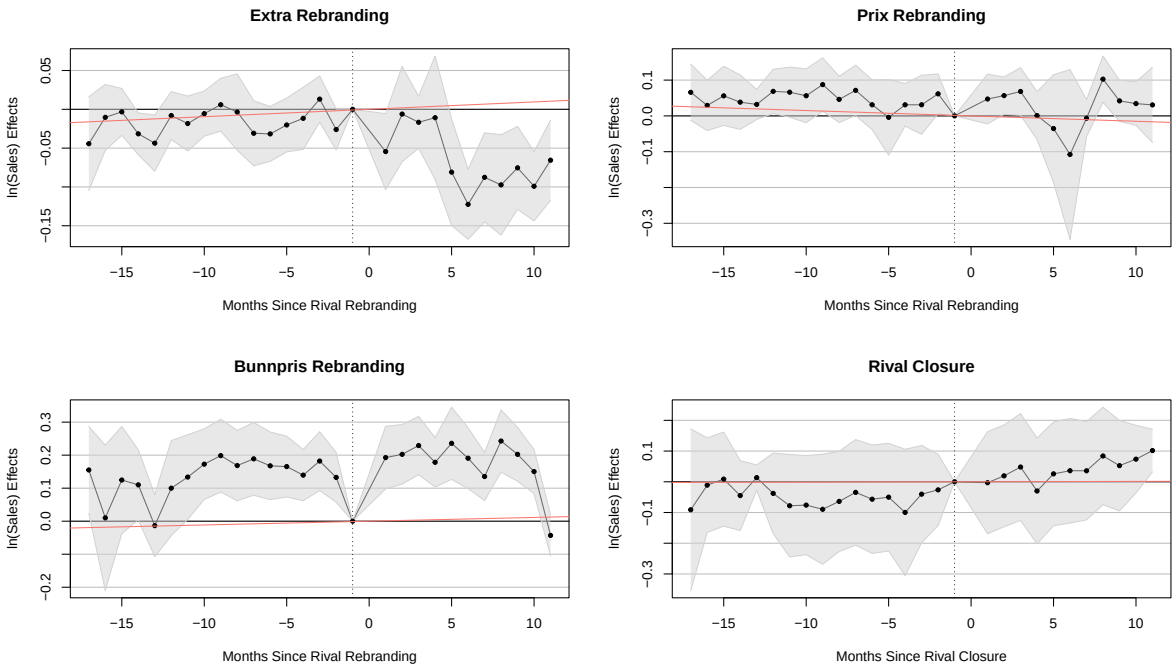


Figure 1.11: Estimated percentage sales effects of rival rebrandings and closures over time, excluding the last period before the event. The horizontal axis displays months relative to the excluded month, $l \in [-17, +11]$, with the vertical line marking $l = -1$—two months before the event. Point estimates are shown with 95% confidence intervals, based on standard errors clustered at the municipality level. A fitted linear pre-trend is displayed and extrapolated into the post-event period for comparison.

The effect of a rival rebranding to Extra now ranges between -5.1% and -9.6% across specifications. As expected, given the high sales immediately before the event, these estimates are smaller than in the unadjusted specification but remain both economically and statistically significant. The ATTs for Prix change little. The estimates for Bunnpris increase substantially due to a negative outlier two periods before the event, while the estimated effect of rival closure becomes smaller and statistically insignificant because of a higher reference point. The S&A estimator is clearly sensitive to the choice of reference point, which the next robustness check addresses.

The parallel-trends assumption is not directly testable, but a long series of pre-event observations allows me to assess whether pre-trends could plausibly account for the observed post-event dynamics. Moreover, by imposing parametric restrictions on the pre-trends, one can extrapolate them into the post-period to estimate the ATTs under the assumption of common deviations from trends (Angrist and Pischke, 2015). Including linear trends relaxes the identifying assumptions: if trends are truly parallel, this specification remains unbiased, while if there are approximately linear differences in trends, the resulting bias could be reduced. Accordingly, I estimate an augmented

version of Equation 1.2:

$$Y_{it} = \gamma_i + \lambda_t + \text{Treated} \cdot \text{Trend} + \sum_{l \geq 0} \mu_l \mathbb{1}\{t - E_i = l\} + \delta' \boldsymbol{Z_{it}} + \epsilon_{it} \tag{1.3}$$

This specification includes a treated-store-specific linear time trend and excludes lead indicators. The linear pre-trend is estimated using all pre-event observations and then extrapolated into the post-period. Consequently, post-event coefficients capture deviations from this extrapolated counterfactual rather than from the final pre-period. This approach reduces sensitivity to idiosyncratic noise in $l = -1$ (or any pre-period) and allows for linear differences in pre-trends between treated and control stores.

Table 1.10: Effect of rival rebrandings on sales: group-specific linear trends and the last period before event excluded

	ln(Sales)					
	(1)	(2)	(3)	(4)	(5)	(6)
Trend – Extra	0.001	0.001	0.002	0.000	0.001	0.001
	(0.001)	(0.001)	(0.001)	(0.001)	(0.001)	(0.001)
T_{Extra}	-0.069**	-0.055*	-0.108***	-0.044	-0.081***	-0.089***
	(0.029)	(0.030)	(0.022)	(0.032)	(0.023)	(0.030)
Trend – Prix	-0.002	-0.002	-0.001	-0.001	-0.002	0.001
	(0.001)	(0.002)	(0.002)	(0.001)	(0.002)	(0.001)
T_{Prix}	-0.003	-0.009	0.015	-0.012	0.028	-0.022
	(0.029)	(0.032)	(0.036)	(0.032)	(0.038)	(0.026)
Trend – Bunnpris	0.002	0.001	0.005	0.005	0.007	0.005
	(0.004)	(0.003)	(0.003)	(0.006)	(0.005)	(0.004)
T_{Bunnpris}	0.058*	0.026	-0.015	-0.016	-0.049	-0.062
	(0.032)	(0.036)	(0.035)	(0.047)	(0.049)	(0.040)
Trend – Closure	0.000	0.000	0.001	-0.001	-0.001	0.003
	(0.002)	(0.002)	(0.002)	(0.002)	(0.004)	(0.002)
T_{Closure}	0.073*	0.091**	0.083*	0.092**	0.111	0.037
	(0.037)	(0.041)	(0.047)	(0.038)	(0.075)	(0.043)
Market Definition	10–1.5 km	10–1 km	10–0.5 km	6–1.5 km	5–0.5 km	Postcode

Notes: Each column reports event-specific ATT estimates and trend coefficients from Equation 1.3 for a given local market definition. The data span all months of 2014–2016 except the last period before the event, and the standard errors shown in parentheses are clustered by municipality. Each ATT and trend is retrieved from separate regressions reported fully in Tables A31–A34 in Appendix 1.B. Controls for municipality-level population and income are included in each regression. *** $p < 0.001$; ** $p < 0.01$; * $p < 0.05$.

The ATTs and treatment-group-specific trends estimated from Equation 1.3, with one pre-period excluded, are reported in Table 1.10. None of the treated groups exhibit statistically significant trends; however, some estimates are large enough in magnitude to meaningfully alter the baseline outcome over time, as illustrated in the

event-study plots. The ATT estimates for Extra rebrandings are consistently negative, ranging from -4.4% to -10.8% across specifications, indicating a somewhat smaller magnitude than in the unadjusted specification (Table 1.8). The ATT estimates for Prix remain close to zero, and while the estimates for Bunnpris are more sensitive to specification, they also remain close to zero on average. The estimates for rival closures remain qualitatively similar to the unadjusted specification, though slightly larger on average. Overall, the main conclusions of Section 1.5 remain robust to the inclusion of treatment-group-specific trends and the exclusion of the final pre-event period.

1.6.3 Heterogeneous Effects across Treatment Arms

Even when the conditions for identifying average treatment effects *on the treated* (parallel trends and no anticipation) are satisfied, selection into treatment arms may still be a concern. Stores selected for closure, for example, likely underperformed and may have exerted less competitive pressure on Kiwi than a typical rival. In that case, a positive ATT on sales from closures should be interpreted as a lower bound on the average competitive effect (ATE) of a representative rival. Similarly, the effects observed from conversions to Extra, Prix, or Bunnpris may not generalize. Counterfactual conversions could have produced different effects from those observed in markets where such conversions actually took place. Coop likely chose the target format to match local conditions and store characteristics to maximize profits. This suggests that the observed sales effects may be stronger in the markets where rebrandings occurred than they would be in other treated markets or in the population at large.

One such store characteristic is size, as illustrated in Figure 1.12, which shows the size distributions of Coop's formats Prix, Extra, and Mega. Extra stores tend to be larger than Prix, although there is substantial overlap.[35] If formats were chosen to fit existing facilities, ICA stores rebranded to Extra would on average be larger than those rebranded to Prix. If the stores that became Prix had instead been rebranded to Extra, they would likely not have generated as strong a business-stealing effect as the ATT estimated here, because the Extra format is less suited to smaller premises. Thus, differences in the ATT for Extra and Prix may partly reflect differences in the store sizes of rebranded rivals.

Taken together, these examples highlight that comparing ATTs across treatment arms requires caution. Even if the ATT can be identified separately for each treatment arm, differences across ATTs are not straightforward to interpret. To regard such differences as evidence of variation in the average effect of different conversion types—whether among all treated groups or in the population at large—one

[35] Bunnpris resembles Prix more closely than Extra in this regard.

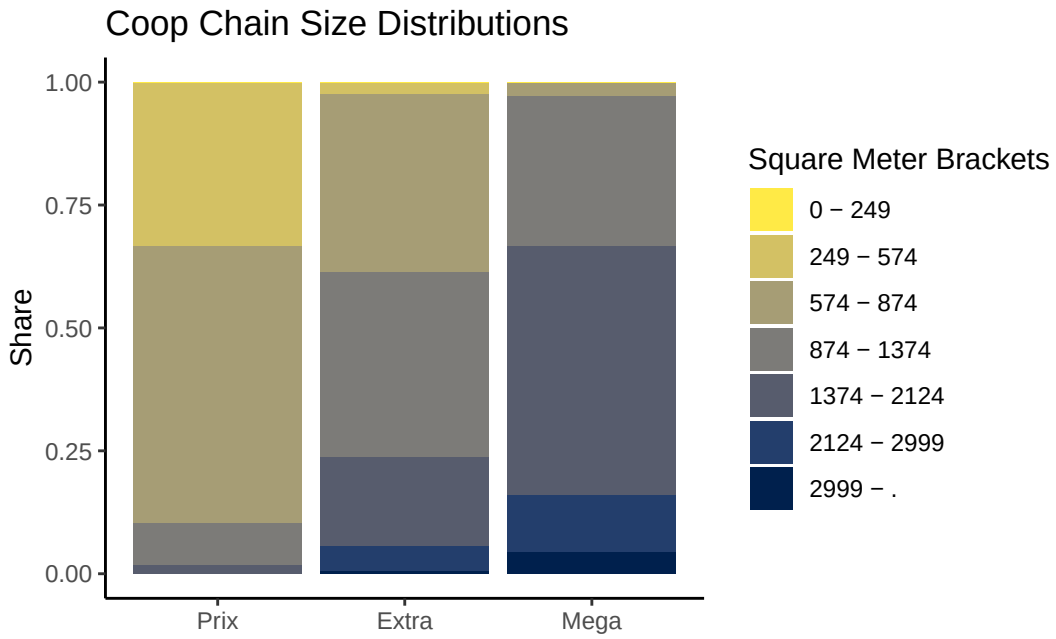


Figure 1.12: Share of stores within each size bracket for Coop's target formats in 2019. While distributions overlap, Coop's Extra stores are routinely larger than their Prix counterparts.

must at minimum assume that treatment effects are homogeneous across treatment groups, i.e., that all treated groups would experience the same effect if exposed to a given conversion. Without this assumption, differences in ATTs may instead reflect heterogeneity in how the same conversion affects distinct groups of treated stores (Callaway, Goodman-Bacon, and Sant'Anna, 2025). The cases discussed illustrate why such heterogeneity may arise, but they are not exhaustive.

1.7 Discussion and Conclusion

This chapter provides an ex post evaluation of a major horizontal merger in Norway's concentrated grocery market: Coop's 2015 acquisition of ICA. While the merger increased national ownership concentration, it simultaneously reshaped local competition through extensive store-format repositioning. The evidence is consistent with consumer gains from the reallocation of complementary, firm-specific assets—ICA's scarce store locations and Coop's soft-discount concept—which facilitated extensive format conversions. Overall, the results underscore the role of firm heterogeneity: when complementary, hard-to-replicate assets are combined, mergers can strengthen both the ability and the incentive to compete, intensifying rivalry even as concentration rises.

The analysis yields several findings consistent with consumer gains despite higher concentration. First, although national ownership concentration rose substantially, the

effect on the sales-weighted local HHI was limited, in part due to divestiture remedies imposed by the Norwegian Competition Authority. Second, I find no evidence of local product-level price responses to rival store conversions, consistent with uniform national pricing in grocery retail (Meile, 2020). Effective store prices and assortment variety exhibit only modest adjustments, consistent with limited local discretion over these margins. Third, despite the absence of local price reactions, national real food prices declined by 3.8–5.2% relative to a synthetic control constructed from neighboring countries. This decline is consistent with both a composition effect—replacing higher-priced formats with lower-priced soft discounters—and intensified competition as the soft-discount segment became more crowded. Fourth, store-level demand estimates show that rival conversions to Coop's Extra format reduced sales for NorgesGruppen's soft-discount incumbent, Kiwi, by 4.4–12.9% across specifications, while conversions to convenience-discount formats generally had no significant effect. Rival store closures, by contrast, increased sales by 2.4–11.1%. Taken together, these results suggest stronger competitive pressure despite rising concentration.

The fall in national food prices and the strong consumer response to Extra rebrandings are consistent with consumer gains from merger-related repositioning, especially where ICA stores were converted to Extra. The central mechanism appears to be the large-scale shift of acquired locations into the soft-discount segment, an outcome that may have been more difficult to achieve through stand-alone expansion.

As discussed in Section 1.2.3, the Norwegian Competition Authority's ex ante assessment emphasized ownership concentration and efficiency claims but did not explicitly consider potential pro-competitive effects arising from store-format conversions. The ex post evidence presented here suggests that mergers combining firm-specific complementarities may generate pro-competitive effects that merit greater attention in merger control. Such effects arise because firms differ in formats, locations, and capabilities. Mergers can realize complementarities among these firm-specific assets that would otherwise require costly investment, facilitating repositioning that benefits consumers directly or strengthening competitive incentives when products become closer substitutes.[36]

The net benefits of divestiture remedies may likewise depend on asymmetries across firms. Divesting stores to Bunnpris mitigated increases in ownership concentration but may also have reduced consumer benefits, given Bunnpris' higher prices, its weaker ability to compete in the large soft-discount segment, and indications that effective store prices at Kiwi tend to rise following Bunnpris rebrandings. For merger reviewers, these findings imply that remedies should be evaluated not

[36]More generally, product repositioning can also harm consumers, either directly through less preferred characteristics or indirectly through its impact on competitive incentives.

only in terms of concentration and scale efficiencies but also in light of whether potential buyers possess assets that are complementary to those being divested. When prospective buyers differ substantially along these dimensions, divestitures may fail to preserve competitive pressure.

While the findings are robust across specifications, several limitations qualify their interpretation. Identification of national price effects relies on comparisons with neighboring countries and cannot fully rule out Norway-specific shocks. The store-level analysis isolates the response of one major rival and may not capture the full industry adjustment, although it is plausible that the other established soft-discount chain was similarly affected by the store conversions. More broadly, causal inference remains challenging because mergers affect markets through multiple channels—ownership concentration, bargaining power, efficiencies, and product positioning—and uniform national pricing complicates the construction of local counterfactuals. Furthermore, the mechanisms highlighted here are most relevant when merging firms control assets that are costly to replicate; where similar complementarities can instead be achieved through contracts, entry, or internal investment, consolidation is unlikely to yield comparable gains.

Future research could extend this analysis in several directions. First, richer industry-wide data would allow tracking substitution patterns and competitive responses across all chains, not only the soft-discount incumbent studied here. Second, alternative quasi-experimental designs could further refine control groups or exploit additional sources of identifying variation to better estimate aggregate effects or isolate specific channels. Finally, the results on Bunnpris rebrandings suggest that the competitive effects of divestiture remedies may depend critically on the identity and characteristics of the buyer. More empirical work is needed to understand how the trade-off between reducing ownership concentration and disrupting potentially efficient asset combinations shapes remedy outcomes.

In sum, mergers can reallocate complementary, firm-specific assets in ways that materially reshape competition and consumer welfare. In this case, the merger enabled rapid repositioning that is consistent with consumer gains despite increased national concentration. While divestiture remedies aim to restore competition, they may fail to replicate the specific complementarities realized by the original buyer. Merger control should therefore weigh the potential gains from reduced concentration against the loss of firm-specific complementarities when designing remedies.

Appendix

1.A Analysis of National Real Food Prices

1.A.1 Structural Break Analysis

I test for a structural break in the evolution of real food prices around the time of the acquisition by estimating the following equation:[37]

$$\ln\left(\frac{P_t^{Food}}{P_t^{All}}\right) = \beta_0 + \beta_1 t + \beta_2 \cdot \text{Post Acquisition} + \beta_3 t \cdot \text{Post Acquisition} + \beta_4 \cdot \text{Month} + \epsilon_t \tag{1.4}$$

where *Post Acquisition* is an indicator for the period after the acquisition (April 2015), and *Month* is a vector of monthly dummies. The time trend t is normalized to represent the number of months after the acquisition, so that β_2 captures the shift in real food prices at the time of the acquisition. The sample is restricted to 2011–2019 to avoid confounding effects from the Great Recession and the COVID-19 pandemic, which could each constitute separate structural breaks. It is also important to note that the ICA takeover occurred gradually between April 2015 and July 2016, during which only a subset of ICA stores was rebranded.

Specification (1) in Table A1 excludes the transition period from April 2015 to June 2016, during which a stable trend cannot be expected. Specification (2) replaces *Post Acquisition* with *Rebranding Share*, which measures the cumulative share of store rebrandings completed (as shown in Figure 1.5) and increases from 0 to 1 over the transition period. This allows inclusion of the full sample period and captures the change in real food prices associated with the complete set of 385 format rebrandings, under the assumption of a constant marginal effect.

For specification (1), real food prices trended slightly upward before the acquisition ($\beta_1 = 0.02\%$), after which the trend shifted downward ($\beta_3 = -0.12\%$), with no significant immediate level change ($\beta_2 = 0.87\%$).[38] The annualized growth rate was 0.24% before the acquisition and declined by 1.43% afterward.[39] Specification (2) yields similar results, with an annualized growth rate of 0.36% before and a reduction of 1.55% after. Both specifications imply a post-acquisition annualized growth rate of -1.19%, and by the end of 2019, the trajectory of real food prices was 5.85% lower

[37] The structural break model is also referred to as an Interrupted Time Series design (Reichardt, 2019).

[38] During the study period, Norwegian grocery retailers and wholesalers negotiate prices and terms biannually, in February and July, which typically entails price increases. This may explain why the effects of the acquisition were not instantaneous.

[39] Calculated as $(1 + \text{monthly growth rate})^{12} - 1$.

Table A1: Structural break analysis: real food prices, 2011–2019

$\ln(P_t^{Food}) - \ln(P_t^{All})$	(1)	(2)
Trend	0.0002*	0.0003**
	(0.0001)	(0.0001)
Post Acquisition	0.0087	
	(0.0068)	
Trend × Post Acquisition	-0.0012***	
	(0.0002)	
Rebranding Share		0.0071
		(0.0047)
Trend × Rebranding Share		-0.0013***
		(0.0001)
Constant	-0.0144***	-0.0128***
	(0.0031)	(0.0032)
Monthly Dummies	Yes	Yes
Observations	93	108

Notes: The sample covers January 2011 to December 2019. Driscoll–Kraay standard errors are reported in parentheses. The transition period (April 2015–June 2016) is excluded in model (1). *** $p < 0.001$; ** $p < 0.01$; * $p < 0.05$.

relative to the pre-acquisition trend. Specification (1) is illustrated in Figure 1.8 in Section 1.5.1.

1.A.2 Comparative Structural Break Analysis

The difference-in-differences approach typically takes the form of a two-way fixed effects regression:

$$\ln\left(\frac{P_{it}^{Food}}{P_{it}^{All}}\right) = \alpha_i + \lambda_t + \alpha_5 \cdot \text{Post Acquisition} \cdot \text{Norway} + \epsilon_{it} \tag{1.5}$$

where α_i and λ_t are fixed effects, and *Norway* is an indicator variable. Column 3 of Table A2 reports the results of this regression, indicating that Norwegian food prices declined by 5.1% after the acquisition relative to the comparison countries. The time fixed effects capture trends of any nonparametric form common across countries. However, with a small cross-section, each fixed effect is estimated from relatively few observations. To address this concern, the specification in column 2 of Table A2 replaces the time fixed effects with a before-after indicator, *Post Acquisition*, and monthly dummies. This alternative specification yields a similar estimated decline in food prices of 5.23%, but assumes no time trend, other than discrete shifts in food prices at the time of the acquisition, which may differ between Norway and the comparison countries.

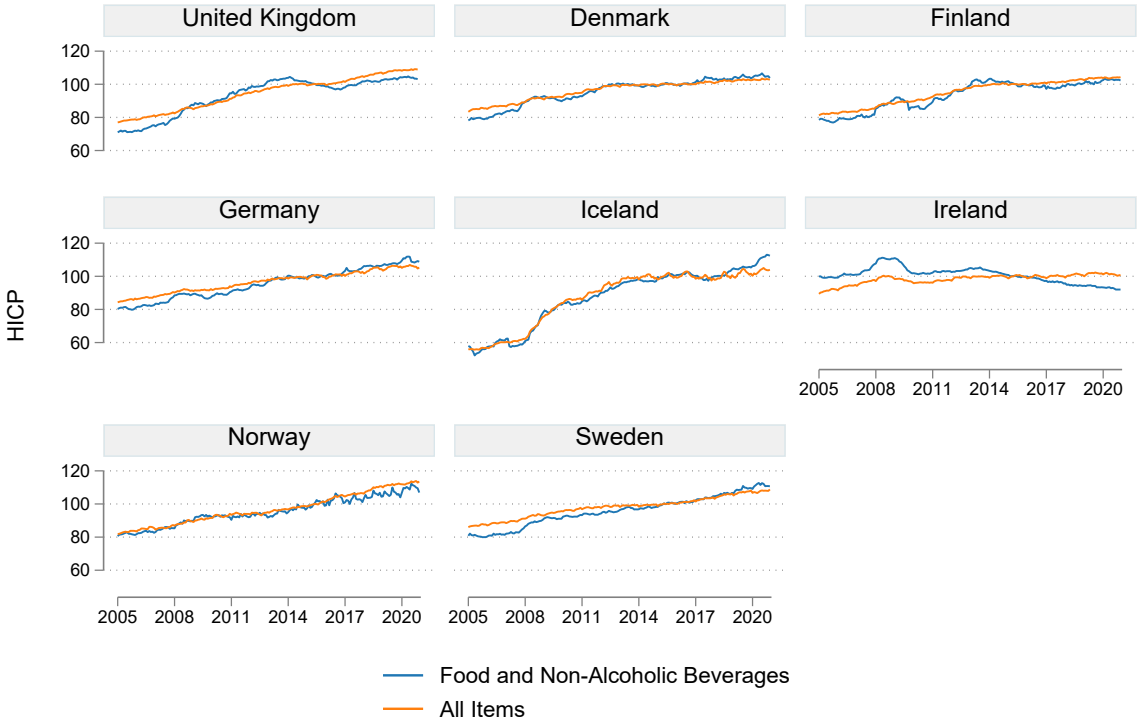


Figure A1: Food and Non-Alcoholic Beverages and All-Items HICP, 2005–2020. Mean index values for 2015 = 100. *Source:* Eurostat.

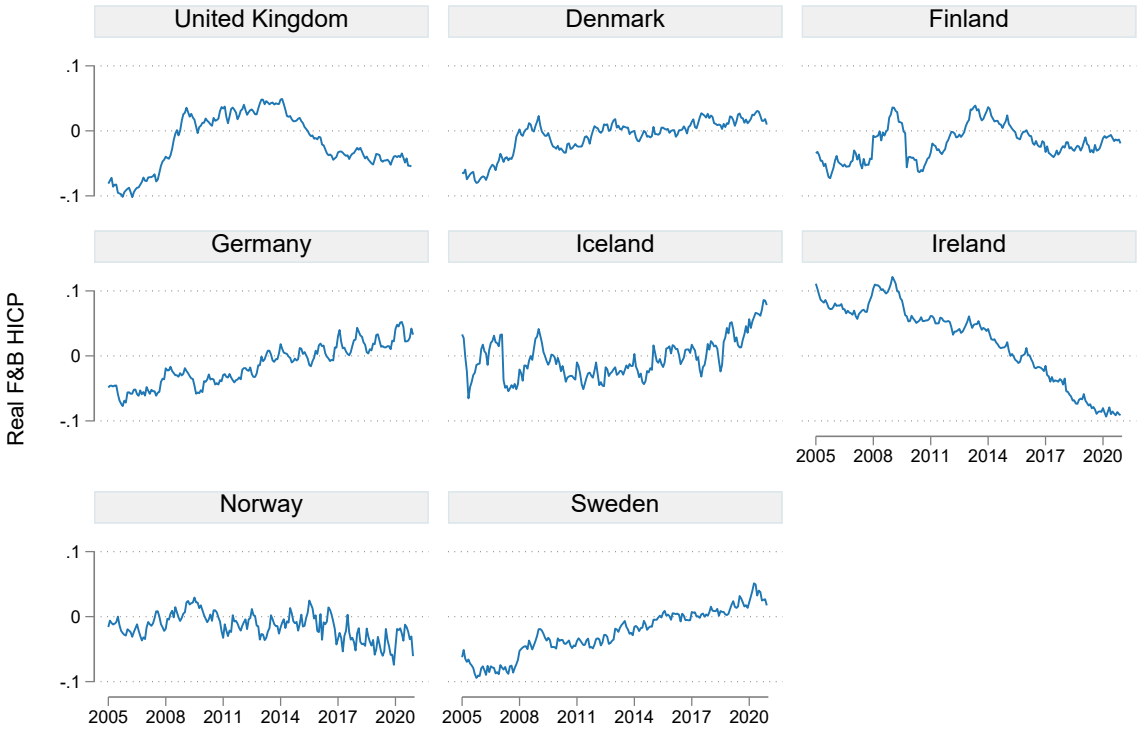


Figure A2: Percentage deviations in food prices from overall prices by country, 2005–2020, calculated as $\ln(P_t^{Food}) - \ln(P_t^{All})$. *Source:* Eurostat.

An alternative approach is to include linear time trends, which can be consistently estimated given the longer time dimension, under the assumption that these trends are indeed linear. By introducing country-specific linear trends, we can dispense with the assumption of common trends and instead assume common deviations from trends (Angrist and Pischke, 2015).[40] I estimate the following equation to examine how real food prices evolved in Norway relative to neighboring countries:

$$\begin{aligned}\ln\left(\frac{P_{it}^{Food}}{P_{it}^{All}}\right) = \alpha_i + \alpha_1 t + \alpha_2 \cdot \text{Post Acquisition} + \alpha_3 t \cdot \text{Post Acquisition} \\ + \alpha_4 t \cdot \text{Norway} + \alpha_5 \cdot \text{Post Acquisition} \cdot \text{Norway} \\ + \alpha_6 t \cdot \text{Post Acquisition} \cdot \text{Norway} + \alpha_7 \cdot \text{Month} \cdot \text{Country} + \epsilon_{it}\end{aligned} \tag{1.6}$$

where α_i are country-specific intercepts and *Country* denotes the vector of country indicators. The specification allows for separate trends and levels for Norway and the control group, both before and after the acquisition. Allowing for country-specific trends and structural breaks within the control group and then comparing Norway with the average control country yields nearly identical results. For simplicity, common trends and breaks are therefore assumed within the control group, while seasonal effects remain country specific. The results, reported in column 1 of Table A2, include tests for structural breaks in both groups to validate the findings in Table A1.

[40] See Nilsen, Sørgard, and Ulsaker (2016) for an application of this approach to evaluating an upstream merger in the Norwegian egg market.

Table A2: Difference-in-differences analysis: real food prices, 2011–2019

$\ln(P_t^{Food}) - \ln(P_t^{All})$	(1)	(2)	(3)	(4)	(5)	(6)
Trend	0.0005***			0.0006***		
	(0.0001)			(0.0000)		
Trend × Norway	-0.0004**			-0.0003**		
	(0.0001)			(0.0001)		
Post Acquisition	-0.0022	0.0301***				
	(0.0030)	(0.0034)				
Post Acquisition × Norway	0.0109	-0.0523***	-0.0510***			
	(0.0081)	(0.0065)	(0.0084)			
Post Acquisition × Trend	-0.0000					
	(0.0001)					
Post Acquisition × Trend × Norway	-0.0012***					
	(0.0002)					
Rebranding Share				-0.0008	0.0279***	
				(0.0026)	(0.0037)	
Rebranding Share × Norway				0.0079	-0.0481***	-0.0483***
				(0.0058)	(0.0074)	(0.0088)
Rebranding Share × Trend				-0.0001		
				(0.0001)		
Rebranding Share × Trend × Norway				-0.0012***		
				(0.0001)		
Constant	-0.0144***	-0.0181***	-0.0234***	-0.0128***	-0.0177***	-0.0224***
	(0.0031)	(0.0038)	(0.0032)	(0.0031)	(0.0038)	(0.0032)
Monthly Dummies	Yes	Yes	No	Yes	Yes	No
Time Fixed Effects	No	No	Yes	No	No	Yes
Country Fixed Effects	Yes	Yes	Yes	Yes	Yes	Yes
Observations	465	465	465	540	540	540

Notes: The panel includes Norway, Sweden, Denmark, Germany, and Iceland, spanning January 2011 to December 2019. Driscoll–Kraay standard errors are reported in parentheses. The transition period (April 2015–June 2016) is excluded from models (1)–(3). *** $p < 0.001$; ** $p < 0.01$; * $p < 0.05$.

The results in column 1 of Table A1 can, in fact, be derived from the structural break regression in Table A2.[41] The difference is that the difference-in-differences specification moderates changes in Norway's trend and level by those observed in the comparison countries. If there had been a structural break common among the comparison countries, regardless of the acquisition, it would be captured by the control group. Coefficient α_5 measures the difference in level shifts after the acquisition between Norway and the control group. Neither Norway nor the control countries exhibit level shifts significantly different from zero or from each other. Coefficient α_6 measures the difference in trend shifts after the acquisition. The absence of a trend

[41] Coefficients $\beta_1 = 0.0002 \approx 0.0005 - 0.0004 = \alpha_1 + \alpha_4$, $\beta_2 = 0.0087 \approx (-0.0022) + 0.0109 = \alpha_2 + \alpha_5$, and $\beta_3 = -0.0012 \approx (-0.0000) - 0.0012 = \alpha_3 + \alpha_6$.

change among the control countries indicates that the observed shift in Norwegian real food prices was not driven by common regional factors ($\beta_3 = \alpha_6 = -0.0012$). However, coefficient $\alpha_4 = -0.0004$ shows that Norway and the control countries had slightly different average pre-acquisition trends (approximately 0.5% annualized growth), violating the common-trends assumption and underscoring the importance of accounting for differing pre-trends.

Columns 4–6 replicate the preceding analysis but replace the *Post Acquisition* indicator with the variable *Rebranding Share*. The results are similar, though the estimated effects in columns 5 and 6 are slightly smaller than their counterparts in columns 2 and 3, at approximately -4.8%.

1.A.3 Synthetic Control Analysis

The preceding analysis assumes either that real food prices follow a linear trend or that the comparison countries share a common trend on average. In practice, real food prices occasionally deviate from linear trends, and the unweighted average of comparison countries does not perfectly parallel Norway's pre-acquisition trajectory. The synthetic control method of Abadie and Gardeazabal (2003) accommodates both non-linear and non-parallel trends by constructing a weighted average of comparison countries that better reproduces Norway's path. The underlying idea is that while outcomes are driven by common factors, differences in factor loadings can generate non-parallel trends. These loadings are unobserved, but only a synthetic control that closely matches Norway's loadings is likely to replicate its pre-acquisition history—especially when the pre-period is long and the data relatively stable (see Abadie (2021) for an in-depth discussion).

The goal is to identify the potentially time-varying effects of the intervention, defined as $\tau_{1t} = Y_{1t}^I - Y_{1t}^N$, where unit 1 is the treated unit and Y_{1t}^I and Y_{1t}^N denote the potential outcomes with and without the intervention, respectively. The counterfactual (no-intervention) outcome is unobserved but imputed as a weighted combination of control units. The synthetic control estimator is therefore

$$\hat{\tau}_{1t} = Y_{1t} - \sum_{j=2}^{J+1} w_j Y_{jt}$$

where the weights w_j are non-negative and sum to one. In the analysis below, the weights are chosen to best reproduce the pre-acquisition history of Norwegian food prices.

The credibility of the synthetic control estimate increases when the model closely reproduces Norway's pre-acquisition outcomes over a long horizon, so I extend the pre-acquisition period to 2005. As emphasized by Abadie (2021), filtering volatile series

helps prevent overfitting. I therefore consider two specifications: one that controls for country-specific monthly effects, and another that applies the Hodrick–Prescott filter with $\lambda = 14400$ (Hodrick and Prescott, 1997).[42] All series are normalized to have mean zero in the pre-acquisition period, thereby controlling for level differences analogous to unit fixed effects in a difference-in-differences framework. To reduce interpolation bias arising from large cross-country discrepancies, I restrict the donor pool to Nordic countries.[43]

Figure A3a plots deseasonalized real food prices for Norway and its synthetic control over 2005–2019, along with their difference. The average post-acquisition decline in real food prices is estimated at -5.2%. The synthetic control tracks Norway's price dynamics reasonably well, although the pre-acquisition fit is imperfect due to substantial volatility. Applying the HP filter ($\lambda = 14400$) improves the pre-acquisition fit, as shown in Figure A3b, and yields an estimated decline of -4.7%. Synthetic control weights are reported in Table A3. Restricting the sample to 2011–2019 produces an almost perfect pre-acquisition fit under both transformations (Figures A4a and A4b), reflecting the stable linear trends in this period, and implies a post-acquisition decline of -3.8%. Across all specifications, the estimated effects are broadly consistent with the difference-in-differences estimates.

[42] This is a standard choice for monthly data and performs well here, smoothing cyclical fluctuations in food prices while preserving larger shifts.

[43] Including Ireland, whose trend differs from Norway's, marginally improves the pre-acquisition fit by offsetting countries with steeper trends. However, Ireland also exhibits a pronounced downward trend relative to Norway, which can introduce interpolation bias (Abadie, 2021).

Table A3: Synthetic control weights

	Deseasonalized 2005–2019	HP-filter 2005–2019	Deseasonalized 2011–2019	HP-filter 2011–2019
Country	Weight	Weight	Weight	Weight
Denmark	0.279	0.313	0.477	0.538
Finland	0	0	0	0
Iceland	0.511	0.687	0.143	0.462
Sweden	0.21	0	0.38	0

1.B Tables and Figures

Table A4: Retail chain segmentation

Retail Chain	Umbrella Group	Market Segment
Joker	NorgesGruppen	General Store
Coop Marked	Coop	General Store
Nærbutikken	NorgesGruppen	General Store
Matkroken	ICA (then Coop)	General Store
Prix	Coop	Convenience Discount
Bunnpris	Bunnpris	Convenience Discount
Rimi	ICA	Convenience Discount
Lidl	Lidl	Hard Discount
Kiwi	NorgesGruppen	Soft Discount
Rema 1000	Reitangruppen	Soft Discount
Extra	Coop	Soft Discount
Eurospar	NorgesGruppen	Supermarket
Meny	NorgesGruppen	Supermarket
Coop Mega	Coop	Supermarket
ICA Supermarked	ICA	Supermarket
Spar	NorgesGruppen	Local Supermarket
obs!	Coop	Hypermarket

Notes: This table shows Norwegian retail chains, their ownership, and market segmentation.

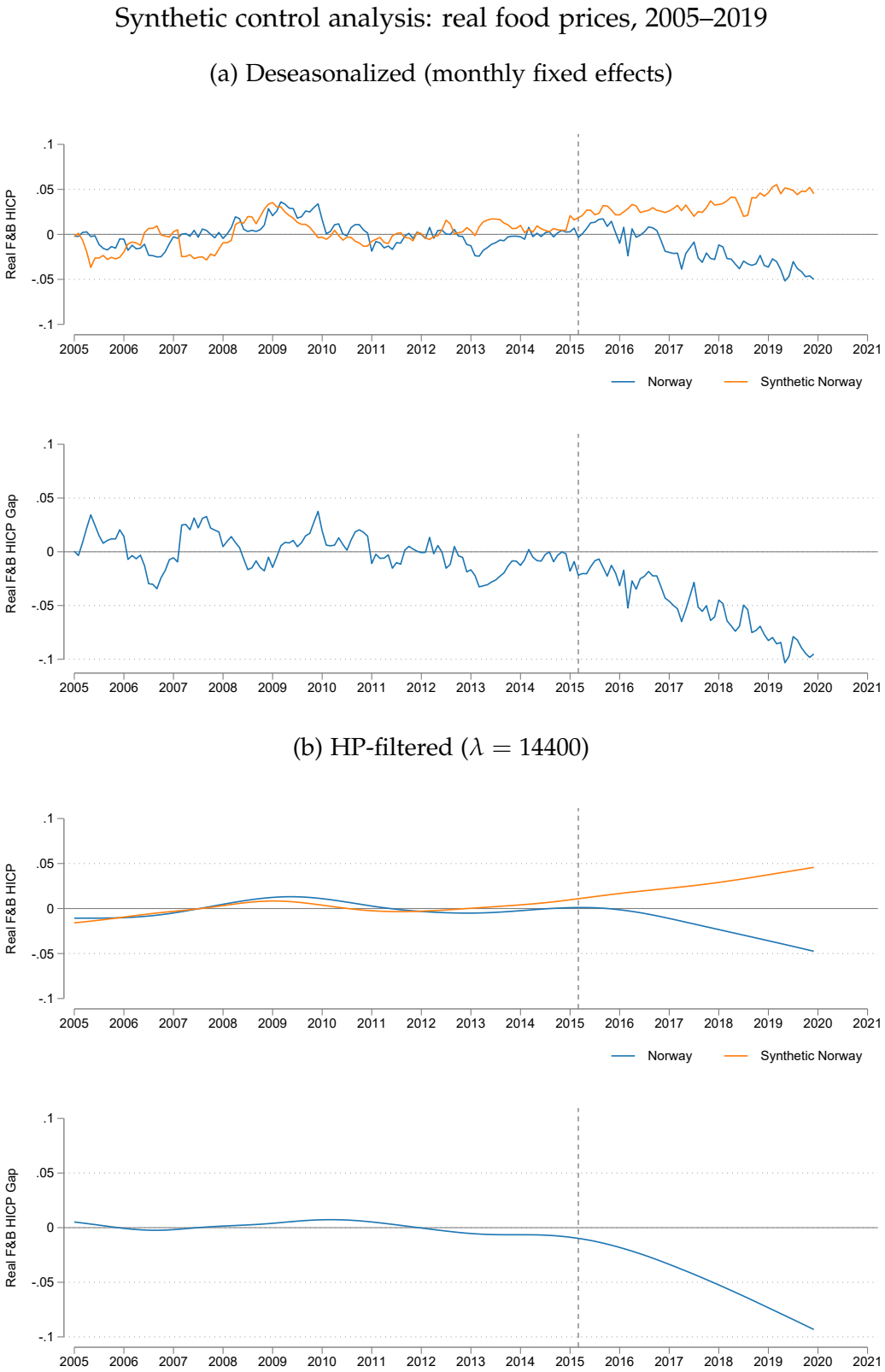


Figure A3: Percentage deviations in food prices relative to overall prices for Norway and its synthetic control, computed as $\ln(P_t^{Food}) - \ln(P_t^{All})$. Panel (a) removes monthly fixed effects (deseasonalization), while Panel (b) applies an HP filter ($\lambda = 14400$). The lower part of each panel shows the gap between Norway and its synthetic control. The vertical line marks March 2015, preceding the first store rebranding. Synthetic control weights (Table A3) minimize pre-acquisition differences. The donor pool is restricted to Nordic countries. *Source:* Eurostat.

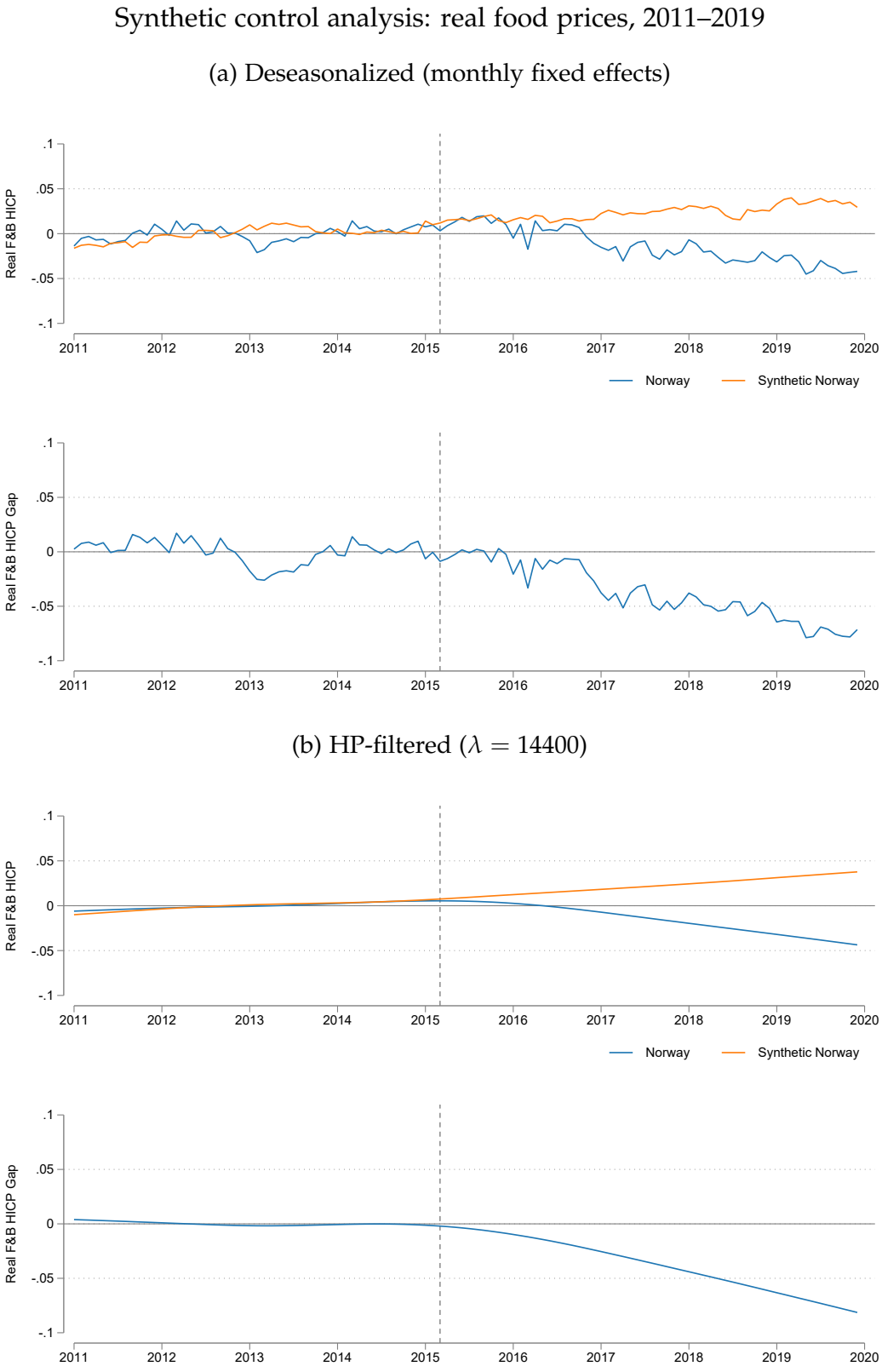


Figure A4: Percentage deviations in food prices relative to overall prices for Norway and its synthetic control, computed as $\ln(P_t^{Food}) - \ln(P_t^{All})$. Panel (a) removes monthly fixed effects (deseasonalization), while Panel (b) applies an HP filter ($\lambda = 14400$). The lower part of each panel shows the gap between Norway and its synthetic control. The vertical line marks March 2015, preceding the first store rebranding. Synthetic control weights (Table A3) minimize pre-acquisition differences. The donor pool is restricted to Nordic countries. *Source:* Eurostat.

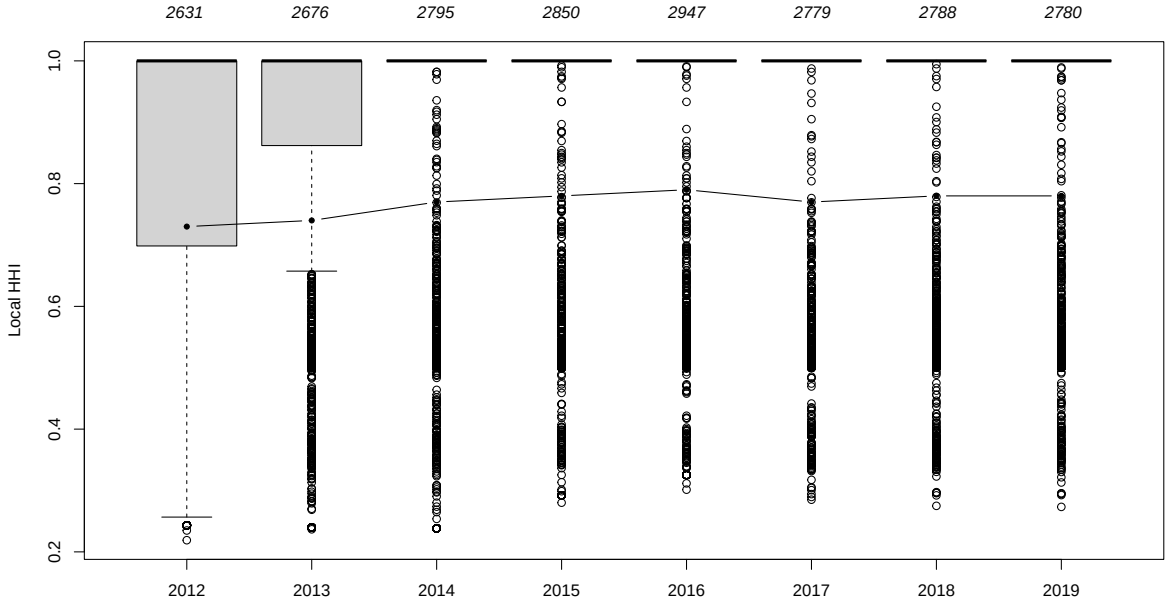


Figure A5: Box plots of the Herfindahl–Hirschman Index (HHI) at the *Basic Statistical Unit (BSU)* level, 2012–2019, calculated using store revenues aggregated at the umbrella chain level. The data cover the vast majority of grocery stores in Norway. The weighted mean, based on BSU sales shares, is shown as black dots connected by lines. The number of local markets in the sample is shown above each box plot. *Source:* Geodata.

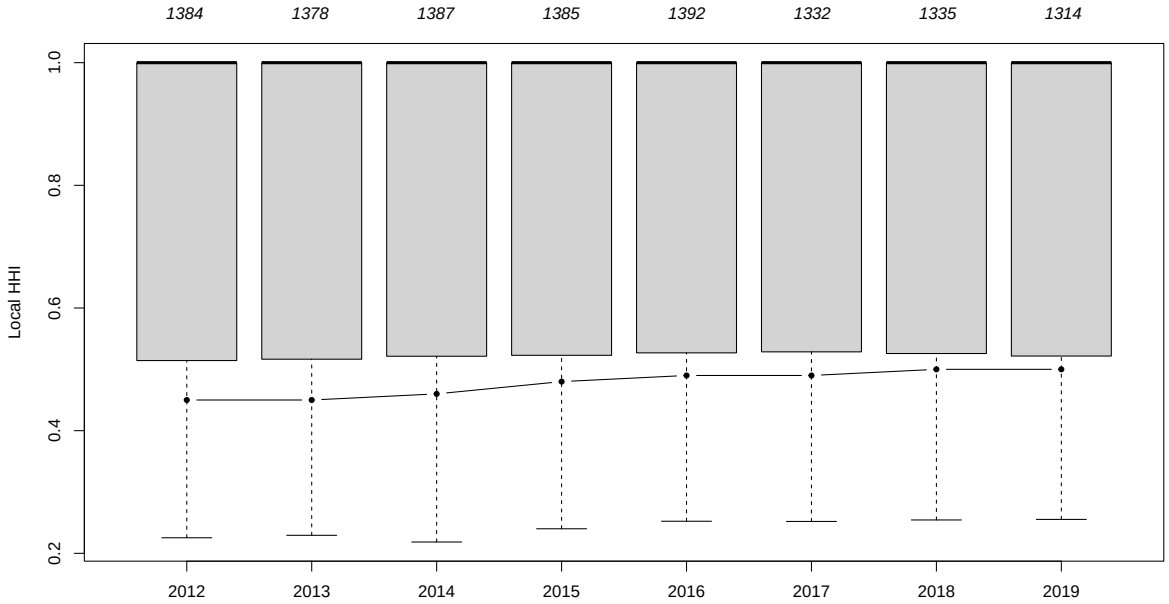


Figure A6: Box plots of the Herfindahl–Hirschman Index (HHI) at the *postal area* level, 2012–2019, calculated using store revenues aggregated at the umbrella chain level. The data cover the vast majority of grocery stores in Norway. The weighted mean, based on postal area sales shares, is shown as black dots connected by lines. The number of local markets in the sample is shown above each box plot. *Source:* Geodata.

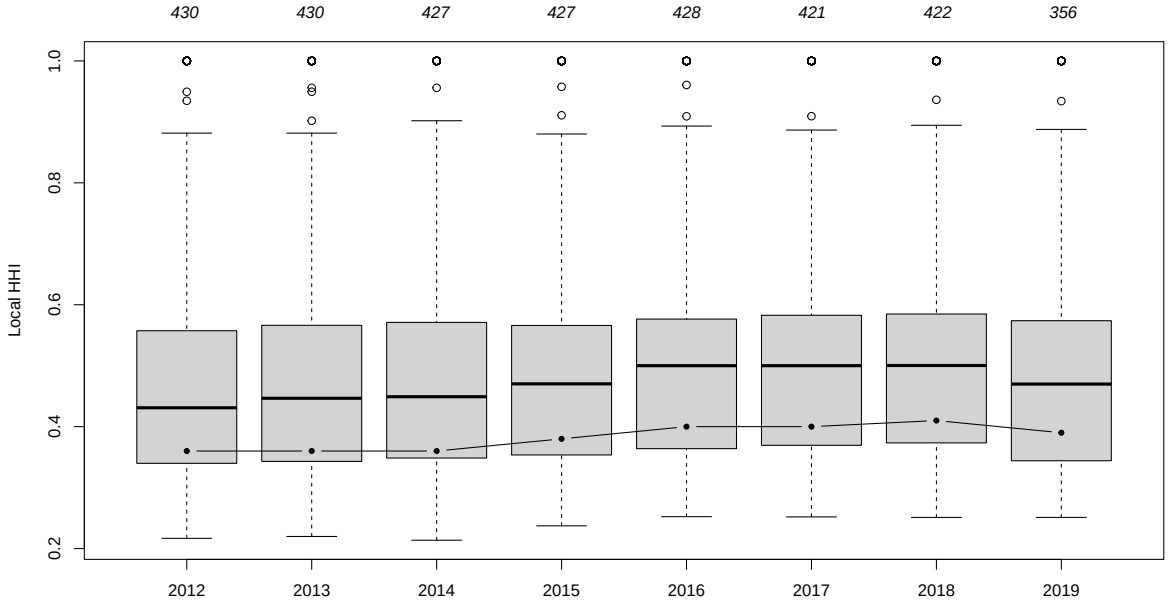


Figure A7: Box plots of the Herfindahl–Hirschman Index (HHI) at the *municipality* level, 2012–2019, calculated using store revenues aggregated at the umbrella chain level. The data cover the vast majority of grocery stores in Norway. The weighted mean, based on municipality sales shares, is shown as black dots connected by lines. The number of local markets in the sample is shown above each box plot. *Source:* Geodata.

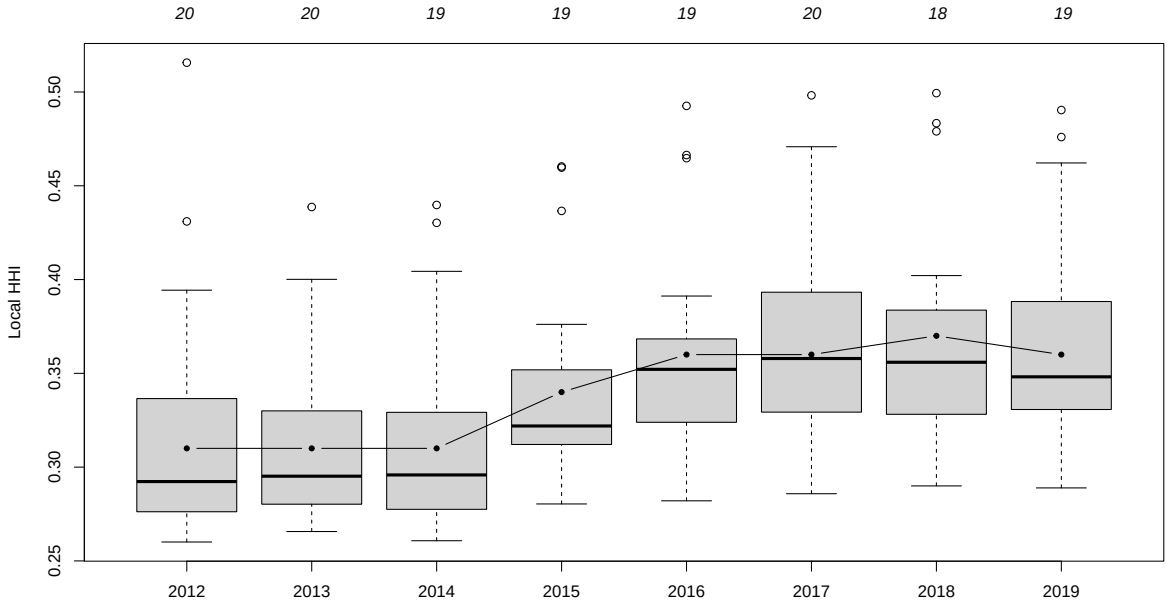


Figure A8: Box plots of the Herfindahl–Hirschman Index (HHI) at the *county* level, 2012–2019, calculated using store revenues aggregated at the umbrella chain level. The data cover the vast majority of grocery stores in Norway. The weighted mean, based on county sales shares, is shown as black dots connected by lines. The number of local markets in the sample is shown above each box plot. *Source:* Geodata.

Table A5: Descriptive statistics: treatment and control groups

	Control (484)			Extra (38)			Prix (20)			Bunnpris (12)			Closed (26)		
	Before	After	%	Before	After	%	Before	After	%	Before	After	%	Before	After	%
Sales (millions NOK)	2.68	2.97	10.82%	3.19	3.19	0.00%	2.81	3.27	16.37%	2.63	3.16	20.15%	2.76	3.28	18.84%
Assortment Size	2166	2266	4.62%	2228	2304	3.41%	2130	2264	6.29%	2157	2300	6.63%	2226	2329	4.63%
Assortment Size (Min. 100 units)	1427	1944	36.23%	1427	1965	37.70%	1383	1945	40.64%	1388	1971	42.00%	1465	1990	35.84%
Assortment Entropy	6.818	6.835	0.25%	6.861	6.864	0.04%	6.805	6.825	0.29%	6.828	6.851	0.34%	6.806	6.816	0.15%
Effective Store Price	48.6	51.9	6.79%	47.6	51.2	7.56%	47.8	51.3	7.32%	48.9	52.5	7.36%	49.6	53.3	7.46%
Local Product Price Deviation	0.0	0.0	0.00%	0.0	0.0	0.00%	0.0	0.0	0.00%	0.0	0.0	0.00%	0.0	0.0	0.00%
Number of Competitors	1.5	1.4	-6.67%	2.6	2.6	0.00%	2.2	2.1	-4.55%	2.8	2.8	0.00%	4.3	3.2	-25.58%
Herfindahl–Hirschman Index	0.62	0.62	0.00%	0.36	0.37	2.78%	0.39	0.41	5.13%	0.34	0.36	5.88%	0.28	0.33	17.86%
Population (in thousands)	124.9	128.2	2.64%	119.6	122.2	2.17%	151.2	130.5	-13.69%	114.1	142.9	25.24%	44	43.2	-1.82%
Population Density (per square km)	406.1	417.7	2.86%	453.7	443.7	-2.20%	390.4	358	-8.30%	334.9	401.1	19.77%	169.5	170.3	0.47%
Median Income (thousands NOK)	488	500.6	2.58%	494.5	503.5	1.82%	466.2	483.8	3.78%	474.2	488.2	2.95%	509.8	518.5	1.71%
Higher Education Share	0.31	0.32	3.23%	0.35	0.35	0.00%	0.29	0.31	6.90%	0.28	0.31	10.71%	0.27	0.28	3.70%
Employees	20.8	19.7	-5.29%	27.8	22.7	-18.35%	24.7	20.6	-16.60%	22.4	21.4	-4.46%	20.4	20.2	-0.98%
Postal Office Present Share	0.28	0.28	0.00%	0.16	0.15	-6.25%	0.27	0.31	14.81%	0	0	–	0.21	0.19	-9.52%

Notes: This table reports mean values of outcome and control variables measured at the store-month level. Values are shown for treatment and control stores before the first (June 2015) and after the last (July 2016) store rebranding in the vicinity of Kiwi. Stores are defined as treated if they shared a postcode with ICA in 2014 and are categorized by the rival format rebranding they experienced. I exclude 32 stores that either experienced multiple format rebrandings or a change in local market concentration. The number of stores is indicated in parentheses next to each group label. Assortment size is measured as the number of unique products sold. Assortment Size (Min. 100 units) is calculated in the same way but restricted to products with at least 100 units sold in every period. Assortment entropy measures variety while accounting for sales shares (see Section 1.5.2 for details). Effective store price is the expenditure-weighted average product price. Local product price deviation is the expenditure-weighted average deviation from the mean product price across stores. The Herfindahl–Hirschman Index is calculated using accounting sales. Demographic variables are measured at the municipality level.

Table A6: Product category expenditure shares

Product Categories	Expenditure Share
Beverages	23.5%
Frozen Food	5.3%
Fresh Pastries	5.1%
Ready-to-eat Meals	11.7%
Fresh Fish and Shellfish	1.4%
Fresh Meat	7.2%
Home Supplies	4.0%
Kiosk Goods	10.9%
Dairy Products	19.9%
Cosmetics	3.8%
Other	7.3%

Notes: This table shows the product categories included in the analysis and their expenditure shares.

Table A7: Effect of Extra on effective store prices

	$\ln(\hat{P}^1_{it})$					
	(1)	(2)	(3)	(4)	(5)	(6)
T_{Extra}	-0.002	0.005	0.007*	-0.001	0.006	0.002
	(0.003)	(0.004)	(0.003)	(0.003)	(0.004)	(0.003)
ln(Population)	-0.056	-0.067	-0.060	-0.031	-0.057	-0.052*
	(0.067)	(0.066)	(0.062)	(0.065)	(0.060)	(0.020)
ln(Median Income)	0.035	0.050	-0.001	0.053	0.001	-0.030
	(0.062)	(0.067)	(0.060)	(0.061)	(0.058)	(0.067)
Market Definition	10–1.5 km	10–1 km	10–0.5 km	6–1.5 km	5–0.5 km	Postcode
Months	36	36	36	36	36	36
Stores	348	396	468	372	492	529
Observations	10,808	12,371	14,955	11,662	15,893	17,351
R^2	0.79	0.81	0.82	0.80	0.83	0.84

Notes: Each column reports the estimated ATT using the estimator of Sun and Abraham (2021) with different local market definitions. The data span all months of 2014–2016 and the demographics are measured at the municipality level. The standard errors shown in parentheses are clustered by Municipality. *** $p < 0.001$; ** $p < 0.01$; * $p < 0.05$.

Event-study plots: log effective store prices ($\ln(\hat{P}^1_{it})$), Extra rebranding

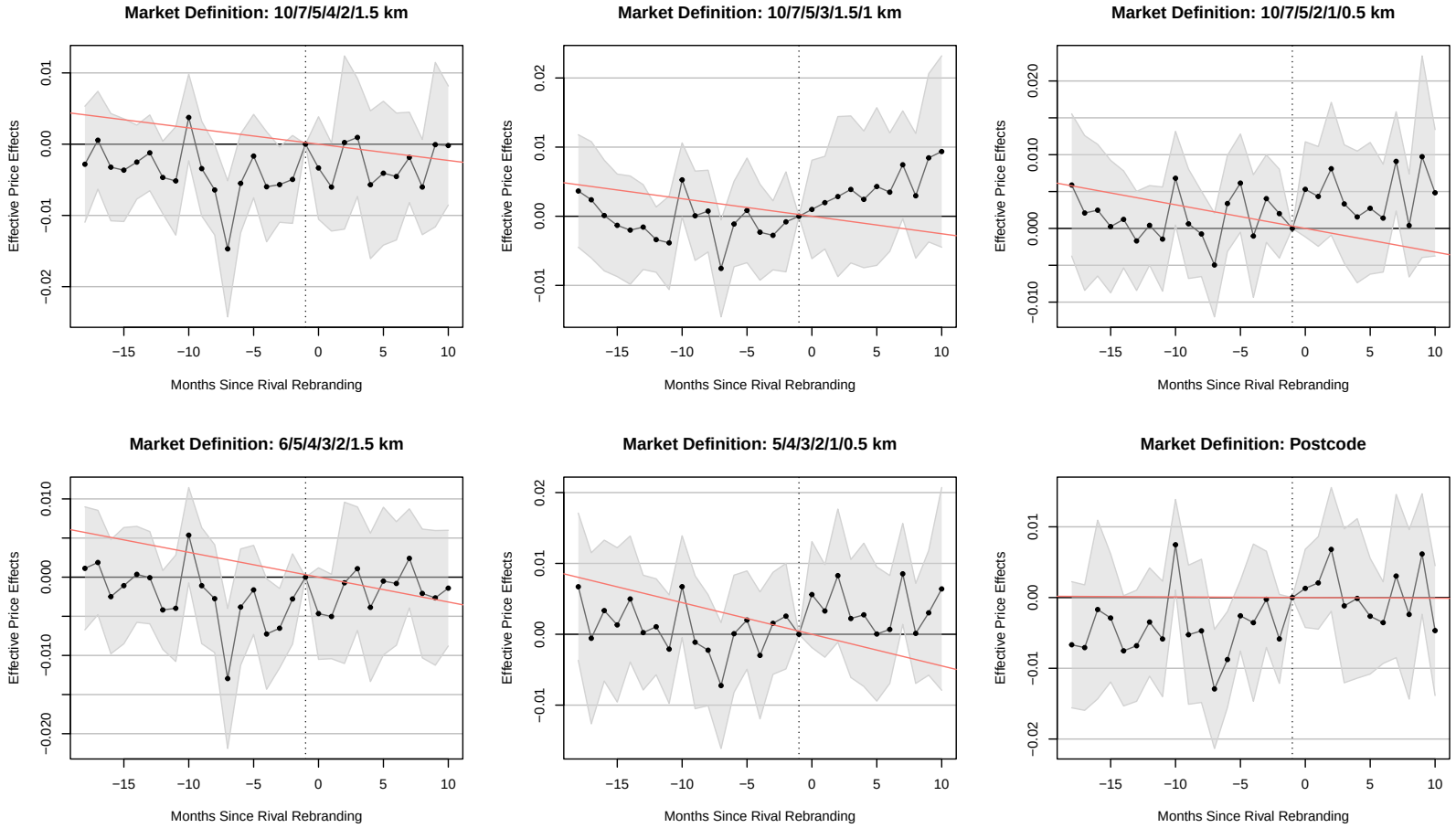


Figure A9: Estimated percentage store price effects of rival rebranding to Extra over time. The horizontal axis displays months relative to the event, $l \in [-18, +10]$, with the vertical line marking $l = -1$—the last month before the event. Point estimates are shown with 95% confidence intervals, based on standard errors clustered at the municipality level. A fitted linear pre-trend is displayed and extrapolated into the post-event period for comparison.

Table A8: Effect of Prix on effective store prices

	$\ln(\hat{P}^1_{it})$					
	(1)	(2)	(3)	(4)	(5)	(6)
T_{Prix}	0.016	0.006	0.001	0.016	-0.002	0.004
	(0.012)	(0.011)	(0.007)	(0.010)	(0.007)	(0.008)
ln(Population)	-0.020	-0.035	-0.052	-0.002	-0.057	-0.049*
	(0.064)	(0.062)	(0.061)	(0.063)	(0.060)	(0.019)
ln(Median Income)	0.025	0.033	-0.013	0.028	-0.018	-0.019
	(0.059)	(0.063)	(0.061)	(0.058)	(0.058)	(0.064)
Market Definition	10–1.5 km	10–1 km	10–0.5 km	6–1.5 km	5–0.5 km	Postcode
Months	36	36	36	36	36	36
Stores	327	372	449	350	479	512
Observations	10,060	11,615	14,365	10,889	15,521	16,705
R^2	0.78	0.79	0.82	0.79	0.82	0.83

Notes: Each column reports the estimated ATT using the estimator of Sun and Abraham (2021) with different local market definitions. The data span all months of 2014–2016 and the demographics are measured at the municipality level. The standard errors shown in parentheses are clustered by Municipality. *** $p < 0.001$; ** $p < 0.01$; * $p < 0.05$.

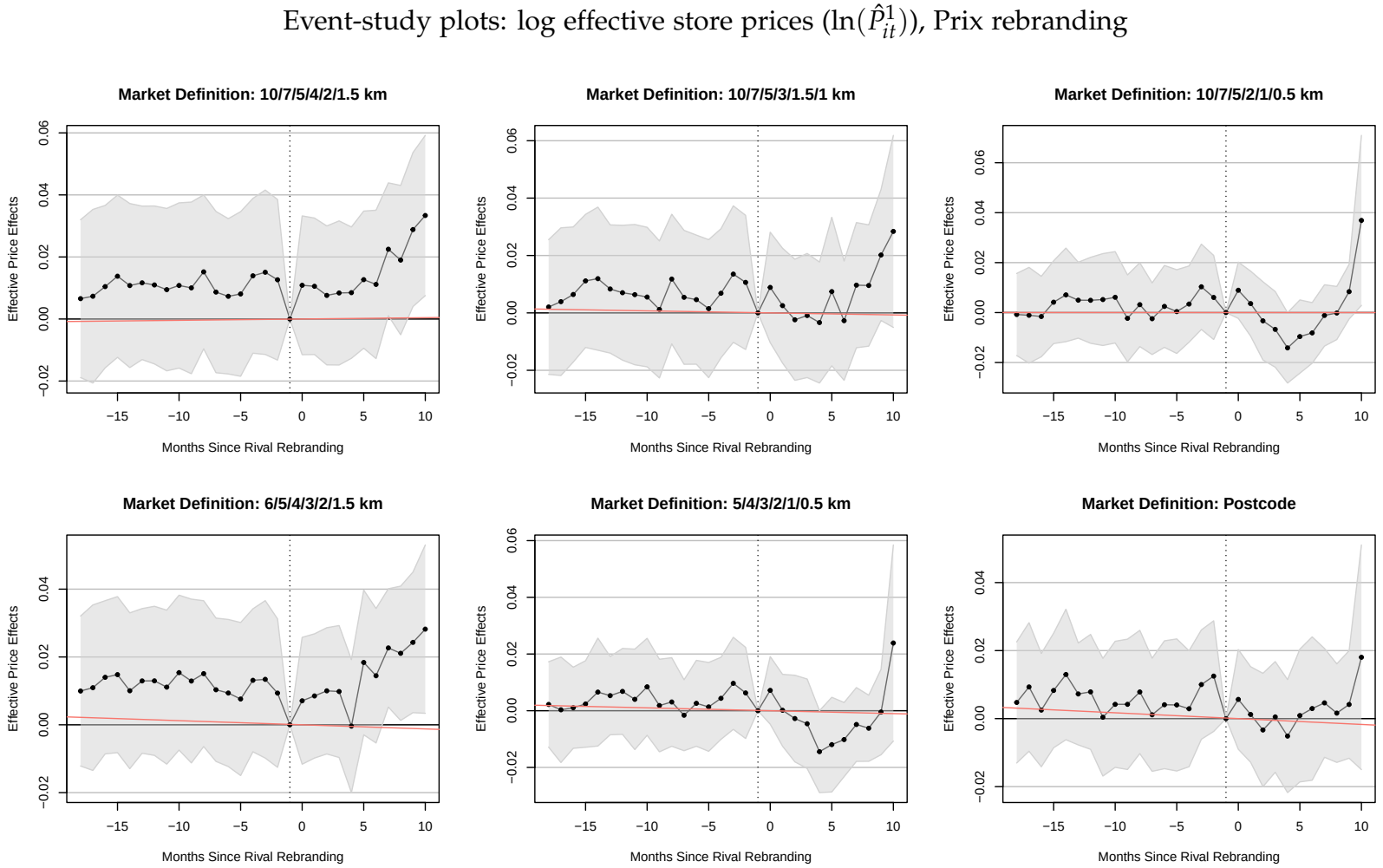


Figure A10: Estimated percentage store price effects of rival rebranding to Prix over time. The horizontal axis displays months relative to the event, $l \in [-18, +10]$, with the vertical line marking $l = -1$—the last month before the event. Point estimates are shown with 95% confidence intervals, based on standard errors clustered at the municipality level. A fitted linear pre-trend is displayed and extrapolated into the post-event period for comparison.

Table A9: Effect of Bunnpris on effective store prices

	$\ln(\hat{P}^1_{it})$					
	(1)	(2)	(3)	(4)	(5)	(6)
T_{Bunnpris}	0.012	0.015	0.018	0.015	0.018	0.013
	(0.016)	(0.013)	(0.010)	(0.017)	(0.010)	(0.009)
ln(Population)	-0.016	-0.028	-0.049	0.007	-0.049	-0.048**
	(0.067)	(0.064)	(0.061)	(0.064)	(0.060)	(0.017)
ln(Median Income)	0.034	0.023	-0.032	0.016	-0.026	-0.040
	(0.060)	(0.065)	(0.058)	(0.060)	(0.056)	(0.063)
Market Definition	10–1.5 km	10–1 km	10–0.5 km	6–1.5 km	5–0.5 km	Postcode
Months	36	36	36	36	36	36
Stores	291	342	424	317	456	489
Observations	9,277	10,951	13,781	10,158	14,903	16,112
R^2	0.76	0.78	0.81	0.77	0.81	0.83

Notes: Each column reports the estimated ATT using the estimator of Sun and Abraham (2021) with different local market definitions. The data span all months of 2014–2016 and the demographics are measured at the municipality level. The standard errors shown in parentheses are clustered by Municipality. *** $p < 0.001$; ** $p < 0.01$; * $p < 0.05$.

Event-study plots: log effective store prices ($\ln(\hat{P}^1_{it})$), Bunnpris rebranding

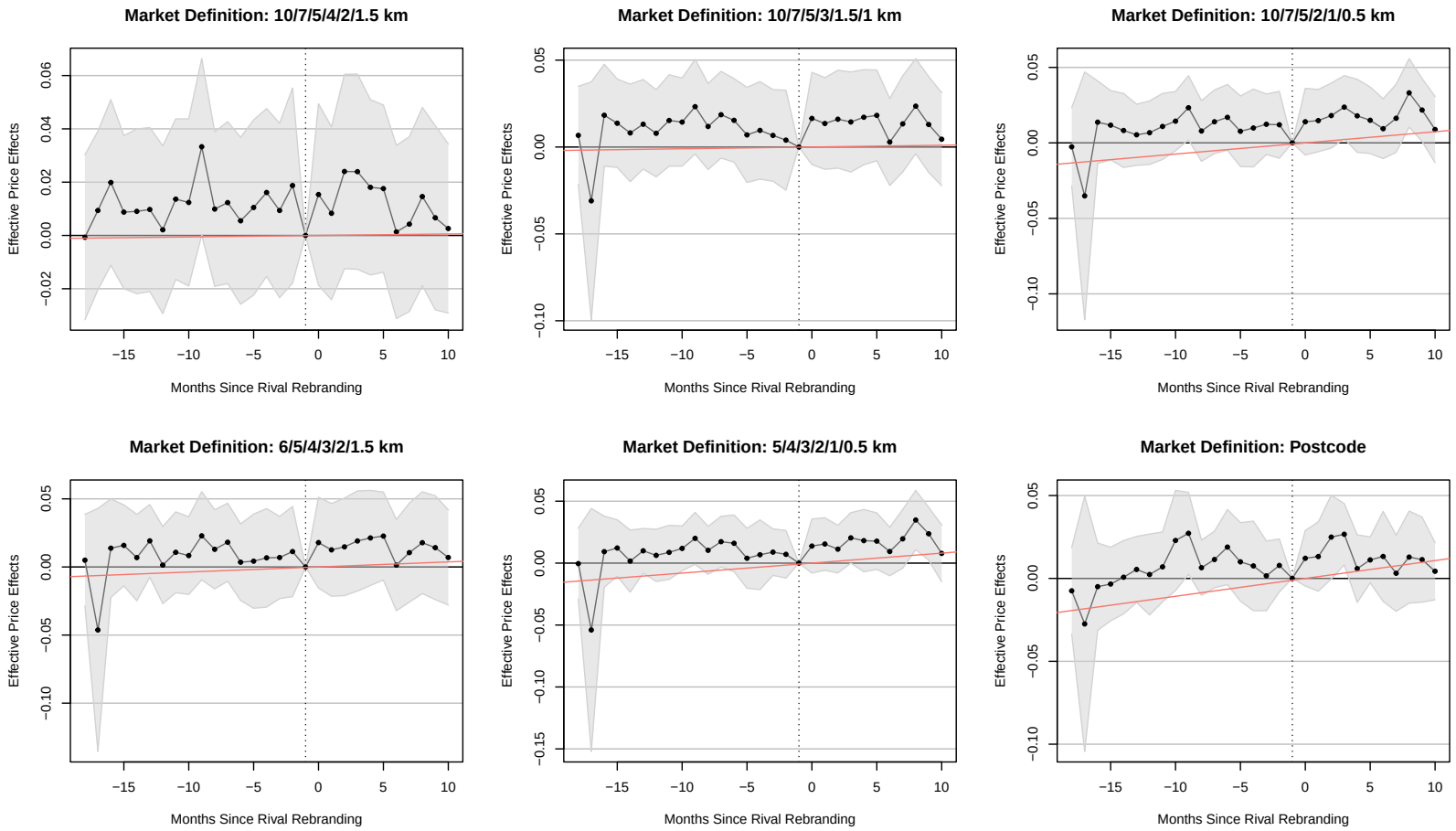


Figure A11: Estimated percentage store price effects of rival rebranding to Bunnpris over time. The horizontal axis displays months relative to the event, $l \in [-18, +10]$, with the vertical line marking $l = -1$—the last month before the event. Point estimates are shown with 95% confidence intervals, based on standard errors clustered at the municipality level. A fitted linear pre-trend is displayed and extrapolated into the post-event period for comparison.

Table A10: Effect of rival closure on effective store prices

	$\ln(\hat{P}^1_{it})$					
	(1)	(2)	(3)	(4)	(5)	(6)
T_{Closure}	0.016	0.015	0.019*	0.016	0.020*	0.005
	(0.011)	(0.010)	(0.010)	(0.010)	(0.010)	(0.008)
ln(Population)	-0.027	-0.045	-0.075	-0.014	-0.066	-0.048**
	(0.067)	(0.067)	(0.066)	(0.065)	(0.063)	(0.019)
ln(Median Income)	0.018	0.028	-0.036	0.021	-0.037	-0.036
	(0.061)	(0.066)	(0.062)	(0.059)	(0.060)	(0.065)
Market Definition	10–1.5 km	10–1 km	10–0.5 km	6–1.5 km	5–0.5 km	Postcode
Months	36	36	36	36	36	36
Stores	314	355	434	337	463	504
Observations	10,040	11,363	14,127	10,815	15,141	16,626
R^2	0.77	0.79	0.81	0.78	0.82	0.83

Notes: Each column reports the estimated ATT using the estimator of Sun and Abraham (2021) with different local market definitions. The data span all months of 2014–2016 and the demographics are measured at the municipality level. The standard errors shown in parentheses are clustered by Municipality. *** $p < 0.001$; ** $p < 0.01$; * $p < 0.05$.

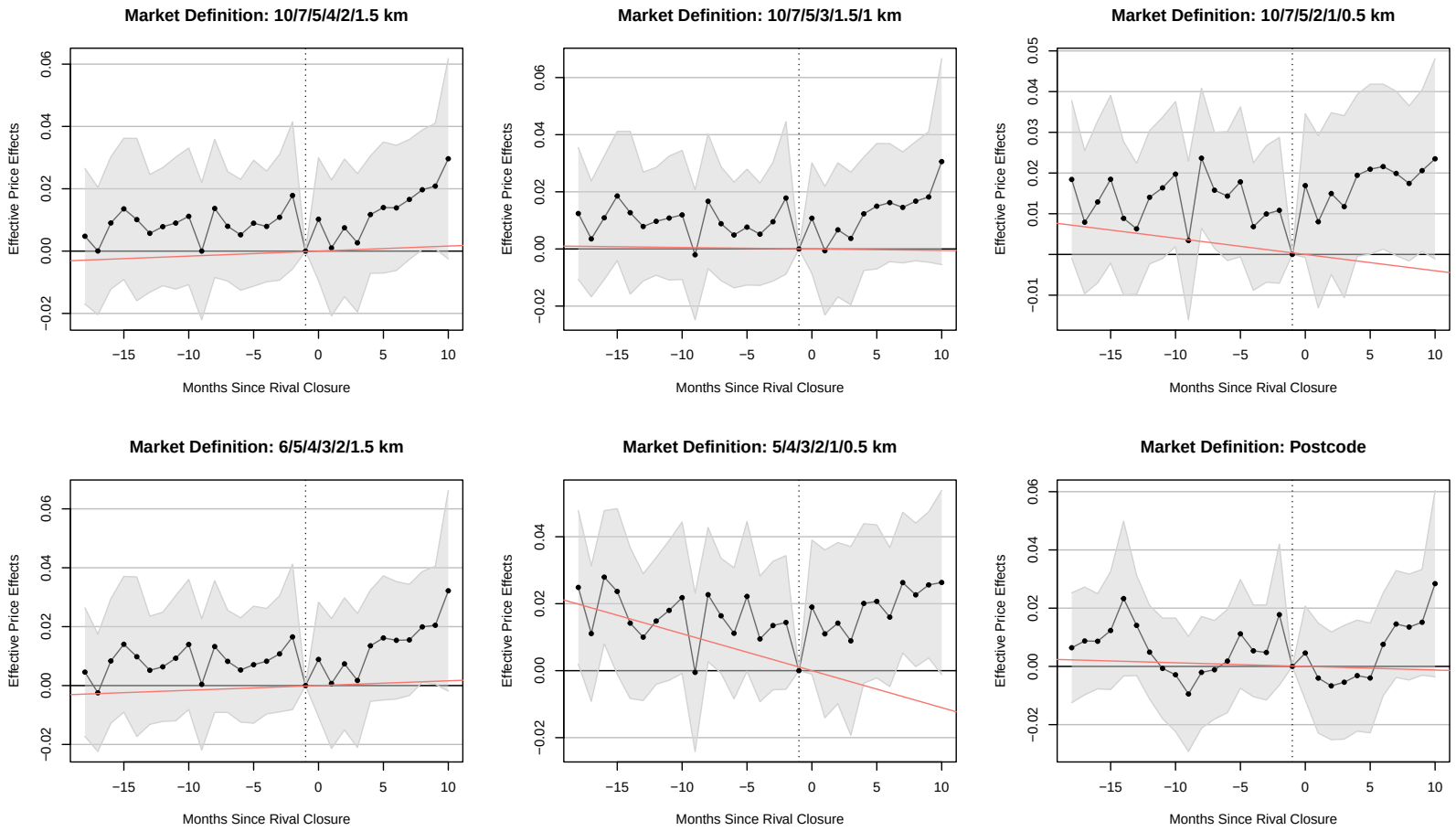


Figure A12: Estimated percentage store price effects of rival closure over time. The horizontal axis displays months relative to the event, $l \in [-18, +10]$, with the vertical line marking $l = -1$—the last month before the event. Point estimates are shown with 95% confidence intervals, based on standard errors clustered at the municipality level. A fitted linear pre-trend is displayed and extrapolated into the post-event period for comparison.

Table A11: Effect of Extra on local product price deviations

	$\hat{P}_{it}^2$					
	(1)	(2)	(3)	(4)	(5)	(6)
T_{Extra}	0.000	-0.001	-0.001	0.000	-0.001	0.000
	(0.000)	(0.000)	(0.000)	(0.000)	(0.000)	(0.000)
ln(Population)	0.040**	0.045***	0.046***	0.045***	0.049***	0.009
	(0.013)	(0.012)	(0.014)	(0.012)	(0.013)	(0.007)
ln(Median Income)	0.001	0.007	0.002	0.003	0.005	0.011
	(0.011)	(0.011)	(0.011)	(0.011)	(0.011)	(0.017)
Market Definition	10–1.5 km	10–1 km	10–0.5 km	6–1.5 km	5–0.5 km	Postcode
Months	36	36	36	36	36	36
Stores	348	396	468	372	492	529
Observations	10,808	12,371	14,955	11,662	15,893	17,351
R^2	0.17	0.17	0.17	0.17	0.17	0.18

Notes: Each column reports the estimated ATT using the estimator of Sun and Abraham (2021) with different local market definitions. The data span all months of 2014–2016 and the demographics are measured at the municipality level. The standard errors shown in parentheses are clustered by Municipality. *** $p < 0.001$; ** $p < 0.01$; * $p < 0.05$.

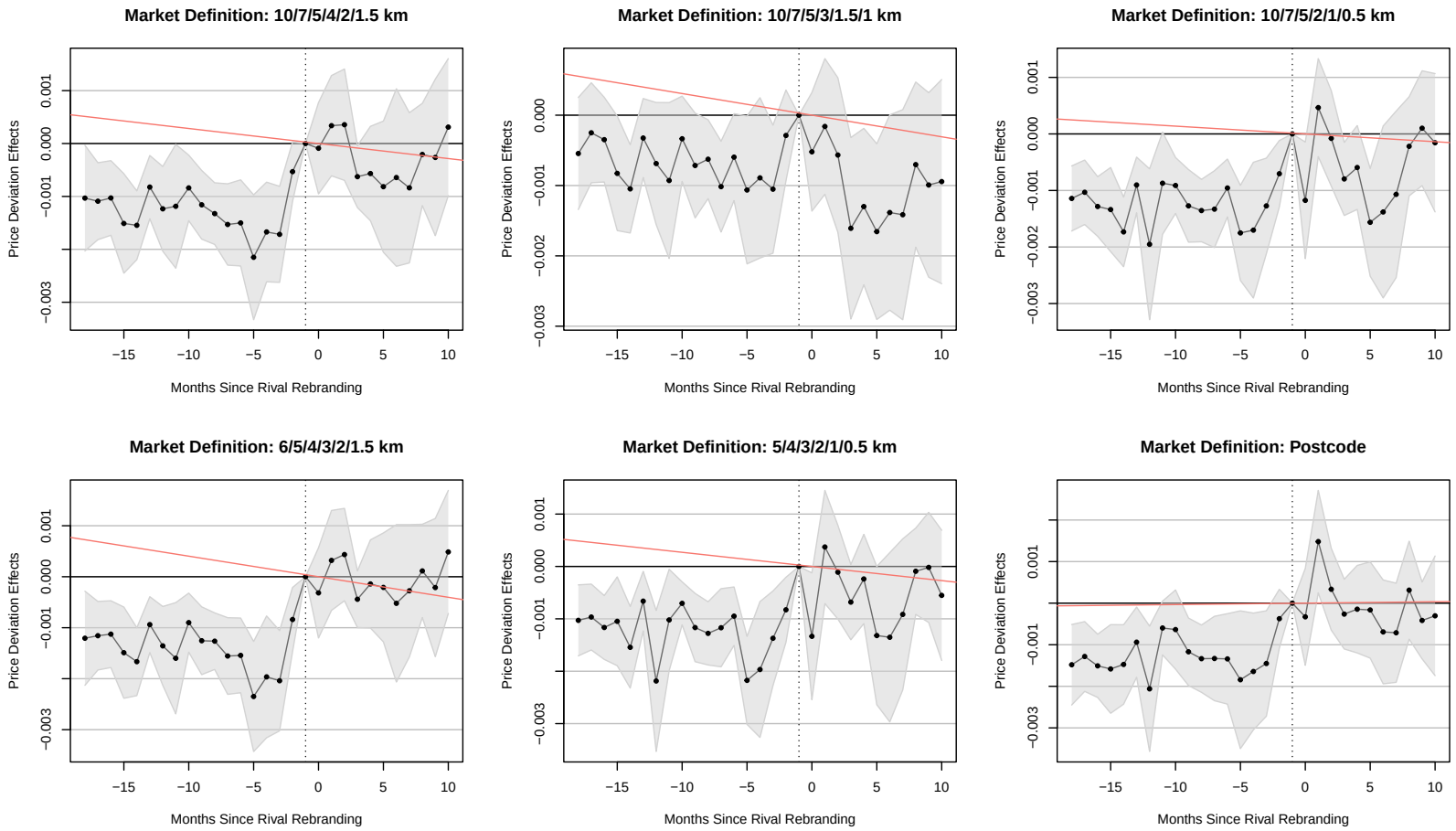


Figure A13: Estimated product price deviation effects of rival rebranding to Extra over time. The horizontal axis displays months relative to the event, $l \in [-18, +10]$, with the vertical line marking $l = -1$—the last month before the event. Point estimates are shown with 95% confidence intervals, based on standard errors clustered at the municipality level. A fitted linear pre-trend is displayed and extrapolated into the post-event period for comparison.

Table A12: Effect of Prix on local product price deviations

	$\hat{P}_{it}^2$					
	(1)	(2)	(3)	(4)	(5)	(6)
T_{Prix}	0.001	0.000	0.001	0.001	0.001	0.000
	(0.001)	(0.001)	(0.001)	(0.001)	(0.001)	(0.001)
ln(Population)	0.041**	0.047***	0.048***	0.044***	0.050***	0.009
	(0.013)	(0.013)	(0.014)	(0.013)	(0.012)	(0.007)
ln(Median Income)	0.002	0.007	0.003	0.004	0.006	0.013
	(0.011)	(0.011)	(0.011)	(0.011)	(0.011)	(0.017)
Market Definition	10–1.5 km	10–1 km	10–0.5 km	6–1.5 km	5–0.5 km	Postcode
Months	36	36	36	36	36	36
Stores	327	372	449	350	479	512
Observations	10,060	11,615	14,365	10,889	15,521	16,705
R^2	0.16	0.16	0.17	0.16	0.17	0.17

Notes: Each column reports the estimated ATT using the estimator of Sun and Abraham (2021) with different local market definitions. The data span all months of 2014–2016 and the demographics are measured at the municipality level. The standard errors shown in parentheses are clustered by Municipality. *** $p < 0.001$; ** $p < 0.01$; * $p < 0.05$.

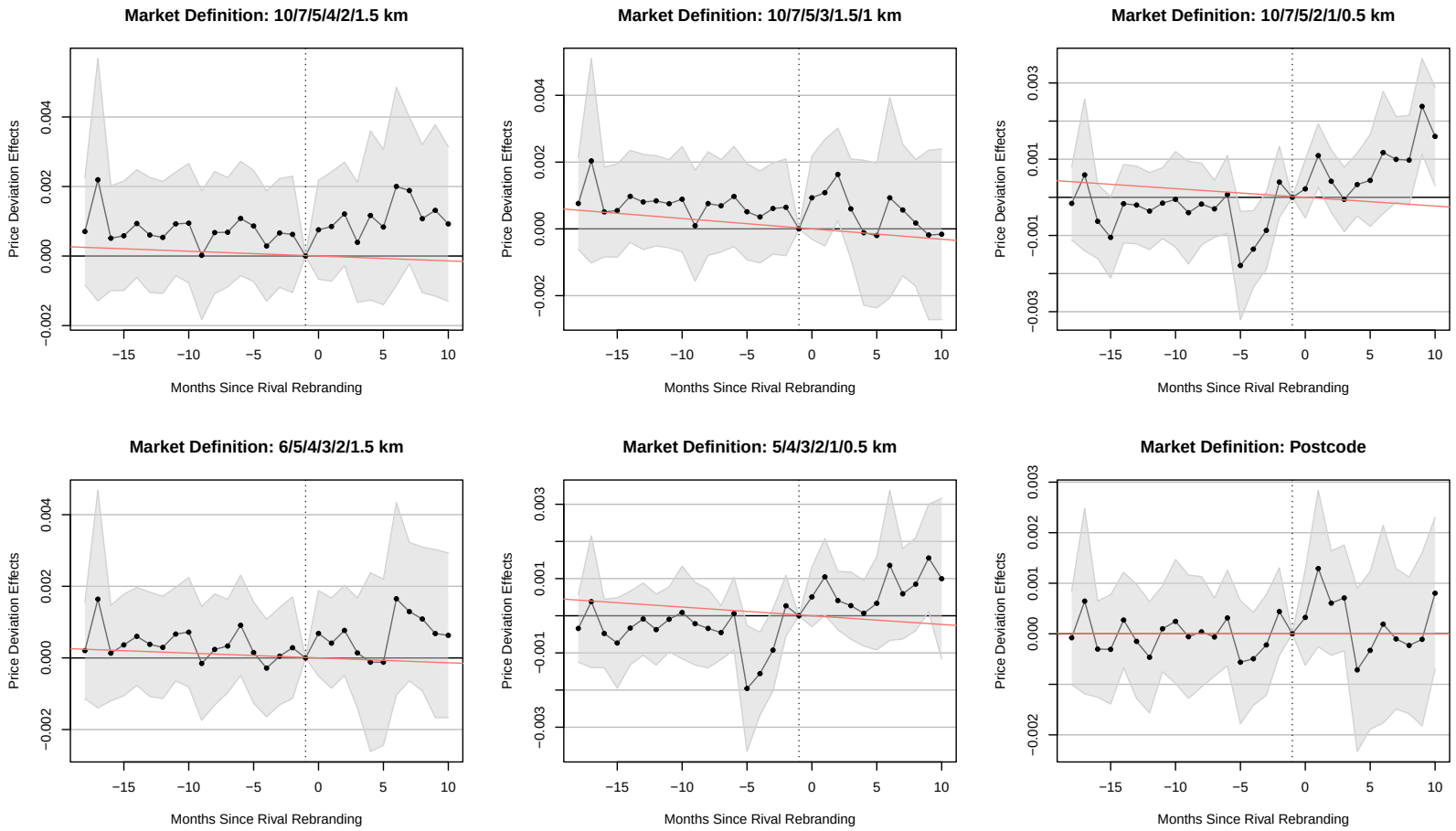


Figure A14: Estimated product price deviation effects of rival rebranding to Prix over time. The horizontal axis displays months relative to the event, $l \in [-18, +10]$, with the vertical line marking $l = -1$—the last month before the event. Point estimates are shown with 95% confidence intervals, based on standard errors clustered at the municipality level. A fitted linear pre-trend is displayed and extrapolated into the post-event period for comparison.

Table A13: Effect of Bunnpris on local product price deviations

	$\hat{P}_{it}^2$					
	(1)	(2)	(3)	(4)	(5)	(6)
T_{Bunnpris}	0.002	0.002	0.001	0.002	0.002	0.000
	(0.001)	(0.001)	(0.001)	(0.001)	(0.001)	(0.001)
ln(Population)	0.042**	0.049***	0.049**	0.048***	0.051***	0.008
	(0.014)	(0.014)	(0.015)	(0.013)	(0.013)	(0.008)
ln(Median Income)	0.001	-0.002	-0.005	-0.003	0.000	0.006
	(0.011)	(0.013)	(0.013)	(0.012)	(0.012)	(0.018)
Market Definition	10–1.5 km	10–1 km	10–0.5 km	6–1.5 km	5–0.5 km	Postcode
Months	36	36	36	36	36	36
Stores	291	342	424	317	456	489
Observations	9,277	10,951	13,781	10,158	14,903	16,112
R^2	0.16	0.16	0.16	0.16	0.17	0.16

Notes: Each column reports the estimated ATT using the estimator of Sun and Abraham (2021) with different local market definitions. The data span all months of 2014–2016 and the demographics are measured at the municipality level. The standard errors shown in parentheses are clustered by Municipality. *** $p < 0.001$; ** $p < 0.01$; * $p < 0.05$.

Event-study plots: local product price deviation index ($\hat{P}^2_{it}$), Bunnpris rebranding

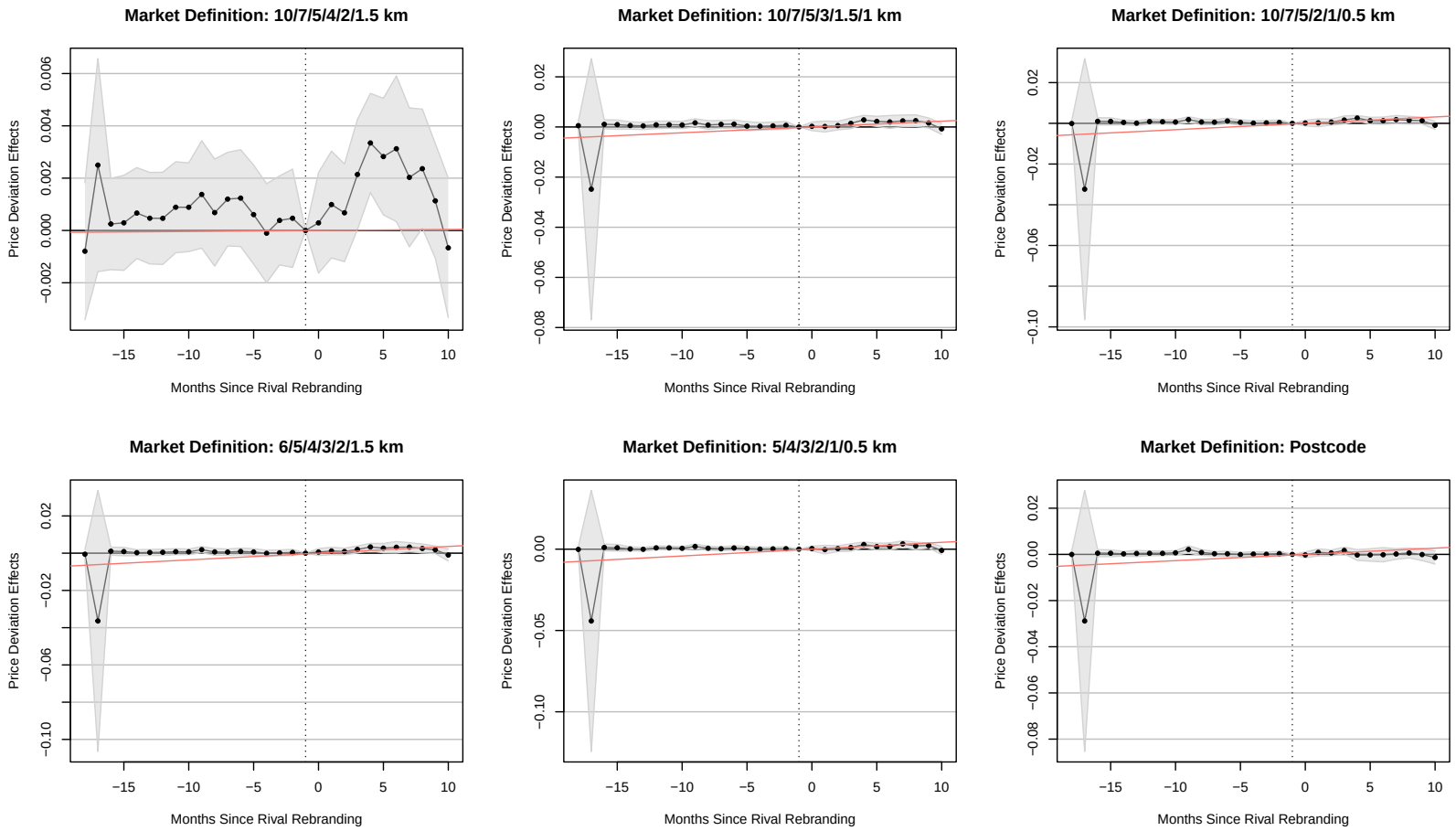


Figure A15: Estimated product price deviation effects of rival rebranding to Bunnpris over time. The horizontal axis displays months relative to the event, $l \in [-18, +10]$, with the vertical line marking $l = -1$—the last month before the event. Point estimates are shown with 95% confidence intervals, based on standard errors clustered at the municipality level. A fitted linear pre-trend is displayed and extrapolated into the post-event period for comparison.

Table A14: Effect of rival closure on local product price deviations

	$\hat{P}_{it}^2$					
	(1)	(2)	(3)	(4)	(5)	(6)
T_{Closure}	0.001	0.001	0.000	0.001	0.000	0.000
	(0.001)	(0.001)	(0.001)	(0.001)	(0.001)	(0.001)
ln(Population)	0.040**	0.046***	0.047***	0.044***	0.049***	0.008
	(0.013)	(0.013)	(0.014)	(0.012)	(0.013)	(0.007)
ln(Median Income)	0.004	0.010	0.004	0.006	0.007	0.009
	(0.011)	(0.011)	(0.012)	(0.011)	(0.011)	(0.017)
Market Definition	10–1.5 km	10–1 km	10–0.5 km	6–1.5 km	5–0.5 km	Postcode
Months	36	36	36	36	36	36
Stores	314	355	434	337	463	504
Observations	10,040	11,363	14,127	10,815	15,141	16,626
R^2	0.16	0.16	0.17	0.16	0.17	0.17

Notes: Each column reports the estimated ATT using the estimator of Sun and Abraham (2021) with different local market definitions. The data span all months of 2014–2016 and the demographics are measured at the municipality level. The standard errors shown in parentheses are clustered by Municipality. *** $p < 0.001$; ** $p < 0.01$; * $p < 0.05$.

Event-study plots: local product price deviation index ($\hat{P}^2_{it}$), rival closure

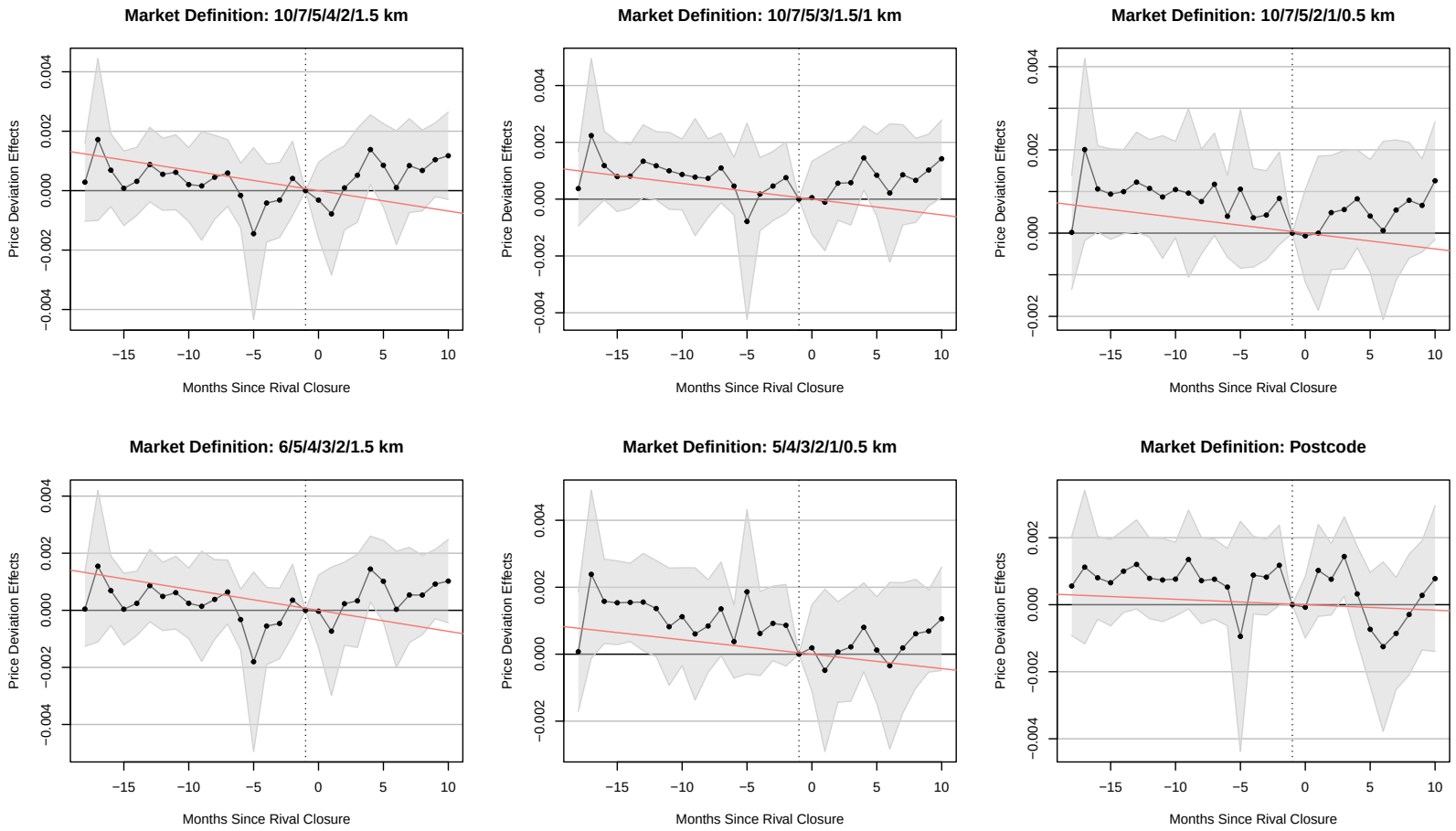


Figure A16: Estimated product price deviation effects of rival closure over time. The horizontal axis displays months relative to the event, $l \in [-18, +10]$, with the vertical line marking $l = -1$—the last month before the event. Point estimates are shown with 95% confidence intervals, based on standard errors clustered at the municipality level. A fitted linear pre-trend is displayed and extrapolated into the post-event period for comparison.

Table A15: Effect of Extra on variety

	ln(Variety)					
	(1)	(2)	(3)	(4)	(5)	(6)
T_{Extra}	-0.009	-0.009*	-0.013***	-0.007	-0.011**	-0.007*
	(0.005)	(0.003)	(0.004)	(0.004)	(0.004)	(0.003)
ln(Population)	-0.017	-0.031	-0.089	0.015	-0.078	0.000
	(0.127)	(0.128)	(0.122)	(0.123)	(0.122)	(0.038)
ln(Median Income)	0.058	0.038	0.036	0.080	0.049	0.052
	(0.098)	(0.096)	(0.089)	(0.096)	(0.089)	(0.113)
Market Definition	10–1.5 km	10–1 km	10–0.5 km	6–1.5 km	5–0.5 km	Postcode
Months	36	36	36	36	36	36
Stores	348	396	468	372	492	529
Observations	10,808	12,371	14,955	11,662	15,893	17,351
R^2	0.96	0.97	0.97	0.97	0.97	0.97

Notes: Each column reports the estimated ATT using the estimator of Sun and Abraham (2021) with different local market definitions. The data span all months of 2014–2016 and the demographics are measured at the municipality level. The standard errors shown in parentheses are clustered by Municipality. *** $p < 0.001$; ** $p < 0.01$; * $p < 0.05$.

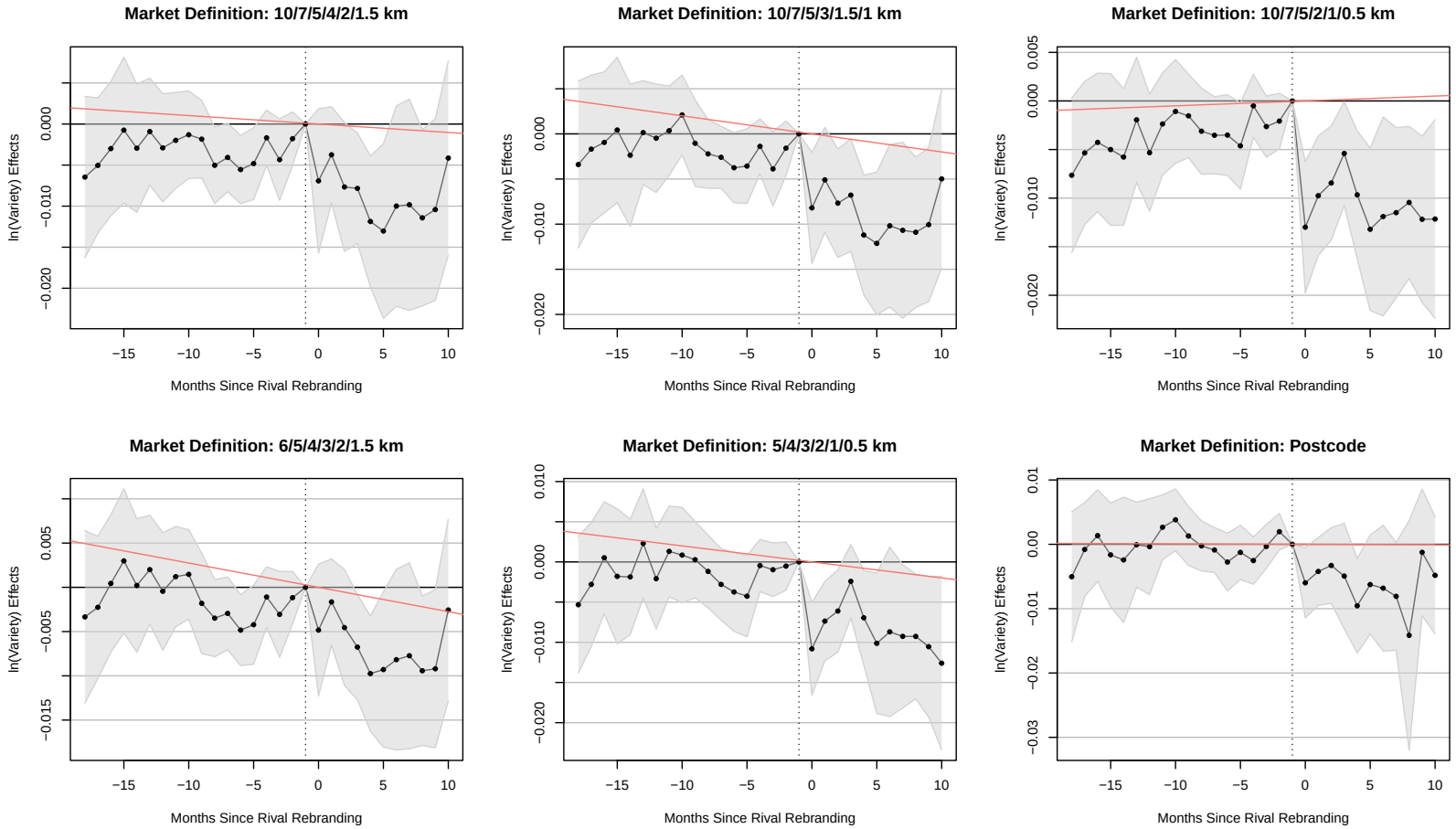


Figure A17: Estimated percentage variety effects of rival rebranding to Extra over time. The horizontal axis displays months relative to the event, $l \in [-18, +10]$, with the vertical line marking $l = -1$—the last month before the event. Point estimates are shown with 95% confidence intervals, based on standard errors clustered at the municipality level. A fitted linear pre-trend is displayed and extrapolated into the post-event period for comparison.

Table A16: Effect of Prix on variety

	ln(Variety)					
	(1)	(2)	(3)	(4)	(5)	(6)
T_{Prix}	-0.001	-0.007	-0.005	-0.002	-0.004	0.000
	(0.004)	(0.004)	(0.004)	(0.004)	(0.003)	(0.004)
ln(Population)	0.015	-0.001	-0.055	0.038	-0.070	0.008
	(0.126)	(0.133)	(0.124)	(0.122)	(0.123)	(0.035)
ln(Median Income)	0.057	0.047	0.047	0.077	0.058	0.065
	(0.099)	(0.099)	(0.090)	(0.097)	(0.090)	(0.109)
Market Definition	10–1.5 km	10–1 km	10–0.5 km	6–1.5 km	5–0.5 km	Postcode
Months	36	36	36	36	36	36
Stores	327	372	449	350	479	512
Observations	10,060	11,615	14,365	10,889	15,521	16,705
R^2	0.96	0.96	0.97	0.96	0.97	0.97

Notes: Each column reports the estimated ATT using the estimator of Sun and Abraham (2021) with different local market definitions. The data span all months of 2014–2016 and the demographics are measured at the municipality level. The standard errors shown in parentheses are clustered by Municipality. *** $p < 0.001$; ** $p < 0.01$; * $p < 0.05$.

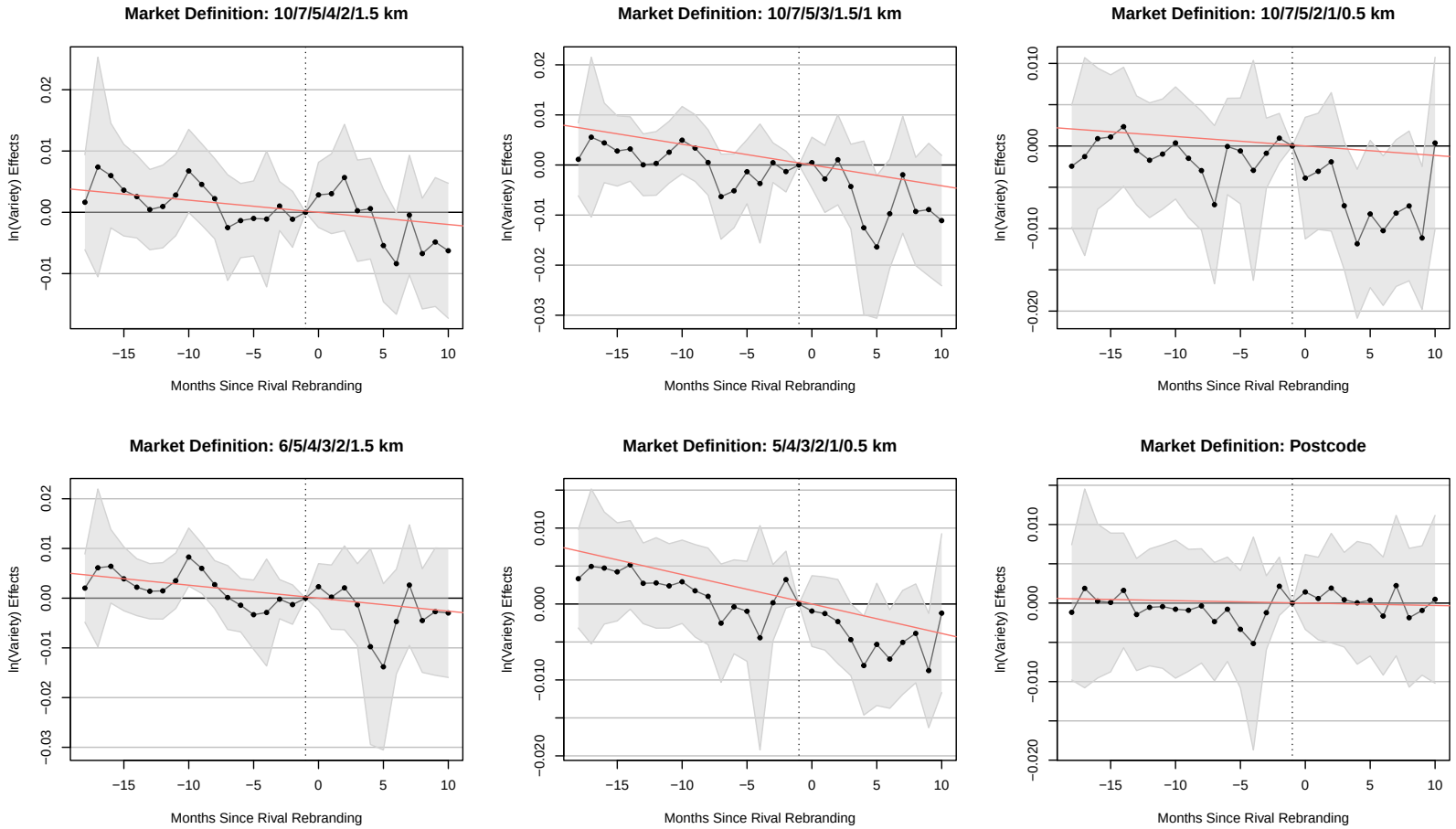


Figure A18: Estimated percentage variety effects of rival rebranding to Prix over time. The horizontal axis displays months relative to the event, $l \in [-18, +10]$, with the vertical line marking $l = -1$—the last month before the event. Point estimates are shown with 95% confidence intervals, based on standard errors clustered at the municipality level. A fitted linear pre-trend is displayed and extrapolated into the post-event period for comparison.

Table A17: Effect of Bunnpris on variety

	ln(Variety)					
	(1)	(2)	(3)	(4)	(5)	(6)
T_{Bunnpris}	0.002	-0.003	-0.003	-0.002	-0.001	0.001
	(0.005)	(0.003)	(0.005)	(0.005)	(0.005)	(0.005)
ln(Population)	0.001	-0.008	-0.077	0.032	-0.080	0.010
	(0.131)	(0.129)	(0.123)	(0.127)	(0.124)	(0.036)
ln(Median Income)	0.080	0.079	0.051	0.110	0.066	0.083
	(0.101)	(0.093)	(0.086)	(0.097)	(0.087)	(0.108)
Market Definition	10–1.5 km	10–1 km	10–0.5 km	6–1.5 km	5–0.5 km	Postcode
Months	36	36	36	36	36	36
Stores	291	342	424	317	456	489
Observations	9,277	10,951	13,781	10,158	14,903	16,112
R^2	0.96	0.96	0.97	0.96	0.97	0.97

Notes: Each column reports the estimated ATT using the estimator of Sun and Abraham (2021) with different local market definitions. The data span all months of 2014–2016 and the demographics are measured at the municipality level. The standard errors shown in parentheses are clustered by Municipality. *** $p < 0.001$; ** $p < 0.01$; * $p < 0.05$.

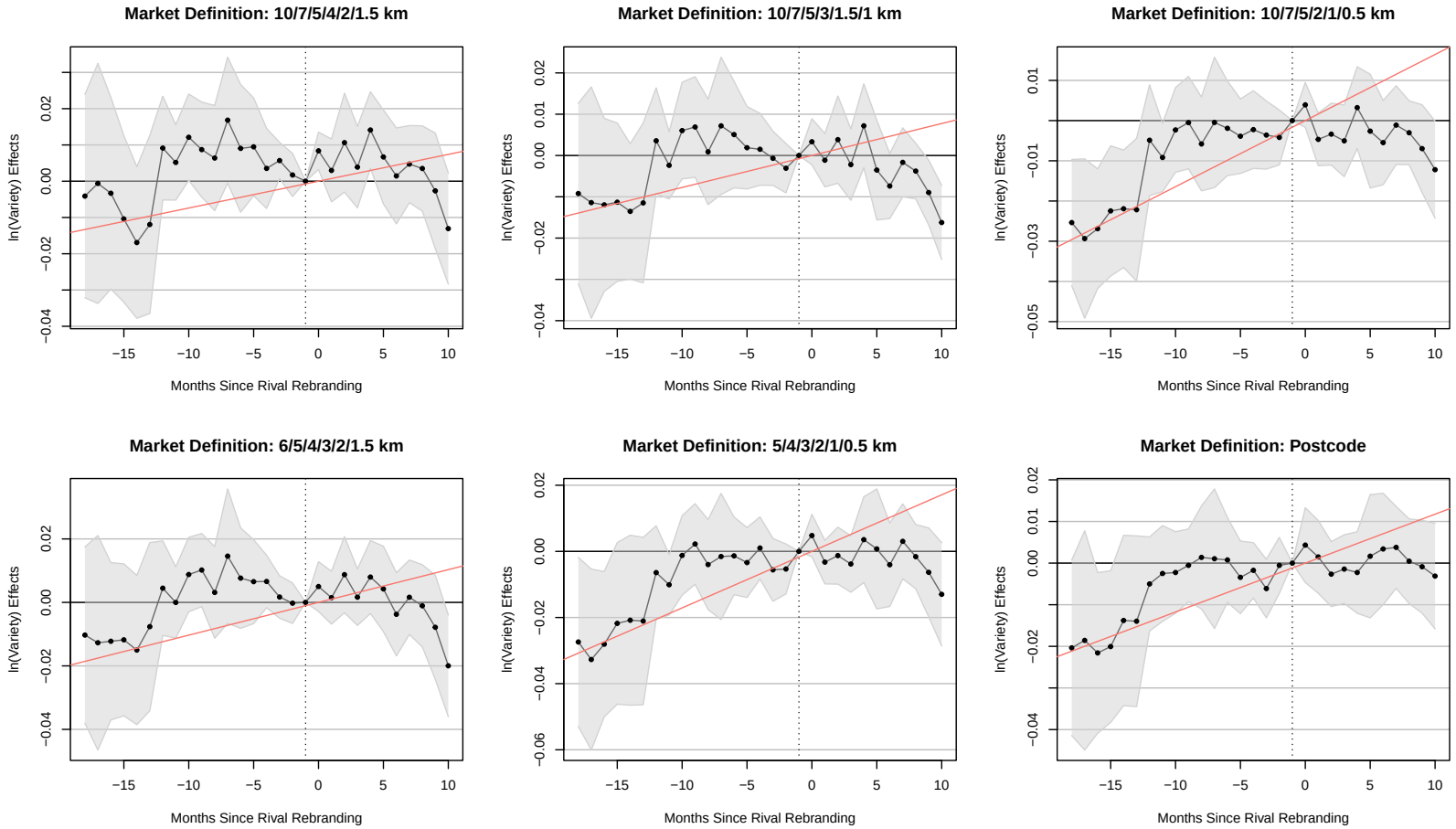


Figure A19: Estimated percentage variety effects of rival rebranding to Bunnpris over time. The horizontal axis displays months relative to the event, $l \in [-18, +10]$, with the vertical line marking $l = -1$—the last month before the event. Point estimates are shown with 95% confidence intervals, based on standard errors clustered at the municipality level. A fitted linear pre-trend is displayed and extrapolated into the post-event period for comparison.

Table A18: Effect of rival closure on variety

	ln(Variety)					
	(1)	(2)	(3)	(4)	(5)	(6)
T_{Closure}	0.006*	0.006	0.007	0.007*	0.004	0.004*
	(0.003)	(0.004)	(0.005)	(0.003)	(0.004)	(0.002)
ln(Population)	-0.038	-0.036	-0.092	-0.011	-0.075	0.004
	(0.131)	(0.129)	(0.127)	(0.128)	(0.125)	(0.037)
ln(Median Income)	0.047	0.055	0.021	0.072	0.031	0.059
	(0.104)	(0.100)	(0.096)	(0.103)	(0.095)	(0.111)
Market Definition	10–1.5 km	10–1 km	10–0.5 km	6–1.5 km	5–0.5 km	Postcode
Months	36	36	36	36	36	36
Stores	314	355	434	337	463	504
Observations	10,040	11,363	14,127	10,815	15,141	16,626
R^2	0.96	0.96	0.97	0.96	0.97	0.97

Notes: Each column reports the estimated ATT using the estimator of Sun and Abraham (2021) with different local market definitions. The data span all months of 2014–2016 and the demographics are measured at the municipality level. The standard errors shown in parentheses are clustered by Municipality. *** $p < 0.001$; ** $p < 0.01$; * $p < 0.05$.

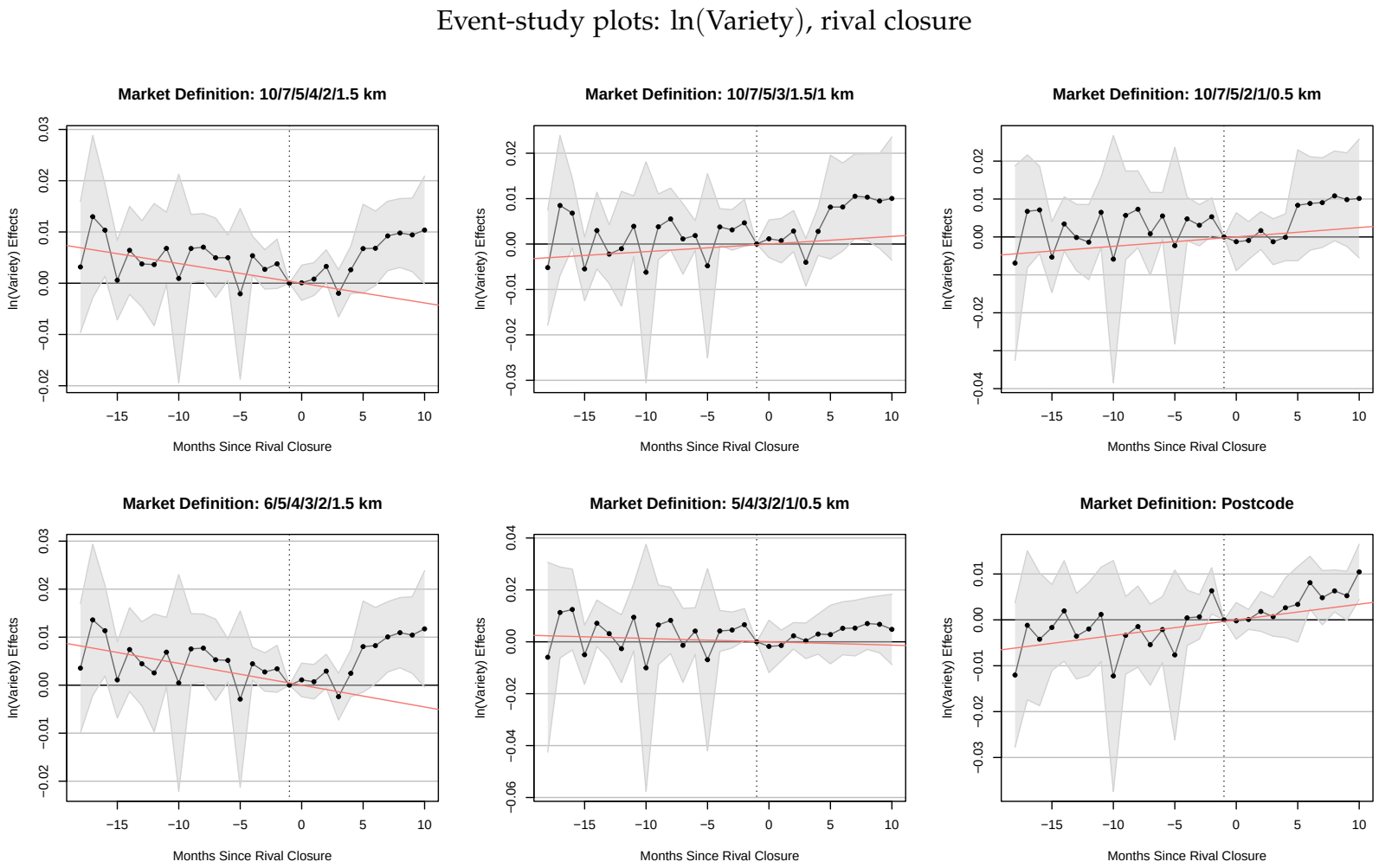


Figure A20: Estimated percentage variety effects of rival closure over time. The horizontal axis displays months relative to the event, $l \in [-18, +10]$, with the vertical line marking $l = -1$—the last month before the event. Point estimates are shown with 95% confidence intervals, based on standard errors clustered at the municipality level. A fitted linear pre-trend is displayed and extrapolated into the post-event period for comparison.

Table A19: Effect of Extra on variety entropy

	Variety Entropy					
	(1)	(2)	(3)	(4)	(5)	(6)
T_{Extra}	-0.007	-0.006	-0.016**	-0.007	-0.020***	-0.010*
	(0.005)	(0.005)	(0.005)	(0.005)	(0.005)	(0.004)
ln(Population)	0.180	0.140	0.108	0.165	0.106	0.097***
	(0.134)	(0.135)	(0.112)	(0.127)	(0.109)	(0.022)
ln(Median Income)	0.170	0.131	0.155	0.177	0.163*	0.196*
	(0.099)	(0.100)	(0.079)	(0.096)	(0.077)	(0.078)
Market Definition	10–1.5 km	10–1 km	10–0.5 km	6–1.5 km	5–0.5 km	Postcode
Months	36	36	36	36	36	36
Stores	348	396	468	372	492	529
Observations	10,808	12,371	14,955	11,662	15,893	17,351
R^2	0.88	0.89	0.90	0.89	0.90	0.90

Notes: Each column reports the estimated ATT using the estimator of Sun and Abraham (2021) with different local market definitions. The data span all months of 2014–2016 and the demographics are measured at the municipality level. The standard errors shown in parentheses are clustered by Municipality. *** $p < 0.001$; ** $p < 0.01$; * $p < 0.05$.

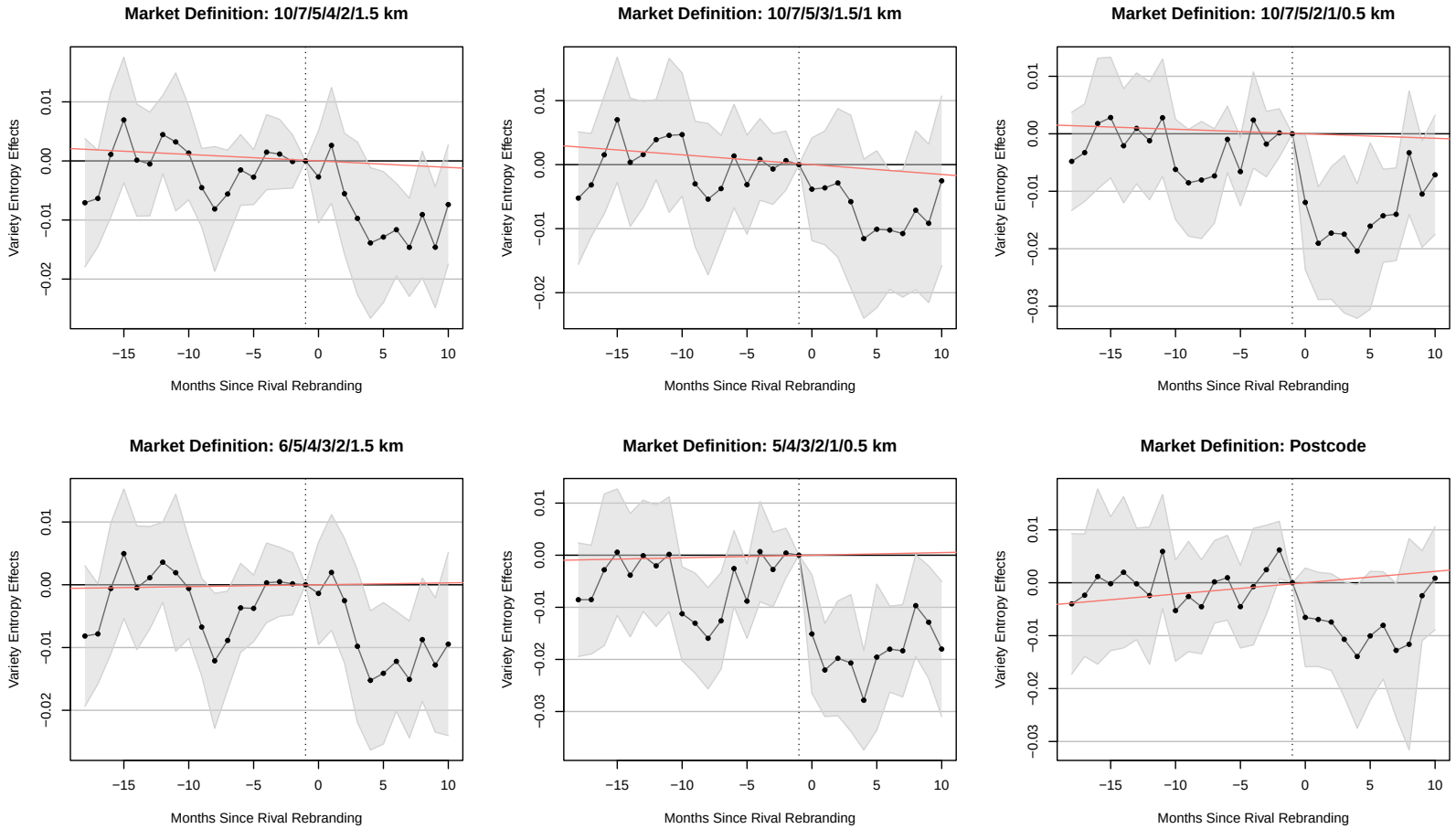


Figure A21: Estimated variety index effects of rival rebranding to Extra over time. The horizontal axis displays months relative to the event, $l \in [-18, +10]$, with the vertical line marking $l = -1$—the last month before the event. Point estimates are shown with 95% confidence intervals, based on standard errors clustered at the municipality level. A fitted linear pre-trend is displayed and extrapolated into the post-event period for comparison.

Table A20: Effect of Prix on variety entropy

	Variety Entropy					
	(1)	(2)	(3)	(4)	(5)	(6)
T_{Prix}	0.003	0.004	0.005	0.003	0.008	0.005
	(0.007)	(0.006)	(0.004)	(0.007)	(0.005)	(0.007)
ln(Population)	0.193	0.160	0.132	0.176	0.110	0.102***
	(0.133)	(0.132)	(0.112)	(0.127)	(0.110)	(0.020)
ln(Median Income)	0.190	0.164	0.176*	0.194*	0.179*	0.214**
	(0.099)	(0.099)	(0.081)	(0.096)	(0.078)	(0.077)
Market Definition	10–1.5 km	10–1 km	10–0.5 km	6–1.5 km	5–0.5 km	Postcode
Months	36	36	36	36	36	36
Stores	327	372	449	350	479	512
Observations	10,060	11,615	14,365	10,889	15,521	16,705
R^2	0.88	0.89	0.89	0.88	0.90	0.90

Notes: Each column reports the estimated ATT using the estimator of Sun and Abraham (2021) with different local market definitions. The data span all months of 2014–2016 and the demographics are measured at the municipality level. The standard errors shown in parentheses are clustered by Municipality. *** $p < 0.001$; ** $p < 0.01$; * $p < 0.05$.

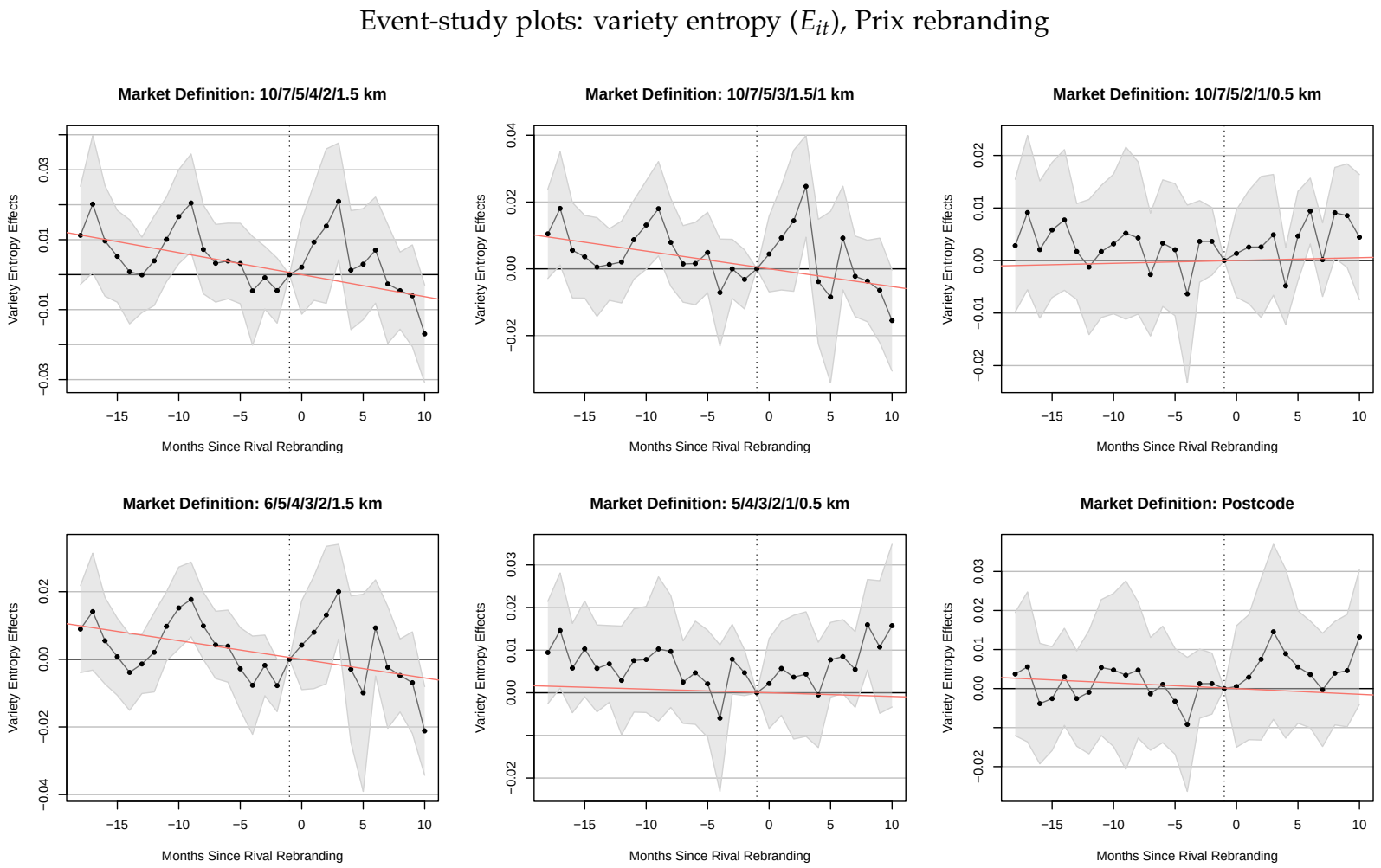


Figure A22: Estimated variety index effects of rival rebranding to Prix over time. The horizontal axis displays months relative to the event, $l \in [-18, +10]$, with the vertical line marking $l = -1$—the last month before the event. Point estimates are shown with 95% confidence intervals, based on standard errors clustered at the municipality level. A fitted linear pre-trend is displayed and extrapolated into the post-event period for comparison.

Table A21: Effect of Bunnpris on variety entropy

	Variety Entropy					
	(1)	(2)	(3)	(4)	(5)	(6)
T_{Bunnpris}	0.020**	-0.002	-0.007	-0.002	-0.009	-0.005
	(0.006)	(0.009)	(0.009)	(0.011)	(0.010)	(0.008)
ln(Population)	0.180	0.150	0.112	0.172	0.101	0.104***
	(0.138)	(0.134)	(0.114)	(0.131)	(0.112)	(0.021)
ln(Median Income)	0.186	0.164	0.168*	0.217*	0.180*	0.226**
	(0.099)	(0.094)	(0.077)	(0.095)	(0.076)	(0.077)
Market Definition	10–1.5 km	10–1 km	10–0.5 km	6–1.5 km	5–0.5 km	Postcode
Months	36	36	36	36	36	36
Stores	291	342	424	317	456	489
Observations	9,277	10,951	13,781	10,158	14,903	16,112
R^2	0.87	0.88	0.89	0.88	0.89	0.90

Notes: Each column reports the estimated ATT using the estimator of Sun and Abraham (2021) with different local market definitions. The data span all months of 2014–2016 and the demographics are measured at the municipality level. The standard errors shown in parentheses are clustered by Municipality. *** $p < 0.001$; ** $p < 0.01$; * $p < 0.05$.

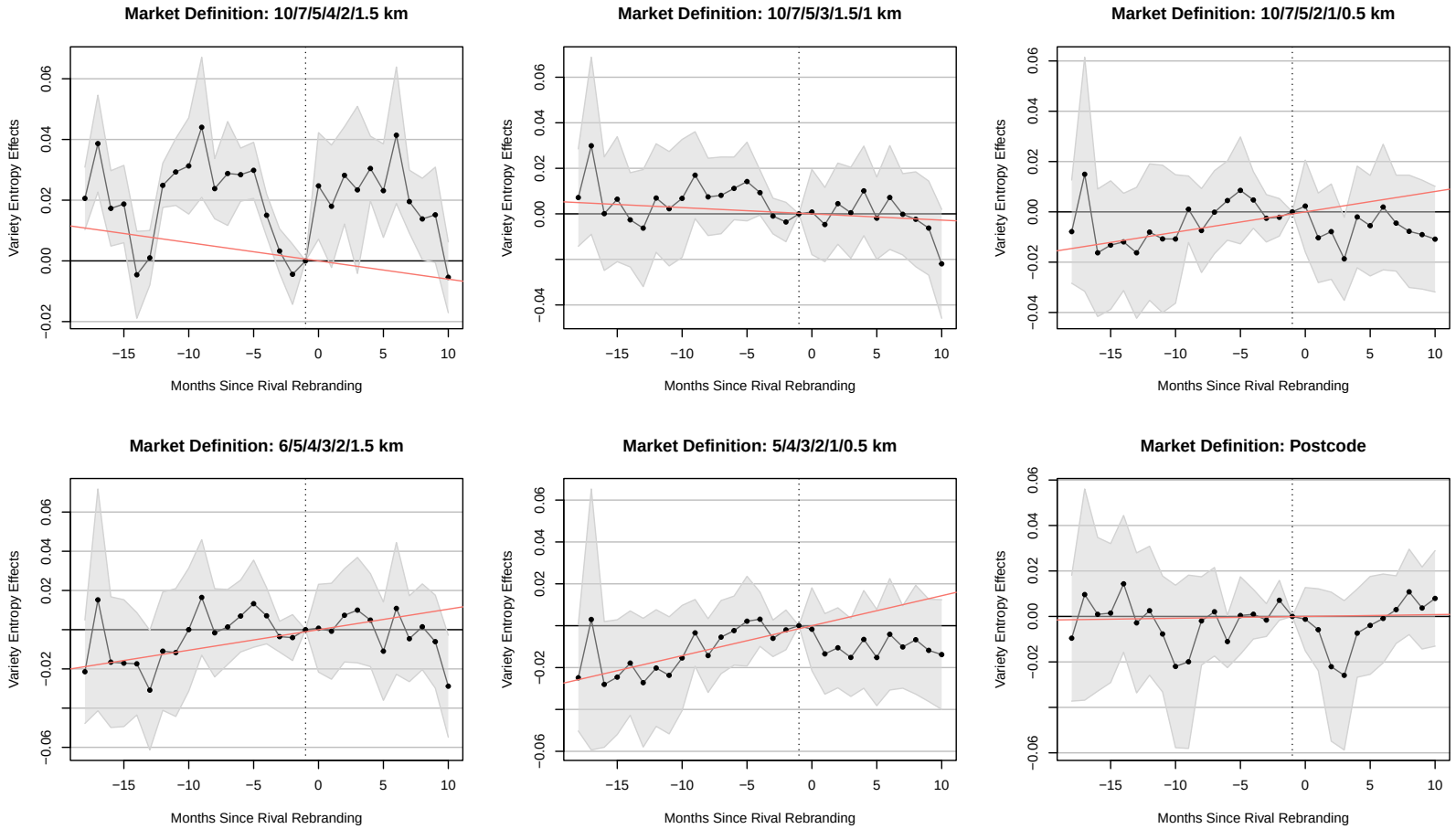


Figure A23: Estimated variety index effects of rival rebranding to Bunnpris over time. The horizontal axis displays months relative to the event, $l \in [-18, +10]$, with the vertical line marking $l = -1$—the last month before the event. Point estimates are shown with 95% confidence intervals, based on standard errors clustered at the municipality level. A fitted linear pre-trend is displayed and extrapolated into the post-event period for comparison.

Table A22: Effect of rival closures on variety entropy

	Variety Entropy					
	(1)	(2)	(3)	(4)	(5)	(6)
T_{Closure}	0.002	0.003	0.001	0.002	-0.002	-0.007
	(0.006)	(0.007)	(0.007)	(0.007)	(0.005)	(0.005)
ln(Population)	0.134	0.133	0.102	0.128	0.127	0.098***
	(0.146)	(0.138)	(0.123)	(0.138)	(0.114)	(0.021)
ln(Median Income)	0.175	0.174	0.142	0.191	0.140	0.201*
	(0.107)	(0.103)	(0.089)	(0.105)	(0.085)	(0.079)
Market Definition	10–1.5 km	10–1 km	10–0.5 km	6–1.5 km	5–0.5 km	Postcode
Months	36	36	36	36	36	36
Stores	314	355	434	337	463	504
Observations	10,040	11,363	14,127	10,815	15,141	16,626
R^2	0.87	0.88	0.89	0.88	0.89	0.90

Notes: Each column reports the estimated ATT using the estimator of Sun and Abraham (2021) with different local market definitions. The data span all months of 2014–2016 and the demographics are measured at the municipality level. The standard errors shown in parentheses are clustered by Municipality. *** $p < 0.001$; ** $p < 0.01$; * $p < 0.05$.

Event-study plots: variety entropy (E_{it}), rival closure

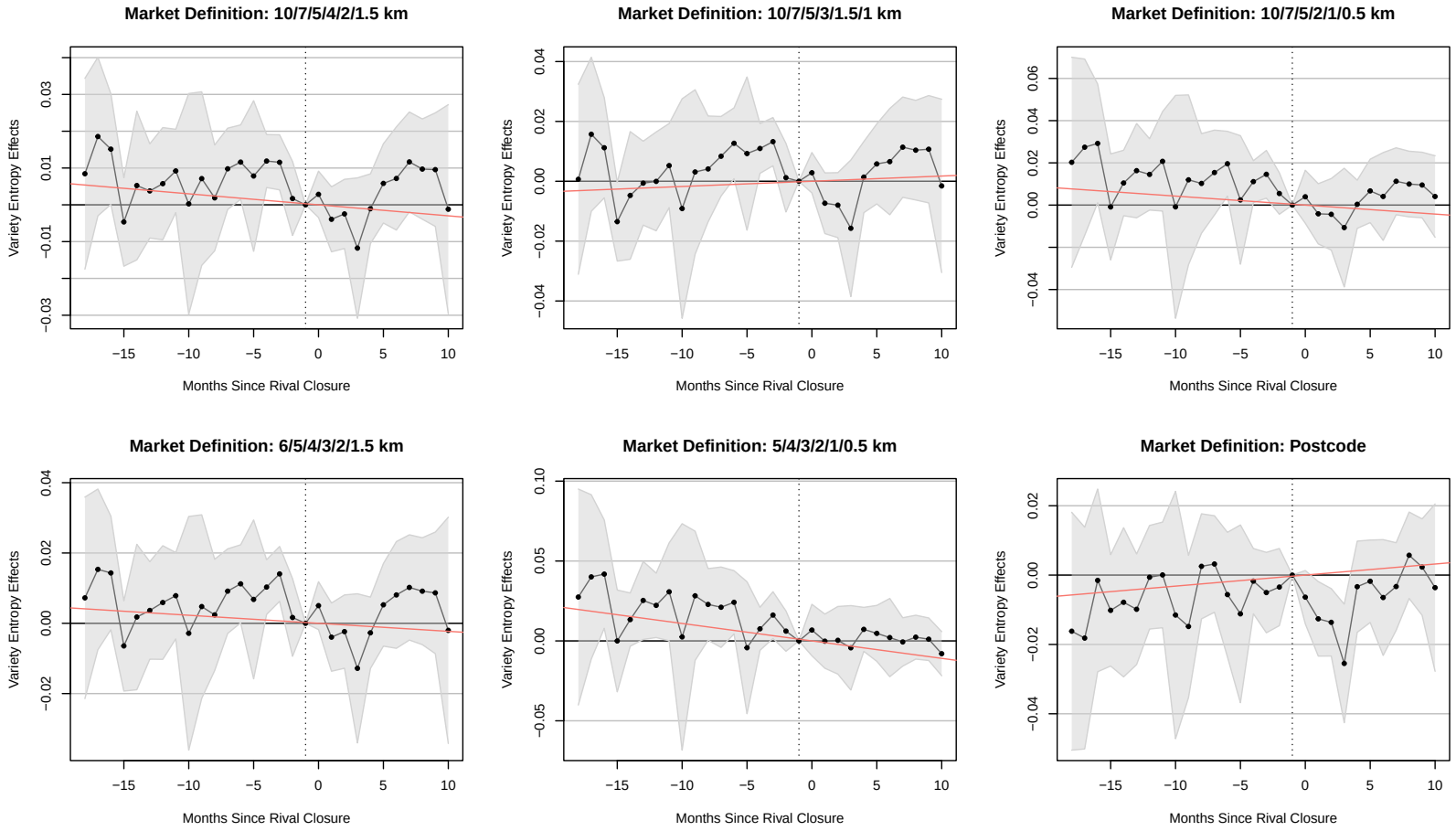


Figure A24: Estimated variety index effects of rival closure over time. The horizontal axis displays months relative to the event, $l \in [-18, +10]$, with the vertical line marking $l = -1$—the last month before the event. Point estimates are shown with 95% confidence intervals, based on standard errors clustered at the municipality level. A fitted linear pre-trend is displayed and extrapolated into the post-event period for comparison.

Table A23: Effect of Extra on sales

	ln(Sales)					
	(1)	(2)	(3)	(4)	(5)	(6)
T_{Extra}	-0.093***	-0.101***	-0.126***	-0.070***	-0.129***	-0.117***
	(0.016)	(0.018)	(0.014)	(0.015)	(0.015)	(0.021)
ln(Population)	-0.109	0.000	-0.108	0.002	-0.149	-0.043
	(0.389)	(0.401)	(0.395)	(0.400)	(0.461)	(0.069)
ln(Median Income)	0.574	0.472	0.164	0.561	0.120	0.357
	(0.351)	(0.335)	(0.308)	(0.342)	(0.320)	(0.271)
Market Definition	10–1.5 km	10–1 km	10–0.5 km	6–1.5 km	5–0.5 km	Postcode
Months	36	36	36	36	36	36
Stores	348	396	468	372	492	529
Observations	10,808	12,371	14,955	11,662	15,893	17,351
R^2	0.83	0.84	0.84	0.83	0.84	0.85

Notes: Each column reports the estimated ATT using the estimator of Sun and Abraham (2021) with different local market definitions. The data span all months of 2014–2016 and the demographics are measured at the municipality level. The standard errors shown in parentheses are clustered by Municipality. *** $p < 0.001$; ** $p < 0.01$; * $p < 0.05$.

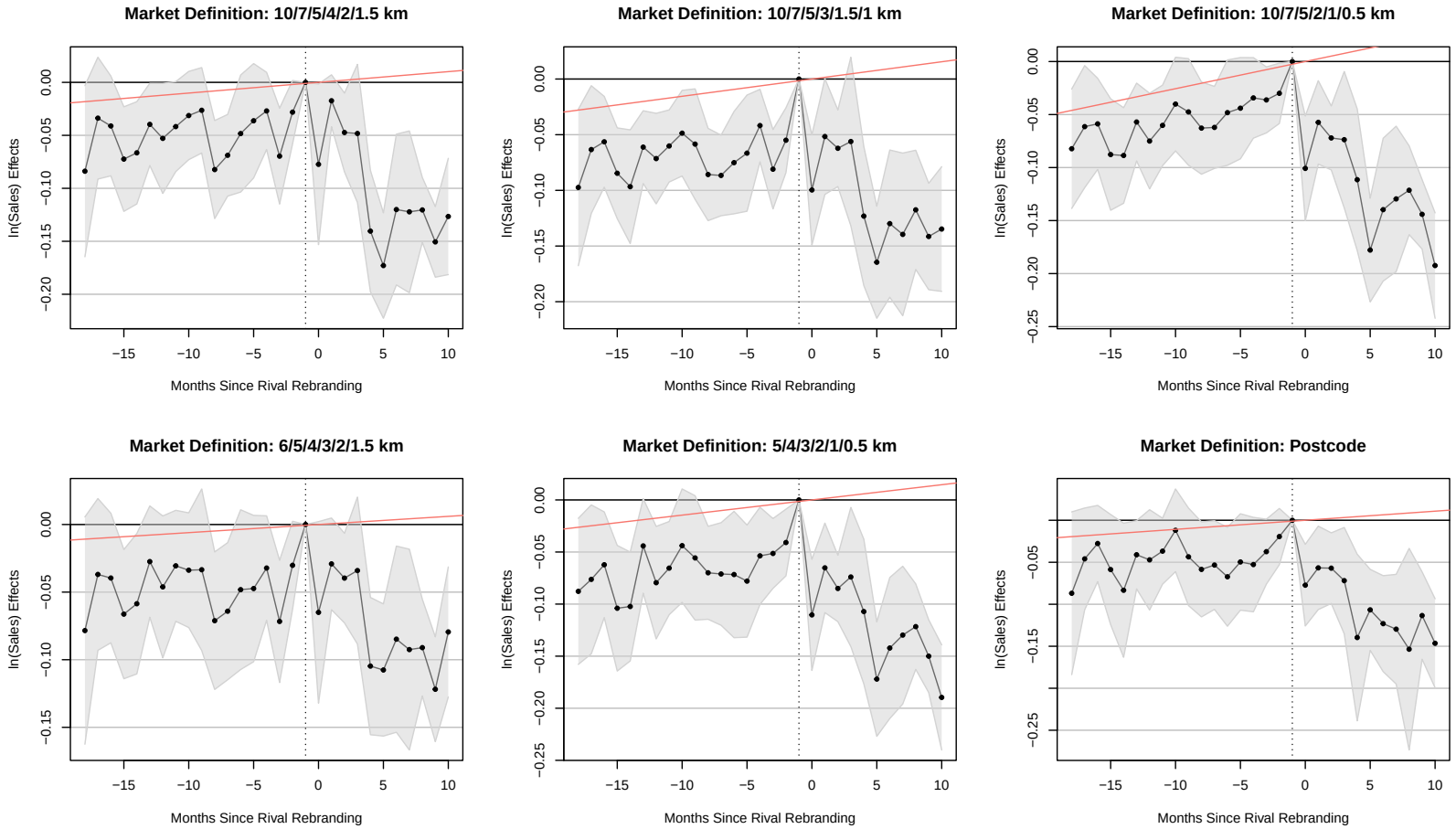


Figure A25: Estimated percentage sales effects of rival rebranding to Extra over time. The horizontal axis displays months relative to the event, $l \in [-18, +10]$, with the vertical line marking $l = -1$—the last month before the event. Point estimates are shown with 95% confidence intervals, based on standard errors clustered at the municipality level. A fitted linear pre-trend is displayed and extrapolated into the post-event period for comparison.

Table A24: Effect of Prix on sales

	ln(Sales)					
	(1)	(2)	(3)	(4)	(5)	(6)
T_{Prix}	-0.010	-0.008	-0.015	-0.025	-0.025	-0.033
	(0.017)	(0.018)	(0.022)	(0.018)	(0.021)	(0.018)
ln(Population)	0.105	0.181	-0.007	0.185	-0.124	-0.003
	(0.378)	(0.383)	(0.407)	(0.391)	(0.471)	(0.057)
ln(Median Income)	0.532	0.429	0.059	0.541	0.058	0.437
	(0.344)	(0.330)	(0.311)	(0.339)	(0.324)	(0.262)
Market Definition	10–1.5 km	10–1 km	10–0.5 km	6–1.5 km	5–0.5 km	Postcode
Months	36	36	36	36	36	36
Stores	327	372	449	350	479	512
Observations	10,060	11,615	14,365	10,889	15,521	16,705
R^2	0.83	0.84	0.84	0.83	0.84	0.85

Notes: Each column reports the estimated ATT using the estimator of Sun and Abraham (2021) with different local market definitions. The data span all months of 2014–2016 and the demographics are measured at the municipality level. The standard errors shown in parentheses are clustered by Municipality. *** $p < 0.001$; ** $p < 0.01$; * $p < 0.05$.

Event-study plots: ln(Sales), Prix rebranding

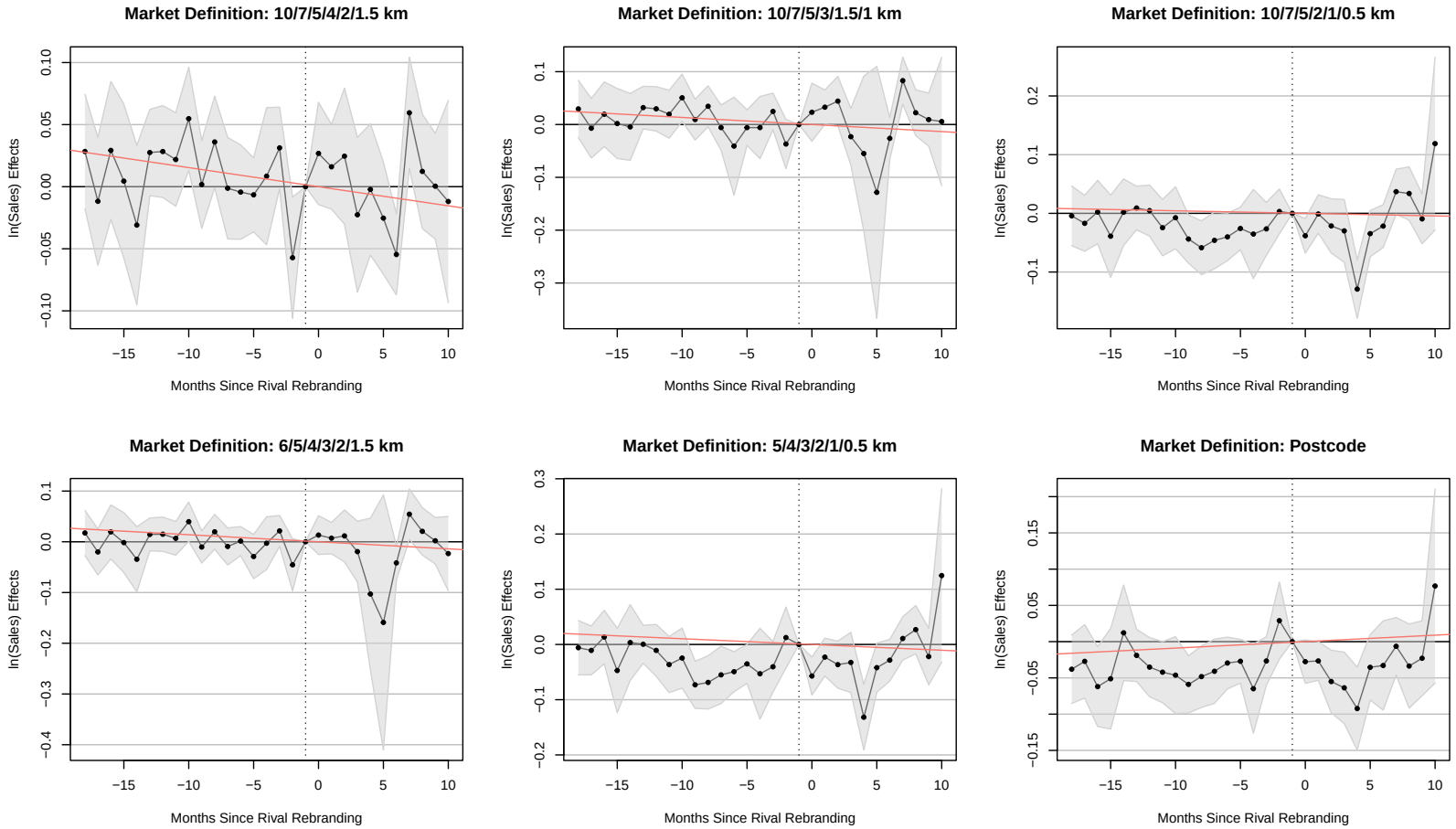


Figure A26: Estimated percentage sales effects of rival rebranding to Prix over time. The horizontal axis displays months relative to the event, $l \in [-18, +10]$, with the vertical line marking $l = -1$—the last month before the event. Point estimates are shown with 95% confidence intervals, based on standard errors clustered at the municipality level. A fitted linear pre-trend is displayed and extrapolated into the post-event period for comparison.

Table A25: Effect of Bunnpris on sales

	ln(Sales)					
	(1)	(2)	(3)	(4)	(5)	(6)
T_{Bunnpris}	0.081**	0.023	0.025	0.037	-0.001	-0.045
	(0.030)	(0.035)	(0.044)	(0.032)	(0.053)	(0.039)
ln(Population)	0.011	0.110	-0.113	0.107	-0.199	-0.011
	(0.391)	(0.394)	(0.399)	(0.400)	(0.469)	(0.058)
ln(Median Income)	0.660	0.453	0.057	0.559	0.058	0.420
	(0.351)	(0.328)	(0.296)	(0.348)	(0.313)	(0.268)
Market Definition	10–1.5 km	10–1 km	10–0.5 km	6–1.5 km	5–0.5 km	Postcode
Months	36	36	36	36	36	36
Stores	291	342	424	317	456	489
Observations	9,277	10,951	13,781	10,158	14,903	16,112
R^2	0.82	0.83	0.83	0.82	0.84	0.85

Notes: Each column reports the estimated ATT using the estimator of Sun and Abraham (2021) with different local market definitions. The data span all months of 2014–2016 and the demographics are measured at the municipality level. The standard errors shown in parentheses are clustered by Municipality. *** $p < 0.001$; ** $p < 0.01$; * $p < 0.05$.

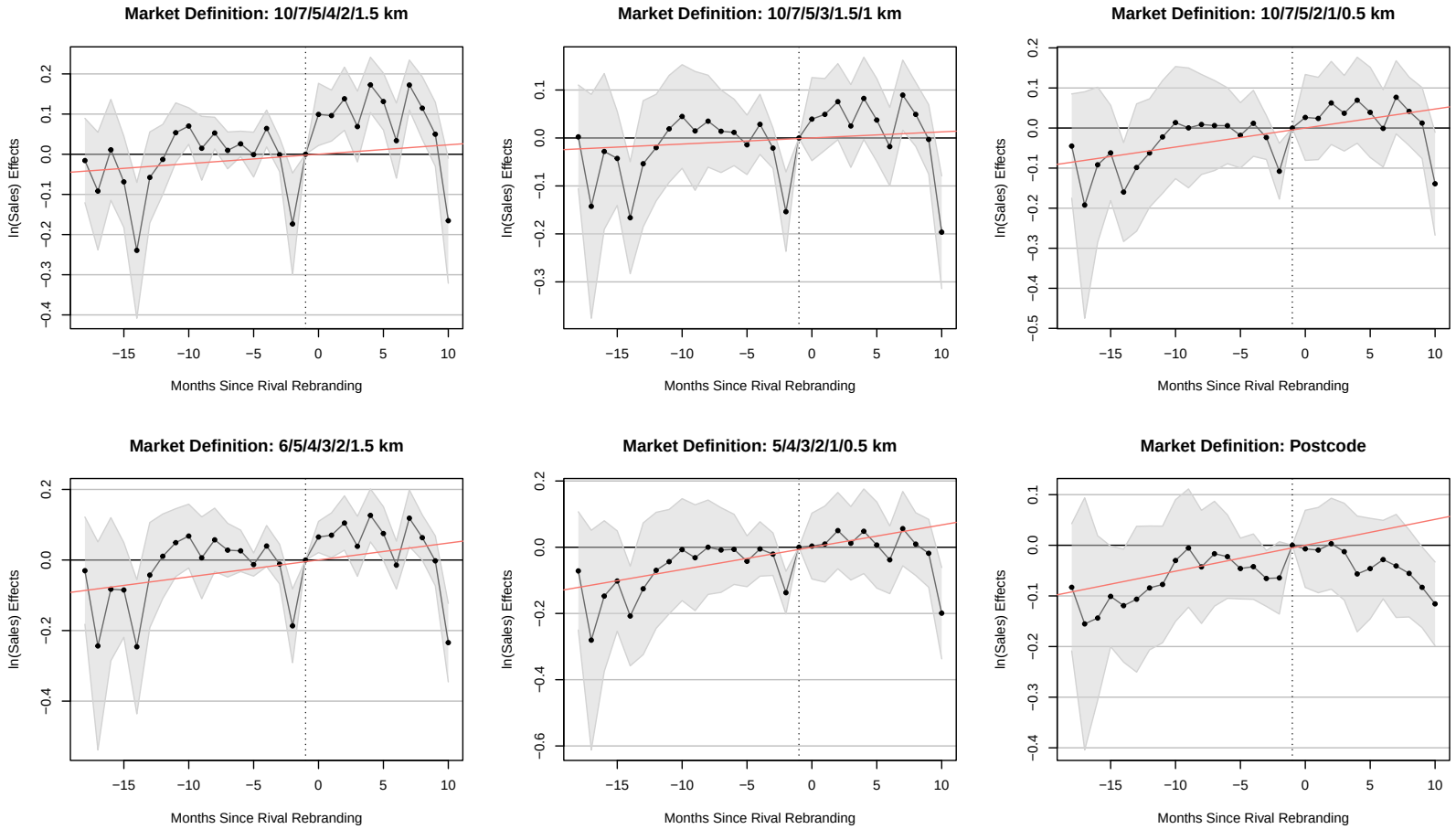


Figure A27: Estimated percentage sales effects of rival rebranding to Bunnpris over time. The horizontal axis displays months relative to the event, $l \in [-18, +10]$, with the vertical line marking $l = -1$—the last month before the event. Point estimates are shown with 95% confidence intervals, based on standard errors clustered at the municipality level. A fitted linear pre-trend is displayed and extrapolated into the post-event period for comparison.

Table A26: Effect of rival closure on sales

	ln(Sales)					
	(1)	(2)	(3)	(4)	(5)	(6)
T_{Closure}	0.078***	0.083***	0.076***	0.082***	0.073**	0.024
	(0.019)	(0.023)	(0.021)	(0.022)	(0.025)	(0.019)
ln(Population)	-0.064	0.063	-0.121	0.011	-0.170	-0.008
	(0.390)	(0.394)	(0.413)	(0.394)	(0.479)	(0.068)
ln(Median Income)	0.558	0.490	0.016	0.531	0.002	0.528
	(0.355)	(0.341)	(0.319)	(0.346)	(0.329)	(0.273)
Market Definition	10–1.5 km	10–1 km	10–0.5 km	6–1.5 km	5–0.5 km	Postcode
Months	36	36	36	36	36	36
Stores	314	355	434	337	463	504
Observations	10,040	11,363	14,127	10,815	15,141	16,626
R^2	0.81	0.83	0.83	0.82	0.84	0.85

Notes: Each column reports the estimated ATT using the estimator of Sun and Abraham (2021) with different local market definitions. The data span all months of 2014–2016 and the demographics are measured at the municipality level. The standard errors shown in parentheses are clustered by Municipality. *** $p < 0.001$; ** $p < 0.01$; * $p < 0.05$.

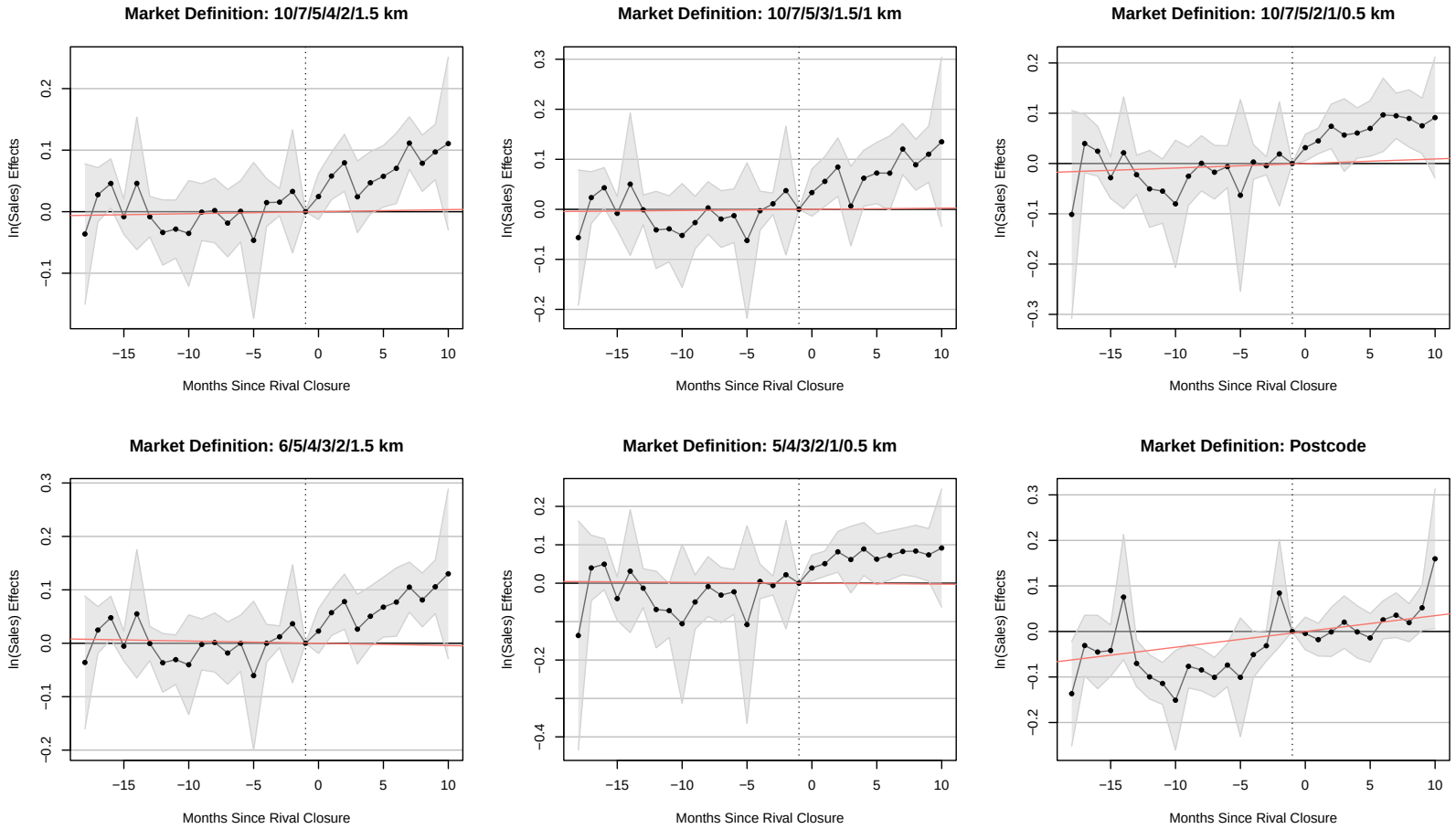


Figure A28: Estimated percentage sales effects of rival closure over time. The horizontal axis displays months relative to the event, $l \in [-18, +10]$, with the vertical line marking $l = -1$—the last month before the event. Point estimates are shown with 95% confidence intervals, based on standard errors clustered at the municipality level. A fitted linear pre-trend is displayed and extrapolated into the post-event period for comparison.

Table A27: Effect of Extra on sales: last period before event excluded

	ln(Sales)					
	(1)	(2)	(3)	(4)	(5)	(6)
T_{Extra}	-0.072***	-0.051**	-0.096***	-0.053**	-0.091***	-0.094***
	(0.017)	(0.018)	(0.015)	(0.018)	(0.013)	(0.022)
ln(Population)	-0.124	-0.017	-0.118	-0.023	-0.165	-0.043
	(0.390)	(0.403)	(0.397)	(0.402)	(0.463)	(0.069)
ln(Median Income)	0.573	0.461	0.158	0.563	0.117	0.356
	(0.350)	(0.335)	(0.309)	(0.342)	(0.321)	(0.271)
Market Definition	10–1.5 km	10–1 km	10–0.5 km	6–1.5 km	5–0.5 km	Postcode
Months	36	36	36	36	36	36
Stores	348	396	468	372	492	529
Observations	10,744	12,306	14,902	11,600	15,850	17,301
R^2	0.83	0.84	0.84	0.83	0.84	0.85

Notes: Each column reports the estimated ATT using the estimator of Sun and Abraham (2021) with different local market definitions. The data span all months of 2014–2016 except the last period before the rival rebranding. The demographics are measured at the municipality level and the standard errors shown in parentheses are clustered by Municipality. *** $p < 0.001$; ** $p < 0.01$; * $p < 0.05$.

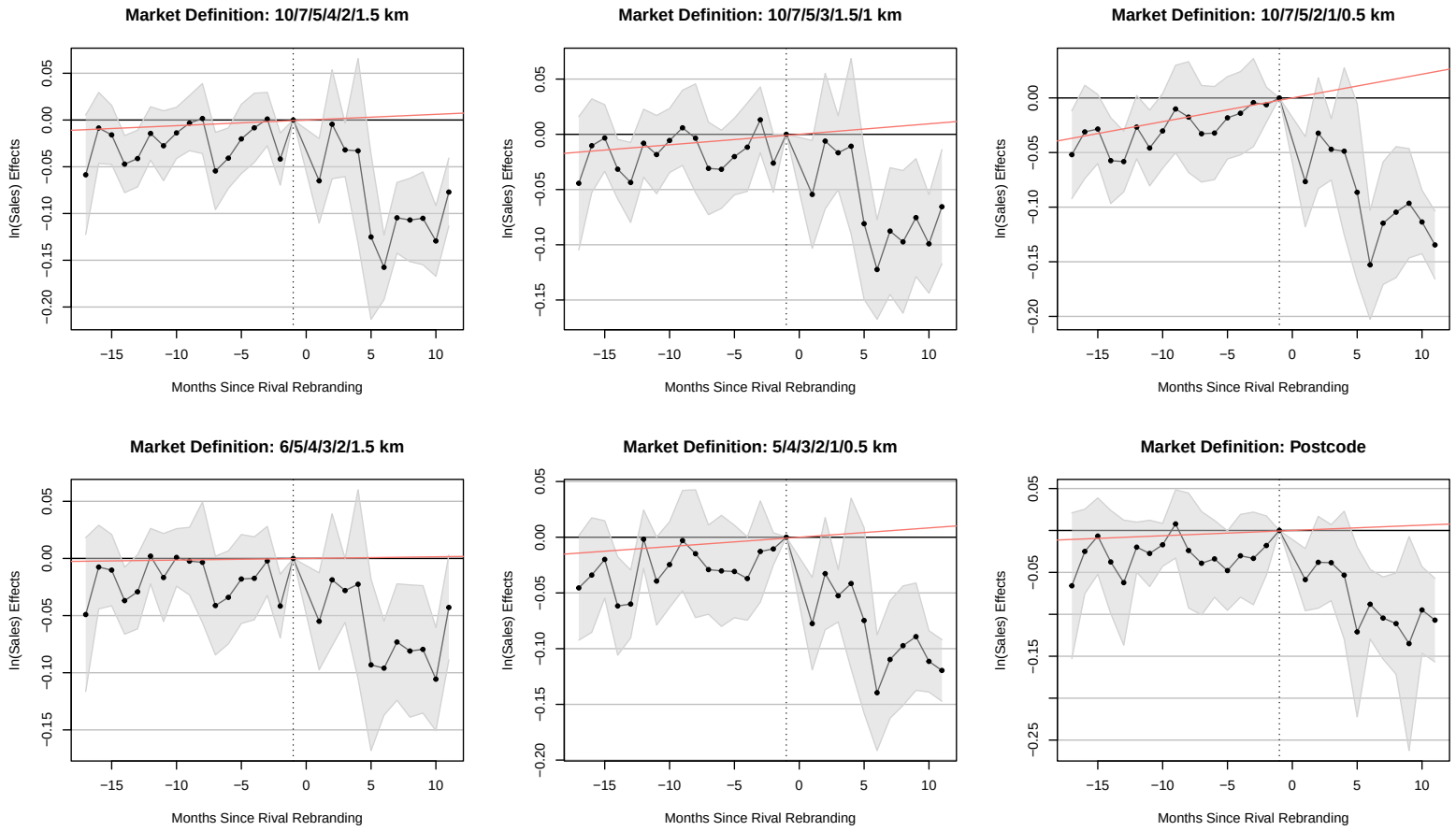


Figure A29: Estimated percentage sales effects of rival rebranding to Extra over time, excluding the last period before rebranding. The horizontal axis displays months relative to the excluded month, $l \in [-17, +11]$, with the vertical line marking $l = -1$—two months before the event. Point estimates are shown with 95% confidence intervals, based on standard errors clustered at the municipality level. A fitted linear pre-trend is displayed and extrapolated into the post-event period for comparison.

Table A28: Effect of Prix on sales: last period before event excluded

	ln(Sales)					
	(1)	(2)	(3)	(4)	(5)	(6)
T_{Prix}	0.034	0.019	-0.030	0.007	-0.045	-0.065*
	(0.031)	(0.026)	(0.026)	(0.033)	(0.036)	(0.030)
ln(Population)	0.096	0.166	-0.011	0.181	-0.120	-0.003
	(0.379)	(0.385)	(0.407)	(0.391)	(0.471)	(0.057)
ln(Median Income)	0.538	0.430	0.064	0.549	0.061	0.442
	(0.345)	(0.331)	(0.312)	(0.340)	(0.324)	(0.263)
Market Definition	10–1.5 km	10–1 km	10–0.5 km	6–1.5 km	5–0.5 km	Postcode
Months	36	36	36	36	36	36
Stores	327	372	449	350	479	512
Observations	10,017	11,574	14,331	10,849	15,491	16,672
R^2	0.83	0.84	0.84	0.83	0.84	0.85

Notes: Each column reports the estimated ATT using the estimator of Sun and Abraham (2021) with different local market definitions. The data span all months of 2014–2016 except the last period before the rival rebranding. The demographics are measured at the municipality level and the standard errors shown in parentheses are clustered by Municipality. *** $p < 0.001$; ** $p < 0.01$; * $p < 0.05$.

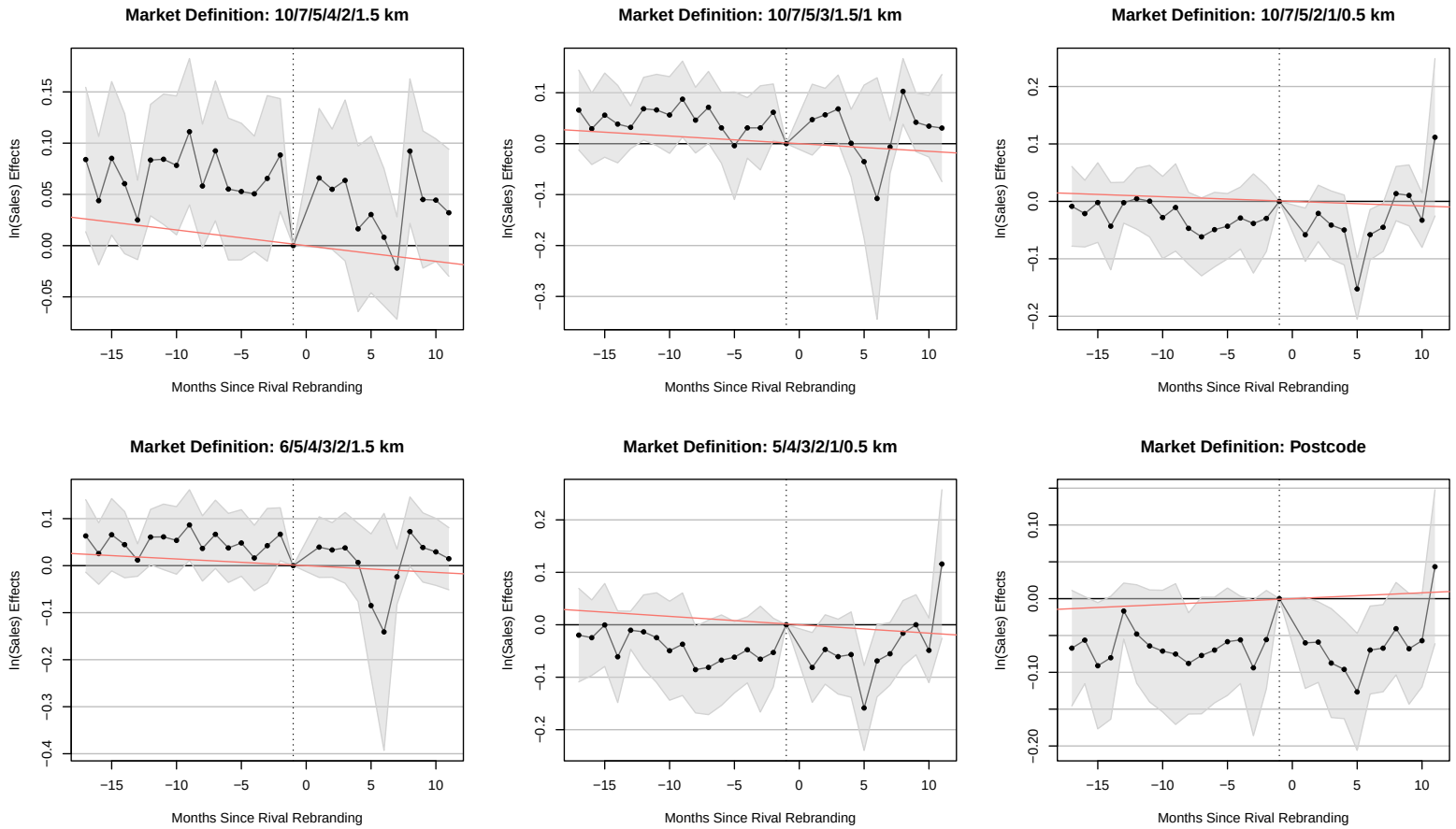


Figure A30: Estimated percentage sales effects of rival rebranding to Prix over time, excluding the last period before rebranding. The horizontal axis displays months relative to the excluded month, $l \in [-17, +11]$, with the vertical line marking $l = -1$—two months before the event. Point estimates are shown with 95% confidence intervals, based on standard errors clustered at the municipality level. A fitted linear pre-trend is displayed and extrapolated into the post-event period for comparison.

Table A29: Effect of Bunnpris on sales: last period before event excluded

	ln(Sales)					
	(1)	(2)	(3)	(4)	(5)	(6)
T_{Bunnpris}	0.255***	0.176***	0.132***	0.223***	0.136**	0.019
	(0.061)	(0.040)	(0.032)	(0.063)	(0.041)	(0.056)
ln(Population)	0.012	0.113	-0.111	0.108	-0.197	-0.011
	(0.391)	(0.394)	(0.398)	(0.400)	(0.468)	(0.058)
ln(Median Income)	0.660	0.454	0.058	0.558	0.059	0.419
	(0.351)	(0.328)	(0.295)	(0.348)	(0.312)	(0.268)
Market Definition	10–1.5 km	10–1 km	10–0.5 km	6–1.5 km	5–0.5 km	Postcode
Months	36	36	36	36	36	36
Stores	291	342	424	317	456	489
Observations	9,270	10,940	13,772	10,151	14,896	16,102
R^2	0.82	0.83	0.83	0.82	0.84	0.85

Notes: Each column reports the estimated ATT using the estimator of Sun and Abraham (2021) with different local market definitions. The data span all months of 2014–2016 except the last period before the rival rebranding. The demographics are measured at the municipality level and the standard errors shown in parentheses are clustered by Municipality. *** $p < 0.001$; ** $p < 0.01$; * $p < 0.05$.

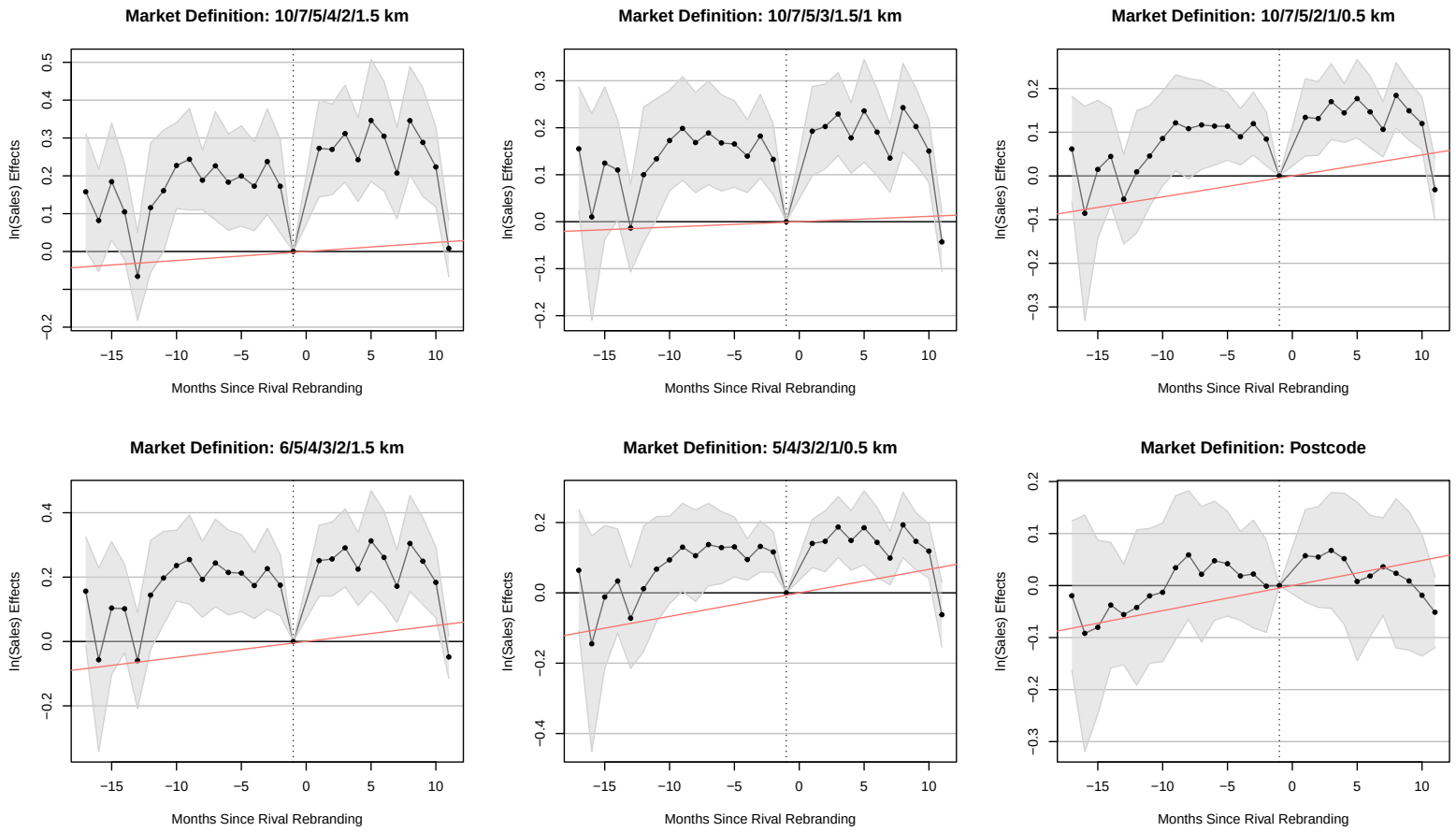


Figure A31: Estimated percentage sales effects of rival rebranding to Bunnpris over time, excluding the last period before rebranding. The horizontal axis displays months relative to the excluded month, $l \in [-17, +11]$, with the vertical line marking $l = -1$—two months before the event. Point estimates are shown with 95% confidence intervals, based on standard errors clustered at the municipality level. A fitted linear pre-trend is displayed and extrapolated into the post-event period for comparison.

Table A30: Effect of rival closure on sales: last period before event excluded

	ln(Sales)					
	(1)	(2)	(3)	(4)	(5)	(6)
T_{Closure}	0.044	0.046	0.056	0.044	0.051	-0.061
	(0.064)	(0.079)	(0.061)	(0.071)	(0.078)	(0.068)
ln(Population)	-0.065	0.062	-0.121	0.010	-0.170	-0.007
	(0.390)	(0.394)	(0.413)	(0.394)	(0.479)	(0.067)
ln(Median Income)	0.555	0.487	0.014	0.529	0.000	0.529
	(0.355)	(0.341)	(0.319)	(0.346)	(0.329)	(0.274)
Market Definition	10–1.5 km	10–1 km	10–0.5 km	6–1.5 km	5–0.5 km	Postcode
Months	36	36	36	36	36	36
Stores	314	355	434	337	463	504
Observations	10,010	11,339	14,108	10,788	15,127	16,601
R^2	0.81	0.83	0.83	0.82	0.84	0.85

Notes: Each column reports the estimated ATT using the estimator of Sun and Abraham (2021) with different local market definitions. The data span all months of 2014–2016 except the last period before the rival rebranding. The demographics are measured at the municipality level and the standard errors shown in parentheses are clustered by Municipality. *** $p < 0.001$; ** $p < 0.01$; * $p < 0.05$.

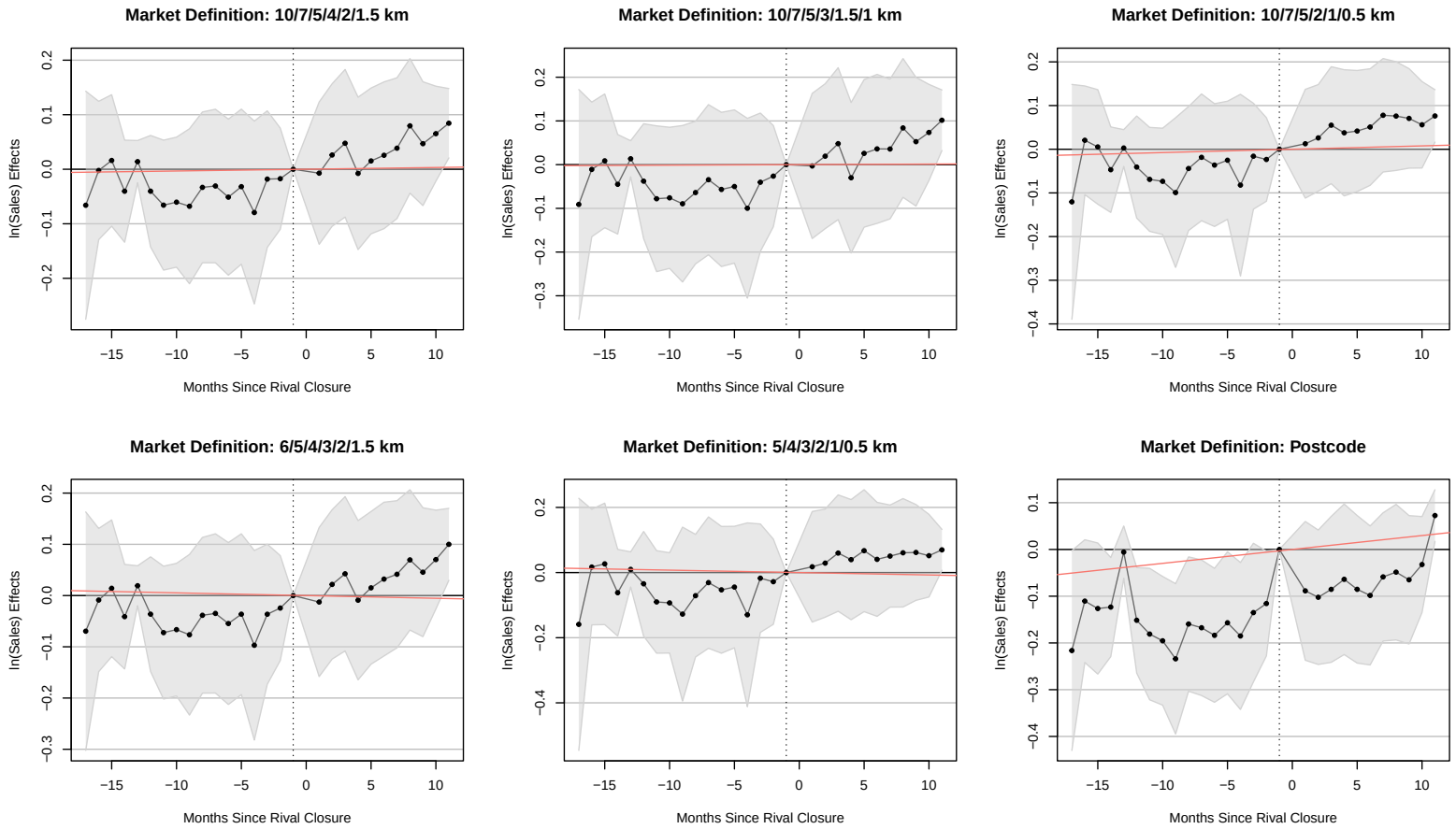


Figure A32: Estimated percentage sales effects of rival closure over time, excluding the last period before closure. The horizontal axis displays months relative to the excluded month, $l \in [-17, +11]$, with the vertical line marking $l = -1$—two months before the event. Point estimates are shown with 95% confidence intervals, based on standard errors clustered at the municipality level. A fitted linear pre-trend is displayed and extrapolated into the post-event period for comparison.

Table A31: Effect of Extra on sales: group-specific linear trends and the last period before event excluded

	ln(Sales)					
	(1)	(2)	(3)	(4)	(5)	(6)
T_{Extra}	-0.069*	-0.055	-0.108***	-0.044	-0.081***	-0.089**
	(0.029)	(0.030)	(0.022)	(0.032)	(0.023)	(0.030)
ln(Population)	-0.057	0.025	-0.084	0.048	-0.142	-0.043
	(0.376)	(0.394)	(0.389)	(0.390)	(0.452)	(0.067)
ln(Median Income)	0.570	0.447	0.156	0.566	0.112	0.326
	(0.345)	(0.331)	(0.306)	(0.338)	(0.316)	(0.269)
Affected × Trend	0.001	0.001	0.002	0.000	0.001	0.001
	(0.001)	(0.001)	(0.001)	(0.001)	(0.001)	(0.001)
Market Definition	10–1.5 km	10–1 km	10–0.5 km	6–1.5 km	5–0.5 km	Postcode
Months	36	36	36	36	36	36
Stores	348	396	468	372	492	529
Observations	10,744	12,306	14,902	11,600	15,850	17,301
R^2	0.83	0.84	0.84	0.83	0.84	0.85

Notes: Each column reports the estimated ATT using the estimator of Sun and Abraham (2021) with different local market definitions and group-specific linear trends. The data span all months of 2014–2016 except the last period before the rival rebranding. The demographics are measured at the municipality level and the standard errors shown in parentheses are clustered by Municipality. *** $p < 0.001$; ** $p < 0.01$; * $p < 0.05$.

Table A32: Effect of Prix on sales: group-specific linear trends and the last period before event excluded

	ln(Sales)					
	(1)	(2)	(3)	(4)	(5)	(6)
T_{Prix}	-0.003	-0.009	0.015	-0.012	0.028	-0.022
	(0.029)	(0.032)	(0.036)	(0.032)	(0.038)	(0.026)
ln(Population)	0.097	0.170	0.018	0.194	-0.094	-0.001
	(0.372)	(0.377)	(0.392)	(0.383)	(0.458)	(0.057)
ln(Median Income)	0.540	0.445	0.094	0.556	0.099	0.447
	(0.340)	(0.327)	(0.306)	(0.334)	(0.318)	(0.259)
Affected × Trend	-0.002	-0.002	-0.001	-0.001	-0.002	0.001
	(0.001)	(0.002)	(0.002)	(0.001)	(0.002)	(0.001)
Market Definition	10–1.5 km	10–1 km	10–0.5 km	6–1.5 km	5–0.5 km	Postcode
Months	36	36	36	36	36	36
Stores	327	372	449	350	479	512
Observations	10,017	11,574	14,331	10,849	15,491	16,672
R^2	0.83	0.84	0.84	0.82	0.84	0.85

Notes: Each column reports the estimated ATT using the estimator of Sun and Abraham (2021) with different local market definitions and group-specific linear trends. The data span all months of 2014–2016 except the last period before the rival rebranding. The demographics are measured at the municipality level and the standard errors shown in parentheses are clustered by Municipality. *** $p < 0.001$; ** $p < 0.01$; * $p < 0.05$.

Table A33: Effect of Bunnpris on sales: group-specific linear trends and the last period before event excluded

	ln(Sales)					
	(1)	(2)	(3)	(4)	(5)	(6)
T_{Bunnpris}	0.058	0.026	-0.015	-0.016	-0.049	-0.062
	(0.032)	(0.036)	(0.035)	(0.047)	(0.049)	(0.040)
ln(Population)	0.012	0.109	-0.113	0.108	-0.199	-0.011
	(0.390)	(0.394)	(0.398)	(0.400)	(0.468)	(0.058)
ln(Median Income)	0.661	0.450	0.057	0.558	0.056	0.418
	(0.351)	(0.328)	(0.295)	(0.347)	(0.313)	(0.268)
Affected × Trend	0.002	0.001	0.005	0.005	0.007	0.005
	(0.004)	(0.003)	(0.003)	(0.006)	(0.005)	(0.004)
Market Definition	10–1.5 km	10–1 km	10–0.5 km	6–1.5 km	5–0.5 km	Postcode
Months	36	36	36	36	36	36
Stores	291	342	424	317	456	489
Observations	9,270	10,940	13,772	10,151	14,896	16,102
R^2	0.82	0.83	0.83	0.82	0.84	0.85

Notes: Each column reports the estimated ATT using the estimator of Sun and Abraham (2021) with different local market definitions and group-specific linear trends. The data span all months of 2014–2016 except the last period before the rival rebranding. The demographics are measured at the municipality level and the standard errors shown in parentheses are clustered by Municipality. *** $p < 0.001$; ** $p < 0.01$; * $p < 0.05$.

Table A34: Effect of rival closure on sales: group-specific linear trends and the last period before event excluded

	ln(Sales)					
	(1)	(2)	(3)	(4)	(5)	(6)
T_{Closure}	0.073	0.091*	0.083	0.092*	0.111	0.037
	(0.037)	(0.041)	(0.047)	(0.038)	(0.075)	(0.043)
ln(Population)	-0.079	0.029	-0.137	-0.016	-0.172	-0.005
	(0.387)	(0.390)	(0.409)	(0.391)	(0.475)	(0.068)
ln(Median Income)	0.561	0.487	0.015	0.525	0.001	0.544*
	(0.355)	(0.340)	(0.318)	(0.345)	(0.329)	(0.273)
Affected × Trend	0.000	0.000	0.001	-0.001	-0.001	0.003
	(0.002)	(0.002)	(0.002)	(0.002)	(0.004)	(0.002)
Market Definition	10–1.5 km	10–1 km	10–0.5 km	6–1.5 km	5–0.5 km	Postcode
Months	36	36	36	36	36	36
Stores	314	355	434	337	463	504
Observations	10,010	11,339	14,108	10,788	15,127	16,601
R^2	0.81	0.83	0.83	0.81	0.84	0.85

Notes: Each column reports the estimated ATT using the estimator of Sun and Abraham (2021) with different local market definitions and group-specific linear trends. The data span all months of 2014–2016 except the last period before the rival rebranding. The demographics are measured at the municipality level and the standard errors shown in parentheses are clustered by Municipality. *** $p < 0.001$; ** $p < 0.01$; * $p < 0.05$.

Chapter 2

The Effect of Cross-Border Shopping on Commodity Tax Revenue: Results from Norway's COVID-19 Border Closings

We use grocery data from Norway and COVID-19 border closings to gauge the effect of cross-border shopping on commodity tax revenue. Detailed store–category-level data identify differential treatment effects that depend on distance to Swedish stores. Economically significant effects extend to up to two hours' drive from the border, and even further for prominent cross-border shopping products such as beer, cigarettes, and soda. Across all products, cross-border shopping decreases tax revenue from VAT by 3.6% at the national level. National commodity tax revenue from carbonated soft drinks (subject to a sugar tax) is reduced by 8.1% and from cigarettes by 11.9%.

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2.1 Introduction

Excise taxes play a key role in limiting the consumption of goods where excessive consumption is seen as harmful to the consumer or to others. Such "sin taxes" are pervasive on, for instance, alcohol, gasoline, and tobacco. Excise taxes, and other commodity taxes such as sales tax and value added tax (VAT), are also an important source of tax revenue. Both of these roles are potentially affected by cross-border shopping, where consumers purchase the good at a lower price in a neighboring country. A rich body of literature documents the empirical relevance of such cross-border shopping across national borders. This is manifested in the number of retailers established in several border areas, in local demand, and in how the number of border crossings responds to relative price changes.[1] A related body of literature examines within-country patterns, and documents the crossing of state or municipal borders to avoid local sales and excise taxes on for instance cigarettes (Lovenheim, 2008) and gasoline (Manuszak and Moul, 2009).[2] A case of much recent policy interest is that of taxes on sugar-sweetened beverages, where city level taxes have been shown to lead to substantial substitution towards out-of-town purchases.[3]

In this chapter, we estimate the impact of cross-border shopping on commodity tax revenue. We use weekly store–category-level data from a Norwegian grocery retail chain and the natural experiment provided by border closures in connection with the COVID-19 pandemic to estimate the quantitative impact of cross-border shopping.

Two features of the setting and data are important in helping us to identify the impact of cross-border shopping on sales and commodity tax revenue. First, as we document at length in Section 2.6, Norway's geography—with large swaths of the country many hours' drive from the border—creates a clear control group against which to gauge the impact of closing the border on the stores in "treated" areas closer to the long Norwegian-Swedish border. Combined with strict border closings, this creates a natural experiment with the kind of autarky flavor often assumed in theoretical work in international economics, but rarely observed in practice; see, for example, Bernhofen and Brown (2005), who perform an empirical study of Japan's opening up to trade in the 1800s, which is a notable study in the same vein. Second, category-level sales from a large number of stores across the country allow us to measure the quantitative effect of cross-border shopping on commodity tax revenue with high precision. Identification is aided by the fact that all the grocery retail chains

[1] See, for instance, Campbell and Lapham (2004), Asplund, Friberg, and Wilander (2007), Chandra, Head, and Tappata (2014) or Baggs, Fung, and Lapham (2018); Leal, Lopez-Laborda, and Rodrigo (2010) provide a survey.

[2] See also Baker, Johnson, and Kueng (2021) who, in a particularly encompassing study, document an important role for cross-border shopping in how US households respond to changes in local sales taxes.

[3] See, e.g., Cawley et al. (2019) or Allcott, Lockwood, and Taubinsky (2019*b*) for an overview.

keep the same prices across all of Norway, which implies that we do not need to control for any endogenous price responses that would be unique to border areas.

We establish that cross-border shopping has a large effect on sales in areas close to Swedish stores, that effects only gradually diminish with driving duration, and that, broadly speaking, effects are stronger and stretch further inland for products with larger price differences and products that are easy to transport and stockpile. We combine our store–category-level estimates of how sales are affected by cross-border shopping with information on the total number of stores in Norway, and their location, to infer the overall loss in tax revenue. In particular, we calculate losses in VAT and excise taxes that are due to cross-border shopping, and how these losses relate to travel duration to the closest Swedish stores. We take the last full year before the COVID-19 pandemic border closures as our benchmark, and use our estimated effects to calculate a counterfactual outcome with a closed border. In border areas the effects are large: due to cross-border shopping VAT revenue is 27% lower in stores that are within 30 minutes' driving distance from a Swedish store, 20.2% lower in stores that are between 30 and 60 minutes away, and 8.5% lower for locations between 60 and 90 minutes away. At the national level we estimate a VAT loss of 3.6% stemming from goods sold in grocery stores as a result of cross-border shopping. As explained in Section 2.5, this number relies on the estimated effects from the stores for which we have access to product-level sales but is then aggregated up to the national level using information on the full set of Norwegian grocery stores.

For several of the products subject to excise taxes, such as carbonated soft drinks (hereafter, "soda"), beer, cigarettes, and snus,[4] excise tax revenue more than doubles in these border areas when we move to a counterfactual situation with no cross-border shopping. Aggregating up to the national level, the effects are still substantial with an estimated loss of excise taxes on, for instance, soda of 8.1% and on cigarettes of 11.9%.

In total we estimate that cross-border shopping from Norway to Sweden implies a tax loss of 2.3 billion NOK through lost VAT and excise taxes on products sold in grocery stores.[5] Nearly one-third of this loss comes through lost excise taxes on five top border-trade product categories: beer, cigarettes, snus, soda, and sweets.

The current research thus highlights that the concerns examined in theoretical models of commodity "tax competition" are quantitatively important. In a setting with two countries and costly cross-border shopping, Kanbur and Keen (1993) show that a smaller country will set lower commodity taxes, and they analyze the case for commodity tax harmonization. Their baseline model has been extended in a number

[4]Snus is a smokeless tobacco product that is placed between the upper lip and gum and commonly used in Norway and Sweden. Statistics Norway (2022*b*) reports that in 2019 15% of the Norwegian population aged between 16 and 74 were daily users of snus.

[5]An important additional loss of VAT and excise taxes comes from sales of wine and other stronger alcoholic beverages that are sold by the state-owned monopoly alcohol retailer Vinmonopolet.

of directions, for instance allowing for one country to decide taxes first (Wang, 1999) and allowing for lower commodity taxes in border regions (Agrawal, 2012).[6]

For VAT revenue, the key motivation for international tax agreements to avoid a "race to the bottom" is clearly tax revenue per se. For goods subject to excise taxes, the tax revenue motivation may be combined with a motivation to limit harm due to externalities (e.g., gasoline taxes to lower greenhouse gas emissions) or "internalities" (e.g., excessive consumption due to self-control problems related to consumption of products such as snus and soda) or a combination of the two (for instance alcohol and cigarettes). For all the products that we examine and which are subject to excise taxes (alcoholic beverages, sugar-sweetened products, and tobacco), there are well-documented studies showing that they are the subject of substantial cross-border shopping (Leal, Lopez-Laborda, and Rodrigo, 2010). With store-level data on quantity of these products, we are able to contribute to the literature by providing evidence of how volumes sold and tax revenue are affected at locations ranging from below 30 minutes up to 180 minutes away from the border.

The numbers clearly point to difficulties for smaller jurisdictions in maintaining higher excise taxes than neighboring jurisdictions. An application of particular interest may be that of soda taxes. Soda taxes (or more broadly excise taxes on non-alcoholic beverages) have been implemented both at a national level (e.g., Chile, France, the UK) and at a city and state level (e.g., Berkeley, Philadelphia, Washington state) and are the subject of a rapidly growing academic literature (see, e.g., Allcott, Lockwood, and Taubinsky, 2019*b* for a survey). Most closely related to the current research are studies that explicitly discuss geographical constraints such as Rojas and Wang (2021), who examine the introduction of these excise taxes in Berkeley and Washington state and Seiler, Tuchman, and Yao (2021), who examine the Philadelphia tax. The strength of the effects also at long distances clearly indicates the important constraint that cross-border shopping imposes on excise taxes at a city level or for smaller jurisdictions. Even in large and mountainous Norway, cross-border shopping was arguably an important reason why the excise tax on non-alcoholic beverages was removed in 2021.

To frame the counterfactual calculations of the effects of cross-border shopping on commodity tax revenue, an analogy from the study of the effects of trade policy might be useful. We examine the difference between a current situation and a situation with no cross-border trade, akin to a move to autarky. The border closings give us an

[6]Much of the related empirical literature on tax competition has focused on characterizing the nature of the strategic interaction. Kanbur and Keen (1993) assume that taxes are strategic complements across jurisdictions, an assumption that also fits well with policy discussions aimed at avoiding a "race to the bottom". A notable aspect of some recent work on within-country tax competition is that it indicates that local taxes are strategic substitutes (downward-sloping reaction functions such that lower taxes in one jurisdiction imply higher taxes in another). Such a pattern is for instance found for local income taxes across Switzerland in Parchet (2019); see Agrawal, Hoyt, and Wilson (2022) for a survey of tax competition and policy choices across local governments.

unusual opportunity to do this. Alternatively, one can examine the effects of marginal changes in taxes (akin to examining the effect of changes in tariff rates). As seen above, it is well established that cross-border shopping responds to changes in relative prices across borders. In principle, sound estimates of pass-through elasticities of VAT and excise taxes together with elasticities of local demand in response to relative price changes could be used to calculate the counterfactual that we examine. This would require assuming that elasticity estimates are valid for a much wider range of prices and taxes than those that underlie their estimation. An additional complication for such an exercise would be that the price elasticity of local demand differs by distance to the border and not necessarily in a linear way. This is examined theoretically and shown empirically (using similar data as in the present study) in Friberg, Steen, and Ulsaker (2022): The response to a price change may be the largest some distance inland (as that is where the extensive margin decision of whether to make the cross-border trip or not bites). With the current natural experiment, we estimate the effect of cross-border shopping directly.

In relation to the literature on trade policy, it may also be useful to highlight the special nature of cross-border shopping in comparison to the welfare effects of international trade in general. In models of international trade, welfare gains are largely a result of specialization according to comparative advantage, with the efficiency gains reflected in lower prices. With cross-border shopping, a very particular form of international trade, consumers also take advantage of lower prices. To the extent that international price differences are the result of excise taxes, that are in place to correct for externalities or internalities, the welfare effects become more complex, however. In particular, individual heterogeneity is likely to play a key role (as for instance in work on the welfare effects of soda taxes by Allcott, Lockwood, and Taubinsky (2019*a*)), and we stop short of trying to model the full welfare effects of cross-border shopping.

On a related note, let us point out that some cross-border shopping would be likely to remain from Norway to Sweden even if all excise taxes were eliminated. For instance, abolishing the excise tax on cigarettes would reduce prices in Norway by around 40% if the tax change was fully passed through to consumer prices. However, in 2019 tobacco prices in Sweden were more than 50% lower than in Norway. For soda, the excise tax in 2019 constituted about 20% of the average price in Norway, while the Swedish prices were close to 40% lower than the Norwegian prices. Removing these two excise taxes alone would have reduced tax revenue from the sample regions by five billion NOK (from only the grocery store sales) but would still come short of completely removing the incentives to buy these products in Sweden.[7] On the one hand, this

[7]The loss of five billion NOK is based on actual tax revenue as reported in the last column of Table 2.9.

example illustrates that large cross-border price differences make independent tax policy difficult because the border trade leakage will be significant. On the other hand, the example also illustrates that it is not necessarily revenue-maximizing to reduce taxes, because revenue loss inland may not make up for gains in regions closer to the border.

Lastly, let us also note that this chapter relates to other research that has documented large effects of COVID-19 related shocks across many outcomes. For instance, several articles use data from credit cards or bank accounts to examine individual-level changes in spending (Bounie, Camara, and Galbraith, 2023; Andersen et al., 2022). Using data from Denmark, Andersen et al. (2022), for instance, document that individual-level health risks and supply restrictions, rather than income risks, were key in explaining individuals' spending cuts. As to the impact of border closings because of COVID-19, we are aware of one paper that examines how cost-of-living expenses vary with access to cross-border shopping (Burstein, Lein, and Vogel, 2024, using Swiss homescan data) and one paper that uses simulations based on a pre-pandemic cross-border study to examine effects on retailers—Baggs, Fung, and Lapham (2021) on Canada, relying on estimates from Baggs, Fung, and Lapham (2018).[8] In contrast we focus on the consequences of cross-border shopping for excise taxes.

In Section 2.2, we look more closely at cross-border shopping and the COVID-19 pandemic. Then, in Section 2.3 we present our data and provide descriptive statistics, before we analyze the effects of cross-border shopping on retail activity in Section 2.4, and discuss the economic impact in Section 2.5. In Section 2.6, we provide a number of robustness tests, and we conclude in Section 2.7.

2.2 Cross-Border Shopping and the COVID-19 Pandemic

Cross-border shopping from Norway into neighboring Sweden features prominently in policy discussions in Norway and has been the subject of many policy reports (see, e.g., Statistics Norway, 2022*a*; Skogli et al., 2020; NHO Mat og Drikke, 2020; Abel, Totland, and Gulseth, 2021). Norway is not a member of the European Union (EU) and maintains tariffs on many food products which, together with a high general price level and high excise taxes, combine to create substantial price differences relative to neighboring Sweden. One way to observe the incentives for cross-border shopping is to examine price-level indices reported by Eurostat at a yearly level for many European countries (Eurostat, 2022*b*). The average across all the 27 EU countries (excluding the

[8]See also Leifman, Dramstad, and Juslin (2022) who use national-level data from 14 European countries to establish that COVID-19 border closures were associated with increasing domestic alcohol sales in inflow countries such as Norway and falling sales in outflow countries.

UK) is set to 100 in each year and Figure 2.1 presents the food price index for a set of European countries. For food the index for Norway is 155.2, thus implying that food prices in Norway are about 50% higher than the average in the EU. Sweden's corresponding price index is 114.8 in the same year. Clearly, a difference of 40.4 percentage points (155.2–114.8) suggests a substantial incentive to cross-border shop for food.[9] While the price difference is large it is notable that it is not unique. We note that, for instance, Switzerland's corresponding price index was 165.8 with a difference of 64.4 percentage points to neighboring Germany, and of 49.9 percentage points to neighboring France. Similarly, Austria's food price index was 41.9 percentage points above that of Czechia.

[9]A set of legal restrictions on cross-border shopping are in place (Norwegian Customs, 2023). A person can bring into the country up to 10 kilos of meat, meat products and cheese, 2 liters of strong beer (ABV above 2.5%), 100 cigarettes or 125 grams of snus. While random checks are possible, border stations are often unmanned and it is unclear how binding these quotas are for a typical cross-border shopper.

Price level index for food (EU27=100) in European countries in 2019

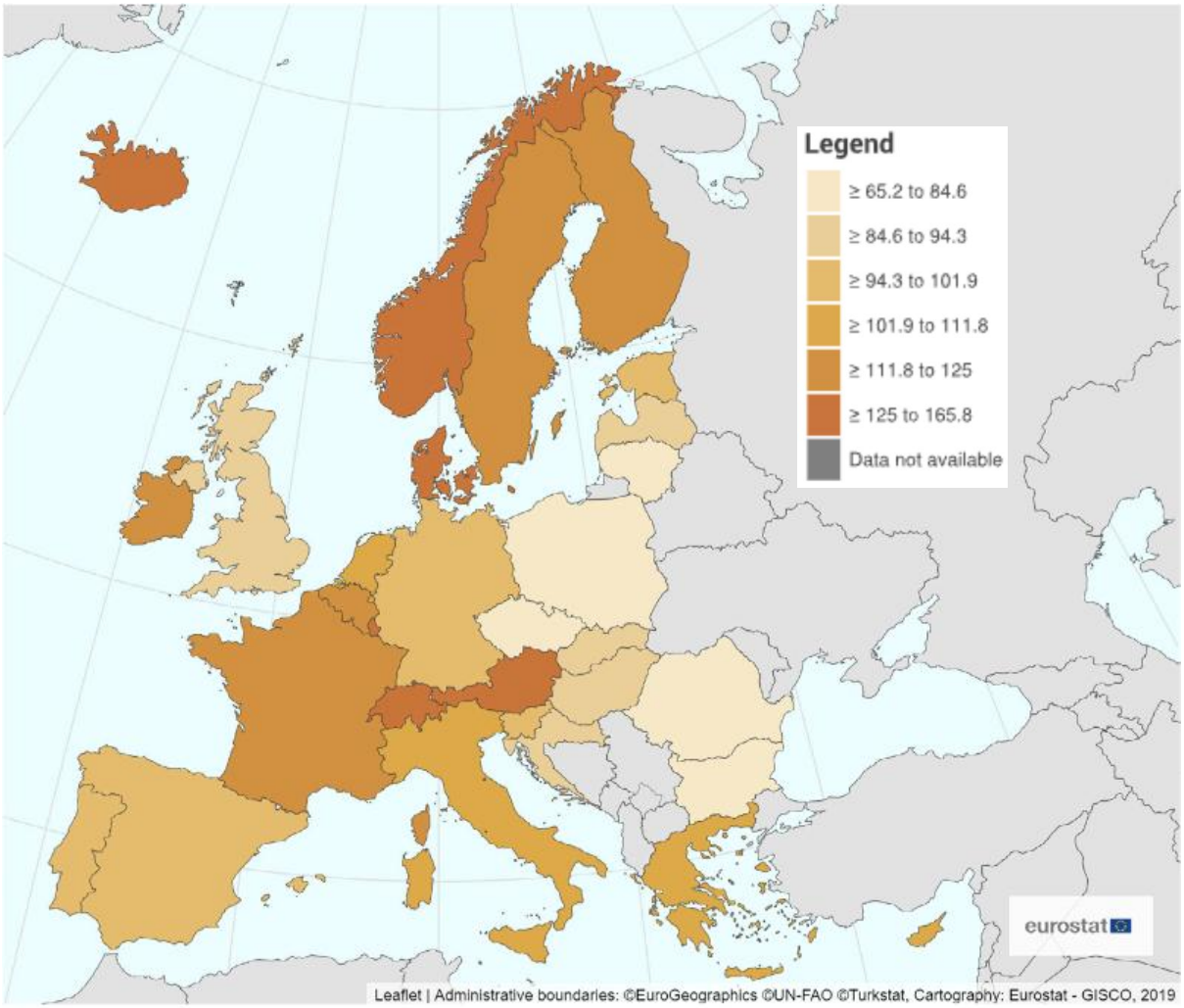


Figure 2.1: This figure shows the purchasing power parities (PPPs), price-level indices, and real expenditures for ESA2010 aggregates. Geopolitical entity (reporting)/time: 2019. Time frequency: annual. National accounts indicator (ESA 2010): price-level indices (EU27_2020 = 100). Analytical categories for PPPs calculation: food. The level of index for some particularly noteworthy countries is given in bold.

Table 2.1 reports the indices for selected categories in 2019 for Norway and Sweden. Price differences are especially marked on products that are protected by agricultural policy (meat, milk, cheese, and eggs) and products subject to excise taxes (alcohol, tobacco, and a "sugar tax"). The corresponding excise tax rates in Sweden were at the time lower (beer, tobacco) or zero (no "sugar tax" in Sweden).[10] VAT rates for groceries are similar in the two countries, 15% in Norway and 12% in Sweden.

A notable feature of the Norwegian grocery market is that all the chains have

[10]For instance in 2019 a liter of beer with ABV of 4.6% was taxed at 22.4 NOK/liter in Norway (8.65 NOK/liter in Sweden), cigarettes were taxed at 2.63 NOK per cigarette in Norway (1.47 NOK + 1% of retail price in Sweden) and snus was taxed at 1070 NOK/kilo in Norway (420 NOK/kilo in Sweden). In these comparisons we use average exchange rate for 2019, NOK/SEK = 0.9306 (Norges Bank, 2022), to express Swedish excise taxes in NOK.

national pricing, and prices are thus not endogenously lower closer to the border (Friberg, Steen, and Ulsaker, 2022): This means that the relevant price comparison is crucially affected by driving duration to the closest Swedish store. Uniform nationwide pricing may at first seem surprising to many economists but is a common feature of grocery retailing, see, for instance, Seaton and Waterson (2013) for UK grocery retailing or DellaVigna and Gentzkow (2019) for US grocery retailing.

Table 2.1: Price level index for selected product categories (EU27 = 100) for Norway and Sweden in 2019

Product category	Norway	Sweden
Food	155.2	114.8
Alcohol	260.4	155.9
Meat	149.1	115.1
Milk, cheese, and eggs	168.7	109.2
Non-alcoholic beverages	177.8	108.3
Tobacco	234.9	112.3

Let us now turn to an overview of the aspects of the COVID-19 pandemic of relevance to the current chapter. As in many countries, the development of the pandemic from March 2020 was very rapid, and we have little reason to assume that there were any important anticipatory measures on the part of consumers with regard to cross-border shopping. On 12 March, the Norwegian government implemented a string of severe infection control measures: schools were closed, cultural and sporting events cancelled, and travel restrictions imposed (Reuters, 2020). The measures effectively closed down much of the economic activity and led to the biggest recorded fall in GDP for mainland Norway, and to a rise in the unemployment rate from 3.8 to 15.3% (NAV, 2021). On 20 April 2020, a partial reopening of Norway was announced, but the pandemic continued to affect everyday life and economic activity throughout 2020, as restrictions were eased and tightened in response to changes in the infection rates.[11]

The pandemic and the control measures affected the grocery sector in several ways. In March 2020, stockpiling led to sharp spikes in sales but also to temporary stock-outs of goods such as toilet paper and flour. Throughout the pandemic, the dramatic increase in remote work, together with closed bars and restaurants, meant that more meals were eaten at home. Especially relevant for our purpose are the

[11]For most of 2020, the restrictions were national in scope. Towards the end of the year, there was some geographical variation in the severity of the restrictions in response to local outbreaks. The regulation *Covid-19-forskriften* (Helse- og omsorgsdepartementet, 2020) contained the valid national legislation at any time. All versions of the regulation are available (in Norwegian) at https://lovdata.no/dokument/SFO/forskrift/2020-03-27-470.

restrictions on international travel. Under regulations imposed in March 2020, any international travelers were required to undergo ten days of quarantine, and for the rest of 2020, visits to countries and areas with high infection rates triggered quarantine upon re-entering Norway (Norwegian Directorate of Health, 2020*a*,*b*). Consequently, cross-border shopping was infeasible for most of 2020, and practically all food and alcohol had to be bought in Norwegian stores. In the next section, we describe in more detail how different counties in Sweden were effectively opened and closed for cross-border shopping during 2020, as travel restrictions were imposed and lifted.

The estimation of the effects of cross-border shopping on local sales in Norway is made simpler by the fact that the relevant restrictions on cross-border shopping were essentially all driven by policy variation in Norway, rather than by policy in neighboring Sweden. Throughout 2020, Sweden largely relied on voluntary measures to limit contagion and enforced light-handed restrictions compared to other European countries.[12] Grocery stores enforced caps on the number of customers who were allowed in store, but these were rarely binding.

2.3 Data and Descriptive Statistics

The main dataset used for estimation of the effects of cross-border shopping consists of weekly sales and volumes at the store–category-level for 2019 and 2020 from a random sample of grocery stores belonging to the largest Norwegian grocery umbrella chain, NorgesGruppen. NorgesGruppen had a market share of 43.7% in 2019 and has retail stores across all market segments and across the entire country (Nielsen, 2020).[13] In addition to the sales data, we have information about the number of distinct store visits per week at the store level. Because we are interested in cross-border shopping across the Swedish border, we oversampled stores located close to the border when selecting our sample of stores. Specifically, 200 stores were drawn randomly from the whole of Norway, and an additional 200 stores were drawn randomly from three counties bordering Sweden (Viken, Innlandet, and Trøndelag). The dataset used for estimation has been created by the authors aggregating up from store-barcode-level data for the full assortment following product categorizations used by NorgesGruppen. Because we are ultimately interested in overall grocery sales, and not just sales in the sampled stores, we also use information about the location and estimated yearly sales amounts for all Norwegian grocery stores in 2019, obtained from the grocery store database, Geodata (2021).

The Norwegian store data contain the location of the Norwegian stores. Data

[12]See, for instance, Andersson et al. (2022) or Ludvigsson (2020).

[13]There are no hard discounters in Norway, and very few hypermarkets. Discount stores, a segment in which NorgesGruppen also has a presence, constitute the largest market segment with a market share of about 60% in 2019 (Nielsen, 2020).

Table 2.2: Stores and population by duration category

	<30	[30, 60)	[60, 90)	[90, 120)	[120, 150)	[150, 180)	>180	Total
Stores								
Number of stores	14	33	55	103	41	27	127	400
Proportion	0.04	0.08	0.14	0.26	0.10	0.07	0.32	1.00
Population								
Population	112,040	324,070	1,003,407	1,030,645	266,896	212,793	1,434,226	4,384,077
Proportion	0.03	0.07	0.23	0.24	0.06	0.05	0.33	1.00

Notes: This table shows the number of stores and the population within each duration category in the week starting on Monday 9 March 2020 (i.e., the week before travel to Sweden became restricted). For each duration category, the population is the population of the municipalities with stores in the store sample. Because not all municipalities have stores in the store sample, the total population is less than the total population of Norway in 2020.

on the locations of all grocery stores in Swedish counties that border Norway were obtained from Delfi Marknadspartner (2021). Using map data from OpenStreetMap, we calculate driving duration in minutes from each Norwegian store to each Swedish store, as outlined in the Appendix.

In our analysis, we explore how changes in the accessibility of cross-border shopping affect grocery store sales in Norway. The source of this variation is regulations that restricted travel to Sweden. Information about which Swedish counties could be visited without triggering quarantine upon re-entering Norway on given dates during 2020 is obtained from the Norwegian Institute of Public Health (2021*a*). We refer to a Swedish county as open for cross-border shopping whenever it could be visited without triggering a quarantine, and closed for cross-border shopping otherwise. Figure A2 in the Appendix plots traffic flows at the main border crossings into Sweden and clearly indicates the impact of the restrictions on travel.

Our main measure of the availability of cross-border shopping is the driving duration to the closest Swedish store, which is located in a border county and is open for cross-border shopping. From the start of our sample period in January 2019 until 17 March 2020, all Swedish border counties were open for cross-border shopping. In the following, we refer to these driving durations as pre-COVID driving durations. In Table 2.2, we report the distribution of pre-COVID driving durations, where we use 30-minute bins (up to 180 minutes) to categorize the stores.

From Tuesday 17 March until 25 July 2020, Norwegians had no access to cross-border shopping in Sweden and the border was effectively closed to any private travel (unless one was willing to undergo ten days of quarantine on the return to Norway). As discussed below, in relation to our examination of the robustness of our empirical estimates, a recent empirical literature has highlighted concerns with difference-in-difference estimates when treatment is heterogeneous in terms of timing (see, e.g., De Chaisemartin and d'Haultfoeuille, 2020). A treatment that puts a stop to cross-border shopping across the whole country on the same day clearly avoids such

concerns. Towards the end of the sample period, there is, however, some variation across locations in border closings. Between 25 July and 9 October 2020, cross-border travel to some Swedish counties was allowed without triggering quarantine, based on local infection rates. For the period from 9 October 2020 until the end of the sample period, the border was then again closed for all cross-border shoppers unless one was willing to undergo quarantine.[14] Figure 2.2 details which Swedish counties were open for cross-border shopping during which weeks. For the summer and early fall period, when some Swedish locations were open for cross-border shopping, we keep track of the driving duration to the closest accessible Swedish store as it varies due to border closures. In Figure A4 in the Appendix, we use maps to illustrate how driving durations to the closest available Swedish store varied during the late summer and early fall of 2020.

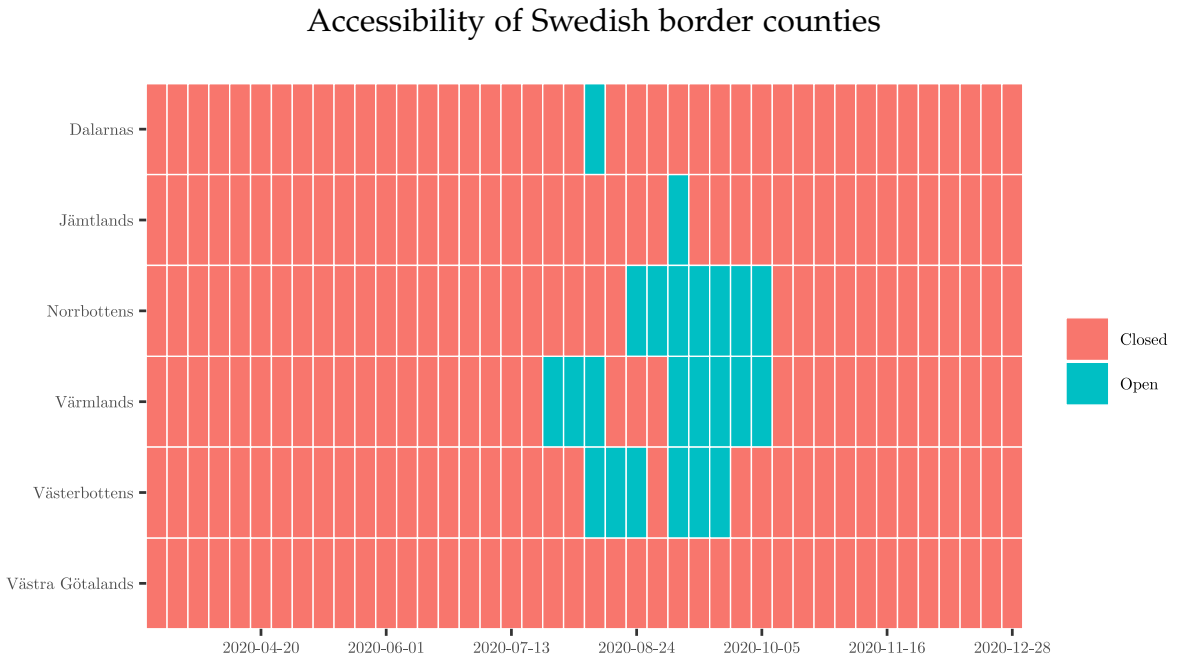


Figure 2.2: This figure indicates which weeks the Swedish border counties were closed to cross-border shopping in 2020. We define a county as closed in a given week if in the majority of the days of the week, visiting the county triggered a quarantine upon returning to Norway. The first week shown is the week starting on Monday 16 March 2020. Travels to Sweden were restricted from Tuesday 17 March 2020.

In our empirical analyses, we include a number of municipality-level control variables that may both affect grocery store activity and be affected by the COVID-19 pandemic. The variables we consider are COVID-19 infection rates (Norwegian Institute of Public Health, 2021*b*), unemployment rates (Norwegian Labour and Welfare

[14]COVID restrictions also implied that cross-border commuting was affected by quarantine rules. Because the wage level is higher in Norway than in Sweden this mainly affected Swedes working in Norway and we do not expect any particularly important effect of this on cross-border shopping since such commuters were likely to shop in Sweden throughout. See Statistics Norway (2021*b*) for a further analysis of cross-border work into Norway.

Administration, 2021), population (Statistics Norway, 2021*c*), and the proportion of residential dwellings in the municipality that are holiday homes (Statistics Norway, 2021*a*). In Table 2.3, we present mean values for these variables, along with the outcome variables we consider in our empirical analysis. We break down the descriptive statistics by year, and by the pre-COVID travel duration category.

Table 2.3: Descriptive statistics

	<30		[30, 60)		[60, 90)		[90, 120)		[120, 150)		[150, 180)		>180	
	2019	2020	2019	2020	2019	2020	2019	2020	2019	2020	2019	2020	2019	2020
Outcome variables														
Store level														
Store sales (millions NOK)	0.97	1.38	0.92	1.20	1.40	1.67	1.60	1.88	1.17	1.31	1.09	1.22	0.84	0.94
Store visits	4591	5071	4179	4479	5891	5870	6703	6627	5104	4927	4350	4189	3734	3603
Basket size (NOK)	206.7	263.8	206.6	253.1	226.1	269.8	230.9	273.7	227.1	266.2	247.4	291.7	229.7	269.0
VAT (thousands NOK)	136	197	131	174	198	240	226	269	167	190	157	178	121	137
Category level														
Ice cream sales (thousands NOK)	15.9	22.4	14.6	19.8	20.0	25.8	21.3	27.6	14.1	17.4	12.7	16.1	11.0	13.8
Milk sales (thousands NOK)	31.0	39.0	28.3	32.5	43.1	47.1	45.3	49.6	35.0	36.2	35.1	36.2	26.5	27.8
Fresh bake sales (thousands NOK)	47.1	53.4	43.2	47.7	59.4	62.9	63.4	67.7	48.4	49.9	40.6	41.2	31.2	32.3
Ready made sales (thousands NOK)	45.5	54.2	38.3	44.6	56.1	61.7	64.5	71.7	44.2	47.7	35.4	38.6	26.6	29.0
Cheese sales (thousands NOK)	42.3	58.9	39.2	50.3	61.9	73.1	75.2	88.8	50.9	55.7	48.6	53.3	34.9	38.7
Meat sales (thousands NOK)	43.6	81.2	46.6	70.7	81.2	104.8	99.7	124.7	66.0	80.8	56.6	68.8	42.0	49.9
Sweets sales (thousands NOK)	39.2	57.9	37.5	49.9	57.8	69.6	66.1	78.5	52.0	58.3	49.9	56.4	38.3	43.6
Sweets sales (kilograms)	167.2	238.3	153.8	193.9	236.2	270.5	273.8	310.4	215.2	228.3	216.2	229.2	163.9	174.6
Soda sales (thousands NOK)	32.3	61.8	35.4	53.3	54.8	67.6	61.1	72.7	52.0	59.0	49.4	56.0	37.8	41.1
Soda sales (liters)	1435	3227	1520	2535	2408	3139	2672	3278	2397	2810	2318	2692	1686	1866
Beer sales (thousands NOK)	46.0	82.2	51.8	79.6	75.5	100.1	85.2	110.9	65.7	81.5	65.9	80.2	53.3	63.4
Beer sales (liters)	782	1368	845	1267	1226	1587	1325	1688	1046	1268	1065	1268	868	987
Cigarette sales (thousands NOK)	17.5	51.7	26.8	58.0	40.9	67.6	42.0	61.4	41.4	57.1	41.9	56.1	32.5	41.2
Cigarette sales (items)	2897	8205	4338	9005	6569	10,434	6760	9491	6713	8877	6917	8904	5308	6459
Snus sales (thousands NOK)	7.5	23.6	12.1	28.2	23.1	38.0	30.0	44.1	23.6	33.7	24.1	33.2	23.0	29.2
Snus sales (kilograms)	1.8	5.5	2.7	6.3	5.1	8.5	6.5	9.5	5.2	7.4	5.5	7.6	5.0	6.3
Control variables														
New Covid-19 cases	0.0	25.8	0.0	23.8	0.0	88.0	0.0	81.3	0.0	9.5	0.0	2.8	0.0	15.4
Holiday home proportion	0.11	0.11	0.14	0.13	0.10	0.10	0.07	0.07	0.17	0.17	0.28	0.28	0.27	0.27
Unemployment rate	0.03	0.06	0.03	0.05	0.02	0.05	0.02	0.05	0.02	0.04	0.02	0.04	0.02	0.04
Population (in thousands)	45.5	45.9	42.5	42.8	134.3	136.1	147.4	149.0	25.8	25.9	11.9	12.2	44.2	43.9

Notes: This table shows the mean value of the outcome and control variables. All variables are measured at the week-store level. The mean values are calculated by year and duration groups, where the duration group is defined by the driving duration to the closest Swedish store when all Swedish counties were open for cross-border shopping.

Some interesting patterns emerge from Table 2.3. First, we see that average weekly sales at the store level are significantly higher in 2020 than in 2019, regardless of the driving duration to Sweden. This indicates that the pandemic had a positive impact on grocery store sales also in regions where cross-border shopping is unlikely to be an important factor. As discussed above, this can be explained by factors such as closed restaurants, remote working, and restrictions on international travel. However, the relative growth is significantly larger for stores located close to Sweden than for stores located further away. We also see that the average number of visits per store and per week is higher in 2020 than in 2019 for the <30 minutes and 30–60 minutes categories, but that the reverse is true for the other duration categories. However, basket size is increasing from 2019 to 2020 in all duration categories. There is substantial variation in the average number of new COVID-19 cases in the municipality in which a store is located. Infection rates are highest in the 60–90 minutes and 90–120 minutes categories. The mean unemployment rate is growing from 2019 to 2020 across the duration groups, with no clear indications that some duration categories are affected more than others. Finally, the mean population in the municipality in which a store is located varies quite a bit between the duration groups, but changes between 2019 and 2020 are relatively small.

Table 2.3 already gives an indication that the COVID-19 pandemic affected grocery store activity in regions close to Sweden differently than in regions further from the border. Our hypothesis is that this difference is driven by travel restrictions that made cross-border shopping more difficult (indeed mostly impossible) during the pandemic. Figure 2.3 provides some initial informal support for our hypothesis. The top panels of Figure 2.3 illustrate how, prior to the pandemic, product categories that are well suited for cross-border shopping (beer, cheese, meat, soda, sweets, and tobacco) account for an increasing share of store level sales as we move away from the border. Conversely, product categories that are less suitable for cross-border shopping (freshly baked products, ice cream, milk, and ready-made food) account for a decreasing share. However, as can be seen from the bottom panels of Figure 2.3, these trends are not prominent in the period from March to July 2020 when all Swedish border regions were closed for cross-border shopping.[15]

[15]As reported in Table A1 in the Appendix, the differences between the sales shares in the >180 minute category and the other duration categories are mostly highly statistically significant in the pre-COVID period, and mostly statistically insignificant in the period during the pandemic.

Category sales shares before (top panels) and during (bottom panels) the COVID-19 pandemic

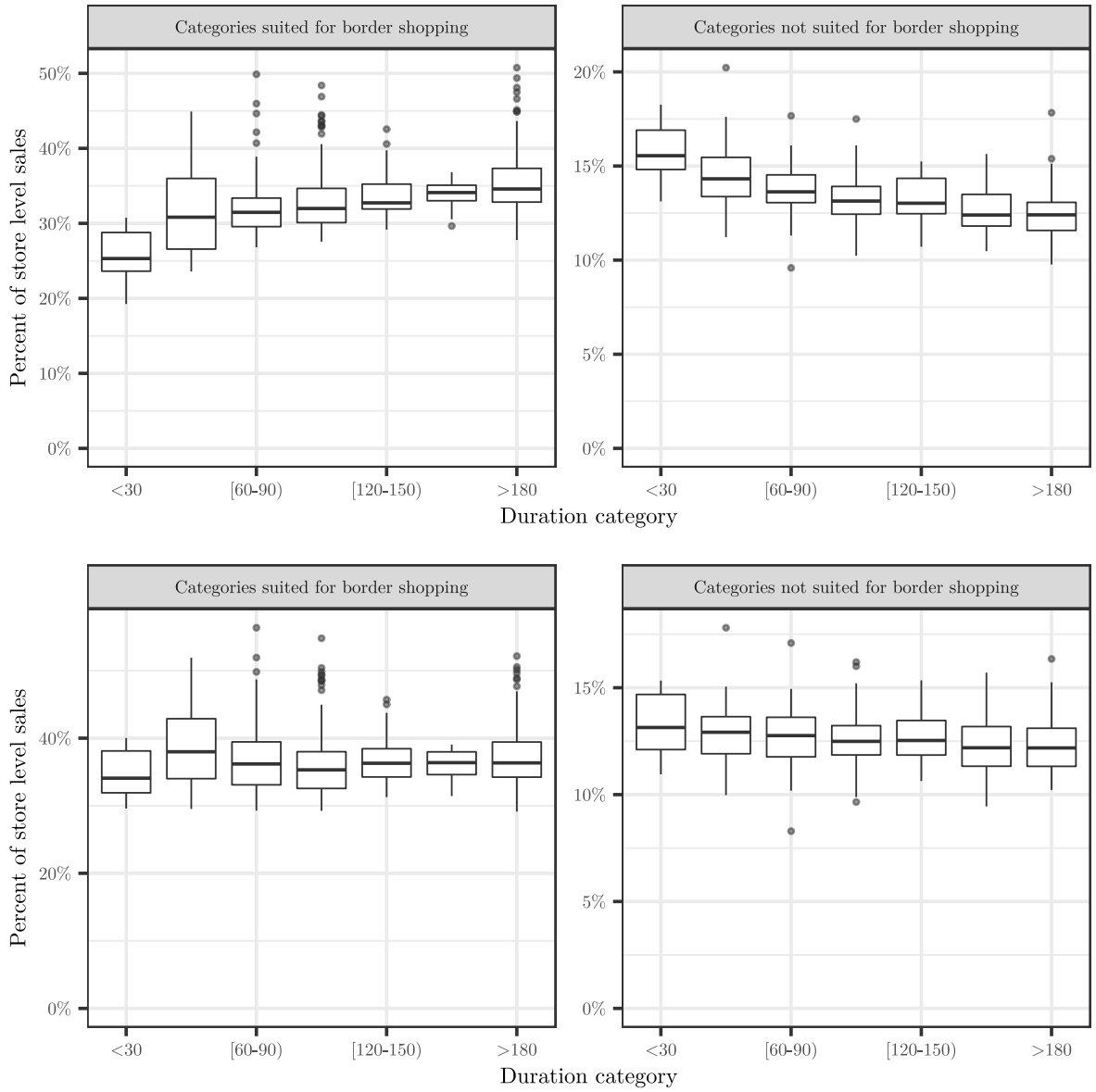


Figure 2.3: The panels on the left-hand side show boxplots of the share of total store-level sales that are accounted for by the categories sweets, soda, cheese, meat, beer, and tobacco. The panels on the right-hand side show boxplots of the share of total store-level sales that are accounted for by the categories ice cream, milk, freshly baked, and ready-made. The period in the top panels is all weeks between the week starting on Monday 31 December 2018 and the week starting on Monday 2 March 2020; in the bottom panels, it is all weeks between the week starting on Monday 16 March 2020 and the week starting on Monday 13 July 2020. During these weeks, all Swedish border counties were closed for cross-border shopping. The lower and upper hinges correspond to the first and third quartiles. The upper whisker extends from the hinge to the largest value no further than 1.5×IQR from the hinge. The lower whisker extends from the hinge to the smallest value at most 1.5×IQR of the hinge.

2.4 The Effect of Cross-Border Shopping on Retail Activity

We are interested in estimating the effect of the availability of cross-border shopping on grocery sales in Norway. Our empirical strategy exploits the fact that travel restrictions due to the COVID-19 pandemic introduced within-store variability with regard to the accessibility of border shopping in Sweden. In addition, because buying groceries in Sweden is not an equally viable alternative to buying groceries locally across Norway, the travel restrictions do not affect all Norwegian stores in the same way. This enables us to estimate the effect of cross-border shopping on retail activity by including both store fixed effects (capturing unobservable time-invariant differences between the stores) and time fixed effects (capturing country-wide shocks that affect all the stores equally).

We follow Friberg, Steen, and Ulsaker (2022) in considering the effect of cross-border shopping on retail activity in Norwegian stores that are located within a 180-minute drive from the closest Swedish grocery store. The variable B_{st} takes the value one if there is an accessible Swedish store less than a 180-minute drive from store s in period t, and takes the value zero otherwise. A Swedish store will be considered accessible if it can be visited without triggering a quarantine stay upon re-entering Norway. In periods where no Swedish stores are accessible, B_{st} will be zero for all stores in the dataset. We assume that cross-border shopping is not a viable alternative for stores located 180 minutes or more from the closest Swedish grocery store. This group of stores will therefore be assumed to be unaffected by the travel restrictions, effectively functioning as a control group in our analysis, by providing a reference point from which we can estimate the effect of the accessibility of cross-border shopping for stores closer to the border.[16] The key identifying assumption that enables a causal interpretation of our results is that the underlying trend in grocery sales is not dependent on the distance to Sweden. In Section 2.6, we provide evidence in support of this assumption.

It seems unlikely that all stores within 180 minutes from a Swedish store are equally affected by cross-border shopping. We therefore interact the binary variable B_{st} with a set of indicator variables D^j_{st}, which capture the driving duration from store s to the closest Swedish store that is accessible in period t. We measure duration in 30-minute bins, indexed by j, between 0 and 180, with a final category capturing cases where the closest available store is more than 180 minutes away or where there are no Swedish stores available because all border counties are closed for cross-border shopping. The

[16] One could of course assume that stores more than 180 minutes from the closest Swedish grocery store are also affected by the availability of cross-border shopping. In Section 2.6 we re-estimate our main model using only stores located at least 300 minutes from Sweden as the control group.

equation we estimate is

$$Y_{st} = \gamma_s + \lambda_t + \sum_j B_{st} D^j_{st} \delta_j + \epsilon_{st}. \tag{2.1}$$

In Equation 2.1, Y_{st} is the outcome of interest (e.g., store-level sales). The variables of interest are δ_j, which estimate the effect of cross-border shopping on sales for the different duration categories. γ_s and λ_t are store and time fixed effects, respectively. The treatment has a geographic dimension and we cluster standard errors on the municipal level.

Table 2.4 reports the results for three different store-level outcome variables: the natural logarithm of total weekly sales, the natural logarithm of the number of weekly customers, and the natural logarithm of the average basket size.

Table 2.4: Effect of cross-border shopping on grocery store activity: main specification

	Sales	Store visits	Basket size
	(1)	(2)	(3)
Duration $<$ 30 $\times$ B	-0.294***	-0.174***	-0.121***
	(0.017)	(0.015)	(0.009)
30 $\leq$ Duration $<$ 60 $\times$ B	-0.207***	-0.135***	-0.073***
	(0.016)	(0.012)	(0.011)
60 $\leq$ Duration $<$ 90 $\times$ B	-0.081***	-0.050*	-0.030***
	(0.023)	(0.020)	(0.007)
90 $\leq$ Duration $<$ 120 $\times$ B	-0.030	-0.009	-0.021***
	(0.015)	(0.014)	(0.006)
120 $\leq$ Duration $<$ 150 $\times$ B	0.004	0.016	-0.011*
	(0.014)	(0.013)	(0.005)
150 $\leq$ Duration $<$ 180 $\times$ B	-0.007	0.005	-0.013
	(0.020)	(0.019)	(0.007)
Observations	42,000	42,000	42,000
Stores	400	400	400

Notes: This table reports results from estimation of Equation 2.1. In Column 1, the dependent variable is the natural logarithm of weekly sales. In Column 2, the dependent variable is the natural logarithm of the number of weekly store visits. In Column 3, the dependent variable is the natural logarithm of the average weekly basket size, defined as weekly sales divided by weekly store visits. The standard errors reported in parentheses are clustered at the municipality level. *** $p < 0.001$; ** $p < 0.01$; * $p < 0.05$.

From Column 1 of Table 2.4, we see that the availability of cross-border shopping is

estimated to lead to a 25% $((\exp(-.294)-1)*100)$ reduction of grocery store sales for stores located within 30 minutes of the closest Swedish store.[17] As one would expect, the size of the effect is reduced as we move further away from the border, eventually becoming insignificant for stores located between 90 and 120 minutes from the closest Swedish store.

Inspecting the relative contributions of store visits and basket size, we see that the contribution of fewer store visits is large close to the border but diminishes as we move away from the border. Conversely, the relative contribution of basket size is larger further away from the border, and the effect is statistically significant at the one percent level even at travel durations between 90 and 120 minutes. A possible explanation for this pattern could be that households living close to the border can cross the border regularly and cover much of their daily grocery needs in Sweden. These households can therefore, to a large extent, replace store visits in Norway with cross-border shopping, whenever the border is open. Households located further away from the border, however, will find it more difficult to substitute store visits in Norway with store visits in Sweden. At these distances, the average basket size bought in Norway could still be sensitive to cross-border shopping, as even infrequent cross-border shopping could cover the needs in categories well suited for cross-border shopping, thus reducing the average basket size in the Norwegian stores.[18]

This explanation can be investigated by estimating the effect of cross-border shopping on the sales in categories that are more or less suitable for (long-distance) cross-border shopping. In Table 2.5, we consider the effect of cross-border shopping in four product categories that are difficult to transport over distance. We see that the estimated effects are relatively large close to the border, but that, as expected, the effects taper off quickly and are statistically insignificant for travel durations above 60 minutes.

[17] Because the effects we estimate often are large in magnitude, the coefficients in our log-level regression models can be poor approximations to the percentage effect of the dependent variable on the outcome. Whenever we refer to a percentage effect in our discussions of the regression results, we therefore carry out the exact calculation in the same manner as we do here.

[18] It may be tempting to think of this decomposition as reflecting the extensive margin (shop in Sweden vs shop in Norway) and the intensive margin (how much to purchase). Note however that they do not rely on traveling during COVID-19 but on how patterns change when households that could travel before now cannot. Our decomposition of sales into store visits and basket size is similar to the decomposition of sales growth in US grocery stores performed by Einav et al. (2021).

Table 2.5: Effect of cross-border shopping on sales in categories not suitable for cross-border shopping

	Ice cream	Milk	Fresh bake	Ready made
	(1)	(2)	(3)	(4)
Duration $< 30 \times B$	-0.123***	-0.220***	-0.151***	-0.142***
	(0.021)	(0.022)	(0.026)	(0.027)
$30 \leq$ Duration $< 60 \times B$	-0.095***	-0.113***	-0.101***	-0.114***
	(0.014)	(0.016)	(0.019)	(0.016)
$60 \leq$ Duration $< 90 \times B$	-0.036	-0.029	-0.042	-0.007
	(0.025)	(0.022)	(0.025)	(0.025)
$90 \leq$ Duration $< 120 \times B$	0.019	-0.003	-0.018	0.021
	(0.017)	(0.015)	(0.015)	(0.015)
$120 \leq$ Duration $< 150 \times B$	0.035	0.029	0.005	0.033*
	(0.019)	(0.016)	(0.018)	(0.017)
$150 \leq$ Duration $< 180 \times B$	-0.024	0.014	0.006	0.000
	(0.024)	(0.020)	(0.020)	(0.022)
Observations	41,995	41,993	41,994	41,993
Stores	400	400	400	400

Notes: This table reports results from estimation of Equation 2.1. Each column shows the result of an estimation where the dependent variable is the weekly sales at store level (in NOK) of the given product category. The standard errors reported in parentheses are clustered at the municipality level. *** $p < 0.001$; ** $p < 0.01$; * $p < 0.05$.

In Table 2.6, we consider seven product categories that should be more suitable for cross-border shopping also at relatively large distances.[19] In all seven categories, we see large effects that are also apparent relatively far from the border. For example, cross-border shopping is estimated to reduce the sales of soda by about 7% as far away as 90–120 minutes from the closest Swedish store. We find the largest effects for beer and tobacco products, which is not surprising given that prices are substantially higher for these categories in Norway than in Sweden, and given that both beer and tobacco are easy to store and transport. Closest to the border, cigarette sales are reduced by about 67%, while beer sales are reduced by about 40%, due to the cross-border shopping. We also see that the effects are substantial even more than 120 minutes from the closest Swedish store.

[19]We consider cheese, meat, sweets and soda (the same four categories as in Friberg, Steen, and Ulsaker (2022)), in addition to beer, cigarettes, and snus.

Table 2.6: Effect of cross-border shopping on sales in categories well suited for cross-border shopping

	Cheese	Meat	Sweets	Soda	Beer	Cigarettes	Snus
	(1)	(2)	(3)	(4)	(5)	(6)	(7)
Duration $< 30 \times B$	-0.281***	-0.519***	-0.349***	-0.660***	-0.510***	-1.102***	-1.096***
	(0.031)	(0.021)	(0.020)	(0.036)	(0.046)	(0.109)	(0.054)
$30 \leq$ Duration $< 60 \times B$	-0.178***	-0.324***	-0.212***	-0.375***	-0.310***	-0.681***	-0.769***
	(0.031)	(0.023)	(0.021)	(0.035)	(0.041)	(0.045)	(0.052)
$60 \leq$ Duration $< 90 \times B$	-0.074**	-0.103***	-0.084**	-0.150***	-0.144***	-0.310***	-0.304***
	(0.024)	(0.027)	(0.028)	(0.025)	(0.026)	(0.038)	(0.042)
$90 \leq$ Duration $< 120 \times B$	-0.038	-0.036	-0.029	-0.076***	-0.091***	-0.141***	-0.138***
	(0.020)	(0.022)	(0.021)	(0.022)	(0.026)	(0.025)	(0.027)
$120 \leq$ Duration $< 150 \times B$	0.004	-0.017	0.006	-0.048*	-0.053**	-0.070**	-0.086***
	(0.019)	(0.020)	(0.017)	(0.019)	(0.019)	(0.022)	(0.025)
$150 \leq$ Duration $< 180 \times B$	-0.014	-0.033	0.003	-0.042	-0.031	-0.045*	-0.048
	(0.023)	(0.022)	(0.025)	(0.025)	(0.026)	(0.020)	(0.030)
Observations	41,994	41,993	41,998	41,997	41,996	41,997	41,927
Stores	400	400	400	400	400	400	400

Notes: This table reports results from estimation of Equation 2.1. Each column shows the result of an estimation where the dependent variable is the weekly sales at store level (in NOK) of the given product category. The standard errors reported in parentheses are clustered at the municipality level. *** $p < 0.001$; ** $p < 0.01$; * $p < 0.05$.

2.5 The Effect of Cross-Border Shopping on Sales and Tax Revenue

2.5.1 Tax Revenue and Cross-Border Shopping: An Overview of Approaches

To frame the exercise that we perform in the following, and to be able to compare with other possible approaches, let us present a stylized view of how demand, prices, and tax revenue are related to cross-border shopping. We can then express the local demand in Norway for good g by household h at location l as depending on the price of the good in Norway (p_N), the price in Sweden translated into NOK at the current exchange rate ($e \times p_S$) and the cost of cross-border shopping (cbc) that varies by distance to Sweden and by household characteristics.[20] The latter includes both direct outlays on transport and opportunity costs, and such costs of cross-border shopping imply that the law of one price need not hold.[21] Thus, formally,

$$q_{ghl} = f(p_N, e \times p_S, cbc_{hl}). \tag{2.2}$$

In any standard model of pricing in imperfectly competitive markets, price can be expressed as a markup (that depends on the elasticity of demand, ϵ) times marginal costs (mc_N) whilst recognizing that excise tax (t_N) and VAT will also affect price. We can thus express price as

$$p_N = (1 + VAT_N)\left(\frac{\epsilon}{\epsilon + 1}\right)(t_N + mc_N). \tag{2.3}$$

The outlined framework can be used to describe several questions of potential interest. One set of questions regards how local demand depends on marginal changes in the relative prices. This is essentially the exercise that is performed in Friberg, Steen, and Ulsaker (2022) where we rely on the shifts in relative price for some product categories (cheese, meat, sweets, soda) generated by exchange rate changes.

A second set of questions pertains to how markups respond to the costs of cross-border shopping. If prices were set at the store or regional level, one would have expected that the non-availability of a set of competitors would lead to higher markups. As discussed, however, all the Norwegian grocery retail chains keep the same prices across all of Norway which implies that such incentives are balanced by areas far from the border. The COVID-19 pandemic clearly affected demand and costs

[20] To simplify this discussion, we disregard that the decision to shop across the border will typically depend on several prices. For instance a user of snus may shop for cheese in Sweden even if the savings on cheese in itself would not have motivated a cross-border trip.

[21] See, for example, Gopinath et al. (2011) for an influential study of reasons for price differences in grocery retailing along the Canada–US border.

in many ways and we do not attempt to tease out the relative contribution of costs and markups to any price changes that we observe.

A third set of possible questions concerns how changes in VAT and excise taxes would affect local demand and the extent of cross-border shopping. Such an analysis hinges on good estimates of the extent to which tax changes are passed-through into retail prices; for example, see Weyl and Fabinger (2013) for theoretical considerations, and for empirical studies, see Ardalan and Kessing (2021) for European beer markets, DeCicca, Kenkel, and Lovenheim (2022) for tobacco, and Allcott, Lockwood, and Taubinsky (2019*a*) for sugar taxes. Whether such effects are desirable or not is not only a matter of tax consequences, but the motivation for excise taxes is also important. A lower excise tax will not only decrease cross-border shopping and have consequences for tax revenue, it will also raise domestic consumption with complex welfare effects that interact with consumer heterogeneity; for example, see Griffith, O'Connell, and Smith (2019) for an analysis of heterogeneity and alcohol taxation, and Allcott, Lockwood, and Taubinsky (2019*a*) for taxes on sugar-sweetened beverages.

In principle, the above kinds of questions could be addressed with the dataset on hand. Note, on the one hand, that there are no important changes in VAT or excise taxes during the period of study that could be used to identify the effects of such changes. On the other hand, the dataset is uniquely suited to examine the overall effect of border closings—not studying the effect of marginal changes but studying the overall effect of cross-border shopping being unavailable on commodity tax revenue, an estimation that we turn to in the following. It deserves to be emphasized, however, that many important questions intersect. For instance, the effect of closing cross-border shopping depends on the extent of price differences when borders are open. In addition, it can be pointed out that a number of factors are kept constant in the experiment that may be endogenous in the longer run—for instance the level of excise taxes and the locations of retail stores.[22]

2.5.2 Tax Revenue and Cross-Border Shopping: An Estimation of the Effect of Closed Borders

In this subsection, we seek to quantify the effect of cross-border shopping on store sales and tax revenue. The detailed data allow for an examination of cross-border shopping of sales at the category level as well as the effect on tax revenue from excise taxes and VAT. As discussed in Section 2.3, several of the product categories most frequently bought in Sweden by Norwegian customers are subject to excise taxes in Norway. In 2019, there were excise taxes for beer, cigarettes, snus, soda, and sweets. These

[22]In the long run we expect the absence of cross-border shopping to lead to the establishment of new stores in Norwegian border areas, an effect documented in response to large exchange rates swings in Canada by Campbell and Lapham (2004).

excise taxes are non-trivial. For instance, cigarettes faced an excise tax of 2.63 NOK per cigarette in 2019, equivalent to around 0.25 euros using the average EUR/NOK exchange rate for 2019 (the average consumer price was 6.17 NOK or about 0.63 euros per cigarette).[23] Of particular interest may be the "sugar tax". In 2019 this excise tax for chocolates and sweets amounted to 20.82 NOK per kilogram (representing 8.7% of the average consumer price); for beverages, the tax was 4.82 NOK per liter (representing 21.4% of the average consumer price).[24]

Because the excise taxes were calculated on the basis of units sold, we first estimate the effect of cross-border shopping on the volume sold in the different categories, and then use these estimates to calculate the estimated loss in excise taxes resulting from cross-border shopping. In addition to category specific excise taxes, all products sold in grocery stores are subject to VAT. Because our sales data include information about VAT, we can estimate the effect of cross-border shopping on VAT revenue directly.[25]

In Table 2.7, we report the estimated effect of cross-border shopping on VAT and on the volume sold in the categories subject to excise taxes. The models are defined by Equation 2.1, and the outcome variables are log-transformed.

[23]The average exchange rate was 9.8527. The exchange rates are available through Norges Bank (2022).

[24]The "sugar tax" has been the subject of considerable discussion in Norway. It was subject to a sharp and unexpected increase in 2018, followed by a partial lowering in 2019. In 2021 the tax was abolished, arguably partly in response to concerns about cross-border shopping as borders reopened. See Table A2 in the Appendix for average consumer prices (per unit) and excise tax rates.

[25]There are two relevant VAT rates for Norwegian grocery stores, 15% for food items and 25% for non-food items, our dataset provides exact values for VAT for each product.

Table 2.7: Effect of cross-border shopping on VAT and category volumes

	VAT	Sweets	Soda	Beer	Cigarettes	Snus
	(1)	(2)	(3)	(4)	(5)	(6)
Duration $< 30 \times B$	-0.314***	-0.373***	-0.789***	-0.521***	-1.099***	-1.100***
	(0.016)	(0.022)	(0.047)	(0.048)	(0.108)	(0.055)
$30 \leq$ Duration $< 60 \times B$	-0.226***	-0.217***	-0.433***	-0.322***	-0.679***	-0.780***
	(0.016)	(0.022)	(0.049)	(0.044)	(0.046)	(0.054)
$60 \leq$ Duration $< 90 \times B$	-0.089***	-0.090**	-0.174***	-0.153***	-0.308***	-0.313***
	(0.023)	(0.028)	(0.024)	(0.027)	(0.038)	(0.042)
$90 \leq$ Duration $< 120 \times B$	-0.034*	-0.034	-0.086***	-0.103***	-0.139***	-0.146***
	(0.015)	(0.021)	(0.021)	(0.028)	(0.025)	(0.028)
$120 \leq$ Duration $< 150 \times B$	0.001	0.008	-0.063**	-0.062**	-0.066**	-0.091***
	(0.014)	(0.018)	(0.021)	(0.021)	(0.022)	(0.025)
$150 \leq$ Duration $< 180 \times B$	-0.008	0.008	-0.052*	-0.046	-0.045*	-0.054
	(0.020)	(0.026)	(0.023)	(0.026)	(0.020)	(0.028)
Observations	41,995	41,998	41,997	41,996	41,997	41,927
Stores	400	400	400	400	400	400

Notes: This table reports results from estimation of Equation 2.1. Columns 2–6 show the result of an estimation where the dependent variable is the weekly sales (in volume) at store level for the given product category. The standard errors reported in parentheses are clustered at the municipality level. *** $p < 0.001$; ** $p < 0.01$; * $p < 0.05$.

As expected, the effect of cross-border shopping on VAT is similar to the effect of cross-border shopping on store sales (as reported in Column 1 of Table 2.4), and the effect of cross-border shopping on the category sales by volume is similar to the effect on category sales in NOK (as reported in Table 2.6), although the estimated effect on soda volume is slightly higher than the estimated effect on sales in NOK. This suggests that lower-priced sodas are more affected by cross-border shopping than premium products, consistent with the notion that it is the more price-sensitive consumers that stock up sufficient volumes of soda from cross-border trips to affect local sales.[26]

Table 2.8 contains our estimates of the effect of cross-border shopping on total sales and VAT, while the estimated effect on category-level sales and excise tax is reported in Table 2.9. To obtain the reported estimates, we proceed as follows. For each duration category, we calculate the mean value of total store sales, category sales volumes, and VAT in our estimation sample in 2019. We then use the estimated effects reported in Table 2.4 (for store level sales) and Table 2.7 (for category volumes and VAT) to calculate the counterfactual closed-border outcome the last full normal year before COVID closures. To get an estimate of the economic impact of cross-border

[26]This is in some contrast to Wang (2015) who, using US household level data for 2002–2004, finds that low-income households are less likely to stockpile soda.

shopping, we then multiply these estimated effects by the total number of grocery stores in each duration category (i.e., not only the stores in our estimation sample).[27]

For instance, stores in the estimation sample within a 30-minute drive from the closest Swedish border sold on average 8.69 metric tons of sweets in 2019 (see Table A3 in the Appendix). For these stores, the estimated effect of cross-border shopping on the log of sweet sales (in metric tons) is -0.373 (see Column 2 of Table 2.7). The counterfactual mean level of sales is therefore given by $8.69/\exp(-.373) = 12.62$, which gives an estimated mean loss due to cross-border shopping equal to $12.62 - 8.69 = 3.93$. Multiplying the actual and counterfactual mean sales volume and the estimated mean sales volume loss by the total number of grocery stores in this duration category (62) gives (subject to rounding) numbers reported in Column 1 of Table 2.9.

Table 2.8: Estimated effect of border trade on store sales and VAT

	<30	[30, 60)	[60, 90)	[90, 120)	[120, 150)	[150, 180)	>180	Total
Store distribution								
Number of stores	62	188	371	762	279	176	1985	3823
Proportion of stores	0.02	0.05	0.10	0.20	0.07	0.05	0.52	1.00
Sales (million NOK)								
Counterfactual	4203	11,091	29,265	65,292	16,904	10,031	86,381	223,166
Actual	3132	9015	26,999	63,369	16,974	9959	86,381	215,828
Loss	1071.6	2076.2	2266.2	1922.7	−70.4	71.5	0.0	7337.8
VAT (million NOK)								
Counterfactual	602	1602	4185	9283	2426	1452	12,531	32,081
Actual	440	1278	3828	8971	2428	1440	12,531	30,916
Loss	162.3	324.0	356.6	311.5	−2.3	12.1	0.0	1164.3

Notes: The duration groups are defined by driving duration to the closest Swedish store when all Swedish counties were open for cross-border shopping.

We see that the economic significance of cross-border shopping is substantial. We commence with the effect on store sales and VAT as reported in Table 2.8. As expected, the effects are strongest close to the border and then gradually diminish with travel duration to the closest Swedish stores. Sales are reduced by 25.5% $(1071.6/4203)$ in stores that are within 30 minutes' driving distance from a Swedish store due to cross-border shopping, and VAT revenue from stores in these locations is reduced by 27% $(162.3/602)$. For locations between 30 and 60 minutes away, the corresponding VAT loss is 20.2%, and for locations between 60 and 90 minutes away, the corresponding VAT loss is 8.5%. These are clearly substantial numbers, and as discussed in the

[27] An underlying assumption of the results reported in Tables 2.8 and 2.9 is that, within each duration category, the stores in our estimation sample are representative of the grocery stores. Because the umbrella chain to which our sample stores belong has stores in all market segments, we believe that this is a reasonable assumption. Furthermore, in Figure A1 in the Appendix, we compare the yearly sales amounts of the estimation stores and the population of stores (within each duration category). We find that the distributions are very similar.

Introduction, numbers of this magnitude serve to motivate minimum levels of VAT within the European single market. Even in a large and mountainous country such as Norway, the VAT loss is non-trivial, also at the national level, with an estimated VAT loss of 3.6% due to cross-border shopping.

Table 2.9: Estimated effect of border trade on sales and tax income

	<30	[30,60)	[60,90)	[90,120)	[120,150)	[150,180)	>180	Total
Sweets								
Sales (metric tons)								
Counterfactual	783	1869	4987	11,221	3098	1963	16,913	40,834
Actual	539	1504	4557	10,846	3122	1978	16,913	39,459
Loss	244	365	429	375	−24	−15	0	1375
Tax (millions NOK)								
Counterfactual	11.0	26.2	69.9	157.2	43.4	27.5	237.0	572.2
Actual	7.6	21.1	63.9	152.0	43.8	27.7	237.0	553.0
Loss	3.4	5.1	6.0	5.3	−0.3	−0.2	0.0	19.3
Soda								
Sales (thousand liters)								
Counterfactual	10,184	22,910	55,269	115,302	37,028	22,354	174,023	437,069
Actual	4626	14,862	46,449	105,844	34,777	21,212	174,023	401,792
Loss	5558	8048	8820	9458	2252	1142	0	35,277
Tax (million NOK)								
Counterfactual	36.1	81.3	196.0	408.9	131.3	79.3	617.2	1550.1
Actual	16.4	52.7	164.7	375.4	123.3	75.2	617.2	1425.0
Loss	19.7	28.5	31.3	33.5	8.0	4.0	0.0	125.1
Beer								
Sales (thousand liters)								
Counterfactual	4245	11,399	27,561	58,211	16,131	10,205	89,595	217,347
Actual	2522	8265	23,651	52,489	15,168	9748	89,595	201,438
Sales loss (thousand liters)	1723	3135	3910	5721	963	458	0	15,909

Continued on next page

Table 2.9: Estimated effect of border trade on sales and tax income (continued)

	<30	[30,60)	[60,90)	[90,120)	[120,150)	[150,180)	>180	Total
Tax (million NOK)								
Counterfactual	94.1	252.7	610.9	1290.3	357.6	226.2	1986.0	4817.9
Actual	55.9	183.2	524.3	1163.5	336.2	216.1	1986.0	4465.2
Loss	38.2	69.5	86.7	126.8	21.3	10.1	0.0	352.7
Cigarettes								
Sales (thousand items)								
Counterfactual	28,046	83,625	172,488	307,781	104,075	66,194	547,895	1,310,104
Actual	9341	42,405	126,734	267,821	97,387	63,308	547,895	1,154,891
Loss	18,705	41,220	45,754	39,961	6688	2885	0	155,213
Tax (million NOK)								
Counterfactual	75.5	225.2	464.6	829.0	280.3	178.3	1475.7	3528.5
Actual	25.2	114.2	341.3	721.3	262.3	170.5	1475.7	3110.5
Loss	50.4	111.0	123.2	107.6	18.0	7.8	0.0	418.0
Snus								
Sales (metric tons)								
Counterfactual	15.76	57.64	135.59	298.57	83.08	53.22	518.77	1162.63
Actual	5.25	26.42	99.19	258.08	75.85	50.44	518.77	1033.99
Loss	10.51	31.22	36.40	40.49	7.23	2.79	0.00	128.65
Tax (million NOK)								
Counterfactual	15.8	57.8	136.0	299.6	83.4	53.4	520.5	1166.5
Actual	5.3	26.5	99.5	258.9	76.1	50.6	520.5	1037.4
Loss	10.5	31.3	36.5	40.6	7.3	2.8	0.0	129.1

Notes: The duration groups are defined by driving duration to the closest Swedish store when all Swedish counties were open for cross-border shopping.

We now turn to an estimation of the loss of sales and excise taxes for products sold in grocery stores due to cross-border shopping. Table 2.9 reports the estimated effects on volumes sold and on commodity tax revenue for different duration intervals from the border.[28]

Again, effects are strongest close to the border and then gradually diminish. In several cases, the estimated foregone tax revenue is very large for stores within a 30-minute drive from the closest Swedish store, and for cigarettes, soda and snus, tax revenue more than doubles in these border areas when we move to a counterfactual situation with no cross-border shopping. However, only 2% of the grocery stores in Norway are located this close to Sweden, whereas more than a third of the grocery stores are located within 120 minutes from the closest Swedish store, and we see substantial effects also at this distance. The effects are largest for beer, soda and tobacco products. For areas located between 90 and 120 minutes from the closest Swedish store, the estimated loss of tax revenue in these categories is close to, or above, 10% of the actual tax revenue. Aggregating up to the national level, the effects are still substantial with an estimated loss of excise taxes on, for instance, soda of 8.1% and on cigarettes of 11.9%. Total estimated loss of tax revenue (VAT and excise taxes) from cross-border shopping at the national level is about 2.3 billion NOK.

Still, whether numbers are seen as large or small at the national level is partly in the eye of the beholder. The impact on some regions and for some product groups is very large but in terms of the overall direct impact of cross-border shopping on tax revenue the consequences for Norway are limited, as our estimate of 2.3 billion NOK corresponds to approximately 0.2% of taxes and levies from non-oil-related income.[29]

2.6 Discussion of the Identification Assumption

The key identification assumption that we make is that the underlying trend in grocery store activity does not vary with the travel duration to Sweden. In this section we provide evidence in support of this underlying assumption, and show that the results found in Section 2.4 are robust to a number of alternative specifications.

Figure 2.4 plots the natural logarithm of average weekly sales for stores in the different pre-COVID duration categories. The time series are normalized by dividing

[28] In the analysis above, we also examined the effect of cross-border shopping on sales of meat and cheese that are not subject to excise taxes but to import tariffs. The motivation for such tariffs is to protect domestic production rather than a combination of externalities and internalities directly linked to consumption. Evaluating the impact of those tariffs could be done in similar ways but would require keeping track of the origin of different products and is arguably of less interest outside Norway than the examination of how cross-border shopping interacts with excise taxes at different distances (travel duration) from the border.

[29] According to the 2019 government budget (Finansdepartementet, 2019, p. 52), revenue from taxes and levies excluding oil-related revenue amounted to 1031.3 billion NOK. As is well known, Norway is a major oil exporter, and taxes and levies from the oil sector contributed a further 312.8 billion NOK.

by the values in the week before the Norwegian lockdown was announced (i.e., the week starting on 2 March 2020). The thick gray line represents the stores in the control group, that is, stores located further than 180 minutes from the closest accessible Swedish store before the travel restrictions were implemented. We plot the average weekly sales in the period when all Swedish counties were open to border trade (from 7 January 2019 until 17 March 2020) and in the period when all Swedish counties were closed to cross-border trade (that is, between 17 March and 25 July 2020). While there is some variation in the levels between the groups, they follow each other quite closely over time, and there is no indication that the trends are different. We also see a clear increase in sales after travels to Sweden became restricted for the duration groups closest to the border.

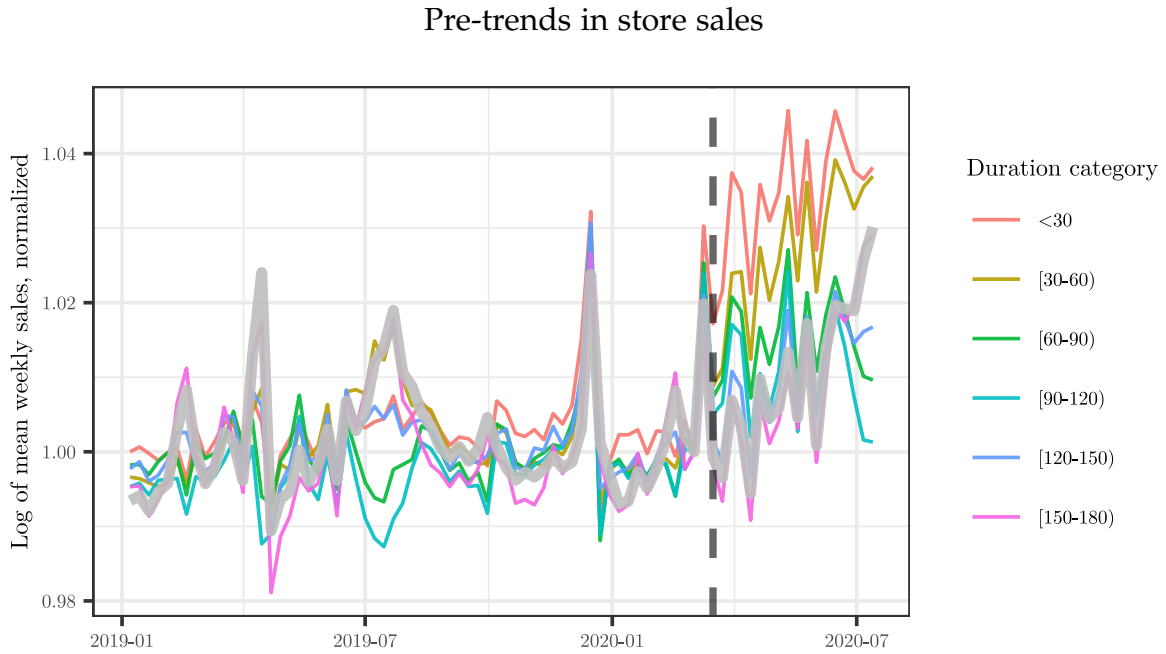


Figure 2.4: The figure shows the natural logarithm of the average weekly sales at the store level (in NOK), for stores in different duration groups. The plotted values have been normalized to the last week before the lockdown was announced (the week starting on 2 March, 2020). The first week included in the figure is the week starting on Monday 7 January 2019. The last week included in the figure is the week starting on Monday 13 July 2020. The dashed vertical line indicates the week travels to Sweden became restricted. The thick gray line represents stores that were located at least 180 minutes from the closest Swedish store.

As a more formal test for common trends, we have estimated the following event study model, where $\tilde{D}_s^j$ indicates the pre-COVID duration category of store s and where we let k be the period the lockdown was announced:

$$
\begin{aligned}
Y_{st} = \gamma_s + \lambda_t + \sum_j 1(t \leq k-6)\tilde{D}_s^j \delta_{-6,j} \\
+ \sum_j \sum_{\tau=-5}^{-2} 1(t = k+\tau)\tilde{D}_s^j \delta_{\tau,j} + \sum_j \sum_{\tau=0}^{3} 1(t = k+\tau)\tilde{D}_s^j \delta_{\tau,j} \\
+ \sum_j 1(t \geq k+4)\tilde{D}_s^j \delta_{4,j} + \epsilon_{st}.
\end{aligned} \tag{2.4}
$$

The parameters of interest are the treatment effects $\delta_{\tau,j}$, which vary according to both pre-COVID duration group and time. We again restrict attention to the period until 25 July 2020. We normalize the effects relative to the week prior to the treatment ($\tau = -1$), and bin the periods at least six weeks before the treatment and the periods at least four weeks after. In Figure 2.5, we plot for each duration category the estimated treatment effects.

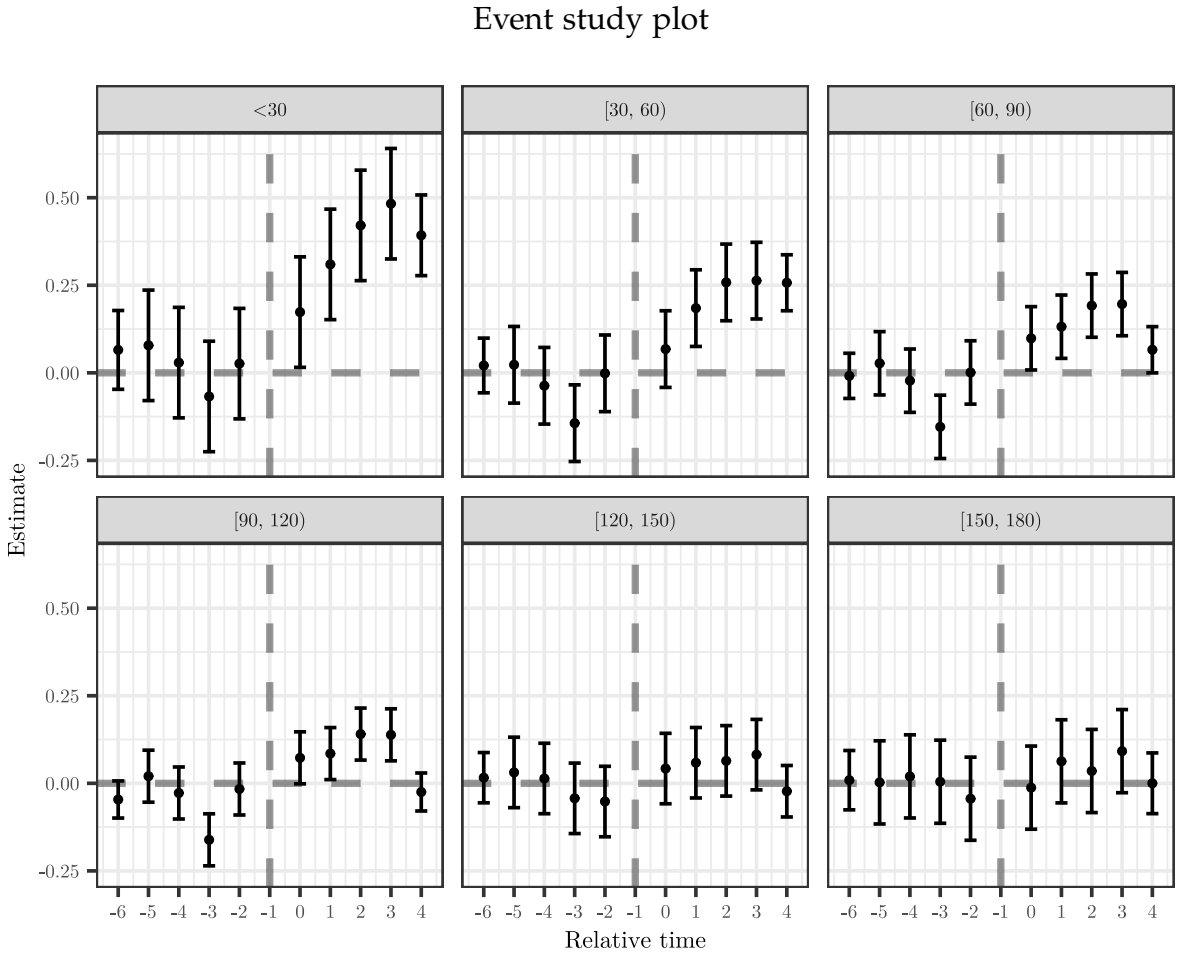


Figure 2.5: The figure shows the estimated treatment effects and 95% confidence intervals from estimation of Equation 2.4. The x-axis shows the time relative to the lockdown was imposed (i.e., the week starting on Monday 9 March 2020).

From Figure 2.5, we see that the leads are generally close to zero and not statistically

significant. An exception is the coefficient for $\tau = -3$, which is negative in several of the duration bins. This can be explained by the fact that this week was a school holiday in parts of Norway. The effect of the border closing is clearly visible through the large positive lags for the duration groups closest to Sweden.

We can also investigate the underlying common trends assumption by adding separate linear time trends for each of the pre-COVID duration categories. If this produces significantly different results from the results reported in Table 2.4, it could reflect diverging underlying trends in sales in different regions rather than the effect of cross-border shopping (Angrist and Pischke, 2008, p. 238). Reassuringly, we see that the results reported in Column 1 of Table 2.10 are very similar to the results in the main specification.

Even though both visual and statistical tests provide support for the common trends assumption, one could still imagine that different regions experienced different shocks that coincided with the closing of the border. Any shocks unrelated to the closing of the border that affected stores differently depending on their distance from the border would bias our results. However, travel restrictions affected stores *within* pre-COVID duration groups differently at different times, because different Swedish counties were open to cross-border shopping at different times during 2020. This variation allows us to include week $\times$ pre-COVID duration group fixed effects in the model. The estimated effects of cross-border shopping now no longer rely on stores from different pre-COVID duration groups having identical underlying developments in sales, but rather that stores in the *same* pre-COVID duration groups share a common underlying trend. The results from this specification are reported in Column 2 of Table 2.10. Compared to the main specification, the estimated effects are somewhat larger in magnitude until the 60–90 duration category, but the general impression is very similar. We also see that the standard errors are slightly larger, which is not surprising given that we are now using only within pre-COVID duration group variation in the accessibility of cross-border shopping to identify the effects.

In our analysis so far, we have considered the effect of cross-border shopping on stores located less than 180 minutes from the closest Swedish store. All stores located 180 minutes or more away have been assumed to be unaffected by the availability of cross-border shopping in Sweden, effectively functioning as a control group. If sales in some of the stores in the control group are in fact negatively affected by the availability of cross-border shopping, the results reported in Table 2.4 would be misleading, with the estimated coefficients biased towards zero. As a robustness check on our results, we therefore estimate our model using only observations where the pre-COVID driving duration to the closest Swedish store is below 180 minutes or above 300 minutes. The control group now consists of stores that are at least 300 minutes away from the closest Swedish store, a distance that should reduce the relevance of

cross-border shopping to a minimum. The results are reported in Column 3 of Table 2.10. The results are very similar to the ones in the main specification.

The results in our main specification are based on an assumption that the effect on grocery store sales in Norway of having an accessible Swedish store a certain distance away is the same before and during the pandemic. That is, we assume that having a Swedish store a 45-minute drive away is the same in September 2019 as in September 2020. However, there could be reasons to believe that this is not the case. For instance, infection rates were typically higher in Sweden than in Norway even in periods where some Swedish counties were open for cross-border shopping, something that could deter Norwegians from cross-border shopping even in periods where Swedish border counties were open. To see whether this implicit assumption affects our results, we re-estimate our model excluding data from the period between the week starting on Monday 27 July 2020, and the week starting on Monday 5 October 2020, a period during which some but not all Swedish border counties were open. We are then left with a pre-period before 17 March 2020 where cross-border shopping was possible and the COVID-19 infection rates were non-existent or very low in both counties, and a post-period where cross-border shopping was not possible. The results are reported in Column 4 of Table 2.10. The point estimates are more or less unchanged from the main specification.

The results in Column 4 are also useful for another reason: several recent methodological papers have shown that the two-way fixed effects regressions we use in this chapter can give biased results if treatment effects are heterogeneous across time or groups, and there is variation in treatment timing (see, e.g., De Chaisemartin and d'Haultfoeuille, 2020; Borusyak, Jaravel, and Spiess, 2024; Callaway and Sant'Anna, 2021; Sun and Abraham, 2021).[30] The results reported in Column 4 of Table 2.10 are based on a restricted time period where there is no variation in treatment timing. This means that we avoid the forbidden comparison of units going into treatment with already treated units (Borusyak, Jaravel, and Spiess, 2024). The fact that the results in this specification are similar to the results in the main specification indicates that they are not biased by heterogeneous treatment effects.[31]

[30]Using the `twowayfeweights` STATA package of De Chaisemartin and d'Haultfoeuille (2020), none of the weights attached to our main regression were negative, satisfying the no sign reversal property.

[31]Several alternative estimators that are robust to heterogeneous treatment effects have been proposed, but most of these estimators do not apply to our natural experiment since treatment is switching. One exception is the estimator of Borusyak, Jaravel, and Spiess (2024), which yields almost identical point estimates to our main specification. Note that one might argue that the specification in Column 4 of Table 2.10 should be the baseline specification as it in a simple way avoids all concerns about treatment heterogeneity across time. We opted for keeping the longer time period, as cross-border shopping can be partly seasonally dependent and we believe it valuable to cover large parts of the year – including both the full summer, fall and early winter. In the end it does not matter much since effects are virtually identical for the short and longer treatment periods. One explanation for the limited differences is that, as seen in Figure 2.2, there are relatively few post-COVID weeks when borders were open, also in summer. Cross-border shopping with Västra Götaland, the region to the south of the capital Oslo and

In our view, the main analysis and robustness checks performed so far provide clear evidence that Norwegian stores close to the border experienced increases in store activity during the COVID-19 pandemic. Our hypothesis is that this increase in store activity was the result of cross-border shopping effectively becoming impossible. An alternative explanation could be that the grocery trade in some or all border regions was affected differently by the COVID-19 pandemic than grocery trade in the rest of Norway. To provide some additional support for our explanation, we estimate a specification where we include several control variables that may both affect grocery demand, and be affected by the pandemic. First, we include a measure of COVID-19 infection rates at the municipality level, specifically, new infections per 1000 inhabitants during the last two weeks.[32] The COVID-19 pandemic also affected the labor market and led to sharp increases in unemployment rates. We therefore include monthly unemployment rates at the municipality level as a control variable. The population dynamics in many Norwegian municipalities also changed during the pandemic. We therefore include population at the municipality-quarterly level as a control variable. Finally, local markets with a large number of holiday homes may have been affected differently by the pandemic than other local markets. First, in the period between 19 March 2020 and 20 April 2020, there was a ban in Norway on staying in holiday homes located outside one's home municipality. Second, after this first period, staying at holiday homes was permitted, while traveling abroad was severely restricted, which could lead to unusually high retail activity in municipalities with a high proportion of holiday homes. To account for these factors, we include an interaction between the proportion of holiday homes in the municipality and an indicator variable, C, which is one in the period after 12 March (when national restrictions were put in place) and zero in the period before 12 March. The results are reported in Column 5 of Table 2.10. Many of these variables affect grocery store sales, but the estimated effects of the interaction terms of interest are little changed compared to the main specification. We do note, however, that the estimated effect of cross-border shopping in the 90–120 minutes duration category is higher and highly statistically significant.

home to several top cross-border shopping stores, was for instance closed during the entire post-COVID period.

[32]That is, the sum of infected individuals in week t and $t-1$ divided by the population of the municipality.

Table 2.10: Effect of border shopping on store sales: robustness checks

	(1)	(2)	(3)	(4)	(5)
Duration $< 30 \times$ B	-0.309***	-0.328***	-0.309***	-0.301***	-0.311***
	(0.020)	(0.074)	(0.017)	(0.016)	(0.018)
$30 \leq$ Duration $< 60 \times$ B	-0.225***	-0.280***	-0.221***	-0.206***	-0.229***
	(0.027)	(0.060)	(0.016)	(0.016)	(0.017)
$60 \leq$ Duration $< 90 \times$ B	-0.101***	-0.199**	-0.090***	-0.077**	-0.109***
	(0.022)	(0.061)	(0.023)	(0.027)	(0.022)
$90 \leq$ Duration $< 120 \times$ B	-0.033	-0.002	-0.042**	-0.034*	-0.059***
	(0.019)	(0.026)	(0.016)	(0.017)	(0.016)
$120 \leq$ Duration $< 150 \times$ B	-0.011	0.022	-0.006	0.010	-0.005
	(0.018)	(0.024)	(0.014)	(0.016)	(0.015)
$150 \leq$ Duration $< 180 \times$ B	-0.018	0.002	-0.017	-0.006	-0.003
	(0.024)	(0.022)	(0.021)	(0.024)	(0.020)
New COVID-19 cases					0.014**
					(0.005)
C $\times$ Holiday home proportion					0.135***
					(0.040)
Unemployment rate					-1.854***
					(0.458)
Population					0.499***
					(0.146)
Observations	42,000	42,000	35,490	37,600	41,016
Stores	400	400	338	400	400
Control group	$>$ 180 min	$>$ 180 min	$>$ 300 min	$>$ 180 min	$>$ 180 min
Linear time trends	Yes	No	No	No	No
Duration group $\times$ week FEs	No	Yes	No	No	No
Time window	Full	Full	Full	Restricted	Full

Notes: This table reports results from estimation of Equation 2.1 with the natural logarithm of sales as the dependent variable. In Column 1, separate linear time trends for duration groups based on the pre-COVID duration to the closest Swedish store are included. The frequency distribution of the groups is reported in Table 2.2. In Column 2, week $\times$ pre-COVID duration group fixed effects are included. In Column 3, we drop observations from stores that, before the COVID-19 restrictions, were located between 180 and 300 minutes from the closest Swedish store. In Column 4, we drop observations from the weeks between and including the week starting on Monday 27 July 2020 and the week starting on Monday 5 October 2020. In these weeks, some, but not all, Swedish border counties were open for cross-border shopping. The variable "New COVID-19 cases" included in Column 5 is the sum of the registered COVID-19 cases in week t and $t-1$ divided by the population of the municipality. C is an indicator variable taking the value one in the weeks after 17 March 2020 (and zero in the weeks before). "Holiday home proportion" is the proportion of residential dwellings that are holiday homes. The unemployment rate is measured at the municipality–month level. Population is measured at the municipality–quarter level. The standard errors reported in parentheses are clustered at the municipality level. *** $p < 0.001$; ** $p < 0.01$; * $p < 0.05$.

For yet additional robustness exercises see the Appendix where we present

pre-treatment plots and specification checks for the other dependent variables considered in Section 2.4: store visits and basket size from Table 2.4, sales in the categories from Tables 2.6 and 2.5, as well as VAT and category volumes from Table 2.7. The results reinforce the impression from this section that our results are very robust to alternative specifications and explanations.

Section 2.E in the Appendix also compares our estimates of lost sales to the surveys of cross-border shopping produced by Norway's national statistical agency (Statistics Norway (2022*a*)), where a random sample of 2000 Norwegians are surveyed about cross-border shopping in each quarter every year. The numbers match up remarkably well, which provides a validation for our results. Importantly, our results also provide validation in the other direction—a well-conducted survey with 2000 respondents sampled nationally paints a picture that is remarkably well in line with observed patterns in this natural experiment. As such, the results should be encouraging for statistical agencies in both Norway and other countries that use surveys to track cross-border shopping.

2.7 Conclusion

Travel restrictions due to the COVID-19 pandemic made cross-border shopping in neighboring Sweden infeasible for Norwegians for most of 2020. In this chapter, we exploit this natural experiment to estimate the effects of cross-border shopping on the grocery retail sector in Norway. The natural experiment is unique in the sense that it mimics the autarky regime often taken as a benchmark in theoretical work on cross-border shopping but rarely observed in practice. The natural experiment allows us to estimate the total effect of cross-border shopping directly, something that is difficult when—as is usually the case in the empirical literature on cross-border shopping—one has to rely only on changes in the relative prices.

Our estimates suggest that cross-border shopping reduces sales in the grocery stores closest to the border by more than a quarter compared to a counterfactual situation with closed borders. For typical cross-border shopping categories such as tobacco, more than half of the sales are lost. By not only estimating the effects of cross-border shopping in the regions closest to the border but also tracking how they gradually diminish as we move away from the border, we are able to quantify the effects at the national level. We estimate that cross-border shopping reduces total grocery sales in Norway by more than 3%, with a corresponding reduction in VAT income.

Let us now comment on the external validity of these results in two dimensions. One dimension concerns the extension to non-COVID periods. COVID-19 was a major shock that affected behavior in many areas. That said, our exploration of the

identification assumption supports the notion that, apart from cross-border shopping, the pandemic affected the treated and control regions in remarkably similar ways. It is also worth noting that Norway's COVID-19 measures were much less restrictive than in most European countries; there was no hard lockdown, for instance. Thus, while necessarily somewhat speculative, we venture to believe that the results are also broadly applicable to more normal times.

Another dimension of external validity regards applicability to other countries. Here we note that border shopping is a significant phenomenon across several borders. Focusing only on Europe, for instance, Switzerland loses sales to several of its neighboring countries, as does Austria, and as do France and Spain for areas close to Andorra. Our results suggest that tax losses can be non-trivial, and clearly illustrate the challenges of independent tax policies when shopping in a neighboring jurisdiction is possible. Excise taxes on products, such as tobacco, alcohol, and soda, can be motivated by both fiscal and public health reasons. Our results demonstrate that a substantial part of the potential tax revenue can disappear through cross-border shopping. For tobacco and soda, we estimate that about 10% of the counterfactual tax income under closed borders is lost due to border shopping. In addition, our results indicate that Norwegians living in the border areas cover a large part of their consumption of products such as tobacco and alcohol through cross-border shopping, dampening the effectiveness of the excise taxes in reducing consumption.

Appendix

2.A Additional Descriptive Statistics



Figure A1: This figure shows the distribution of yearly store-level sales in million NOK in 2019 in the different duration bins up to 180 minutes from Sweden, for stores included in the estimation sample as well as for stores that are excluded. For all stores, the source of the yearly sales figure is from Geodata (2021).

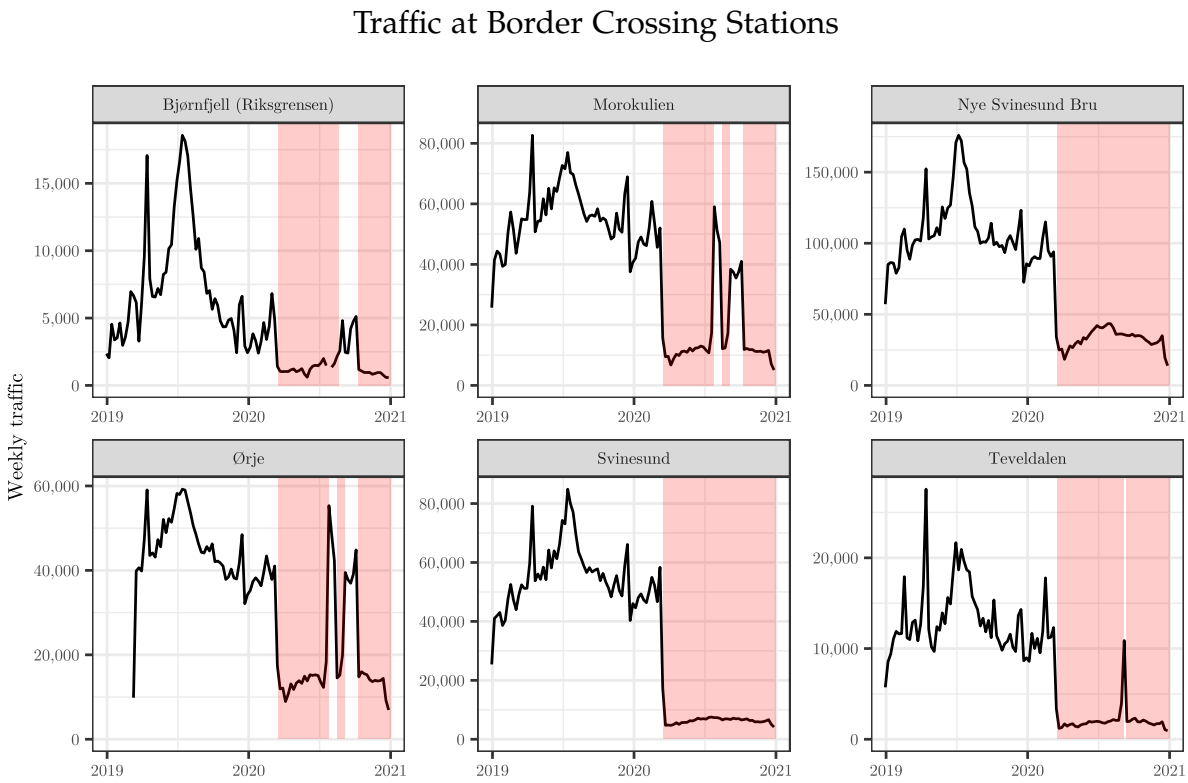


Figure A2: The figure shows weekly traffic of small vehicles (<5.6 meters) at the six border crossing stations with most traffic (excluding the traffic station "Nye Ørje Vest" which was closed for parts of 2020). Red shading indicates that visiting the bordering Swedish county triggered quarantine upon re-entering Norway. As can be seen from the figures, there were some border crossings even in periods where travel to Sweden triggered quarantine stays upon entering Norway. These border crossings would trigger a quarantine stay unless satisfying some limited exceptions to the quarantine rules (e.g., for persons living in Sweden and working in Norway or for persons needing to attend to properties in Sweden). For the traffic station "Ørje" there are no data for the first part of 2019.

Table A1: Share of store-level sales for categories suited for cross-border shopping (CBS) and categories not suited for CBS.

	Pre COVID-19 pandemic		During COVID-19 pandemic	
	Not suited for CBS	Suited for CBS	Not suited for CBS	Suited for CBS
	(1)	(2)	(3)	(4)
< 30	0.0321***	-0.0960***	0.0098**	-0.0235
	(0.0036)	(0.0121)	(0.0037)	(0.0137)
$[30, 60)$	0.0211***	-0.0407***	0.0055*	0.0147
	(0.0025)	(0.0084)	(0.0026)	(0.0095)
$[60, 90)$	0.0123***	-0.0296***	0.0033	-0.0005
	(0.0021)	(0.0069)	(0.0021)	(0.0079)
$[90, 120)$	0.0071***	-0.0236***	0.0019	-0.0087
	(0.0017)	(0.0057)	(0.0017)	(0.0065)
$[120, 150)$	0.0076**	-0.0184*	0.0040	-0.0050
	(0.0023)	(0.0077)	(0.0024)	(0.0088)
$[150, 180)$	0.0017	-0.0161	0.0004	-0.0108
	(0.0027)	(0.0091)	(0.0028)	(0.0103)
Observations	400	400	400	400

Notes: This table reports results from models where the dependent variable is the total store-level sales share of categories not suited for CBS and suited for CBS, respectively. Explanatory variables are duration-category indicator variables; the > 180 minutes category is the omitted base group. Standard errors are in parentheses. * $p < 0.05$, ** $p < 0.01$, *** $p < 0.001$.

Table A2: Average unit prices and excise taxes

	2019	2020
Sweets		
Average price (NOK per kilo)	239.26	252.56
Tax (NOK per kilo)	20.82	21.22
Soda		
Average price (NOK per liter)	22.50	21.60
Tax (NOK per liter)	4.82	4.91
Beer		
Average price (NOK per liter)	62.46	64.21
Tax (NOK per liter)	22.40	22.83
Cigarettes		
Average price (NOK per item)	6.17	6.42
Tax (NOK per item)	2.63	2.68
Snus		
Average price (NOK per kilo)	4542.7	4547.6
Tax (NOK per kilo)	1070.0	1090.0

Notes: The table shows the average unit prices and excise taxes. The average prices are calculated as the total sales expenditure (including VAT and excise taxes) divided by total units.

Table A3: Average yearly sales volume

	<30		[30, 60)		[60, 90)		[90, 120)		[120, 150)		[150, 180)		>180	
	2019	2020	2019	2020	2019	2020	2019	2020	2019	2020	2019	2020	2019	2020
Sweets (metric tons)	8.69	12.63	8.00	10.28	12.28	14.34	14.23	16.45	11.19	12.10	11.24	12.15	8.52	9.25
Soda (thousand liters)	74.61	171.03	79.05	134.37	125.20	166.37	138.90	173.74	124.65	148.93	120.52	142.66	87.67	98.90
Beer (thousand liters)	40.68	72.51	43.96	67.17	63.75	84.10	68.88	89.43	54.37	67.19	55.39	67.20	45.14	52.33
Cigarettes (thousand items)	150.7	434.8	225.6	477.3	341.6	553.0	351.5	502.9	349.1	470.5	359.7	471.9	276.0	342.3
Snus (metric tons)	0.09	0.29	0.14	0.33	0.27	0.45	0.34	0.51	0.27	0.39	0.29	0.40	0.26	0.34

Note: The table shows average yearly sales volumes at the store–category-level. The duration groups are defined by driving duration to the closest Swedish store when all Swedish counties were open for cross-border shopping.

2.B Driving Durations: Additional Details

The Norwegian store data contain the location of the Norwegian stores. Data on the locations of all grocery stores in Swedish counties that border on Norway were obtained from Delfi Marknadspartner (2021). We accessed OpenStreetMap data for Norway and Sweden through Geofabrik (2021). We used the OSRM routing engine to calculate driving durations and driving distances. See http://project-osrm.org for more information about the routing engine. Figure A3 plots the location of the Norwegian and Swedish stores. In this figure we also plot the driving duration to the closest Swedish store across Norwegian municipalities.

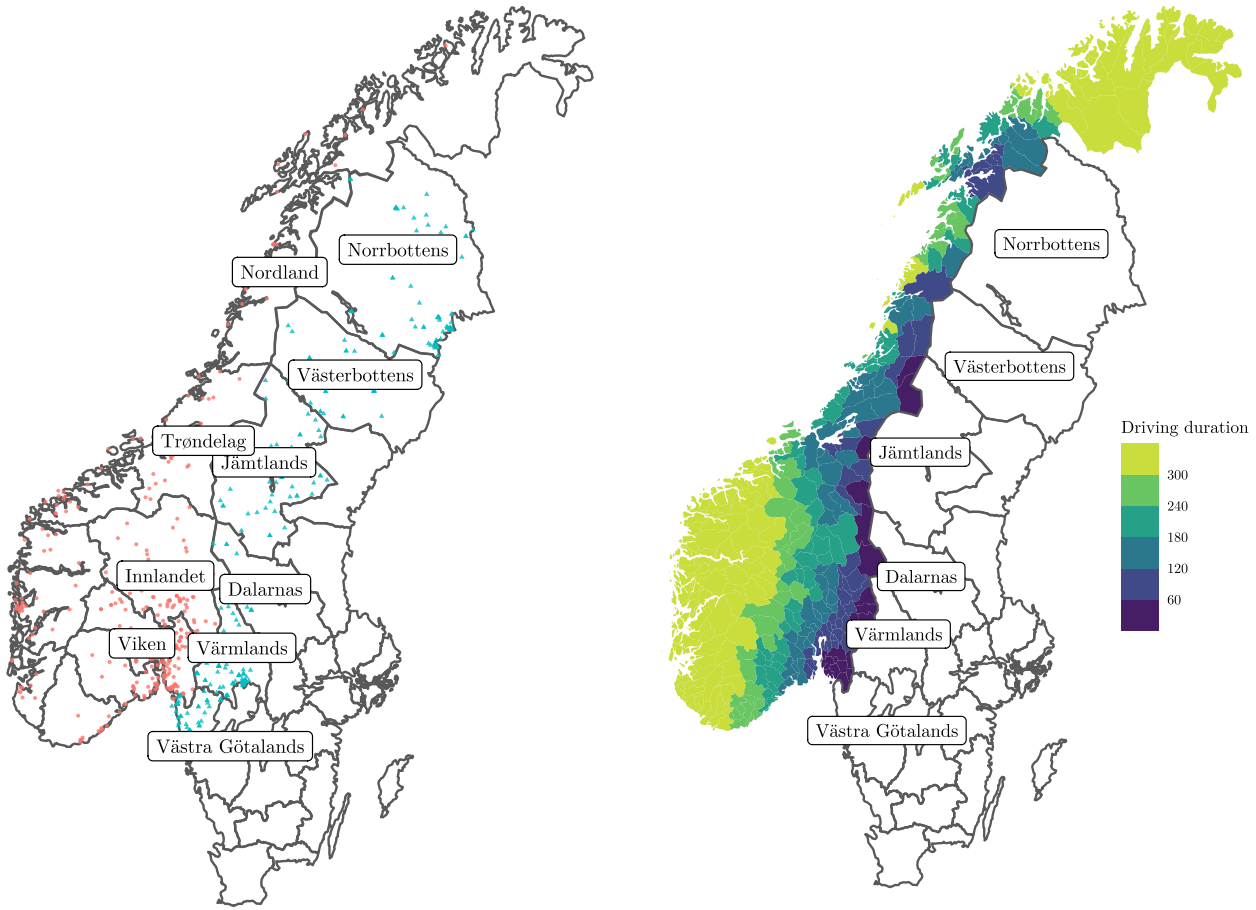
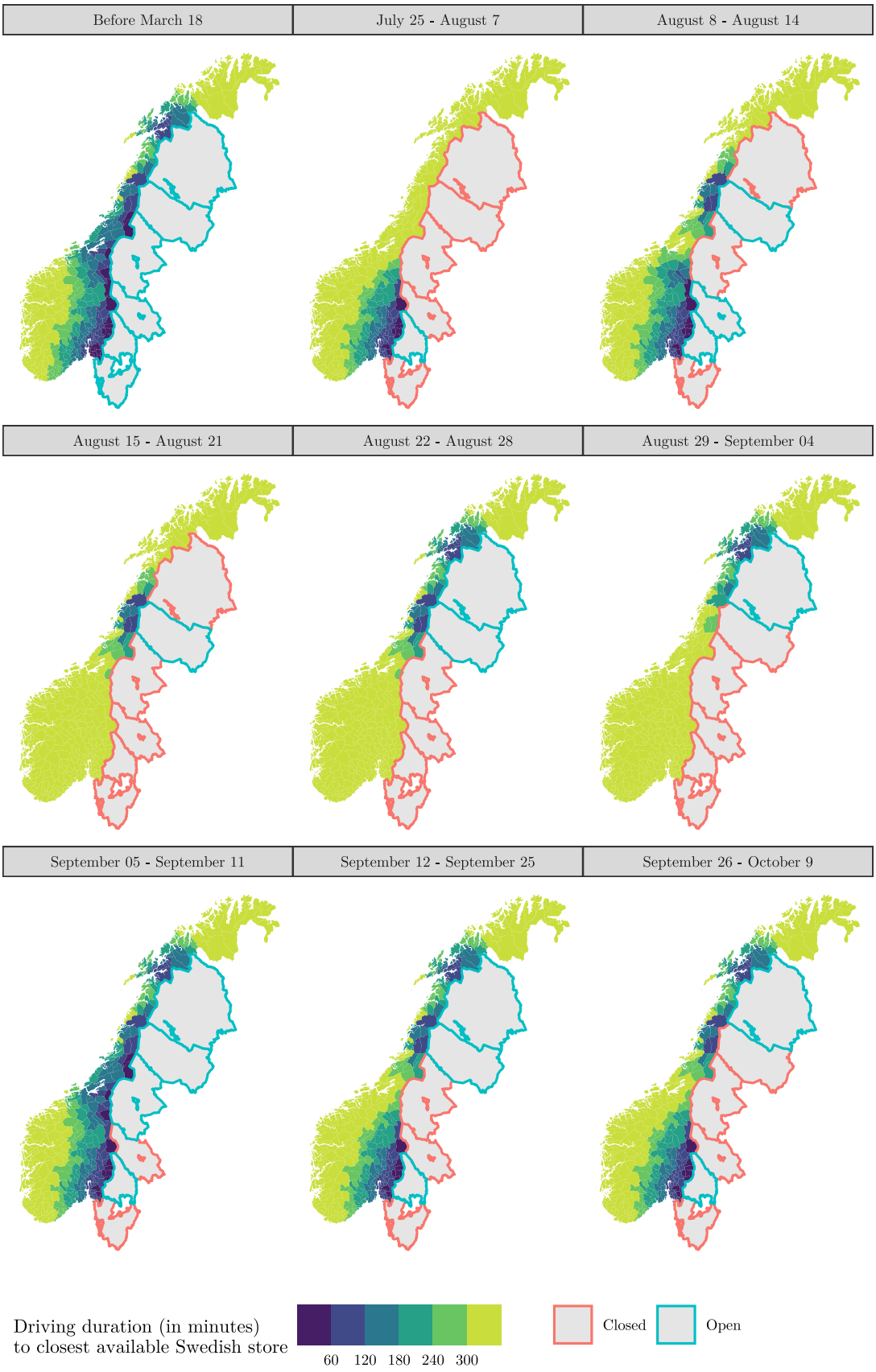


Figure A3: Left panel: The red points are the 400 Norwegian grocery stores in the sample. The turquoise points are the 403 Swedish grocery stores that are located in the 10 labor market regions that border Norway. Right panel: Driving duration in minutes from centroid of Norwegian municipalities to closest Swedish grocery store.

Figure A4: Driving Duration to Closest Accessible Swedish Store



2.C Examination of Pre-Trends

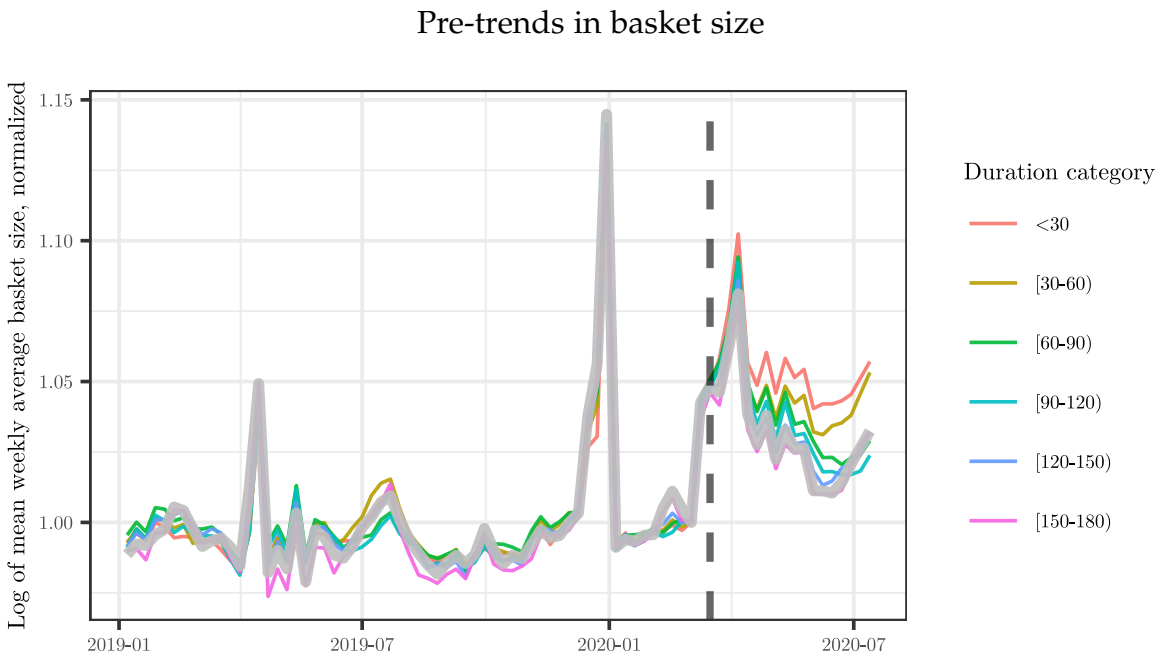


Figure A5: The figure shows the natural logarithm of the average basket size (in NOK) at the store level, for stores in different duration groups. The plotted values have been normalized to the last week before the lockdown was announced (the week starting on 2 March, 2020). The first week included in the figure is the week starting on Monday, January 7, 2019. The last week included in the figure is the week starting on Monday July 13 2020. The dashed vertical line indicates the week travels to Sweden became restricted. The thick gray line represents stores that were located at least 180 minutes from the closest Swedish store.

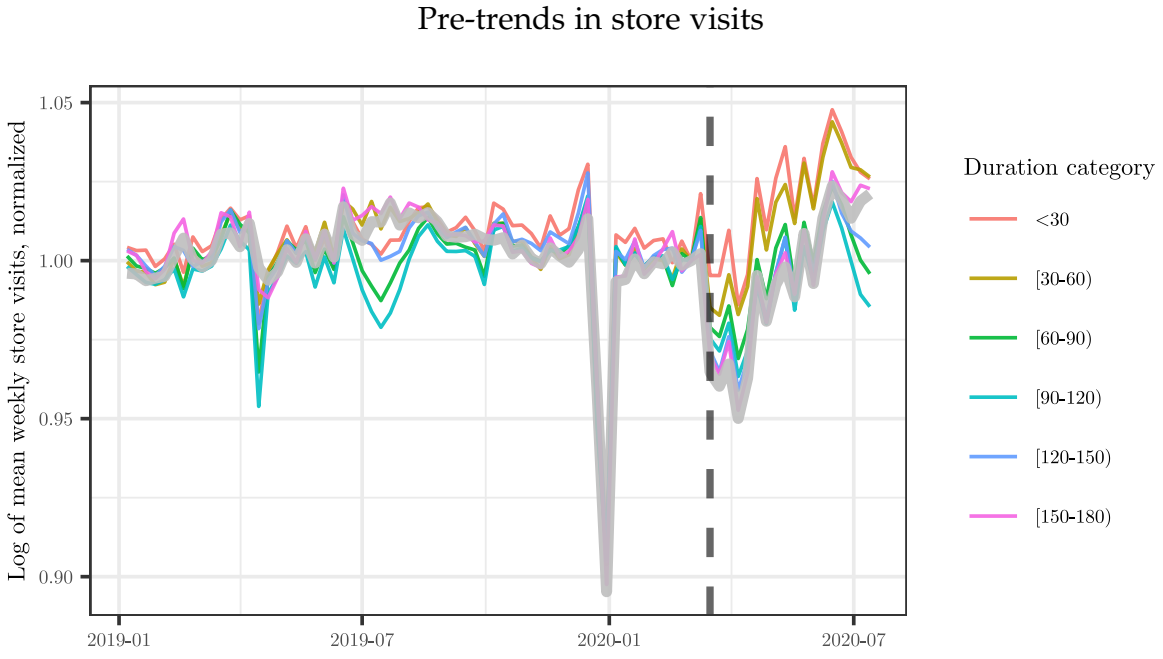


Figure A6: The figure shows the natural logarithm of the average number of weekly store visits at the store level, for stores in different duration groups. The plotted values have been normalized to the last week before the lockdown was announced (the week starting on 2 March, 2020). The first week included in the figure is the week starting on Monday, January 7, 2019. The last week included in the figure is the week starting on Monday July 13 2020. The dashed vertical line indicates the week travels to Sweden became restricted. The thick gray line represents stores that were located at least 180 minutes from the closest Swedish store.

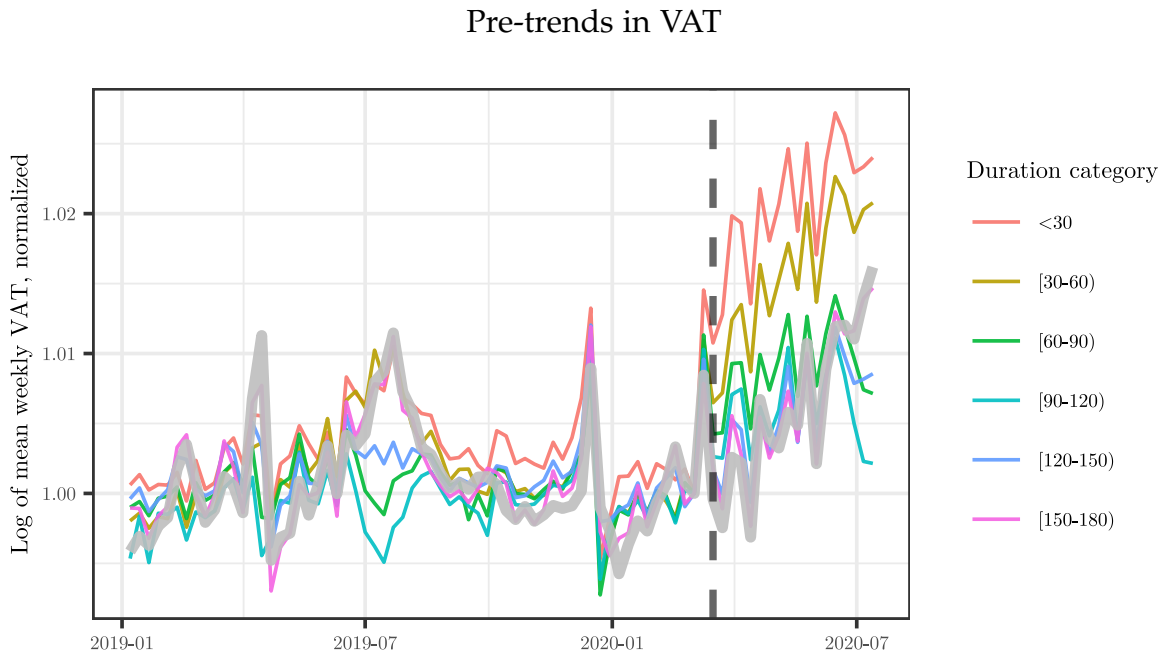


Figure A7: The figure shows the natural logarithm of the average VAT amount (in NOK) at the store level, for stores in different duration groups. The plotted values have been normalized to the last week before the lockdown was announced (the week starting on 2 March, 2020). The first week included in the figure is the week starting on Monday, January 7, 2019. The last week included in the figure is the week starting on Monday July 13 2020. The dashed vertical line indicates the week travels to Sweden became restricted. The thick gray line represents stores that were located at least 180 minutes from the closest Swedish store.

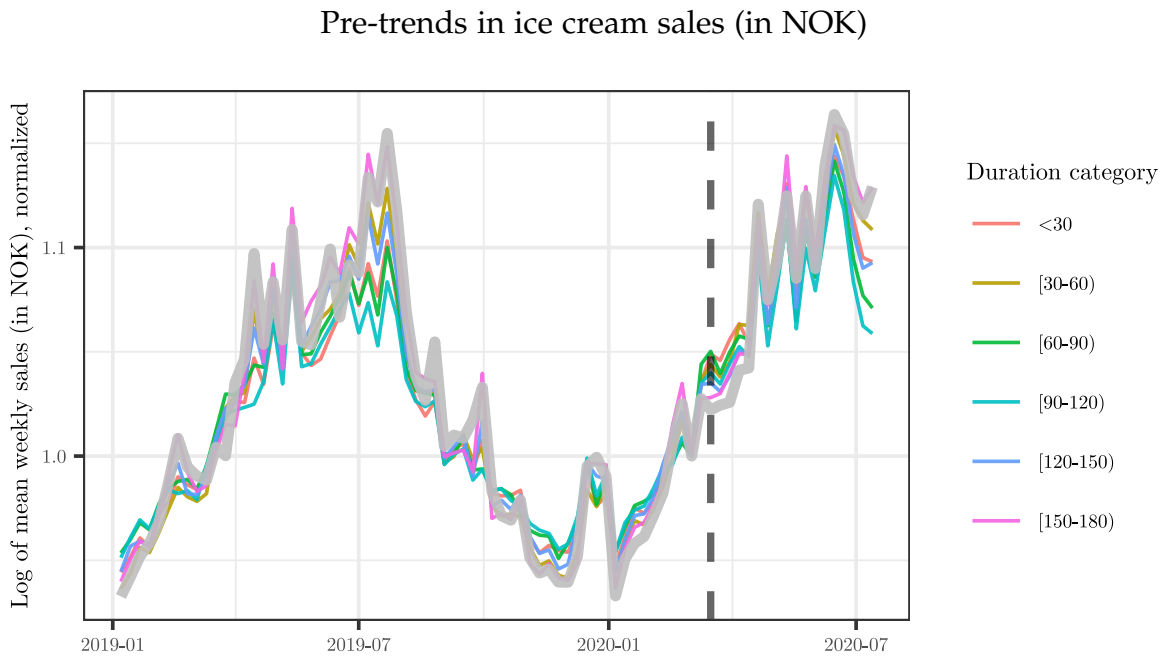


Figure A8: The figure shows the natural logarithm of the average weekly sales of ice cream (in NOK) at the store level, for stores in different duration groups. The plotted values have been normalized to the last week before the lockdown was announced (the week starting on 2 March, 2020). The first week included in the figure is the week starting on Monday, January 7, 2019. The last week included in the figure is the week starting on Monday July 13 2020. The dashed vertical line indicates the week travels to Sweden became restricted. The thick gray line represents stores that were located at least 180 minutes from the closest Swedish store.

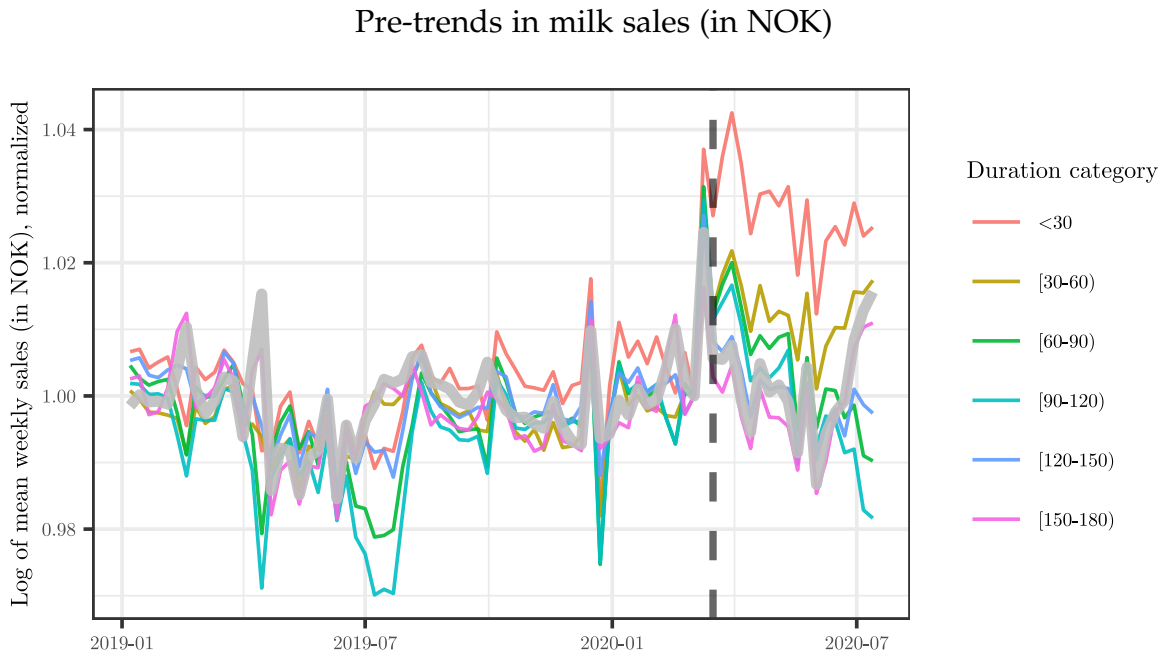


Figure A9: The figure shows the natural logarithm of the average weekly sales of milk (in NOK) at the store level, for stores in different duration groups. The plotted values have been normalized to the last week before the lockdown was announced (the week starting on 2 March, 2020). The first week included in the figure is the week starting on Monday, January 7, 2019. The last week included in the figure is the week starting on Monday July 13 2020. The dashed vertical line indicates the week travels to Sweden became restricted. The thick gray line represents stores that were located at least 180 minutes from the closest Swedish store.

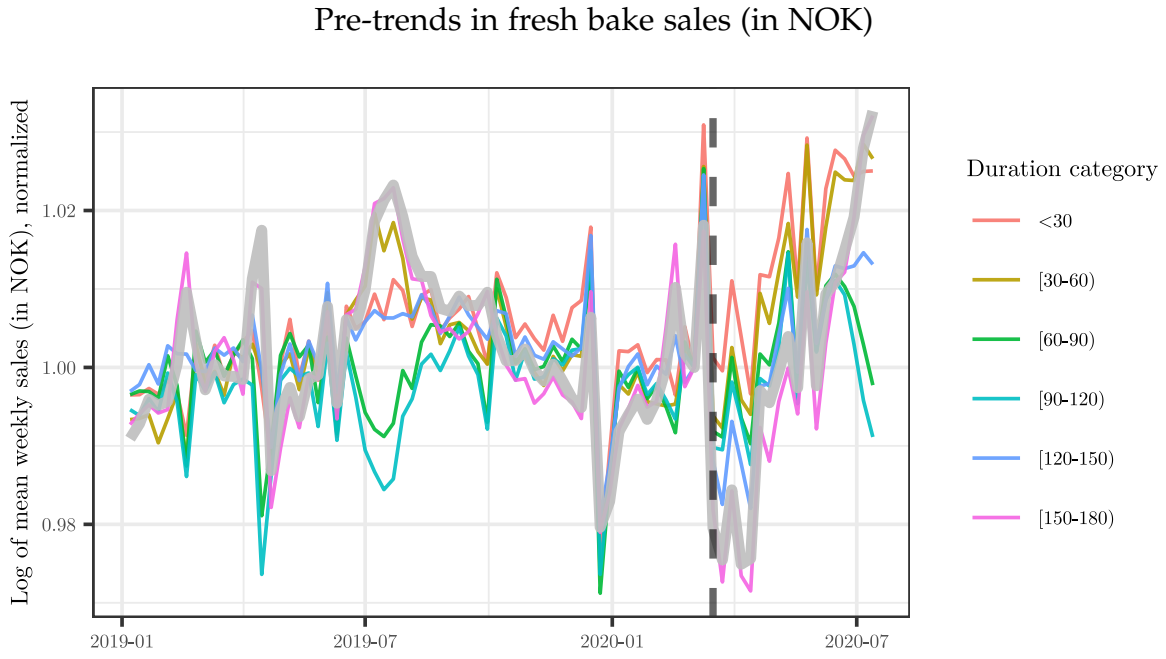


Figure A10: The figure shows the natural logarithm of the average weekly sales of freshly baked goods (in NOK) at the store level, for stores in different duration groups. The plotted values have been normalized to the last week before the lockdown was announced (the week starting on 2 March, 2020). The first week included in the figure is the week starting on Monday, January 7, 2019. The last week included in the figure is the week starting on Monday July 13 2020. The dashed vertical line indicates the week travels to Sweden became restricted. The thick gray line represents stores that were located at least 180 minutes from the closest Swedish store.

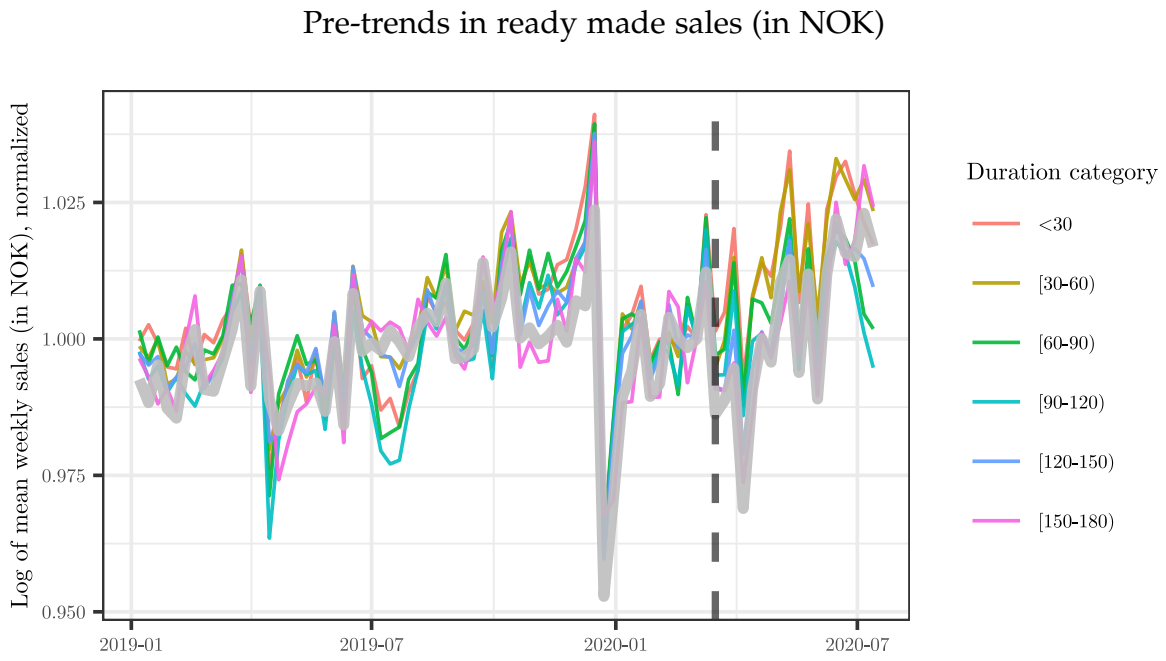


Figure A11: The figure shows the natural logarithm of the average weekly sales of fresh ready-made (in NOK) at the store level, for stores in different duration groups. The plotted values have been normalized to the last week before the lockdown was announced (the week starting on 2 March, 2020). The first week included in the figure is the week starting on Monday, January 7, 2019. The last week included in the figure is the week starting on Monday July 13 2020. The dashed vertical line indicates the week travels to Sweden became restricted. The thick gray line represents stores that were located at least 180 minutes from the closest Swedish store.

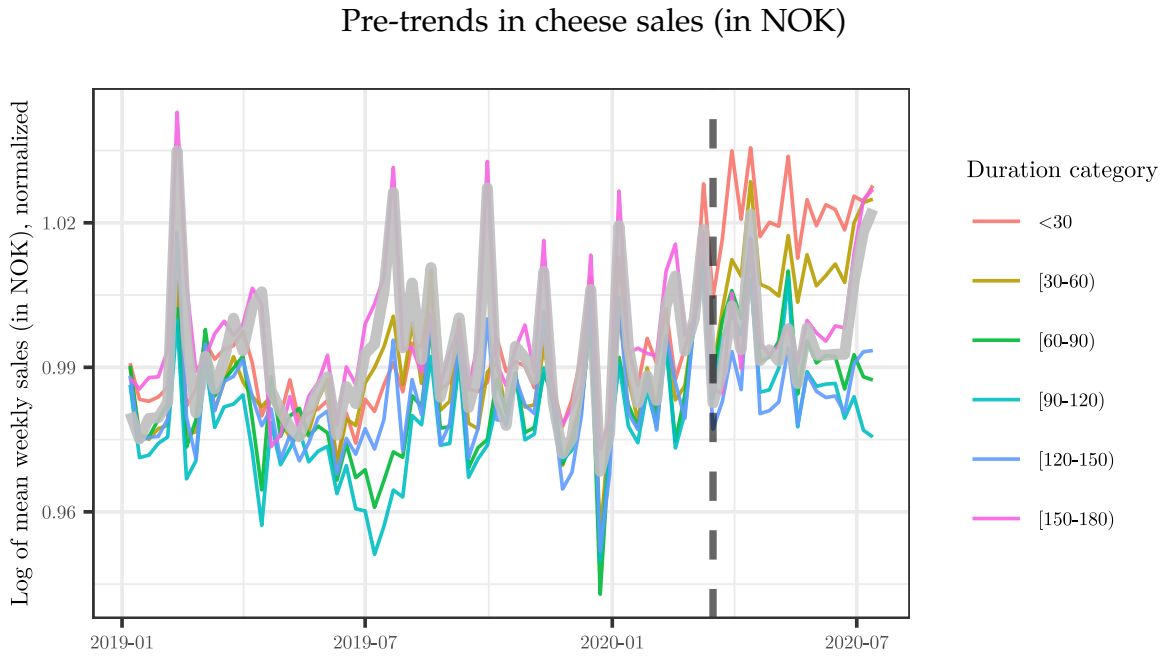


Figure A12: The figure shows the natural logarithm of the average weekly sales of cheese (in NOK) at the store level, for stores in different duration groups. The plotted values have been normalized to the last week before the lockdown was announced (the week starting on 2 March, 2020). The first week included in the figure is the week starting on Monday, January 7, 2019. The last week included in the figure is the week starting on Monday July 13 2020. The dashed vertical line indicates the week travels to Sweden became restricted. The thick gray line represents stores that were located at least 180 minutes from the closest Swedish store.

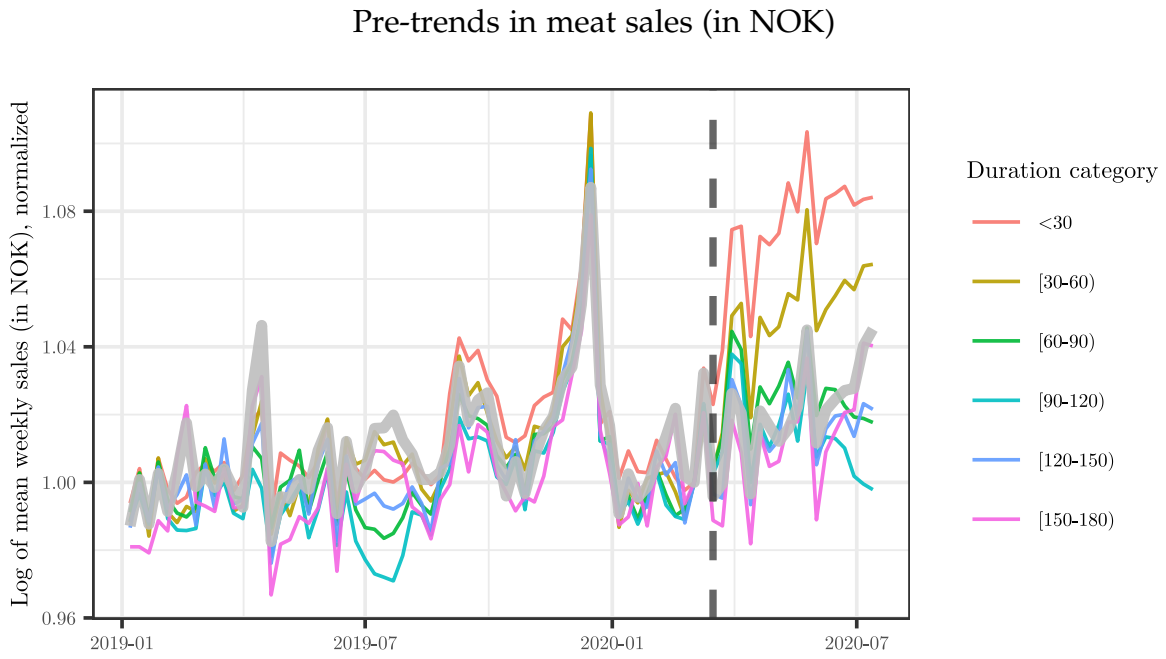


Figure A13: The figure shows the natural logarithm of the average weekly sales of meat (in NOK) at the store level, for stores in different duration groups. The plotted values have been normalized to the last week before the lockdown was announced (the week starting on 2 March, 2020). The first week included in the figure is the week starting on Monday, January 7, 2019. The last week included in the figure is the week starting on Monday July 13 2020. The dashed vertical line indicates the week travels to Sweden became restricted. The thick gray line represents stores that were located at least 180 minutes from the closest Swedish store.

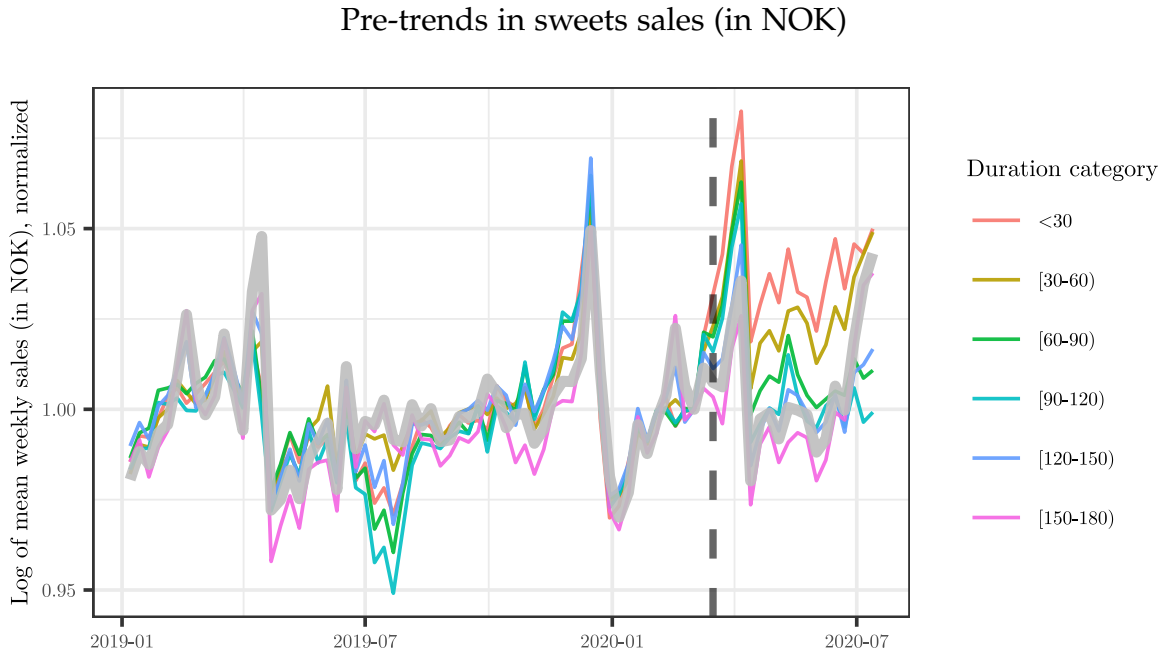


Figure A14: The figure shows the natural logarithm of the average weekly sales of sweets (in NOK) at the store level, for stores in different duration groups. The plotted values have been normalized to the last week before the lockdown was announced (the week starting on 2 March, 2020). The first week included in the figure is the week starting on Monday, January 7, 2019. The last week included in the figure is the week starting on Monday July 13 2020. The dashed vertical line indicates the week travels to Sweden became restricted. The thick gray line represents stores that were located at least 180 minutes from the closest Swedish store.

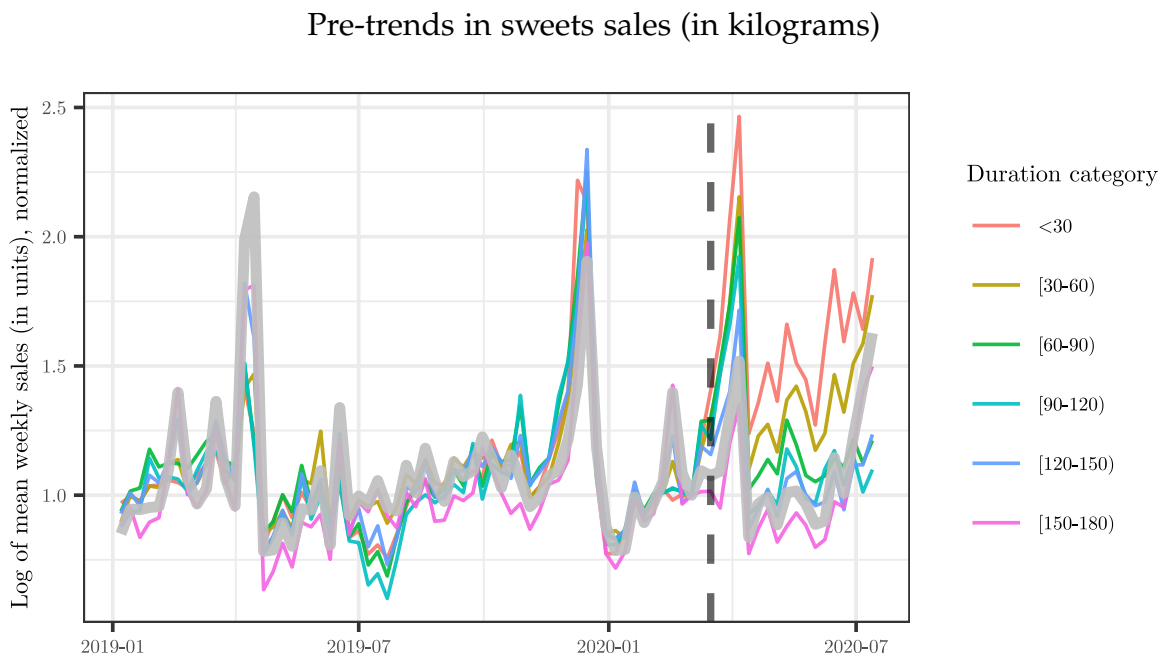


Figure A15: The figure shows the natural logarithm of the average weekly sales of sweets (in kilograms) at the store level, for stores in different duration groups. The plotted values have been normalized to the last week before the lockdown was announced (the week starting on 2 March, 2020). The first week included in the figure is the week starting on Monday, January 7, 2019. The last week included in the figure is the week starting on Monday July 13 2020. The dashed vertical line indicates the week travels to Sweden became restricted. The thick gray line represents stores that were located at least 180 minutes from the closest Swedish store.

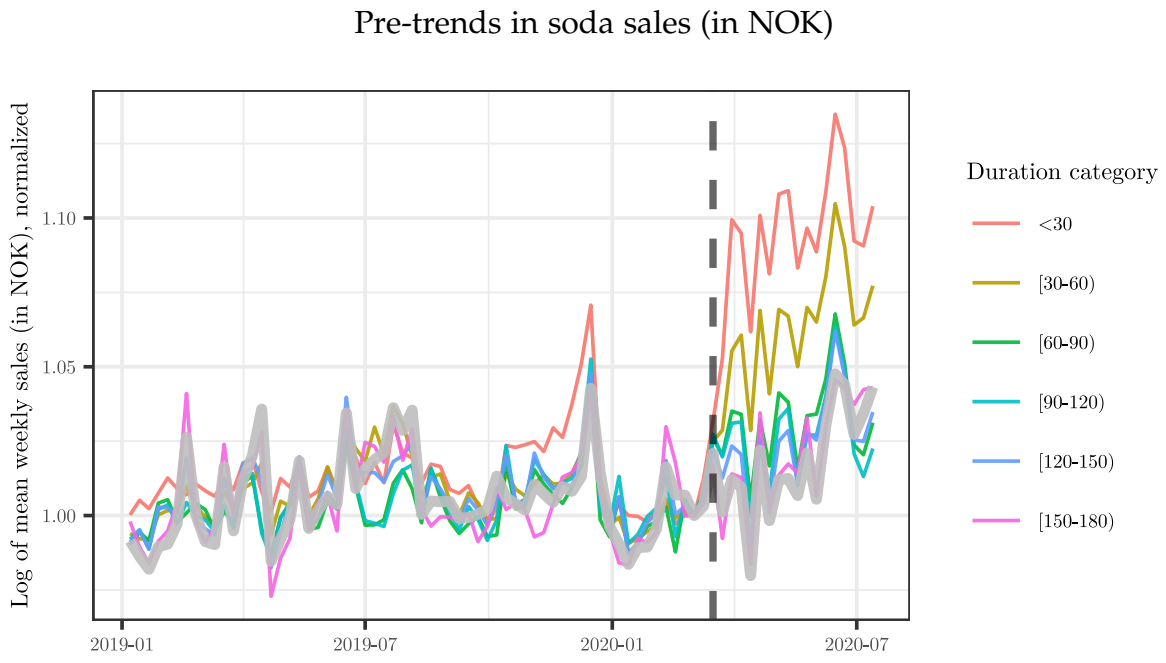


Figure A16: The figure shows the natural logarithm of the average weekly sales of soda (in NOK) at the store level, for stores in different duration groups. The plotted values have been normalized to the last week before the lockdown was announced (the week starting on 2 March, 2020). The first week included in the figure is the week starting on Monday, January 7, 2019. The last week included in the figure is the week starting on Monday July 13 2020. The dashed vertical line indicates the week travels to Sweden became restricted. The thick gray line represents stores that were located at least 180 minutes from the closest Swedish store.

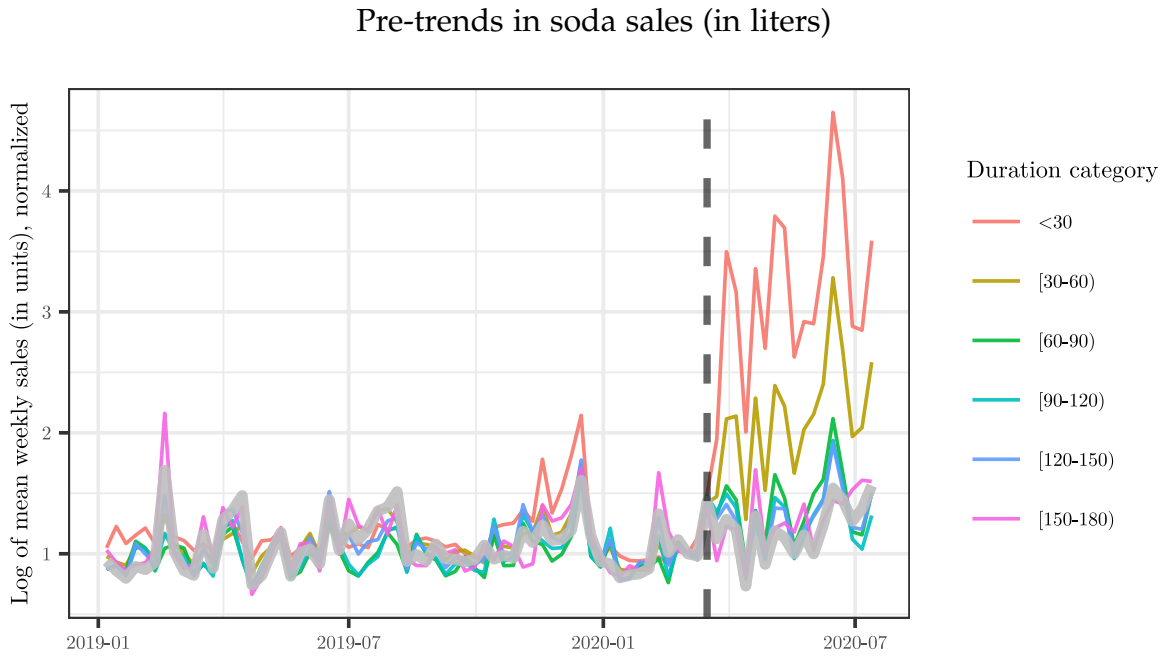


Figure A17: The figure shows the natural logarithm of the average weekly sales of soda (in liters) at the store level, for stores in different duration groups. The plotted values have been normalized to the last week before the lockdown was announced (the week starting on 2 March, 2020). The first week included in the figure is the week starting on Monday, January 7, 2019. The last week included in the figure is the week starting on Monday July 13 2020. The dashed vertical line indicates the week travels to Sweden became restricted. The thick gray line represents stores that were located at least 180 minutes from the closest Swedish store.

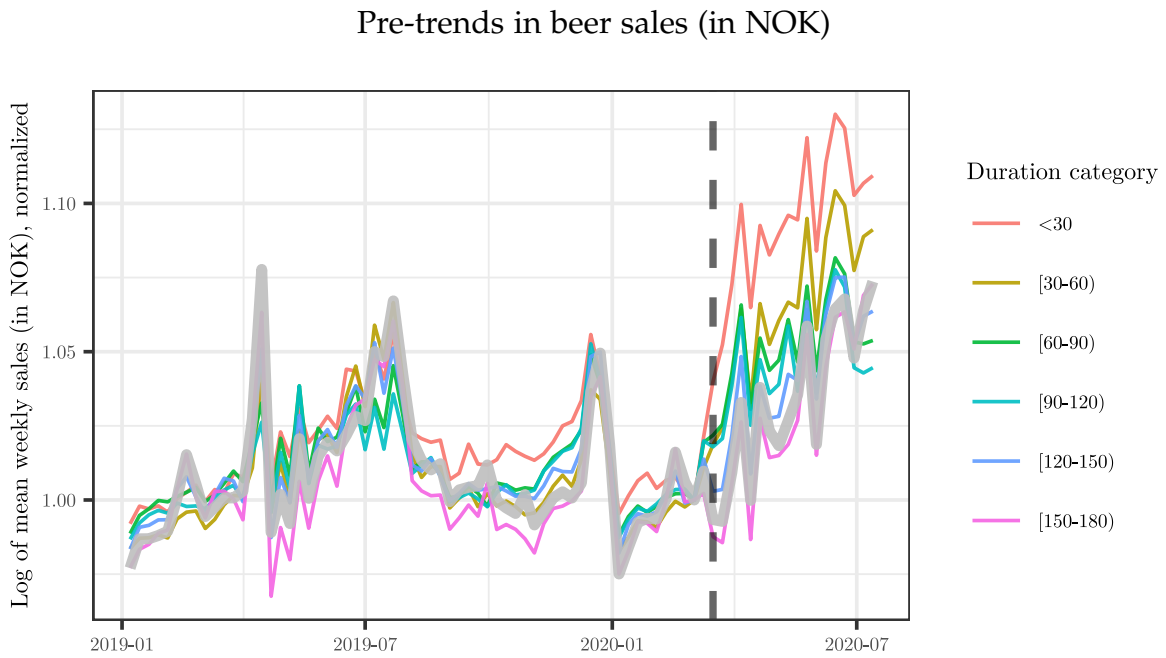


Figure A18: The figure shows the natural logarithm of the average weekly sales of beer (in NOK) at the store level, for stores in different duration groups. The plotted values have been normalized to the last week before the lockdown was announced (the week starting on 2 March, 2020). The first week included in the figure is the week starting on Monday, January 7, 2019. The last week included in the figure is the week starting on Monday July 13 2020. The dashed vertical line indicates the week travels to Sweden became restricted. The thick gray line represents stores that were located at least 180 minutes from the closest Swedish store.

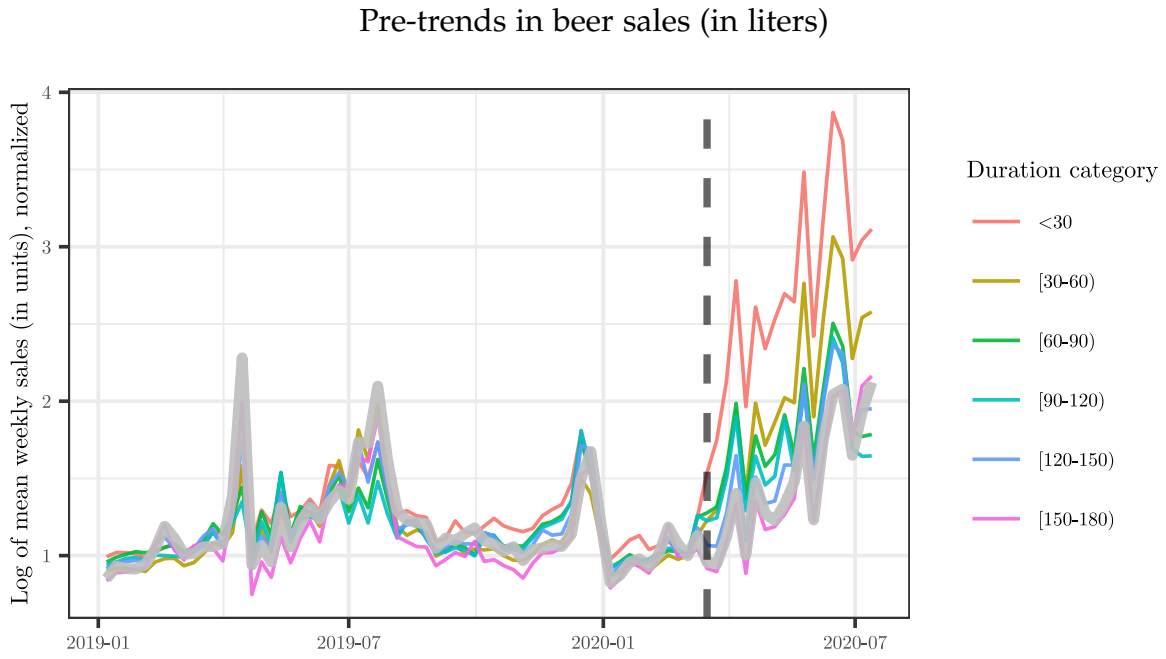


Figure A19: The figure shows the natural logarithm of the average weekly sales of beer (in liters) at the store level, for stores in different duration groups. The plotted values have been normalized to the last week before the lockdown was announced (the week starting on 2 March, 2020). The first week included in the figure is the week starting on Monday, January 7, 2019. The last week included in the figure is the week starting on Monday July 13 2020. The dashed vertical line indicates the week travels to Sweden became restricted. The thick gray line represents stores that were located at least 180 minutes from the closest Swedish store.

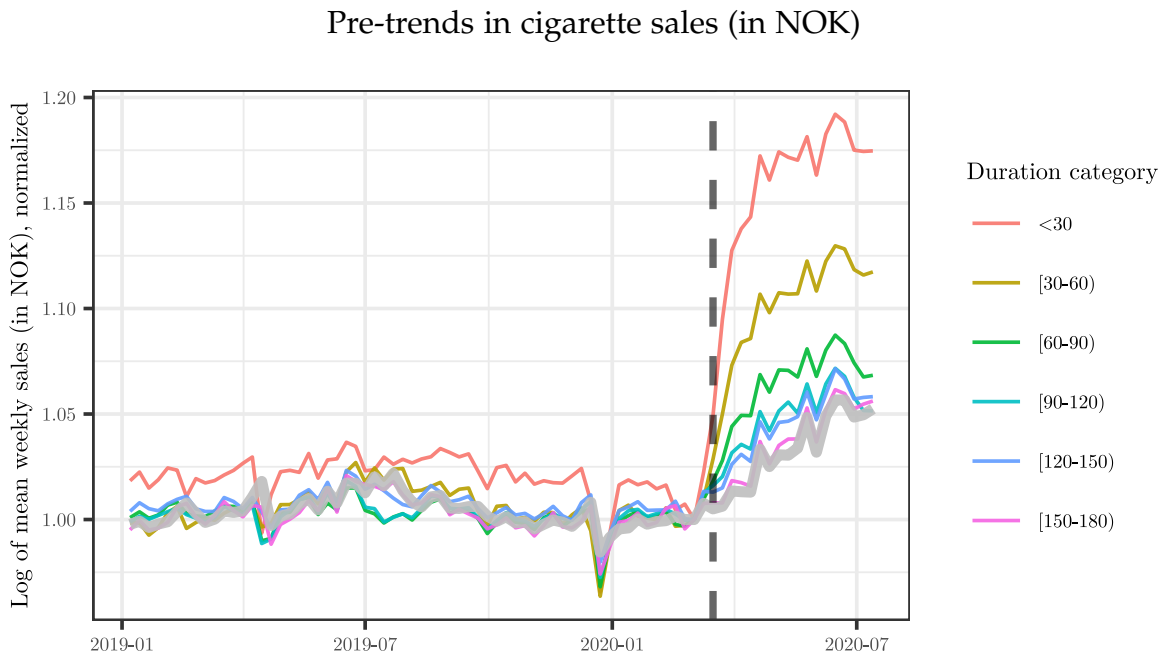


Figure A20: The figure shows the natural logarithm of the average weekly sales of cigarettes (in NOK) at the store level, for stores in different duration groups. The plotted values have been normalized to the last week before the lockdown was announced (the week starting on 2 March, 2020). The first week included in the figure is the week starting on Monday, January 7, 2019. The last week included in the figure is the week starting on Monday July 13 2020. The dashed vertical line indicates the week travels to Sweden became restricted. The thick gray line represents stores that were located at least 180 minutes from the closest Swedish store.

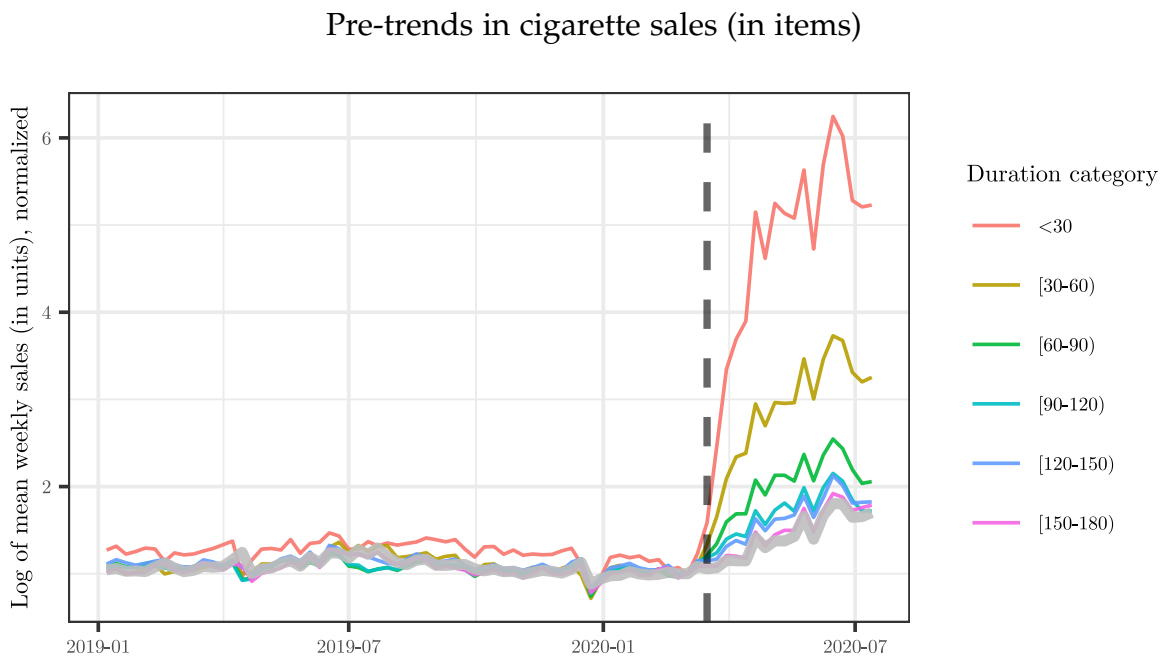


Figure A21: The figure shows the natural logarithm of the average weekly sales of cigarettes (in items) at the store level, for stores in different duration groups. The plotted values have been normalized to the last week before the lockdown was announced (the week starting on 2 March, 2020). The first week included in the figure is the week starting on Monday, January 7, 2019. The last week included in the figure is the week starting on Monday July 13 2020. The dashed vertical line indicates the week travels to Sweden became restricted. The thick gray line represents stores that were located at least 180 minutes from the closest Swedish store.

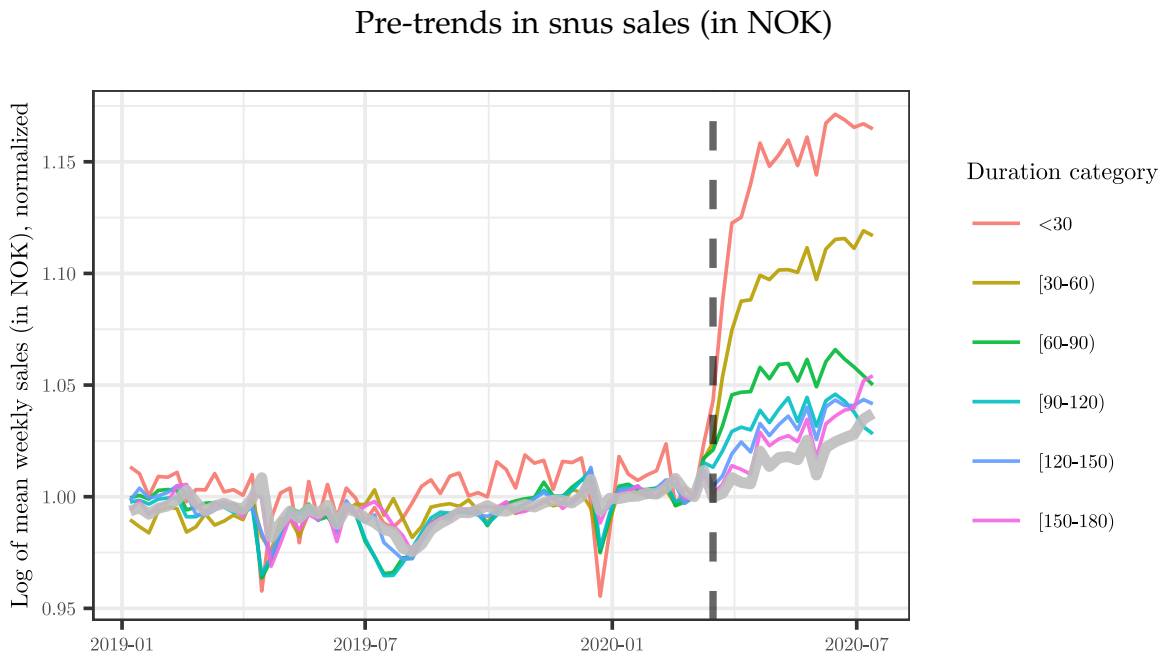


Figure A22: The figure shows the natural logarithm of the average weekly sales of snus (in NOK) at the store level, for stores in different duration groups. The plotted values have been normalized to the last week before the lockdown was announced (the week starting on 2 March, 2020). The first week included in the figure is the week starting on Monday, January 7, 2019. The last week included in the figure is the week starting on Monday July 13 2020. The dashed vertical line indicates the week travels to Sweden became restricted. The thick gray line represents stores that were located at least 180 minutes from the closest Swedish store.

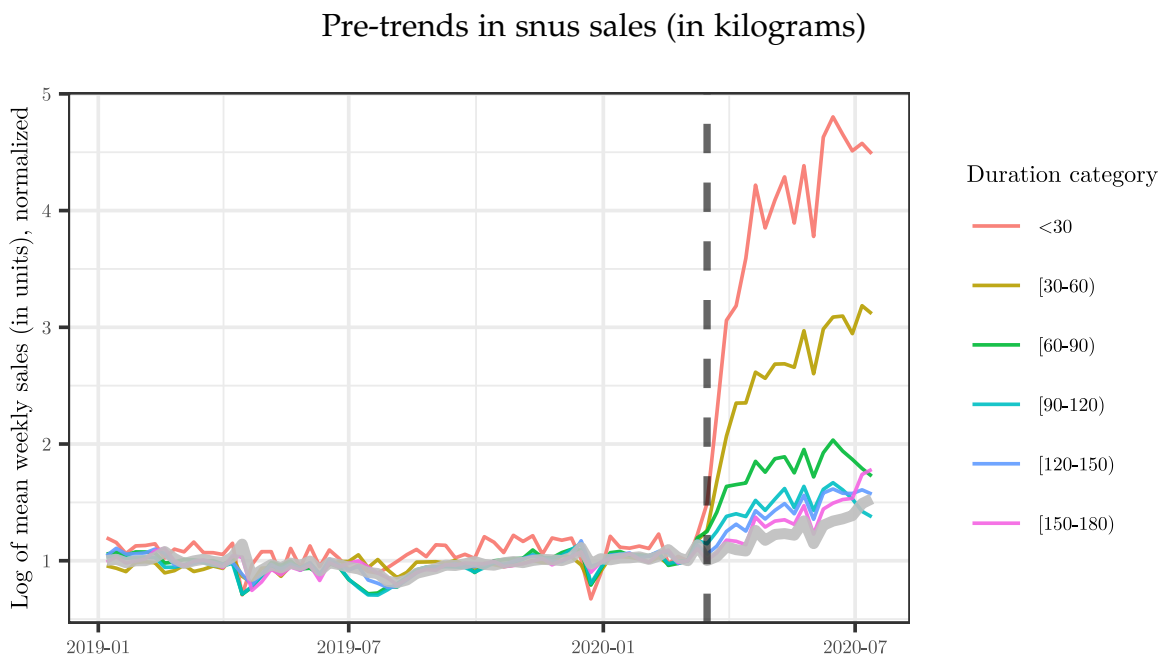


Figure A23: The figure shows the natural logarithm of the average weekly sales of snus (in kilograms) at the store level, for stores in different duration groups. The plotted values have been normalized to the last week before the lockdown was announced (the week starting on 2 March, 2020). The first week included in the figure is the week starting on Monday, January 7, 2019. The last week included in the figure is the week starting on Monday July 13 2020. The dashed vertical line indicates the week travels to Sweden became restricted. The thick gray line represents stores that were located at least 180 minutes from the closest Swedish store.

2.D Additional Regression Results

Table A4: Effect of cross-border shopping on store visits: robustness checks

	(1)	(2)	(3)	(4)	(5)
Duration $<$ 30 $\times$ B	-0.189***	-0.217***	-0.191***	-0.182***	-0.189***
	(0.025)	(0.058)	(0.017)	(0.016)	(0.016)
30 $\leq$ Duration $<$ 60 $\times$ B	-0.158***	-0.202***	-0.151***	-0.136***	-0.154***
	(0.020)	(0.049)	(0.013)	(0.013)	(0.013)
60 $\leq$ Duration $<$ 90 $\times$ B	-0.068***	-0.163***	-0.062**	-0.041	-0.075***
	(0.020)	(0.049)	(0.021)	(0.025)	(0.019)
90 $\leq$ Duration $<$ 120 $\times$ B	-0.003	0.008	-0.023	-0.011	-0.034*
	(0.016)	(0.022)	(0.015)	(0.016)	(0.014)
120 $\leq$ Duration $<$ 150 $\times$ B	0.006	0.024	0.003	0.022	0.007
	(0.016)	(0.020)	(0.014)	(0.015)	(0.013)
150 $\leq$ Duration $<$ 180 $\times$ B	-0.012	0.003	-0.007	0.010	0.007
	(0.020)	(0.019)	(0.020)	(0.023)	(0.018)
New COVID-19 cases					0.001
					(0.004)
C $\times$ Holiday home proportion					0.099**
					(0.033)
Unemployment rate					-1.591***
					(0.379)
Population					0.474***
					(0.130)
Observations	42,000	42,000	35,490	37,600	41,016
Stores	400	400	338	400	400
Control group	$>$ 180 min	$>$ 180 min	$>$ 300 min	$>$ 180 min	$>$ 180 min
Linear time trends	Yes	No	No	No	No
Duration group $\times$ week FEs	No	Yes	No	No	No
Time window	Full	Full	Full	Restricted	Full

Notes: This table reports results from estimation of Equation 2.1 with the natural logarithm of store visits as the dependent variable. In Column 1, separate linear time trends for duration groups based on the pre-COVID duration to the closest Swedish store are included. The frequency distribution of the groups is reported in Table 2.2. In Column 2, week $\times$ pre-COVID duration group fixed effects are included. In Column 3, we drop observations from stores that, before the COVID-19 restrictions, were located between 180 and 300 minutes from the closest Swedish store. In Column 4, we drop observations from the weeks between and including the week starting on Monday 27 July 2020 and the week starting on Monday 5 October 2020. In these weeks, some, but not all, Swedish border counties were open for cross-border shopping. The variable "New COVID-19 cases" included in Column 5 is the sum of registered COVID-19 cases in week t and $t-1$ divided by the population of the municipality. C is an indicator variable taking the value one in the weeks after 17 March 2020 (and zero in the weeks before). "Holiday home proportion" is the proportion of residential dwellings that are holiday homes. The unemployment rate is measured at the municipality–month level. Population is measured at the municipality–quarter level. The standard errors reported in parentheses are clustered at the municipality level. *** $p < 0.001$; ** $p < 0.01$; * $p < 0.05$.

Table A5: Effect of cross-border shopping on basket size: robustness checks

	(1)	(2)	(3)	(4)	(5)
Duration $<$ 30 $\times$ B	-0.120***	-0.111***	-0.118***	-0.119***	-0.122***
	(0.013)	(0.028)	(0.010)	(0.009)	(0.011)
30 $\leq$ Duration $<$ 60 $\times$ B	-0.068***	-0.077***	-0.070***	-0.070***	-0.074***
	(0.011)	(0.021)	(0.011)	(0.011)	(0.010)
60 $\leq$ Duration $<$ 90 $\times$ B	-0.033***	-0.036	-0.028***	-0.036***	-0.034***
	(0.006)	(0.020)	(0.008)	(0.008)	(0.007)
90 $\leq$ Duration $<$ 120 $\times$ B	-0.030***	-0.010	-0.018**	-0.024***	-0.025***
	(0.006)	(0.008)	(0.007)	(0.006)	(0.006)
120 $\leq$ Duration $<$ 150 $\times$ B	-0.017***	-0.002	-0.009	-0.012	-0.012*
	(0.005)	(0.007)	(0.007)	(0.007)	(0.006)
150 $\leq$ Duration $<$ 180 $\times$ B	-0.006	-0.001	-0.010	-0.015*	-0.011
	(0.008)	(0.008)	(0.008)	(0.008)	(0.007)
New COVID-19 cases					0.013***
					(0.002)
C $\times$ Holiday home proportion					0.037*
					(0.015)
Unemployment rate					-0.264
					(0.147)
Population					0.025
					(0.038)
Observations	42,000	42,000	35,490	37,600	41,016
Stores	400	400	338	400	400
Control group	$>$ 180 min	$>$ 180 min	$>$ 300 min	$>$ 180 min	$>$ 180 min
Linear time trends	Yes	No	No	No	No
Duration group $\times$ week FEs	No	Yes	No	No	No
Time window	Full	Full	Full	Restricted	Full

Notes: This table reports results from estimation of Equation 2.1 with the natural logarithm of basket size as the dependent variable. In Column 1, separate linear time trends for duration groups based on the pre-COVID duration to the closest Swedish store are included. The frequency distribution of the groups is reported in Table 2.2. In Column 2, week $\times$ pre-COVID duration group fixed effects are included. In Column 3, we drop observations from stores that, before the COVID-19 restrictions, were located between 180 and 300 minutes from the closest Swedish store. In Column 4, we drop observations from the weeks between and including the week starting on Monday 27 July 2020 and the week starting on Monday 5 October 2020. In these weeks, some, but not all, Swedish border counties were open for cross-border shopping. The variable "New COVID-19 cases" included in Column 5 is the sum of registered COVID-19 cases in week t and $t-1$ divided by the population of the municipality. C is an indicator variable taking the value one in the weeks after 17 March 2020 (and zero in the weeks before). "Holiday home proportion" is the proportion of residential dwellings that are holiday homes. The unemployment rate is measured at the municipality–month level. Population is measured at the municipality–quarter level. The standard errors reported in parentheses are clustered at the municipality level. *** $p < 0.001$; ** $p < 0.01$; * $p < 0.05$.

Table A6: Effect of cross-border shopping on VAT: robustness checks

	(1)	(2)	(3)	(4)	(5)
Duration $< 30 \times$ B	-0.327***	-0.345***	-0.329***	-0.321***	-0.331***
	(0.019)	(0.076)	(0.016)	(0.016)	(0.017)
$30 \leq$ Duration $< 60 \times$ B	-0.241***	-0.292***	-0.241***	-0.225***	-0.247***
	(0.027)	(0.060)	(0.016)	(0.016)	(0.017)
$60 \leq$ Duration $< 90 \times$ B	-0.107***	-0.203**	-0.099***	-0.087**	-0.118***
	(0.022)	(0.061)	(0.023)	(0.028)	(0.023)
$90 \leq$ Duration $< 120 \times$ B	-0.035	-0.003	-0.047**	-0.039*	-0.063***
	(0.019)	(0.027)	(0.016)	(0.017)	(0.016)
$120 \leq$ Duration $< 150 \times$ B	-0.014	0.022	-0.010	0.006	-0.008
	(0.018)	(0.025)	(0.014)	(0.015)	(0.014)
$150 \leq$ Duration $< 180 \times$ B	-0.019	0.002	-0.019	-0.007	-0.005
	(0.024)	(0.022)	(0.021)	(0.024)	(0.021)
New COVID-19 cases					0.014**
					(0.005)
C $\times$ Holiday home proportion					0.134**
					(0.041)
Unemployment rate					-1.839***
					(0.462)
Population					0.493***
					(0.143)
Observations	41,995	41,995	35,489	37,595	41,012
Stores	400	400	338	400	400
Control group	$>$ 180 min	$>$ 180 min	$>$ 300 min	$>$ 180 min	$>$ 180 min
Linear time trends	Yes	No	No	No	No
Duration group $\times$ week FEs	No	Yes	No	No	No
Time window	Full	Full	Full	Restricted	Full

Notes: This table reports results from estimation of Equation 2.1 with the natural logarithm of VAT as the dependent variable. In Column 1, separate linear time trends for duration groups based on the pre-COVID duration to the closest Swedish store are included. The frequency distribution of the groups is reported in Table 2.2. In Column 2, week $\times$ pre-COVID duration group fixed effects are included. In Column 3, we drop observations from stores that, before the COVID-19 restrictions, were located between 180 and 300 minutes from the closest Swedish store. In Column 4, we drop observations from the weeks between and including the week starting on Monday 27 July 2020 and the week starting on Monday 5 October 2020. In these weeks, some, but not all, Swedish border counties were open for cross-border shopping. The variable "New COVID-19 cases" included in Column 5 is the sum of registered COVID-19 cases in week t and $t-1$ divided by the population of the municipality. C is an indicator variable taking the value one in the weeks after 17 March 2020 (and zero in the weeks before). "Holiday home proportion" is the proportion of residential dwellings that are holiday homes. The unemployment rate is measured at the municipality–month level. Population is measured at the municipality–quarter level. The standard errors reported in parentheses are clustered at the municipality level. *** $p < 0.001$; ** $p < 0.01$; * $p < 0.05$.

Table A7: Effect of cross-border shopping on ice cream sales (in NOK): robustness checks

	(1)	(2)	(3)	(4)	(5)
Duration $<30\times$ B	-0.105*	-0.229*	-0.128***	-0.129***	-0.135***
	(0.053)	(0.112)	(0.021)	(0.023)	(0.022)
$30\leq$ Duration $<60\times$ B	-0.130***	-0.215**	-0.099***	-0.091***	-0.119***
	(0.026)	(0.065)	(0.015)	(0.015)	(0.014)
$60\leq$ Duration $<90\times$ B	-0.074*	-0.252***	-0.032	-0.011	-0.073**
	(0.029)	(0.065)	(0.025)	(0.029)	(0.023)
$90\leq$ Duration $<120\times$ B	0.015	0.018	0.018	0.022	-0.018
	(0.025)	(0.032)	(0.018)	(0.017)	(0.020)
$120\leq$ Duration $<150\times$ B	0.019	0.056	0.036	0.042*	0.028
	(0.026)	(0.035)	(0.019)	(0.020)	(0.019)
$150\leq$ Duration $<180\times$ B	-0.047	0.002	-0.022	-0.019	-0.010
	(0.030)	(0.034)	(0.025)	(0.027)	(0.025)
New COVID-19 cases					0.039***
					(0.011)
C $\times$ Holiday home proportion					0.192***
					(0.043)
Unemployment rate					-3.041***
					(0.517)
Population					0.923***
					(0.234)
Observations	41,995	41,995	35,486	37,595	41,002
Stores	400	400	338	400	400
Control group	>180 min	>180 min	>300 min	>180 min	>180 min
Linear time trends	Yes	No	No	No	No
Duration group $\times$ week FEs	No	Yes	No	No	No
Time window	Full	Full	Full	Restricted	Full

Notes: This table reports results from estimation of Equation 2.1 with the natural logarithm of ice cream sales as the dependent variable. In Column 1, separate linear time trends for duration groups based on the pre-COVID duration to the closest Swedish store are included. The frequency distribution of the groups is reported in Table 2.2. In Column 2, week $\times$ pre-COVID duration group fixed effects are included. In Column 3, we drop observations from stores that, before the COVID-19 restrictions, were located between 180 and 300 min from the closest Swedish store. In Column 4, we drop observations from the weeks between and including the week starting on Monday 27 July 2020 and the week starting on Monday 5 October 2020. In these weeks, some, but not all, Swedish border counties were open for cross-border shopping. The variable "New COVID-19 cases" included in Column 5 is the sum of registered COVID-19 cases in week t and $t-1$ divided by the population of the municipality. C is an indicator variable taking the value one in the weeks after 17 March 2020 (and zero in the weeks before). "Holiday home proportion" is the proportion of residential dwellings that are holiday homes. The unemployment rate is measured at the municipality–month level. Population is measured at the municipality–quarter level. The standard errors reported in parentheses are clustered at the municipality level. *** $p<0.001$; ** $p<0.01$; * $p<0.05$.

Table A8: Effect of cross-border shopping on milk sales (in NOK): robustness checks

	(1)	(2)	(3)	(4)	(5)
Duration $< 30 \times$ B	-0.238***	-0.194**	-0.230***	-0.227***	-0.238***
	(0.017)	(0.060)	(0.021)	(0.023)	(0.022)
$30 \leq$ Duration $< 60 \times$ B	-0.129***	-0.167**	-0.123***	-0.109***	-0.135***
	(0.022)	(0.052)	(0.014)	(0.017)	(0.016)
$60 \leq$ Duration $< 90 \times$ B	-0.055**	-0.102	-0.035	-0.023	-0.054**
	(0.019)	(0.053)	(0.021)	(0.027)	(0.020)
$90 \leq$ Duration $< 120 \times$ B	-0.013	0.020	-0.011	-0.005	-0.030
	(0.014)	(0.020)	(0.014)	(0.018)	(0.017)
$120 \leq$ Duration $< 150 \times$ B	0.001	0.030	0.021	0.035	0.018
	(0.016)	(0.019)	(0.015)	(0.018)	(0.015)
$150 \leq$ Duration $< 180 \times$ B	-0.016	0.020	0.007	0.016	0.016
	(0.020)	(0.017)	(0.019)	(0.024)	(0.019)
New COVID-19 cases					0.011*
					(0.005)
C $\times$ Holiday home proportion					0.142***
					(0.038)
Unemployment rate					-1.478***
					(0.420)
Population					0.757***
					(0.165)
Observations	41,993	41,993	35,485	37,593	40,996
Stores	400	400	338	400	400
Control group	$>$ 180 min	$>$ 180 min	$>$ 300 min	$>$ 180 min	$>$ 180 min
Linear time trends	Yes	No	No	No	No
Duration group $\times$ week FEs	No	Yes	No	No	No
Time window	Full	Full	Full	Restricted	Full

Notes: This table reports results from estimation of Equation 2.1 with the natural logarithm of milk sales as the dependent variable. In Column 1, separate linear time trends for duration groups based on the pre-COVID duration to the closest Swedish store are included. The frequency distribution of the groups is reported in Table 2.2. In Column 2, week $\times$ pre-COVID duration group fixed effects are included. In Column 3, we drop observations from stores that, before the COVID-19 restrictions, were located between 180 and 300 min from the closest Swedish store. In Column 4, we drop observations from the weeks between and including the week starting on Monday 27 July 2020 and the week starting on Monday 5 October 2020. In these weeks, some, but not all, Swedish border counties were open for cross-border shopping. The variable "New COVID-19 cases" included in Column 5 is the sum of registered COVID-19 cases in week t and $t-1$ divided by the population of the municipality. C is an indicator variable taking the value one in the weeks after 17 March 2020 (and zero in the weeks before). "Holiday home proportion" is the proportion of residential dwellings that are holiday homes. The unemployment rate is measured at the municipality–month level. Population is measured at the municipality–quarter level. The standard errors reported in parentheses are clustered at the municipality level. *** $p < 0.001$; ** $p < 0.01$; * $p < 0.05$.

Table A9: Effect of cross-border shopping on fresh bake sales (in NOK): robustness checks

	(1)	(2)	(3)	(4)	(5)
Duration $< 30 \times$ B	-0.168***	-0.202**	-0.167***	-0.161***	-0.171***
	(0.029)	(0.072)	(0.027)	(0.027)	(0.028)
$30 \leq$ Duration $< 60 \times$ B	-0.122***	-0.240***	-0.116***	-0.104***	-0.127***
	(0.027)	(0.066)	(0.020)	(0.020)	(0.021)
$60 \leq$ Duration $< 90 \times$ B	-0.076***	-0.232***	-0.051*	-0.028	-0.075**
	(0.022)	(0.066)	(0.025)	(0.030)	(0.023)
$90 \leq$ Duration $< 120 \times$ B	-0.035*	-0.025	-0.030	-0.021	-0.052**
	(0.016)	(0.030)	(0.017)	(0.018)	(0.019)
$120 \leq$ Duration $< 150 \times$ B	-0.021	0.012	-0.005	0.012	-0.006
	(0.019)	(0.030)	(0.019)	(0.020)	(0.020)
$150 \leq$ Duration $< 180 \times$ B	-0.006	0.039	-0.004	0.005	0.010
	(0.023)	(0.025)	(0.021)	(0.025)	(0.020)
New COVID-19 cases					0.015*
					(0.006)
C $\times$ Holiday home proportion					0.163***
					(0.040)
Unemployment rate					-2.097***
					(0.458)
Population					0.694**
					(0.236)
Observations	41,994	41,994	35,486	37,594	40,999
Stores	400	400	338	400	400
Control group	$>$ 180 min	$>$ 180 min	$>$ 300 min	$>$ 180 min	$>$ 180 min
Linear time trends	Yes	No	No	No	No
Duration group $\times$ week FEs	No	Yes	No	No	No
Time window	Full	Full	Full	Restricted	Full

Notes: This table reports results from estimation of Equation 2.1 with the natural logarithm of fresh bake sales as the dependent variable. In Column 1, separate linear time trends for duration groups based on the pre-COVID duration to the closest Swedish store are included. The frequency distribution of the groups is reported in Table 2.2. In Column 2, week $\times$ pre-COVID duration group fixed effects are included. In Column 3, we drop observations from stores that, before the COVID-19 restrictions, were located between 180 and 300 min from the closest Swedish store. In Column 4, we drop observations from the weeks between and including the week starting on Monday 27 July 2020 and the week starting on Monday 5 October 2020. In these weeks, some, but not all, Swedish border counties were open for cross-border shopping. The variable "New COVID-19 cases" included in Column 5 is the sum of registered COVID-19 cases in week t and $t-1$ divided by the population of the municipality. C is an indicator variable taking the value one in the weeks after 17 March 2020 (and zero in the weeks before). "Holiday home proportion" is the proportion of residential dwellings that are holiday homes. The unemployment rate is measured at the municipality–month level. Population is measured at the municipality–quarter level. The standard errors reported in parentheses are clustered at the municipality level. *** $p < 0.001$; ** $p < 0.01$; * $p < 0.05$.

Table A10: Effect of cross-border shopping on ready-made sales (in NOK): robustness checks

	(1)	(2)	(3)	(4)	(5)
Duration $<30\times$ B	-0.146***	-0.235***	-0.163***	-0.147***	-0.162***
	(0.030)	(0.068)	(0.029)	(0.027)	(0.026)
$30\leq$ Duration $<60\times$ B	-0.119***	-0.208***	-0.134***	-0.115***	-0.139***
	(0.024)	(0.059)	(0.018)	(0.018)	(0.013)
$60\leq$ Duration $<90\times$ B	-0.039*	-0.156**	-0.021	0.006	-0.039
	(0.019)	(0.049)	(0.026)	(0.031)	(0.021)
$90\leq$ Duration $<120\times$ B	0.006	-0.018	0.003	0.020	-0.014
	(0.016)	(0.024)	(0.017)	(0.019)	(0.013)
$120\leq$ Duration $<150\times$ B	0.010	0.027	0.017	0.037	0.021
	(0.017)	(0.025)	(0.019)	(0.020)	(0.015)
$150\leq$ Duration $<180\times$ B	-0.007	0.032	-0.015	-0.005	0.004
	(0.020)	(0.020)	(0.024)	(0.027)	(0.021)
New COVID-19 cases					0.011*
					(0.005)
C $\times$ Holiday home proportion					0.166***
					(0.040)
Unemployment rate					-1.859***
					(0.344)
Population					0.432***
					(0.117)
Observations	41,993	41,993	35,485	37,593	40,996
Stores	400	400	338	400	400
Control group	>180 min	>180 min	>300 min	>180 min	>180 min
Linear time trends	Yes	No	No	No	No
Duration group $\times$ week FEs	No	Yes	No	No	No
Time window	Full	Full	Full	Restricted	Full

Notes: This table reports results from estimation of Equation 2.1 with the natural logarithm of ready-made sales as the dependent variable. In Column 1, separate linear time trends for duration groups based on the pre-COVID duration to the closest Swedish store are included. The frequency distribution of the groups is reported in Table 2.2. In Column 2, week $\times$ pre-COVID duration group fixed effects are included. In Column 3, we drop observations from stores that, before the COVID-19 restrictions, were located between 180 and 300 min from the closest Swedish store. In Column 4, we drop observations from the weeks between and including the week starting on Monday 27 July 2020 and the week starting on Monday 5 October 2020. In these weeks, some, but not all, Swedish border counties were open for cross-border shopping. The variable "New COVID-19 cases" included in Column 5 is the sum of registered COVID-19 cases in week t and $t-1$ divided by the population of the municipality. C is an indicator variable taking the value one in the weeks after 17 March 2020 (and zero in the weeks before). "Holiday home proportion" is the proportion of residential dwellings that are holiday homes. The unemployment rate is measured at the municipality–month level. Population is measured at the municipality–quarter level. The standard errors reported in parentheses are clustered at the municipality level. *** $p<0.001$; ** $p<0.01$; * $p<0.05$.

Table A11: Effect of cross-border shopping on cheese sales (in NOK): robustness checks

	(1)	(2)	(3)	(4)	(5)
Duration $<$ 30 $\times$ B	-0.292***	-0.314***	-0.292***	-0.279***	-0.298***
	(0.028)	(0.079)	(0.031)	(0.029)	(0.031)
30 $\leq$ Duration $<$ 60 $\times$ B	-0.195***	-0.286***	-0.189***	-0.169***	-0.201***
	(0.039)	(0.069)	(0.031)	(0.030)	(0.031)
60 $\leq$ Duration $<$ 90 $\times$ B	-0.099***	-0.195**	-0.080**	-0.068*	-0.099***
	(0.021)	(0.066)	(0.024)	(0.029)	(0.023)
90 $\leq$ Duration $<$ 120 $\times$ B	-0.049*	-0.002	-0.047*	-0.041	-0.065**
	(0.019)	(0.030)	(0.020)	(0.022)	(0.022)
120 $\leq$ Duration $<$ 150 $\times$ B	-0.027	-0.016	-0.004	0.020	-0.003
	(0.020)	(0.031)	(0.019)	(0.021)	(0.020)
150 $\leq$ Duration $<$ 180 $\times$ B	-0.026	-0.055*	-0.021	-0.007	-0.009
	(0.027)	(0.022)	(0.024)	(0.028)	(0.024)
New COVID-19 cases					0.014**
					(0.005)
C $\times$ Holiday home proportion					0.168***
					(0.046)
Unemployment rate					-1.757***
					(0.489)
Population					0.692**
					(0.234)
Observations	41,994	41,994	35,486	37,594	40,999
Stores	400	400	338	400	400
Control group	$>$ 180 min	$>$ 180 min	$>$ 300 min	$>$ 180 min	$>$ 180 min
Linear time trends	Yes	No	No	No	No
Duration group $\times$ week FEs	No	Yes	No	No	No
Time window	Full	Full	Full	Restricted	Full

Notes: This table reports results from estimation of Equation 2.1 with the natural logarithm of cheese sales as the dependent variable. In Column 1, separate linear time trends for duration groups based on the pre-COVID duration to the closest Swedish store are included. The frequency distribution of the groups is reported in Table 2.2. In Column 2, week $\times$ pre-COVID duration group fixed effects are included. In Column 3, we drop observations from stores that, before the COVID-19 restrictions, were located between 180 and 300 min from the closest Swedish store. In Column 4, we drop observations from the weeks between and including the week starting on Monday 27 July 2020 and the week starting on Monday 5 October 2020. In these weeks, some, but not all, Swedish border counties were open for cross-border shopping. The variable "New COVID-19 cases" included in Column 5 is the sum of registered COVID-19 cases in week t and $t-1$ divided by the population of the municipality. C is an indicator variable taking the value one in the weeks after 17 March 2020 (and zero in the weeks before). "Holiday home proportion" is the proportion of residential dwellings that are holiday homes. The unemployment rate is measured at the municipality–month level. Population is measured at the municipality–quarter level. The standard errors reported in parentheses are clustered at the municipality level. *** $p < 0.001$; ** $p < 0.01$; * $p < 0.05$.

Table A12: Effect of cross-border shopping on meat sales (in NOK): robustness checks

	(1)	(2)	(3)	(4)	(5)
Duration $<$ 30 $\times$ B	-0.443***	-0.390***	-0.533***	-0.524***	-0.538***
	(0.022)	(0.084)	(0.022)	(0.022)	(0.021)
30 $\leq$ Duration $<$ 60 $\times$ B	-0.292***	-0.299***	-0.338***	-0.322***	-0.347***
	(0.036)	(0.075)	(0.024)	(0.024)	(0.022)
60 $\leq$ Duration $<$ 90 $\times$ B	-0.115***	-0.231***	-0.111***	-0.106**	-0.143***
	(0.025)	(0.063)	(0.028)	(0.033)	(0.026)
90 $\leq$ Duration $<$ 120 $\times$ B	-0.053*	-0.003	-0.047*	-0.036	-0.078***
	(0.023)	(0.035)	(0.022)	(0.026)	(0.019)
120 $\leq$ Duration $<$ 150 $\times$ B	-0.010	0.038	-0.027	-0.015	-0.024
	(0.026)	(0.035)	(0.020)	(0.023)	(0.019)
150 $\leq$ Duration $<$ 180 $\times$ B	-0.033	-0.037	-0.042	-0.024	-0.026
	(0.028)	(0.031)	(0.023)	(0.028)	(0.023)
New COVID-19 cases					0.021***
					(0.006)
C $\times$ Holiday home proportion					0.173***
					(0.046)
Unemployment rate					-2.015***
					(0.571)
Population					-0.330*
					(0.142)
Observations	41,993	41,993	35,485	37,593	40,996
Stores	400	400	338	400	400
Control group	$>$ 180 min	$>$ 180 min	$>$ 300 min	$>$ 180 min	$>$ 180 min
Linear time trends	Yes	No	No	No	No
Duration group $\times$ week FEs	No	Yes	No	No	No
Time window	Full	Full	Full	Restricted	Full

Notes: This table reports results from estimation of Equation 2.1 with the natural logarithm of meat sales as the dependent variable. In Column 1, separate linear time trends for duration groups based on the pre-COVID duration to the closest Swedish store are included. The frequency distribution of the groups is reported in Table 2.2. In Column 2, week $\times$ pre-COVID duration group fixed effects are included. In Column 3, we drop observations from stores that, before the COVID-19 restrictions, were located between 180 and 300 min from the closest Swedish store. In Column 4, we drop observations from the weeks between and including the week starting on Monday 27 July 2020 and the week starting on Monday 5 October 2020. In these weeks, some, but not all, Swedish border counties were open for cross-border shopping. The variable "New COVID-19 cases" included in Column 5 is the sum of registered COVID-19 cases in week t and $t-1$ divided by the population of the municipality. C is an indicator variable taking the value one in the weeks after 17 March 2020 (and zero in the weeks before). "Holiday home proportion" is the proportion of residential dwellings that are holiday homes. The unemployment rate is measured at the municipality–month level. Population is measured at the municipality–quarter level. The standard errors reported in parentheses are clustered at the municipality level. *** $p < 0.001$; ** $p < 0.01$; * $p < 0.05$.

Table A13: Effect of cross-border shopping on sweets sales (in NOK): robustness checks

	(1)	(2)	(3)	(4)	(5)
Duration $<30\times$ B	-0.371***	-0.374***	-0.375***	-0.360***	-0.372***
	(0.026)	(0.088)	(0.020)	(0.023)	(0.019)
$30\leq$ Duration $<60\times$ B	-0.237***	-0.311***	-0.237***	-0.214***	-0.245***
	(0.030)	(0.080)	(0.022)	(0.022)	(0.034)
$60\leq$ Duration $<90\times$ B	-0.115***	-0.279***	-0.102***	-0.073*	-0.123***
	(0.026)	(0.079)	(0.028)	(0.033)	(0.026)
$90\leq$ Duration $<120\times$ B	-0.036	-0.024	-0.051*	-0.036	-0.056*
	(0.023)	(0.036)	(0.022)	(0.022)	(0.025)
$120\leq$ Duration $<150\times$ B	-0.009	0.017	-0.014	0.011	-0.025
	(0.022)	(0.032)	(0.018)	(0.020)	(0.019)
$150\leq$ Duration $<180\times$ B	-0.014	-0.015	-0.016	0.013	-0.009
	(0.030)	(0.028)	(0.027)	(0.030)	(0.033)
New COVID-19 cases					0.017*
					(0.008)
C $\times$ Holiday home proportion					0.166***
					(0.049)
Unemployment rate					-2.138***
					(0.504)
Population					0.620***
					(0.185)
Observations	41,998	41,998	35,488	37,598	41,011
Stores	400	400	338	400	400
Control group	>180 min	>180 min	>300 min	>180 min	>180 min
Linear time trends	Yes	No	No	No	No
Duration group $\times$ week FEs	No	Yes	No	No	No
Time window	Full	Full	Full	Restricted	Full
Tax control	No	No	No	No	Yes

Notes: This table reports results from estimation of Equation 2.1 with the natural logarithm of sweet sales (in NOK) as the dependent variable. In Column 1, separate linear time trends for duration groups based on the pre-COVID duration to the closest Swedish store are included. The frequency distribution of the groups is reported in Table 2.2. In Column 2, week $\times$ pre-COVID duration group fixed effects are included. In Column 3, we drop observations from stores that, before the COVID-19 restrictions, were located between 180 and 300 min from the closest Swedish store. In Column 4, we drop observations from the weeks between and including the week starting on Monday 27 July 2020 and the week starting on Monday 5 October 2020. In these weeks, some, but not all, Swedish border counties were open for cross-border shopping. The variable "New COVID-19 cases" included in Column 5 is the sum of registered COVID-19 cases in week t and $t-1$ divided by the population of the municipality. C is an indicator variable taking the value one in the weeks after 17 March 2020 (and zero in the weeks before). "Holiday home proportion" is the proportion of residential dwellings that are holiday homes. The unemployment rate is measured at the municipality–month level. Population is measured at the municipality–quarter level. The standard errors reported in parentheses are clustered at the municipality level. *** $p<0.001$; ** $p<0.01$; * $p<0.05$.

Table A14: Effect of cross-border shopping on sweets sales (in kilograms): robustness checks

	(1)	(2)	(3)	(4)	(5)
Duration $<$ 30 $\times$ B	-0.397***	-0.459***	-0.405***	-0.382***	-0.404***
	(0.038)	(0.095)	(0.023)	(0.024)	(0.029)
30 $\leq$ Duration $<$ 60 $\times$ B	-0.243***	-0.344***	-0.249***	-0.219***	-0.255***
	(0.032)	(0.084)	(0.023)	(0.023)	(0.038)
60 $\leq$ Duration $<$ 90 $\times$ B	-0.111***	-0.288***	-0.115***	-0.084*	-0.128***
	(0.027)	(0.084)	(0.029)	(0.033)	(0.026)
90 $\leq$ Duration $<$ 120 $\times$ B	-0.030	-0.037	-0.062**	-0.043	-0.056*
	(0.023)	(0.038)	(0.022)	(0.023)	(0.025)
120 $\leq$ Duration $<$ 150 $\times$ B	-0.004	0.008	-0.019	0.010	-0.019
	(0.022)	(0.033)	(0.019)	(0.021)	(0.020)
150 $\leq$ Duration $<$ 180 $\times$ B	-0.006	-0.025	-0.018	0.019	-0.006
	(0.033)	(0.030)	(0.027)	(0.030)	(0.034)
New COVID-19 cases					0.018*
					(0.008)
C $\times$ Holiday home proportion					0.171***
					(0.051)
Unemployment rate					-2.107***
					(0.529)
Population					0.654***
					(0.195)
Observations	41,998	41,998	35,488	37,598	41,011
Stores	400	400	338	400	400
Control group	$>$ 180 min	$>$ 180 min	$>$ 300 min	$>$ 180 min	$>$ 180 min
Linear time trends	Yes	No	No	No	No
Duration group $\times$ week FEs	No	Yes	No	No	No
Time window	Full	Full	Full	Restricted	Full
Tax control	No	No	No	No	Yes

Notes: This table reports results from estimation of Equation 2.1 with the natural logarithm of sweet sales (in kilograms) as the dependent variable. In Column 1, separate linear time trends for duration groups based on the pre-COVID duration to the closest Swedish store are included. The frequency distribution of the groups is reported in Table 2.2. In Column 2, week $\times$ pre-COVID duration group fixed effects are included. In Column 3, we drop observations from stores that, before the COVID-19 restrictions, were located between 180 and 300 min from the closest Swedish store. In Column 4, we drop observations from the weeks between and including the week starting on Monday 27 July 2020 and the week starting on Monday 5 October 2020. In these weeks, some, but not all, Swedish border counties were open for cross-border shopping. The variable "New COVID-19 cases" included in Column 5 is the sum of registered COVID-19 cases in week t and $t-1$ divided by the population of the municipality. C is an indicator variable taking the value one in the weeks after 17 March 2020 (and zero in the weeks before). "Holiday home proportion" is the proportion of residential dwellings that are holiday homes. The unemployment rate is measured at the municipality–month level. Population is measured at the municipality–quarter level. The standard errors reported in parentheses are clustered at the municipality level. *** $p < 0.001$; ** $p < 0.01$; * $p < 0.05$.

Table A15: Effect of cross-border shopping on soda sales (in NOK): robustness checks

	(1)	(2)	(3)	(4)	(5)
Duration $<30\times$ B	-0.657***	-0.585***	-0.673***	-0.667***	-0.699***
	(0.023)	(0.101)	(0.036)	(0.033)	(0.036)
$30\leq$ Duration $<60\times$ B	-0.394***	-0.364***	-0.388***	-0.372***	-0.399***
	(0.047)	(0.084)	(0.036)	(0.033)	(0.050)
$60\leq$ Duration $<90\times$ B	-0.166***	-0.223**	-0.159***	-0.148***	-0.158***
	(0.026)	(0.084)	(0.025)	(0.030)	(0.026)
$90\leq$ Duration $<120\times$ B	-0.078***	-0.010	-0.086***	-0.079***	-0.076**
	(0.023)	(0.044)	(0.023)	(0.022)	(0.024)
$120\leq$ Duration $<150\times$ B	-0.055*	-0.004	-0.057**	-0.044*	-0.056**
	(0.022)	(0.040)	(0.020)	(0.021)	(0.020)
$150\leq$ Duration $<180\times$ B	-0.044	-0.033	-0.052*	-0.045	-0.029
	(0.027)	(0.040)	(0.026)	(0.027)	(0.030)
New COVID-19 cases					0.012*
					(0.006)
C $\times$ Holiday home proportion					0.145**
					(0.050)
Unemployment rate					-1.698***
					(0.493)
Population					0.160
					(0.168)
Observations	41,997	41,997	35,488	37,597	41,008
Stores	400	400	338	400	400
Control group	>180 min	>180 min	>300 min	>180 min	>180 min
Linear time trends	Yes	No	No	No	No
Duration group $\times$ week FEs	No	Yes	No	No	No
Time window	Full	Full	Full	Restricted	Full
Tax control	No	No	No	No	Yes

Notes: This table reports results from estimation of Equation 2.1 with the natural logarithm of soda sales (in NOK) as the dependent variable. In Column 1, separate linear time trends for duration groups based on the pre-COVID duration to the closest Swedish store are included. The frequency distribution of the groups is reported in Table 2.2. In Column 2, week $\times$ pre-COVID duration group fixed effects are included. In Column 3, we drop observations from stores that, before the COVID-19 restrictions, were located between 180 and 300 min from the closest Swedish store. In Column 4, we drop observations from the weeks between and including the week starting on Monday 27 July 2020 and the week starting on Monday 5 October 2020. In these weeks, some, but not all, Swedish border counties were open for cross-border shopping. The variable "New COVID-19 cases" included in Column 5 is the sum of registered COVID-19 cases in week t and $t-1$ divided by the population of the municipality. C is an indicator variable taking the value one in the weeks after 17 March 2020 (and zero in the weeks before). "Holiday home proportion" is the proportion of residential dwellings that are holiday homes. The unemployment rate is measured at the municipality–month level. Population is measured at the municipality–quarter level. The standard errors reported in parentheses are clustered at the municipality level. *** $p<0.001$; ** $p<0.01$; * $p<0.05$.

Table A16: Effect of cross-border shopping on soda sales (in liters): robustness checks

	(1)	(2)	(3)	(4)	(5)
Duration $<30 \times$ B	-0.774***	-0.633***	-0.801***	-0.791***	-0.832***
	(0.027)	(0.106)	(0.048)	(0.044)	(0.053)
$30 \leq$ Duration $<60 \times$ B	-0.422***	-0.365***	-0.444***	-0.429***	-0.449***
	(0.058)	(0.087)	(0.048)	(0.046)	(0.062)
$60 \leq$ Duration $<90 \times$ B	-0.174***	-0.190*	-0.182***	-0.175***	-0.173***
	(0.027)	(0.085)	(0.024)	(0.029)	(0.027)
$90 \leq$ Duration $<120 \times$ B	-0.078**	0.001	-0.095***	-0.086***	-0.088***
	(0.025)	(0.044)	(0.023)	(0.021)	(0.024)
$120 \leq$ Duration $<150 \times$ B	-0.054*	-0.008	-0.072**	-0.058*	-0.073**
	(0.026)	(0.043)	(0.022)	(0.022)	(0.023)
$150 \leq$ Duration $<180 \times$ B	-0.038	-0.039	-0.061*	-0.058*	-0.040
	(0.026)	(0.047)	(0.024)	(0.024)	(0.028)
New COVID-19 cases					0.016*
					(0.007)
C $\times$ Holiday home proportion					0.143**
					(0.052)
Unemployment rate					-1.468**
					(0.526)
Population					0.152
					(0.152)
Observations	41,997	41,997	35,488	37,597	41,008
Stores	400	400	338	400	400
Control group	>180 min	>180 min	>300 min	>180 min	>180 min
Linear time trends	Yes	No	No	No	No
Duration group $\times$ week FEs	No	Yes	No	No	No
Time window	Full	Full	Full	Restricted	Full
Tax control	No	No	No	No	Yes

Notes: This table reports results from estimation of Equation 2.1 with the natural logarithm of soda sales (in liters) as the dependent variable. In Column 1, separate linear time trends for duration groups based on the pre-COVID duration to the closest Swedish store are included. The frequency distribution of the groups is reported in Table 2.2. In Column 2, week $\times$ pre-COVID duration group fixed effects are included. In Column 3, we drop observations from stores that, before the COVID-19 restrictions, were located between 180 and 300 min from the closest Swedish store. In Column 4, we drop observations from the weeks between and including the week starting on Monday 27 July 2020 and the week starting on Monday 5 October 2020. In these weeks, some, but not all, Swedish border counties were open for cross-border shopping. The variable "New COVID-19 cases" included in Column 5 is the sum of registered COVID-19 cases in week t and $t-1$ divided by the population of the municipality. C is an indicator variable taking the value one in the weeks after 17 March 2020 (and zero in the weeks before). "Holiday home proportion" is the proportion of residential dwellings that are holiday homes. The unemployment rate is measured at the municipality–month level. Population is measured at the municipality–quarter level. The standard errors reported in parentheses are clustered at the municipality level. *** $p<0.001$; ** $p<0.01$; * $p<0.05$.

Table A17: Effect of cross-border shopping on beer sales (in NOK): robustness checks

	(1)	(2)	(3)	(4)	(5)
Duration $<30\times$ B	-0.499***	-0.539***	-0.522***	-0.524***	-0.521***
	(0.047)	(0.095)	(0.047)	(0.051)	(0.048)
$30\leq$ Duration $<60\times$ B	-0.317***	-0.389***	-0.322***	-0.312***	-0.323***
	(0.039)	(0.096)	(0.042)	(0.040)	(0.050)
$60\leq$ Duration $<90\times$ B	-0.160***	-0.334***	-0.149***	-0.145***	-0.164***
	(0.026)	(0.079)	(0.027)	(0.029)	(0.027)
$90\leq$ Duration $<120\times$ B	-0.094**	-0.033	-0.099***	-0.106***	-0.106***
	(0.028)	(0.058)	(0.029)	(0.024)	(0.030)
$120\leq$ Duration $<150\times$ B	-0.051*	0.027	-0.059**	-0.058**	-0.063*
	(0.025)	(0.052)	(0.021)	(0.020)	(0.025)
$150\leq$ Duration $<180\times$ B	-0.033	0.002	-0.037	-0.033	-0.032
	(0.035)	(0.039)	(0.028)	(0.029)	(0.039)
New COVID-19 cases					0.018*
					(0.008)
C $\times$ Holiday home proportion					0.181***
					(0.050)
Unemployment rate					-2.465***
					(0.545)
Population					0.276
					(0.173)
Observations	41,996	41,996	35,487	37,596	41,005
Stores	400	400	338	400	400
Control group	>180 min	>180 min	>300 min	>180 min	>180 min
Linear time trends	Yes	No	No	No	No
Duration group $\times$ week FEs	No	Yes	No	No	No
Time window	Full	Full	Full	Restricted	Full
Tax control	No	No	No	No	Yes

Notes: This table reports results from estimation of Equation 2.1 with the natural logarithm of beer sales (in NOK) as the dependent variable. In Column 1, separate linear time trends for duration groups based on the pre-COVID duration to the closest Swedish store are included. The frequency distribution of the groups is reported in Table 2.2. In Column 2, week $\times$ pre-COVID duration group fixed effects are included. In Column 3, we drop observations from stores that, before the COVID-19 restrictions, were located between 180 and 300 min from the closest Swedish store. In Column 4, we drop observations from the weeks between and including the week starting on Monday 27 July 2020 and the week starting on Monday 5 October 2020. In these weeks, some, but not all, Swedish border counties were open for cross-border shopping. The variable "New COVID-19 cases" included in Column 5 is the sum of registered COVID-19 cases in week t and $t-1$ divided by the population of the municipality. C is an indicator variable taking the value one in the weeks after 17 March 2020 (and zero in the weeks before). "Holiday home proportion" is the proportion of residential dwellings that are holiday homes. The unemployment rate is measured at the municipality–month level. Population is measured at the municipality–quarter level. The standard errors reported in parentheses are clustered at the municipality level. *** $p<0.001$; ** $p<0.01$; * $p<0.05$.

Table A18: Effect of cross-border shopping on beer sales (in liters): robustness checks

	(1)	(2)	(3)	(4)	(5)
Duration $<$ 30 $\times$ B	-0.508***	-0.581***	-0.544***	-0.533***	-0.529***
	(0.048)	(0.094)	(0.051)	(0.052)	(0.050)
30 $\leq$ Duration $<$ 60 $\times$ B	-0.324***	-0.411***	-0.344***	-0.324***	-0.330***
	(0.040)	(0.098)	(0.047)	(0.042)	(0.049)
60 $\leq$ Duration $<$ 90 $\times$ B	-0.162***	-0.350***	-0.168***	-0.154***	-0.168***
	(0.026)	(0.077)	(0.032)	(0.030)	(0.027)
90 $\leq$ Duration $<$ 120 $\times$ B	-0.101***	-0.045	-0.122***	-0.119***	-0.114***
	(0.028)	(0.063)	(0.034)	(0.026)	(0.031)
120 $\leq$ Duration $<$ 150 $\times$ B	-0.057*	0.028	-0.079**	-0.070**	-0.070**
	(0.025)	(0.054)	(0.026)	(0.022)	(0.024)
150 $\leq$ Duration $<$ 180 $\times$ B	-0.047	-0.007	-0.062*	-0.048	-0.047
	(0.033)	(0.039)	(0.031)	(0.030)	(0.036)
New COVID-19 cases					0.016*
					(0.008)
C $\times$ Holiday home proportion					0.199***
					(0.051)
Unemployment rate					-2.307***
					(0.517)
Population					0.295
					(0.171)
Observations	41,996	41,996	35,487	37,596	41,005
Stores	400	400	338	400	400
Control group	$>$ 180 min	$>$ 180 min	$>$ 300 min	$>$ 180 min	$>$ 180 min
Linear time trends	Yes	No	No	No	No
Duration group $\times$ week FEs	No	Yes	No	No	No
Time window	Full	Full	Full	Restricted	Full
Tax control	No	No	No	No	Yes

Notes: This table reports results from estimation of Equation 2.1 with the natural logarithm of beer sales (in liters) as the dependent variable. In Column 1, separate linear time trends for duration groups based on the pre-COVID duration to the closest Swedish store are included. The frequency distribution of the groups is reported in Table 2.2. In Column 2, week $\times$ pre-COVID duration group fixed effects are included. In Column 3, we drop observations from stores that, before the COVID-19 restrictions, were located between 180 and 300 min from the closest Swedish store. In Column 4, we drop observations from the weeks between and including the week starting on Monday 27 July 2020 and the week starting on Monday 5 October 2020. In these weeks, some, but not all, Swedish border counties were open for cross-border shopping. The variable "New COVID-19 cases" included in Column 5 is the sum of registered COVID-19 cases in week t and $t-1$ divided by the population of the municipality. C is an indicator variable taking the value one in the weeks after 17 March 2020 (and zero in the weeks before). "Holiday home proportion" is the proportion of residential dwellings that are holiday homes. The unemployment rate is measured at the municipality–month level. Population is measured at the municipality–quarter level. The standard errors reported in parentheses are clustered at the municipality level. *** $p < 0.001$; ** $p < 0.01$; * $p < 0.05$.

Table A19: Effect of cross-border shopping on cigarette sales (in NOK): robustness checks

	(1)	(2)	(3)	(4)	(5)
Duration $< 30 \times$ B	-1.060***	-0.863***	-1.115***	-1.131***	-1.107***
	(0.093)	(0.159)	(0.109)	(0.124)	(0.094)
$30 \leq$ Duration $< 60 \times$ B	-0.610***	-0.509***	-0.695***	-0.696***	-0.638***
	(0.052)	(0.091)	(0.046)	(0.048)	(0.053)
$60 \leq$ Duration $< 90 \times$ B	-0.257***	-0.287**	-0.322***	-0.342***	-0.259***
	(0.035)	(0.095)	(0.039)	(0.041)	(0.037)
$90 \leq$ Duration $< 120 \times$ B	-0.101***	-0.002	-0.153***	-0.167***	-0.108***
	(0.023)	(0.044)	(0.028)	(0.022)	(0.022)
$120 \leq$ Duration $< 150 \times$ B	-0.044*	0.046	-0.082***	-0.086***	-0.042*
	(0.020)	(0.037)	(0.023)	(0.023)	(0.018)
$150 \leq$ Duration $< 180 \times$ B	-0.030	0.000	-0.057*	-0.054*	-0.019
	(0.019)	(0.034)	(0.022)	(0.023)	(0.018)
New COVID-19 cases					0.011*
					(0.005)
C $\times$ Holiday home proportion					0.206**
					(0.068)
Unemployment rate					-0.497
					(0.420)
Population					0.006
					(0.154)
Observations	41,997	41,997	35,488	37,597	41,007
Stores	400	400	338	400	400
Control group	$>$ 180 min	$>$ 180 min	$>$ 300 min	$>$ 180 min	$>$ 180 min
Linear time trends	Yes	No	No	No	No
Duration group $\times$ week FEs	No	Yes	No	No	No
Time window	Full	Full	Full	Restricted	Full
Tax control	No	No	No	No	Yes

Notes: This table reports results from estimation of Equation 2.1 with the natural logarithm of cigarette sales (in NOK) as the dependent variable. In Column 1, separate linear time trends for duration groups based on the pre-COVID duration to the closest Swedish store are included. The frequency distribution of the groups is reported in Table 2.2. In Column 2, week $\times$ pre-COVID duration group fixed effects are included. In Column 3, we drop observations from stores that, before the COVID-19 restrictions, were located between 180 and 300 min from the closest Swedish store. In Column 4, we drop observations from the weeks between and including the week starting on Monday 27 July 2020 and the week starting on Monday 5 October 2020. In these weeks, some, but not all, Swedish border counties were open for cross-border shopping. The variable "New COVID-19 cases" included in Column 5 is the sum of registered COVID-19 cases in week t and $t-1$ divided by the population of the municipality. C is an indicator variable taking the value one in the weeks after 17 March 2020 (and zero in the weeks before). "Holiday home proportion" is the proportion of residential dwellings that are holiday homes. The unemployment rate is measured at the municipality–month level. Population is measured at the municipality–quarter level. The standard errors reported in parentheses are clustered at the municipality level. *** $p < 0.001$; ** $p < 0.01$; * $p < 0.05$.

Table A20: Effect of cross-border shopping on cigarette sales (items): robustness checks

	(1)	(2)	(3)	(4)	(5)
Duration $< 30 \times$ B	-1.057***	-0.860***	-1.112***	-1.128***	-1.105***
	(0.093)	(0.156)	(0.109)	(0.124)	(0.094)
$30 \leq$ Duration $< 60 \times$ B	-0.608***	-0.505***	-0.691***	-0.693***	-0.635***
	(0.052)	(0.091)	(0.046)	(0.048)	(0.054)
$60 \leq$ Duration $< 90 \times$ B	-0.257***	-0.287**	-0.320***	-0.340***	-0.258***
	(0.035)	(0.095)	(0.039)	(0.041)	(0.037)
$90 \leq$ Duration $< 120 \times$ B	-0.100***	-0.001	-0.151***	-0.165***	-0.106***
	(0.023)	(0.044)	(0.028)	(0.023)	(0.022)
$120 \leq$ Duration $< 150 \times$ B	-0.042*	0.045	-0.078**	-0.082***	-0.040*
	(0.020)	(0.037)	(0.024)	(0.024)	(0.019)
$150 \leq$ Duration $< 180 \times$ B	-0.029	0.004	-0.056*	-0.054*	-0.018
	(0.019)	(0.034)	(0.023)	(0.023)	(0.018)
New COVID-19 cases					0.011*
					(0.005)
C $\times$ Holiday home proportion					0.204**
					(0.068)
Unemployment rate					-0.467
					(0.421)
Population					0.014
					(0.155)
Observations	41,997	41,997	35,488	37,597	41,007
Stores	400	400	338	400	400
Control group	$>$ 180 min	$>$ 180 min	$>$ 300 min	$>$ 180 min	$>$ 180 min
Linear time trends	Yes	No	No	No	No
Duration group $\times$ week FEs	No	Yes	No	No	No
Time window	Full	Full	Full	Restricted	Full
Tax control	No	No	No	No	Yes

Notes: This table reports results from estimation of Equation 2.1 with the natural logarithm of cigarette sales (items) as the dependent variable. In Column 1, separate linear time trends for duration groups based on the pre-COVID duration to the closest Swedish store are included. The frequency distribution of the groups is reported in Table 2.2. In Column 2, week $\times$ pre-COVID duration group fixed effects are included. In Column 3, we drop observations from stores that, before the COVID-19 restrictions, were located between 180 and 300 min from the closest Swedish store. In Column 4, we drop observations from the weeks between and including the week starting on Monday 27 July 2020 and the week starting on Monday 5 October 2020. In these weeks, some, but not all, Swedish border counties were open for cross-border shopping. The variable "New COVID-19 cases" included in Column 5 is the sum of registered COVID-19 cases in week t and $t-1$ divided by the population of the municipality. C is an indicator variable taking the value one in the weeks after 17 March 2020 (and zero in the weeks before). "Holiday home proportion" is the proportion of residential dwellings that are holiday homes. The unemployment rate is measured at the municipality–month level. Population is measured at the municipality–quarter level. The standard errors reported in parentheses are clustered at the municipality level. *** $p < 0.001$; ** $p < 0.01$; * $p < 0.05$.

Table A21: Effect of cross-border shopping on snus sales (NOK): robustness checks

	(1)	(2)	(3)	(4)	(5)
Duration $<30\times$ B	-1.056***	-0.937**	-1.132***	-1.096***	-1.125***
	(0.064)	(0.282)	(0.055)	(0.045)	(0.064)
$30\leq$ Duration $<60\times$ B	-0.648***	-0.420***	-0.805***	-0.795***	-0.722***
	(0.048)	(0.107)	(0.053)	(0.063)	(0.053)
$60\leq$ Duration $<90\times$ B	-0.257***	-0.288**	-0.338***	-0.340***	-0.271***
	(0.038)	(0.094)	(0.043)	(0.049)	(0.038)
$90\leq$ Duration $<120\times$ B	-0.106***	-0.005	-0.173***	-0.163***	-0.133***
	(0.025)	(0.047)	(0.031)	(0.027)	(0.024)
$120\leq$ Duration $<150\times$ B	-0.065**	0.016	-0.119***	-0.102**	-0.072***
	(0.021)	(0.037)	(0.026)	(0.031)	(0.018)
$150\leq$ Duration $<180\times$ B	-0.041	-0.004	-0.081*	-0.059	-0.028
	(0.027)	(0.035)	(0.031)	(0.036)	(0.027)
New COVID-19 cases					0.009
					(0.006)
C $\times$ Holiday home proportion					0.265***
					(0.064)
Unemployment rate					-1.829***
					(0.491)
Population					0.108
					(0.180)
Observations	41,927	41,927	35,417	37,527	40,928
Stores	400	400	338	400	400
Control group	>180 min	>180 min	>300 min	>180 min	>180 min
Linear time trends	Yes	No	No	No	No
Duration group $\times$ week FEs	No	Yes	No	No	No
Time window	Full	Full	Full	Restricted	Full
Tax control	No	No	No	No	Yes

Notes: This table reports results from estimation of Equation 2.1 with the natural logarithm of snus sales (NOK) as the dependent variable. In Column 1, separate linear time trends for duration groups based on the pre-COVID duration to the closest Swedish store are included. The frequency distribution of the groups is reported in Table 2.2. In Column 2, week $\times$ pre-COVID duration group fixed effects are included. In Column 3, we drop observations from stores that, before the COVID-19 restrictions, were located between 180 and 300 min from the closest Swedish store. In Column 4, we drop observations from the weeks between and including the week starting on Monday 27 July 2020 and the week starting on Monday 5 October 2020. In these weeks, some, but not all, Swedish border counties were open for cross-border shopping. The variable "New COVID-19 cases" included in Column 5 is the sum of registered COVID-19 cases in week t and $t-1$ divided by the population of the municipality. C is an indicator variable taking the value one in the weeks after 17 March 2020 (and zero in the weeks before). "Holiday home proportion" is the proportion of residential dwellings that are holiday homes. The unemployment rate is measured at the municipality–month level. Population is measured at the municipality–quarter level. The standard errors reported in parentheses are clustered at the municipality level. *** $p<0.001$; ** $p<0.01$; * $p<0.05$.

Table A22: Effect of cross-border shopping on snus sales (kilograms): robustness checks

	(1)	(2)	(3)	(4)	(5)
Duration $<30\times$ B	-1.062***	-0.961***	-1.135***	-1.100***	-1.124***
	(0.066)	(0.275)	(0.055)	(0.046)	(0.066)
$30\leq$ Duration $<60\times$ B	-0.658***	-0.421***	-0.815***	-0.807***	-0.730***
	(0.049)	(0.106)	(0.054)	(0.065)	(0.053)
$60\leq$ Duration $<90\times$ B	-0.262***	-0.289**	-0.346***	-0.351***	-0.274***
	(0.038)	(0.094)	(0.042)	(0.049)	(0.038)
$90\leq$ Duration $<120\times$ B	-0.109***	-0.005	-0.179***	-0.172***	-0.135***
	(0.025)	(0.048)	(0.030)	(0.027)	(0.024)
$120\leq$ Duration $<150\times$ B	-0.068**	0.018	-0.124***	-0.109***	-0.075***
	(0.021)	(0.038)	(0.026)	(0.031)	(0.018)
$150\leq$ Duration $<180\times$ B	-0.044	0.001	-0.086**	-0.068*	-0.032
	(0.026)	(0.035)	(0.029)	(0.034)	(0.026)
New COVID-19 cases					0.009
					(0.006)
C $\times$ Holiday home proportion					0.260***
					(0.064)
Unemployment rate					-1.741***
					(0.481)
Population					0.071
					(0.177)
Observations	41,927	41,927	35,417	37,527	40,928
Stores	400	400	338	400	400
Control group	>180 min	>180 min	>300 min	>180 min	>180 min
Linear time trends	Yes	No	No	No	No
Duration group $\times$ week FEs	No	Yes	No	No	No
Time window	Full	Full	Full	Restricted	Full
Tax control	No	No	No	No	Yes

Notes: This table reports results from estimation of Equation 2.1 with the natural logarithm of snus sales (kilograms) as the dependent variable. In Column 1, separate linear time trends for duration groups based on the pre-COVID duration to the closest Swedish store are included. The frequency distribution of the groups is reported in Table 2.2. In Column 2, week $\times$ pre-COVID duration group fixed effects are included. In Column 3, we drop observations from stores that, before the COVID-19 restrictions, were located between 180 and 300 min from the closest Swedish store. In Column 4, we drop observations from the weeks between and including the week starting on Monday 27 July 2020 and the week starting on Monday 5 October 2020. In these weeks, some, but not all, Swedish border counties were open for cross-border shopping. The variable "New COVID-19 cases" included in Column 5 is the sum of registered COVID-19 cases in week t and $t-1$ divided by the population of the municipality. C is an indicator variable taking the value one in the weeks after 17 March 2020 (and zero in the weeks before). "Holiday home proportion" is the proportion of residential dwellings that are holiday homes. The unemployment rate is measured at the municipality–month level. Population is measured at the municipality–quarter level. The standard errors reported in parentheses are clustered at the municipality level. *** $p<0.001$; ** $p<0.01$; * $p<0.05$.

2.E A Comparison with a Survey-Based Estimate of Cross-Border Shopping

One way of establishing that results on lost sales due to cross-border shopping are credible is to compare to other sources of information about Norwegian cross-border shopping. In this chapter we estimated a counterfactual sales loss in 2019 of approximately 7.3 billion NOK, including excise taxes and VAT. One external comparison for this number comes from the survey of cross-border shopping conducted by Statistics Norway (2022*a*), where a random sample of 2000 Norwegians are surveyed about cross-border shopping in each quarter every year. Our counterfactual estimates based on the border closures provide an opportunity to validate the results from the 2019 pre-closure survey of cross-border shopping and vice versa. Based on survey responses, Statistics Norway (2022*a*) estimates that the cross-border grocery trade amounted to around 10.6 billion NOK in 2019.[33] The difference between the two estimates is thus relatively minor, which is reassuring. Let us nevertheless briefly discuss reasons for the potential discrepancies.

We estimate that grocery stores lost about 7.3 billion to cross-border shopping in 2019. However, groceries are also sold through kiosks, gasoline stations and variety discount stores. In 2019 more than 13 billion NOK worth of grocery sales originated from these sales channels.[34] Assuming the same sales loss due to cross-border shopping for these channels as what we found for our grocery stores (3.4%), this suggests that grocery sales of about 488 million NOK are disregarded when we rely only on grocery stores to estimate the loss from cross-border shopping.

More conceptually interesting however is that while we estimate the value of the sales lost in Norway through cross-border shopping, Statistics Norway estimates the value of the goods bought in Sweden. Given the large price differences between

[33]According to Statistics Norway (2020), Norwegians spent 16,041 million NOK in Sweden in 2019, but this included spending on gasoline, restaurant meals and products from all retail segments. In the 2019 survey, respondents were also asked which product categories they bought in Sweden. The largest category was food and non-food items typically sold in grocery stores (33.7%), followed by alcohol (18.2%) and tobacco (15.9%), soda (10.6%) and sweets (6.7%), with a range of other products such as clothes and electronics making up the remaining 15%. If we disregard alcohol (in Norway and Sweden wine and liquor are exclusively retailed in state-controlled retail monopolies and only beer can be sold in grocery stores) and "other products", we estimate that 66% of the cross-border shopping in Sweden is groceries, giving the number 10,587 million NOK.

[34]The Norwegian gasoline stations sold grocery products for 7033 million NOK and Norwegian kiosks for another 4196 million NOK (Dagligvarehandelen, 2021). The discount chain Normal offers the same personal care products as the grocery chains and had a turnover of 944 million NOK in 2019 (Kapital, 2021). Finally, Europris, the largest Norwegian variety discount chain, had a total turnover of 6234 million NOK in 2019 (Europris, 2020). Europris has a product overlap with the Norwegian grocery chains of more than 35% (Evensen, Steen, and Ulsaker, 2021). If we use this as a rough estimate for the share of Europris' turnover stemming from groceries, it suggests that they had a "grocery" turnover of 2182 million NOK in 2019. In total these four channels thus represented grocery sales of an estimated 14.4 billion NOK in 2019.

the two countries, it is not surprising that we see a discrepancy between these two estimates. The price indices reported in Table 2.1 suggest that, overall, food prices were about 35% (155/114.8) more expensive in Norway than in Sweden in 2019.[35] Furthermore, price differences tend to be greater than 35% for the categories most frequently bought in Sweden by Norwegian customers. For instance, non-alcoholic beverages were 64% more expensive in Norway in 2019 and tobacco was 109% more expensive. In response to lower prices, we expect that Norwegian customers buy substantially more when shopping in Sweden than they would if facing Norwegian prices. If we account for sales through gasoline stations, kiosks and variety discount stores, as well as the larger price differences for key cross-border shopping categories, the difference between our estimate of a loss of 7.3 billion NOK in Norway and a cross-border shopping value of 10.6 billion NOK, as estimated by Statistics Norway, is explained by an own-price elasticity of cross-border shopping goods in the range between -4 to -2.

The elasticities outlined in the discussion above are well within the bounds of estimates from the international literature, even if typical estimates are less elastic (see, e.g., Fogarty, 2010, for beer and Cornelsen et al., 2015, for meat, dairy and sweets). For soda, Allcott, Lockwood, and Taubinsky (2019*b*, p. 216) report own-price elasticities across a large number of studies ranging from -0.13 to -3.9, with Allcott, Lockwood, and Taubinsky (2019*a*) using an estimated own-price elasticity of soda of -1.4.

In conclusion, this validation exercise finds that our results match up well with the questionnaire of cross-border shopping conducted by Norway's national statistical agency. Perhaps even more importantly our results also provide validation in the other direction—a well conducted survey with 2000 respondents sampled nationally paints a picture that is remarkably well in line with observed patterns in this natural experiment. As such the results should be encouraging for statistical agencies in both Norway and other countries that use surveys to track cross-border shopping.

[35]This is well in line with Pettersen (2020), who estimates that the price differences between Norway and Sweden and Denmark were about 32% in 2018.

Chapter 3

Do Informed Consumers Pay Less? Evidence from a Survey with Linked Grocery Purchase Data

Abstract: We examine how consumer price knowledge affects prices paid in grocery markets, combining survey-based price recall data from over 2000 Norwegian households with 18 months of linked transaction histories. Better-informed consumers pay lower prices by timing purchases to coincide with sales: a 10 percentage-point increase in price knowledge is associated with a 1.3 percent reduction in prices paid. This provides direct empirical support for the central mechanism in Varian's (1980) model of sales. Consumers who actively seek price information have higher price knowledge, suggesting that policies improving price accessibility may effectively enhance competition.

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3.1 Introduction

Consumer knowledge of prices across retailers is crucial for effective market competition. As emphasized in Stigler (1961), acquiring price information requires effort, as consumers must allocate attention and often engage in various forms of search. The prevailing intuition, both among economists and policymakers, is that reducing search costs leads to more informed consumers, thereby intensifying competition and driving prices down.

A central modeling approach in the literature on costly search is to distinguish between two types of consumers—one with zero search costs ("informed") and one with some positive search cost ("uninformed")—see, for example, Salop and Stiglitz (1977), Varian (1980), and Stahl (1989).[1] These models predict that a higher share of informed consumers intensifies competition and drives prices down—a result that has become central to economic research on search costs and market outcomes. Such models have, for instance, been used to generate predictions for the effects of mandatory price disclosure in grocery markets (Ater and Rigbi, 2023), price dispersion in gasoline retailing (Pennerstorfer et al., 2020), the role of temporary sales in grocery markets (Villas-Boas, 1995), and the impact of reduced search costs as consumers gain access to the internet (Brown and Goolsbee, 2002). They have also been tested in laboratory settings to better understand price dispersion and consumer behavior (Morgan, Orzen, and Sefton, 2006).[2]

While theory predicts that lower search costs lead to more informed consumers and stronger price competition, empirical evidence on how price knowledge affects actual consumer outcomes remains limited. This chapter provides direct evidence on how consumer price knowledge affects prices paid in the grocery market by asking: do more informed consumers systematically pay lower prices? Using an online survey, we measure consumers' knowledge of grocery prices and link their responses to 18 months of individual-level purchasing history from Norway's largest grocery retail group. This unique combination of survey data and purchasing histories allows us to test the prediction that informed consumers pay lower prices and investigate whether price knowledge is associated with particular shopping behaviors, attitudes, or demographics. The final sample includes more than 2000 households and over 70,000 recalled prices across two grocery chains and 24 products.

[1] While the intuition that lower search costs intensify competition is attractive, models that feature this outcome are surprisingly intricate. Indeed, the formal literature on the interaction between search costs and competition started with a paradox—the result in Diamond (1971) that introducing even minuscule search costs into a model of Bertrand competition with homogeneous products leads all firms to charge monopoly prices instead of pricing at marginal cost.

[2] See also empirical work documenting that patterns of price dispersion match predictions from search models, including Sorensen (2000) and, for a survey, Honka, Hortaçsu, and Wildenbeest (2019). Relatedly, structural models have been used to recover the distribution of search costs from observed prices and quantities (see, e.g., Hong and Shum, 2006).

Theoretical models such as Varian (1980) predict that when consumers become more informed, firms have stronger incentives to lower prices through temporary discounts and promotions to attract price-sensitive shoppers. A key assumption behind this prediction is that informed consumers exploit price variation by timing purchases to coincide with sales rather than paying regular prices. To our knowledge, we are the first to provide direct empirical support for this mechanism, namely that consumers with higher price knowledge systematically pay lower prices, not because they buy inherently cheaper products or shop at inherently cheaper stores, but because they are more likely to purchase products when the price is low. This suggests that informed consumers impose a competitive constraint on retailers by responding to price fluctuations, reinforcing the theoretical link between consumer information and price competition. The effects we find are economically significant: for instance, moving from relatively low price knowledge (an average percentage absolute deviation between recalled and true prices of 40%) to relatively high price knowledge (an average percentage absolute deviation between recalled and true prices of 10%) is associated with a 4 percentage-point decrease in the average price paid. This relationship holds even after controlling for demographic characteristics, shopping behavior, and stated price consciousness.

We further show that the returns to price knowledge are heterogeneous. The reduction in prices paid is significantly larger for consumers classified as price conscious, reflecting a higher motivation to act on price knowledge. Consistent with Stigler's (1961) prediction that the returns to search increase with price dispersion, we find that the effect is amplified for shopping baskets characterized by higher price volatility. We also observe stronger returns for non-perishable products, where the scope for intertemporal substitution is greater due to the possibility of stockpiling (Hendel and Nevo, 2006, 2013).

Finally, we assess the yearly monetary savings a nuclear family can expect on a representative shopping basket. These estimates capture only direct savings, disregarding the additional competitive pressure that a larger share of informed consumers would exert on equilibrium prices. We find that a 30-percentage-point increase in price knowledge is associated with meaningful but modest average savings of 2.1 percent (3629 NOK)—partly because high-spending products in our sample exhibit less price volatility—but the savings increase to 2.6 percent (4428 NOK) for consumers who purchase products with high price fluctuations.

Having established the relationship between price knowledge and prices paid, we go on to investigate the drivers of price knowledge. We find that, overall, product characteristics and observed consumer behaviors are more important than demographic characteristics and other self-reported variables in explaining consumers' price knowledge. For example, a high number of substitutes strongly predicts poorer

price recall accuracy, while prior purchasing experience and spending on a product are associated with higher recall. Still, we also find that consumers who are more active in gathering information about prices—by comparing prices in stores, reading flyers, or consulting price comparison surveys in the media—have higher price knowledge. This association also holds after controlling for differences in demographics, shopping behavior, and consumers' reported price sensitivity, that is, how important prices are in purchasing decisions.

Our findings have implications for policy design. First, the relationship between price knowledge and the prices consumers pay suggests that policies that successfully raise consumers' price knowledge can lead to lower prices and stronger competition. Second, we find that consumers who actively engage in price-gathering activities tend to be more knowledgeable. This matters because if price knowledge were solely driven by traits such as cognitive ability or shopping patterns, reducing search costs through better access to price information might have limited effect. Instead, our results indicate that price knowledge is shaped by access to information and that interventions aimed at improving the accessibility and visibility of price information—such as price disclosure regulations or comparison tools—may be effective in raising consumer knowledge and affecting market outcomes.[3]

These policy implications relate closely to a large empirical literature studying the effects of price transparency regulations, which are often motivated by the prediction that better-informed consumers intensify competition and discipline prices.

For instance, Ater and Rigbi (2023) study a price transparency regulation in the Israeli food retail industry that required supermarkets to report daily prices to an online repository used by price-comparison websites, and find that average prices declined by 4 to 5 percent following the reform. In a different context, Brown (2019) analyzes the introduction of a price transparency website for medical imaging procedures and finds that prices fell due to both demand-side responses (patients switching to lower-cost providers) and supply-side responses (providers lowering prices). Studies of price transparency in gasoline markets find more mixed effects (Rossi and Chintagunta, 2016; Luco, 2019; Montag, Sagimuldina, and Winter, 2024).

While this line of literature typically focuses on market-level outcomes—such as reduced average prices or lower price dispersion—our study complements it by providing direct micro-level evidence for the link between price knowledge and prices paid.

Clearly, the role of price knowledge may differ between product markets: compare, for instance, one-time, long-term purchases (e.g., pension savings plans), simple products with little physical differentiation (e.g., gasoline), and markets where

[3]Of course, such comparison tools may also have unintended effects, such as making it easier for competitors to learn about each other's prices.

consumers regularly purchase many differentiated products and must consider an entire basket of prices (e.g., grocery retailing). Given the large share of household budgets devoted to groceries, grocery retailing is an especially important setting. The focus on temporary sales as a source of price variation is highly relevant in retail grocery markets. Anderson et al. (2017), for instance, document that 95% of price changes in a large US supermarket chain are due to temporary sales. Clearly, we cannot be certain whether the magnitude of the effects translates to markets other than groceries, but we expect the fundamental mechanism at play—more informed consumers taking greater advantage of temporary sales—to manifest itself in other markets as well. Nakamura and Steinsson (2008), for instance, show the importance of temporary sales in a broader sample using product-level price data gathered by the Bureau of Labor Statistics to construct the US Consumer Price Index.

While public policy is a key motivation for our interest in the links between price knowledge and discount utilization, we note that the interaction between price knowledge and behavior is also of interest from other perspectives. Firms may act to affect the amount of information that consumers have—for instance via informative advertising (Milyo and Waldfogel, 1999), obfuscation (Ellison and Ellison, 2009; Ellison and Wolitzky, 2012), by "educating" consumers about competitors' price obfuscation (Gabaix and Laibson, 2006), or via platform design (Dinerstein et al., 2018). Again, our results buttress the fundamental mechanism at play in such models.

We also contribute to the marketing literature on price knowledge and consumer behavior. Prior studies in this field have primarily measured price recall and examined its implications for pricing strategies and the marketing mix (Winer, 1986; Dickson and Sawyer, 1990; Loy, Ceynowa, and Kuhn, 2020), but have generally lacked data linking recall to actual purchase behavior. We address this gap by systematically combining survey-based price knowledge data with transaction histories from the same consumers. Our results show that much of the variation in price knowledge is explained by differences in shopping behavior, highlighting the value of behavioral data for understanding who is more or less informed.

The remainder of the chapter is organized as follows. Section 3.2 introduces the data and outlines the empirical framework. Section 3.3 examines the relationship between price knowledge and discount utilization. Section 3.4 investigates the determinants of price knowledge. Section 3.5 concludes.

3.2 Data and Empirical Framework

3.2.1 Data Sources

We use four primary sources of data. The first is a digital survey, distributed with support from NorgesGruppen—Norway's largest grocery umbrella chain—to 50,000 members of the customer loyalty program Trumf, of whom 2505 responded. After being asked about demographics, the respondents were asked to recall the current shelf price, accounting for sales but ignoring individual discounts, for 18 out of 24 representative grocery products at two grocery chains belonging to the umbrella chain: a leading discount chain and a leading supermarket chain in Norway.[4] The selected products cover the major categories of groceries and should be familiar to the vast majority of grocery shoppers in Norway. Finally, they were asked to state their degree of agreement with 10 different statements designed to elicit their subjective degree of price consciousness. The complete survey and more details can be found in Appendix 3.C. Out of the 2505 respondents, 2285 made at least one price recall, resulting in 77,646 price recall responses. Note that there is one unique recall for every individual $\times$ chain $\times$ product combination, whereas the other survey questions only vary across individuals. The data are unbalanced primarily because each respondent was only asked about 18 out of the 24 products included in the survey, but also because of some missing responses.

The second data source comprises the daily purchasing history of each respondent at the grocery umbrella chain, derived from loyalty card data. Covering the period from 1 January 2023 to 12 June 2024, this dataset captures spending and quantities for the products in the survey and their close substitutes—a total of more than 700 items. The dimensions of the daily transaction data are individual $\times$ chain $\times$ product $\times$ date, and the data are highly unbalanced since most people do not buy every product, every day, at both chains.

The third data source is weekly purchasing frequency, spending, and the number of unique products purchased at both grocery chains, including products not included in the daily data. The weekly purchasing data have the following dimensions: individual $\times$ chain $\times$ week.

The fourth data source consists of the modal (most frequently paid) price for each chain $\times$ product $\times$ municipality $\times$ date combination for the products included in the daily transaction data, which almost always corresponds to the shelf price (hereafter denoted as P^{Shelf}).

In order to relate the sales and price data to the survey data, we aggregate various

[4] By asking for specific numbers, we situate ourselves in the part of the price-knowledge literature focusing on explicit (rather than implicit) price knowledge (see Evanschitzky, Kenning, and Vogel, 2004).

variables based on the purchasing histories, such as spending, purchasing frequencies, and prices paid, across time and product categories before merging. When appropriate, we further aggregate the data to the individual level, for instance by computing the average error rates for every chain $\times$ product combination for which the individual recalled prices, or by calculating the individual's overall discount utilization. Similarly, we calculate average price recall error rates for the different chains and products to see how the respondents' price knowledge varies across them, and whether they tend to over- or underestimate the prices of certain chains or products.

For comprehensive descriptive statistics and details on data cleaning, we refer the reader to Appendix 3.A.[5] Here, to assess the external validity of our results, we focus on the composition of the survey sample. The survey is based on a random sample of Trumf members. Trumf is the largest loyalty program in the Norwegian grocery industry, with nearly 3 million members from an adult population of 4.5 million (about 80% of Norway's total population of 5.6 million is over 18 years old; see Statistics Norway, 2024*a*), suggesting that Trumf members are broadly representative of the Norwegian population.[6] Our survey sample of 2505 respondents is also broadly representative in terms of demographics, albeit with some noteworthy deviations. Compared to the general adult population in 2024, respondents are more likely to have higher education (71% vs. 38%), to be over 40 years of age (77% vs. 64%), and to be female (62% vs. 50%).[7] The overrepresentation of highly educated respondents aligns with documented patterns in European survey data, where participation rates typically correlate positively with educational attainment (Spitzer, 2020). The older age profile may reflect that retirees have more time to respond to surveys, and the overrepresentation of women may reflect, on average, greater interest in grocery shopping and prices, consistent with women spending more time on household work—including purchasing goods and services (Statistics Norway, 2024*c*)—and with women more frequently being the household's main grocery shopper.[8] Including demographic controls in the regressions helps address these imbalances.

[5]The cleaning process involves removing extreme outliers in recall accuracy. In Appendices 3.B.5 and 3.B.6, we replicate the main analyses on the full sample using indicators for correct price recalls within fixed error bounds, instead of the continuous recall accuracy measure introduced below. These checks confirm that our results regarding both discount utilization and the drivers of price knowledge are robust to the cleaning process.

[6]In fact, the umbrella chain reports that it stopped tracking non-members' bank card activity after finding that their purchasing patterns closely mirrored those of Trumf members.

[7]Population benchmarks from Statistics Norway (2024*a*) for age and gender and Statistics Norway (2024*b*) for education.

[8]In our sample, there is a positive correlation between being the household's main grocery shopper and being a woman (Spearman's $\rho = 0.24$).

3.2.2 Discussion of Hypotheses and Empirical Framework

Conceptually, we can think of two sources of price dispersion: across retailers and within retailers over time. The latter is the focus of Varian (1980), who shows how a mixed-strategy equilibrium with random occurrences of low-price periods (temporary sales) allows retailers to increase profits by attracting informed shoppers, in essence, trading off incentives to attract informed consumers against incentives to extract surplus from uninformed shoppers. Models similar to Varian's remain central to the modeling of information frictions and search behavior, and recent examples in this class include Fabra and Reguant (2020), which examines consumer heterogeneity beyond search costs; Shelegia and Wilson (2021), which modifies how ties are resolved by consumers when firms post identical prices; and Myatt and Ronayne (2026), which allows for asymmetric costs across firms.[9]

The central mechanism in Varian (1980), and the search literature that has followed, is that informed consumers act upon their price knowledge and take advantage of temporary sales. While, as discussed in the introduction, the previous literature is broadly consistent with the predictions from search models, it does not necessarily follow that variation in consumer price knowledge is key for driving consumer responses to temporary sales. Observed responses to temporary sales might instead reflect that some consumers are more willing to act upon observed price differences than others, rather than reflecting differences in knowledge per se. For instance, Hendel and Nevo (2013) show how temporary sales of storable products are profitable as they allow retailers to sell at low prices to more price-sensitive consumers who are willing to stockpile, while at the same time keeping high non-sale prices, allowing them to charge high prices to non-stockpiling customers.

The key hypothesis we want to evaluate is that more informed consumers pay lower prices by timing their purchases to coincide with temporary sales. Note that we focus on the demand-side mechanism: the effect of an individual's price knowledge on their discount utilization. A key result in the class of models that motivate our study is that informed consumers also provide a positive externality for other consumers in the market—strengthening competition. Our setting does not allow us to examine such equilibrium effects. Hence, we examine the quantitative importance of the key demand-side mechanism in this class of models, rather than trying to quantify equilibrium pricing responses. Figure 3.1 provides an overview of our strategy to identify the effect of consumer price knowledge on discount utilization.

[9]Related work also examines the "rockets and feathers" phenomenon, e.g., Tappata (2009), which introduces uncertainty over input costs in a search model similar to that of Varian (1980). In this model, expected gains from searching are lower when costs are expected to be high, giving rise to asymmetric price responses to cost shocks. We expect such mechanisms to be particularly relevant in, for example, gasoline markets with large fluctuations in input costs and no temporary sales.

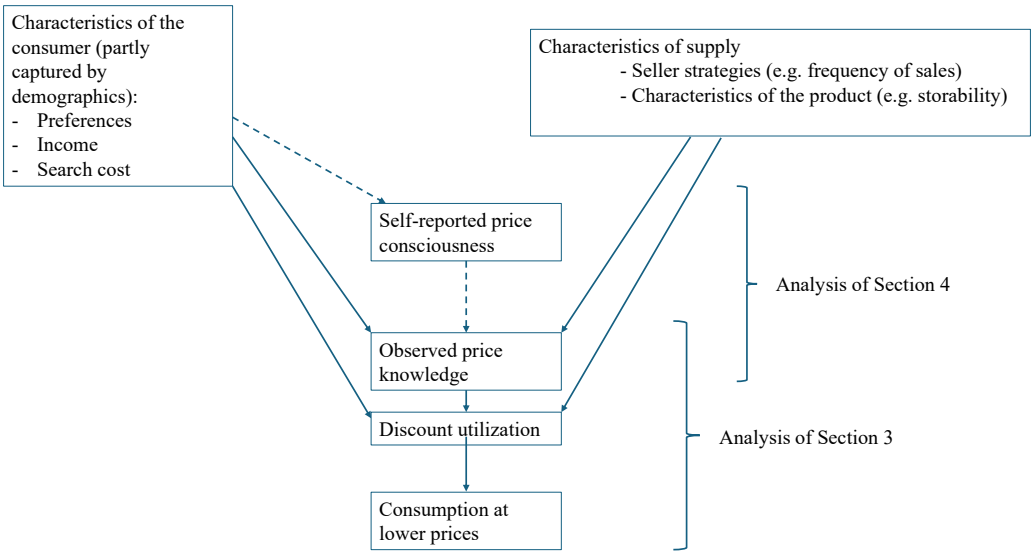


Figure 3.1: An overview of the links between price knowledge and discount utilization.

In Section 3.3, we use OLS to estimate variants of the following specification:[10]

$$DU_i = \alpha + \beta\, PK_i + \mathbf{X}_i'\boldsymbol{\gamma} + \varepsilon_i \tag{3.1}$$

where DU_i denotes observed discount utilization for individual i. We detail the construction of the variables further below, but note that DU_i is calculated using each individual's purchasing history over the preceding 18 months for more than 700 products: the products included in the questionnaire as well as their close substitutes. PK_i denotes the individual's observed price knowledge, based on their estimates of the shelf prices of 18 products from our survey. Our hypothesis is that the estimated β is positive, implying that consumers with greater price knowledge respond more strongly to temporary sales. We further include a rich set of demographic controls $\mathbf{X}_i$.

To what extent do our estimates of β capture a causal effect of price knowledge on discount utilization, as opposed to an association driven by confounding factors?

First, note the timing of the survey and the purchase history—the survey was unanticipated by respondents, and the price knowledge that they demonstrate on the day of the survey is set in relation to their use of temporary sales over a period of 18 months prior to the survey. One potential threat to identification would be reverse causality: discount utilization may increase price knowledge, and individuals might happen to know prices because they recently shopped at that price. Both the much larger set of products used to calculate discount utilization (more than 700 vs. 24) and the long time period of 18 months suggest that this is unlikely to be a prominent

[10]We assess the validity of the linearity assumption using a binned scatterplot in Section 3.3 and by replacing PK_i with quintile indicators in Appendix 3.B.4.

concern.[11]

Second, we do not expect measurement error in the dependent variable to be a material concern. Discount utilization is based on recorded sales, where the loyalty card is linked to payment cards.[12]

The extent to which consumers can purchase products on temporary sale clearly depends on product characteristics and the competitive environment in which they make their purchases. We ask about the prices of 24 products in total, which represent items that are arguably typical of a grocery retail store: relatively low-priced consumer goods such as dairy products (milk, cheese, butter), vegetables (avocado, cucumber), fish and meat, potato chips, and carbonated soft drinks. We examine differences in discount utilization using sales across all close substitutes for the products we ask about. Our interpretation is that we capture the extent of price knowledge and its effect on discount utilization for a set of typical grocery products, reporting an average effect rather than focusing on how discount utilization varies with product characteristics. We make two exceptions along the heterogeneity dimension: first, we examine whether discount utilization depends on price variability; second, we examine heterogeneity with respect to product storability. Other things equal, we expect the gains from price knowledge to be higher for storable products such as detergent (the focus of Hendel and Nevo, 2006) or soft drinks (the focus of Hendel and Nevo, 2013).

Another concern is omitted variable bias—specifically, underlying preferences for purchasing on temporary sales. We therefore report estimates from specifications both with and without a rich set of controls. First, we directly target these preferences by controlling for demographics and a subjective measure of price sensitivity conditional on price knowledge. Demographics can clearly have direct effects on both price knowledge and discount utilization—for instance, a person with full-time employment may find it less attractive both to spend time gathering price information and to act upon temporary sales when they become aware of them. Since these variables may be imperfect proxies for the underlying preferences, we additionally control for past shopping behaviors (e.g., number of store visits) and a self-reported measure of price learning effort to further mitigate omitted variable bias in a separate specification. However, these measures also absorb what we consider valid variation in price knowledge—for example, consumers who exert greater effort in gathering price information tend to have higher knowledge. This may affect the precision of our estimates, as identification then relies on residual variation in price knowledge, driven,

[11]To test this formally, we also estimate Equation 3.1 using a measure of discount utilization constructed only from products not included in the survey. The results are reported in Appendix 3.B.3.

[12]Payment cards are the dominant way to purchase in retail settings in Norway. In a survey commissioned by the Norwegian central bank, only 2% of payments by consumers in retail settings were made with cash (see Norges Bank, 2024, Chart 21).

for instance, by differences in attention, memory, or exposure to environmental cues and distractions.

Beyond the average effects, we extend the main analysis by examining heterogeneity in the returns to price knowledge to better understand the conditions under which it is most effective. Specifically, we investigate whether the link between knowledge and discount utilization is stronger for consumers whose shopping baskets exhibit high price volatility and for those with higher subjective price consciousness. We also explore the role of product characteristics by distinguishing between perishable and non-perishable goods. Finally, to assess the economic significance of our findings, we quantify the expected monetary savings associated with higher price knowledge.

In Section 3.4, we turn our attention to the determinants of price knowledge. If price knowledge affects the prices consumers pay, understanding what drives it is a prerequisite for assessing whether and how it can be improved. We estimate regressions of the following form:

$$PK_{ijc} = \alpha + \mathbf{X}_i'\boldsymbol{\beta} + \mathbf{Z}_j'\boldsymbol{\delta} + \mathbf{S}_{jc}'\boldsymbol{\phi} + \mathbf{W}_{ic}'\boldsymbol{\theta} + \mathbf{Q}_{ijc}'\boldsymbol{\kappa} + \eta_c + \nu_{ijc}. \tag{3.2}$$

We thus regress the price knowledge of consumer i for product j in chain c on individual characteristics ($\mathbf{X}_i$), product characteristics ($\mathbf{Z}_j$), chain-specific product characteristics ($\mathbf{S}_{jc}$), individual–chain shopping behavior ($\mathbf{W}_{ic}$), individual–product–chain shopping behavior ($\mathbf{Q}_{ijc}$), and a chain effect (η_c).

3.2.3 Defining Some Key Variables

Price Knowledge

The binary distinction between informed and uninformed consumers in many search models is clearly a simplification. To test whether price knowledge affects prices paid, we therefore construct a continuous measure of price knowledge based on consumers' ability to accurately recall current prices for a range of store × product combinations. This measure is then aggregated across store × product combinations to derive an overall price knowledge score for each consumer.

Our primary measure of price knowledge, PK, is based on the Percentage Absolute Deviation (PAD) of the surveyed price recall of individual i for product j at chain c relative to the true shelf price in the home municipality m on survey response date t:

$$\mathrm{PK}_{ijc} = -\underbrace{\left|\frac{P_{ijc}^{\mathrm{Recall}} - P_{jcmt}^{\mathrm{Shelf}}}{P_{jcmt}^{\mathrm{Shelf}}}\right|}_{PAD_{ijc}} \tag{3.3}$$

The PAD measure has previously been used in the marketing literature on price

knowledge (see, e.g., Dickson and Sawyer, 1990). By surveying respondents outside of purchasing situations, as opposed to studies that survey consumers post-purchase, in-store (see Conover, 1986; McGoldrick and Marks, 1987; Dickson and Sawyer, 1990; Wakefield and Inman, 1993; Le Boutillier, Le Boutillier, and Neslin, 1994), we aim to capture the long-term price memory of the respondents (see Evanschitzky, Kenning, and Vogel, 2004 for a discussion of short- vs. long-term memory and survey timing). PK_{ijc} is non-positive, and an increase in PK_{ijc} represents an increase in price knowledge towards zero, i.e., perfect information.

To obtain a measure of the individual's overall price knowledge, we calculate the average percentage error the individual made in the survey:

$$\mathrm{PK}_i = \frac{1}{K_i} \sum_{k=1}^{K_i} \mathrm{PK}_{ijc} \tag{3.4}$$

where K_i is the number of product $\times$ chain combinations for which the individual recalled prices in the survey.

Price Sensitivity, Learning Effort, and Consciousness

The survey includes questions designed to capture different aspects of how consumers relate to grocery prices, using statements answered on a 5-point Likert scale. The exact statements are reported in Appendix 3.C, and the distributions of responses are shown in Figure A2. The first three questions measure the extent to which respondents incorporate prices into their purchasing decisions, including the importance of price for store and product choice and whether they try to buy on sales and promotions. We use these items to construct an index of *Price Sensitivity*, which throughout the chapter is interpreted as consumers' underlying preferences for responding to price incentives, reflecting tastes rather than price beliefs or knowledge. Questions 4–7 capture behaviors related to monitoring and comparing grocery prices, such as checking shelf prices and reading grocery chain flyers, and are used to construct an index of *Price Learning Effort*. To summarize these two dimensions in a single measure, we additionally construct an aggregate index of subjective *Price Consciousness* based on questions 1–7. Following the broad use of the term in the marketing literature (see, e.g., Alford and Biswas, 2002; Loy, Ceynowa, and Kuhn, 2020), we interpret this aggregate as capturing a general tendency to care about prices.

Indices are constructed as weighted averages of the underlying Likert items. We obtain weights from a simple factor model, $\textit{Likert Response}_i = \tau_i + \lambda_i \textit{Factor} + \epsilon_i$, where i indexes the item, τ_i is an intercept, λ_i is the factor loading, ϵ_i is a residual, and *Factor* denotes either Price Sensitivity, Price Learning Effort, or Price Consciousness. Factor loadings are estimated using diagonally weighted least squares and reported in Table A3 in Appendix 3.A. Each item's weight is its factor loading divided by the sum

of loadings for the corresponding factor.[13]

Discount Utilization

The products in Varian (1980) are assumed to be homogeneous, making it straightforward to compare prices paid by different consumer groups. However, in grocery retailing, products and stores are differentiated, and consumers may pay lower prices not only due to price knowledge but also due to preferences for lower-priced stores or products. Our detailed data on purchasing history allow us to calculate an individual-level price measure that strips out variation stemming from differences in the overall price level of the store and products.

To test the prediction that more informed consumers pay lower prices on average, and to quantify the gains from price knowledge, we calculate *Discount Utilization* (DU): the percentage difference between (i) the average of the daily shelf prices in the consumer's home municipality m over an event window e of length E ($\bar{P}^{\text{Shelf}}_{jcme}$), and (ii) the average price paid by consumer i on day t for product j at chain c (P^{Paid}_{ijct}):

$$\text{DU}(E)_{ijct} = \frac{\bar{P}^{\text{Shelf}}_{jcme} - P^{\text{Paid}}_{ijct}}{\bar{P}^{\text{Shelf}}_{jcme}} \tag{3.5}$$

Thus, we interpret "discount" broadly as any price paid below the average, and not only low prices due to promotional sales. A positive value indicates that the consumer paid less on that day than what someone who times their purchases uniformly over the event window would expect to pay (i.e., buying on any day with equal probability).

Of course, the scope for paying lower prices depends on the event window. Since traditional sales apply to everyone who buys a product on the same day, the only way to obtain intraday bargains is to use personal coupons or a quantity or expiration-date discount. Conversely, over longer periods like 1–3 months, one can pay less than the average shelf price by timing purchases with sales. We have 18 months of purchasing history, but Equation 3.5 could capture unbalanced spending over time combined with general inflation, so we focus on a three-month event window ($E = Q$). This limits the effects of inflation and leaves ample room for timing sales.[14]

In order to calculate the overall tendency of a consumer to buy specific products at lower prices, we average Equation 3.5 over time:

$$\text{DU}(E)_{ijc} = \frac{1}{T_{ijc}} \sum_{t=1}^{T_{ijc}} \text{DU}(E)_{ijct} \tag{3.6}$$

where T_{ijc} is the number of times that an individual i is observed purchasing product

[13] Using simple unweighted averages yields very similar results.

[14] The results are similar using 1, 3, or 18 months ($E \in \{M, Q, FP\}$), but are strongly muted using daily discount utilization ($E = D$), as expected.

j at chain c in the data. This measures how much less an individual tends to pay for specific products than someone buying the same product but timing their purchases uniformly over the event windows e that the individual's purchases fall into.

To obtain an overall measure of a consumer's ability to identify and act on bargains, we further average Equation 3.6 across product $\times$ chain combinations:

$$\text{DU}(E)_i^{\text{Simple}} = \frac{1}{N_i} \sum_{n=1}^{N_i} \text{DU}(E)_{ijc} \tag{3.7}$$

where N_i is the number of product $\times$ chain combinations that we observe individual i purchasing in the data. This gives us a simple measure that quantifies an individual's tendency to exploit price variation over time for all the different products that they buy across the sample of more than 700 products, and not only the 18 products for which they were later asked to provide price recalls.

Note that we do not average over every transaction, which would overweight frequently purchased products. The reason is that one could identify a sale once and then act on it several times on consecutive days. We are primarily interested in the lower prices one can achieve because of greater price knowledge. However, we also calculate a sales-weighted version of Equation 3.7,[15] which allows us to calculate the percentage savings that the consumer makes by buying their current basket of groceries on sale relative to buying the same basket uniformly over time. Since sales weights are also sensitive to purchasing frequency, we focus on simple discount utilization (Equation 3.7) when testing and discussing the theoretical predictions of models such as Varian's (1980) model of sales concerning price knowledge and average prices paid, and we instead focus on sales-weighted discount utilization (Equation 3.8) when calculating the monetary gains associated with price knowledge.

Price Variation and Scope for Discount Utilization

The scope for buying products on sale critically depends on the sales activity and overall price variation of the products. To capture and control for this feature of grocery prices, we calculate the coefficient of variation for each product j $\times$ chain c $\times$ municipality m combination over the sample period:

$$CV_{jcm} = \frac{\sigma_{jcm}}{\mu_{jcm}} \tag{3.9}$$

where σ_{jcm} is the empirical standard deviation of price and μ_{jcm} is the product's mean

[15] Formally, we compute

$$\text{DU}(E)_i^{\text{Sales-Weighted}} = \sum_{n=1}^{N_i} \frac{S_{ijc}}{S_i} \text{DU}(E)_{ijc} \tag{3.8}$$

where S_{ijc} is the consumer's spending on each chain $\times$ product combination and S_i is the consumer's total spending, over the sample period.

price over the sample period. There is more room for discount utilization in products with high coefficients of variation, but the relation between knowledge of a product's price and its variation is less clear.

At the individual level, it is useful to consider the overall price variation in the consumer's shopping basket. Consumers who primarily buy products with little price variation over time are more limited in their opportunity to find bargains, and vice versa. We calculate the average price variation of the products that individual i buys in the following way:

$$CV_{i[\text{Simple}]} = \frac{1}{N_i} \sum_{n=1}^{N_i} CV_{jcm} \tag{3.10}$$

where N_i is the number of product $\times$ chain combinations that we observe individual i purchasing in the data.[16]

3.3 Price Knowledge and Discount Utilization

Before turning to regression analysis of the relation between price knowledge, discount utilization, and other variables, let us describe the variation in these variables. The empirical distribution of individual price knowledge, summarized by PK, is shown in the left panel of Figure 3.2. The mean PK is -0.27, indicating that, on average, respondents over- or underestimate the true shelf price by 27%. The interquartile range spans from -0.32 to -0.22. We find that 17% of recalled prices are within 5% of the true shelf price, and 50% are within 20%. The distribution shows no clear mass points that would suggest discrete consumer information types. Moreover, no respondent recalls prices perfectly—the maximum PK is -0.04.[17]

The distribution of discount utilization, computed as the simple average across products, is shown in the right panel of Figure 3.2. A value of zero implies behavior analogous to the uninformed consumers in Varian's (1980) model—those who time purchases randomly with respect to sales. Most consumers, however, exhibit positive discount utilization: the interquartile range is 0.01 to 0.06, and the mean is 0.04, indicating that consumers tend to concentrate purchases on days when prices are below average.

[16]Similarly, the price variation of the consumer's average shopping basket is given by:

$$CV_{i[\text{Sales-Weighted}]} = \sum_{n=1}^{N_i} \frac{S_{ijc}}{S_i} CV_{jcm} \tag{3.11}$$

where S_{ijc} is the consumer's spending on each chain $\times$ product combination and S_i is the consumer's total spending over the sample period.

[17]We also note that among all recalled prices included in the analysis, 31% were provided by respondents who had purchased the product at the relevant chain during the sample period. For these cases, the average time since purchase is 4.7 months, and the median is 2.7 months.

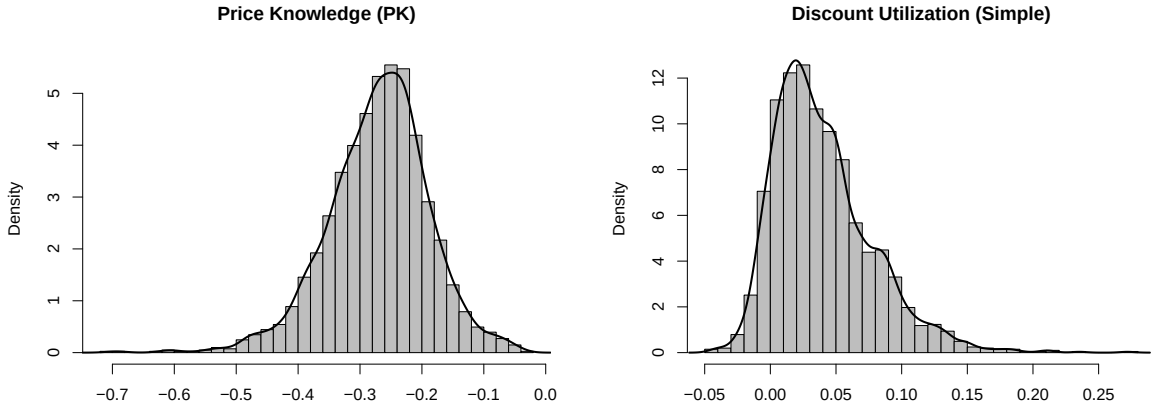


Figure 3.2: Histograms and density estimates of price knowledge and discount utilization. Histogram bin sizes are chosen using Scott's rule, and density bandwidths are chosen using the Sheather–Jones plug-in estimator.

The Spearman rank correlation between PK and DU is 0.31. To obtain a nonparametric estimate of the expected value of discount utilization conditional on price knowledge, we present a binned scatterplot in Figure 3.3.[18] The figure provides clear evidence that respondents with greater price knowledge purchase more on sale and pay lower prices on average, consistent with Varian's (1980) model of sales.

3.3.1 Linear Regression Analysis

To parameterize the relationship between discount utilization and price knowledge, and to control for confounding variables, we present OLS estimates of Equation 3.1 with different sets of controls in Table 3.1.[19]

[18]The implementation follows Cattaneo et al. (2025). Each point is the average discount utilization among the respondents with price knowledge within the corresponding bin. The bins are chosen to minimize the integrated mean squared error, and the 95% confidence bands are calculated using piecewise polynomials.

[19]We report heteroskedasticity-robust standard errors. Following Abadie et al. (2022), we do not cluster by municipality because the sampling design was not clustered.

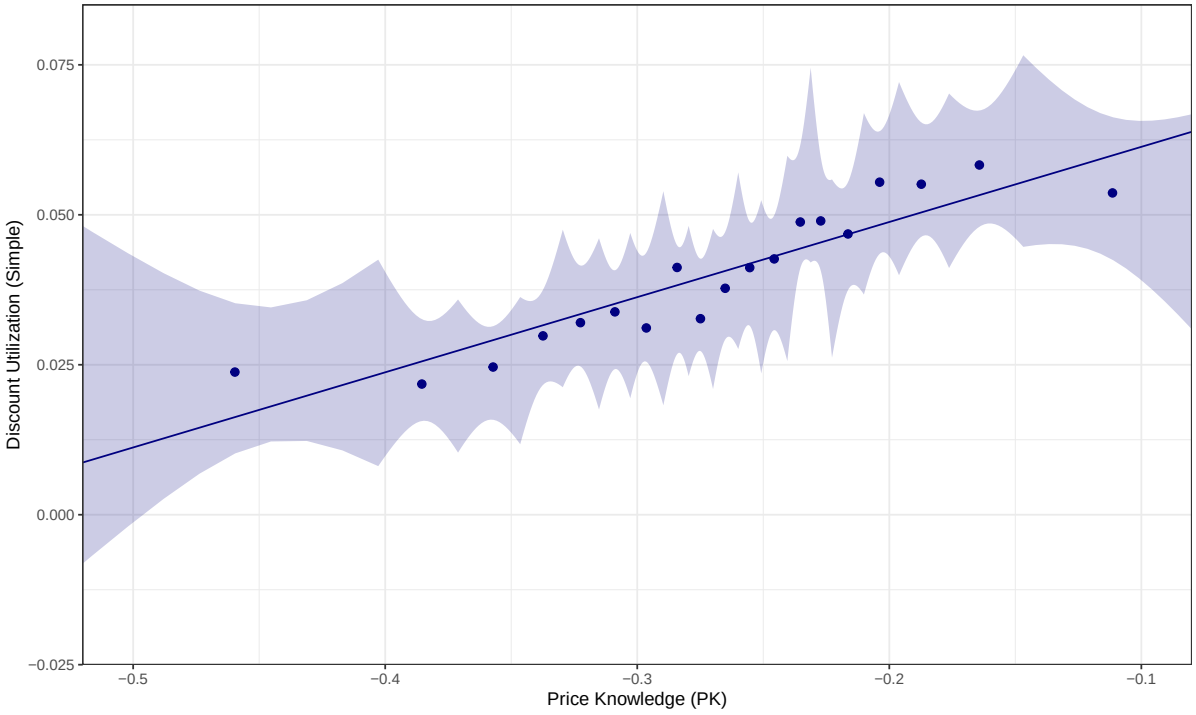


Figure 3.3: Estimate of the expected value of discount utilization conditional on price knowledge using a binned scatterplot with a linear fit. The number of bins is chosen to minimize the integrated mean squared error. Confidence bands at 95% are calculated using piecewise polynomials.

Table 3.1: Discount utilization regressions

	Discount Utilization		
	(1)	(2)	(3)
Price Knowledge	0.125*** (0.011)	0.121*** (0.010)	0.087*** (0.009)
Male		0.000 (0.002)	0.001 (0.002)
Single Person Household		0.004 (0.003)	0.004 (0.003)
Has Children		0.004* (0.002)	0.005** (0.002)
Full-Time Employment		-0.005* (0.002)	-0.003 (0.002)
Higher Education		0.001 (0.002)	0.002 (0.002)
Rural		-0.006*** (0.002)	-0.005** (0.002)
Household Income > 950′ NOK		-0.002 (0.002)	-0.001 (0.002)
Age: 40-59 Years		0.007*** (0.002)	0.007*** (0.002)
Age: Above 60 Years		0.008** (0.003)	0.009*** (0.002)
Price Sensitivity		0.007*** (0.001)	0.003* (0.001)
Price Learning Effort			0.008*** (0.001)
# Frequented Grocery Chains			0.000 (0.001)
Mainly Uses Discount Stores			-0.003 (0.002)
Average Weekly Store Visits			0.003*** (0.001)
Main Grocery Shopper			0.002 (0.002)
Average Weekly Variety (per 10 products)			-0.003*** (0.001)
3 Customer Loyalty Program Memberships			0.005*** (0.001)
Basket Coefficient of Variation			1.216*** (0.074)
Constant	0.074*** (0.003)	0.041*** (0.006)	-0.116*** (0.010)
Observations	2,028	1,935	1,928
Dependent variable mean	0.04	0.04	0.04
R^2	0.07	0.12	0.35

Notes: This table reports results from estimation of Equation 3.1. The dependent variable, individual-level discount utilization, is the percentage difference between the quarterly average of daily modal (shelf) prices and the average price paid on the purchase day, averaged across purchase days and then across product × chain combinations (simple weights). Price knowledge is calculated as the negative average percentage absolute deviation between the respondent's recalled prices and true shelf prices. The average weekly variety is calculated as the average number of unique products purchased per week over the sample period. Price sensitivity and learning effort are calculated as the factor-weighted averages of Likert-scale survey items 1–3 and 4–7, respectively. Basket CV is calculated as the coefficient of variation of shelf prices over the sample period, averaged across all products in the consumer's purchase history, using simple weights. Column 1 reports the simple regression on price knowledge (PK), Column 2 adds controls for demographics and price sensitivity, and Column 3 adds controls for shopping habits and price learning effort. The standard errors shown in parentheses are robust to heteroskedasticity (HC1). *** $p < 0.001$; ** $p < 0.01$; * $p < 0.05$.

The regression of discount utilization on price knowledge, presented in Column 1, shows that a 10 percentage-point increase in price knowledge is associated with a 1.3 percentage-point increase in discount utilization. This change in price knowledge is well within the variation in the data; the average PK in the sample is -27%, the standard deviation is 8%, and 50% of respondents have a PK between -32% and -22%. An individual with high price knowledge (e.g., $PK_i = -10\%$) pays, on average, 4% less for groceries than someone with low price knowledge (e.g., $PK_i = -40\%$).

The regression constant is also of theoretical interest. Assuming a linear extrapolation to $PK_i = 0$ is reasonable, it yields the expected discount utilization of a perfectly informed consumer. At 7.4%, the discount utilization of a perfectly informed consumer exceeds both the sample average of 4% and the reference level associated with "uninformed" consumers in Varian's (1980) model, which is 0%.

Some consumers may have a greater propensity to exploit temporary sales, independent of price knowledge. If consumers with higher conditional price sensitivity are also more inclined to acquire price information, this would confound the relationship between price knowledge and discount utilization—more price-sensitive consumers might have paid lower prices even in the absence of superior price knowledge. To address this concern, Column 2 in Table 3.1 adds controls for demographics and the price sensitivity index, which proxy for stable traits and preferences that may jointly affect price knowledge and discount utilization. Reassuringly, the coefficient on PK is not materially altered, suggesting that price knowledge is an independent prerequisite for identifying and acting on temporary sales.

In terms of explanatory power, the R^2 increases modestly by 5 percentage points as we move from Column 1 to Column 2, and the coefficients generally have the expected signs: individuals with children and those without full-time employment pay lower prices, as do older respondents and those living in urban areas. The price sensitivity index ranges from 1 to 5, and a one-unit increase (equal to the interquartile range) is associated with a 0.7% reduction in prices paid. Income, conditional on price sensitivity and other demographics, shows no ceteris paribus effect on discount utilization.

The price sensitivity index is constructed to capture consumers' underlying preference for responding to price incentives. However, the index relies on subjective interpretations of three survey questions and may only imperfectly capture the individual's preferences. Even if the price sensitivity index is the best single proxy at our disposal, a broader set of control variables may better capture the underlying trait.

Column 3 adds variables related to consumers' shopping habits and price learning effort, further increasing R^2 by 23 percentage points. Frequent store visits are associated with higher discount utilization, consistent with having more opportunities

to act on sales. Buying many different products per week is associated with lower discount utilization, perhaps because regularly purchasing a wide variety of products implies that some are bought outside of sales.[20] Membership in all major customer loyalty programs is associated with paying 0.5% lower prices. A substantial share of the increase in R^2 is driven by the inclusion of the average coefficient of variation of the products purchased by the respondent: a 0.01 standard deviation increase in basket price variation is associated with 1.2% lower prices paid. This reflects that the scope for finding bargains depends on price variation over time.[21] Finally, a one-unit increase in the price learning effort index is associated with 0.8% lower prices paid, while price sensitivity continues to independently predict higher discount utilization.

The ceteris paribus association between price knowledge and discount utilization is now somewhat muted but remains economically and statistically significant. One interpretation is that we have further mitigated omitted variable bias by better capturing underlying preferences for exploiting temporary sales. For example, shopping frequency, loyalty program membership, and engagement in price information gathering may all reflect a general interest in exploiting temporary sales that the price sensitivity index does not capture. On the other hand, as we now rely on variation in price knowledge unrelated to learning effort—for instance, variation driven by attention, memory, or environmental cues and distractions—we also absorb what we consider valid variation in price knowledge. As this may affect the precision of our estimates, we view the estimates in Columns 1 and 2 as our primary results.[22]

Taken together, Columns 2–3 suggest that the association between price knowledge and discount utilization is not driven by underlying tastes for shopping on sale, and is therefore consistent with a causal effect of price knowledge on discount utilization and price responsiveness.

[20] If we instead count the number of different products purchased over the entire sample period, the sign flips. A consumer might buy many different products over time precisely because they take advantage of discounts when they arise.

[21] The coefficient on PK is minimally affected by the inclusion of Basket CV, despite its substantial contribution to R^2. More generally, to the extent that variables affecting the scope for discount utilization (e.g., basket price variation, shopping frequency, and product variety) are driven by underlying preferences unrelated to price, we do not expect their inclusion to alter the ceteris paribus relationship between PK and DU.

[22] As discussed in Section 3.2.2, one potential concern is that discount utilization may itself increase price knowledge, and that individuals may know the prices of the surveyed products simply because they recently purchased them at a discounted price. In Appendix 3.B.3, we therefore report results from estimating Equation 3.1 using a measure of discount utilization constructed only from products not included in the survey. The association between price knowledge and discount utilization remains robust and highly statistically significant when the survey products are excluded. If anything, the estimated coefficient on price knowledge is slightly larger in this specification than in the baseline results.

3.3.2 Effect Heterogeneity

We now consider several extensions of the main specification, presented in Table 3.2, to examine heterogeneity across products and respondents and thereby assess the conditions under which price knowledge yields the highest returns. Column 1 reproduces Column 1 of Table 3.1 for ease of comparison.

Table 3.2: Discount utilization regressions: effect heterogeneity

Discount Utilization:	Simple Weights			Perishable	Non-Perishable	Sales Weights
	(1)	(2)	(3)	(4)	(5)	(6)
PK	0.125***	0.087***	-0.003	0.117***	0.133***	0.070***
	(0.011)	(0.010)	(0.044)	(0.012)	(0.013)	(0.010)
High Basket CV		0.041***				
		(0.006)				
PK × High Basket CV		0.053**				
		(0.020)				
Price Consciousness			0.021***			
			(0.004)			
PK × Price Consciousness			0.028*			
			(0.012)			
Constant	0.074***	0.050***	-0.011	0.067***	0.081***	0.040***
	(0.003)	(0.003)	(0.014)	(0.003)	(0.004)	(0.003)
Observations	2,028	2,028	1,961	2,028	2,028	2,028
Dependent variable mean	0.04	0.04	0.04	0.04	0.05	0.02
R^2	0.07	0.20	0.14	0.05	0.06	0.04

Notes: This table reports discount utilization regressions examining heterogeneity in the relationship between price knowledge and discount utilization across products and respondents. The dependent variable, individual-level discount utilization, is the percentage difference between the quarterly average of daily modal (shelf) prices and the average price paid on the purchase day, averaged across purchase days and then across product × chain combinations. Price knowledge (PK) is calculated as the negative average percentage absolute deviation between the respondent's recalled prices and true shelf prices. Price consciousness is calculated as the factor-weighted averages of Likert-scale survey items 1–7. High basket CV is an indicator for above-median basket price variation, calculated as the coefficient of variation of shelf prices over the sample period, averaged across all products in the consumer's purchase history, using simple weights. Column 1 reports the baseline regression of DU, calculated using simple weights, on PK. Column 2 interacts PK with an indicator for above-median basket price variation. Column 3 interacts PK with the price consciousness index. Columns 4–5 calculates discount utilization separately for perishable and non-perishable products, respectively. Column 6 calculates discount utilization using sales weights. The standard errors shown in parentheses are robust to heteroskedasticity (HC1). *** $p < 0.001$; ** $p < 0.01$; * $p < 0.05$.

Basket Price Variation If price knowledge drives discount utilization, its impact should be most pronounced for consumers purchasing products with high price

volatility—precisely where the opportunity for identifying bargains is greatest. Column 2 of Table 3.2 presents the results of interacting price knowledge with an indicator for a shopping basket with above-median price variation. The interaction-term coefficient is positive and statistically significant, confirming that the link between price knowledge and discount utilization is stronger for individuals with higher basket volatility—a 10 percentage-point increase in price knowledge is associated with 0.9 and 1.4 percentage-point increases in discount utilization for the low- and high-volatility consumers, respectively.

This finding aligns with the fundamental prediction of Stigler (1961): that the returns to search are increasing in price dispersion. Interpreting dispersion here as variation over time, this result confirms that the marginal value of acquiring price knowledge is greatest when the underlying distribution of prices is widest.[23]

Price Consciousness While price knowledge provides the ability to identify bargains, acting on this information requires motivation. We therefore examine whether the link between price knowledge and discount utilization is stronger for consumers who are fundamentally more concerned with prices.

For this analysis, we utilize the aggregate *Price Consciousness* index constructed in Section 3.2.3. While our previous specifications used distinct components (such as price sensitivity and learning effort), here we employ the composite index drawing on a broader set of survey items to capture a consumer's general tendency to care about prices. Using a single proxy allows for a tractable and easily interpretable interaction model that tests whether price-conscious consumers benefit more from being informed.

Column 3 of Table 3.2 presents the regression estimates including the interaction between price knowledge and price consciousness.[24] We find that the interaction term is positive and statistically significant at the 5% level, confirming that the effect of price knowledge on discount utilization is increasing in price consciousness. Our interpretation is that knowledge translates into lower prices paid more effectively when the consumer is highly motivated to use it.

To quantify this heterogeneity, we compute the marginal effect of price knowledge at different levels of price consciousness, reported in Table 3.3. The results show that the return to price knowledge is substantially higher for highly price-conscious consumers. For a consumer at the first quartile of price consciousness, a 10 percentage-point increase in price knowledge is associated with a 0.92 percentage-point increase in discount utilization. For a consumer at the third quartile, approximately

[23]The expected minimum price paid for a product after sampling n sellers when prices are uniformly distributed on $p \in [0, b]$ is equal to $E[p^{\text{Min}}] = \frac{b}{n+1}$ (see Stigler, 1961, p. 215). The gain from more search n is then $\frac{\partial |E[p^{\text{Min}}]|}{\partial n} = \frac{b}{(n+1)^2}$, which is increasing in price dispersion b: $\frac{\partial^2 |E[p^{\text{Min}}]|}{\partial n \partial b} = \frac{1}{(n+1)^2} > 0$.

[24]The estimating equation is $DU_i = \alpha + \delta \cdot PK_i + \beta_1 \cdot PC_i + \beta_2 \cdot PK_i \cdot PC_i + \epsilon_i$

one unit higher on the price consciousness index, this effect rises to 1.21 percentage points—corresponding to an increase in efficacy of roughly 30%.[25]

Table 3.3: Marginal effects of PK on DU conditional on price consciousness

	Min	1st Quartile	Median	Mean	3rd Quartile	Max
Price Consciousness	1	3.36	3.91	3.81	4.38	5
Effect of Price Knowledge	0.025	0.092^{***}	0.107^{***}	0.105^{***}	0.121^{***}	0.138^{***}
	(0.032)	(0.012)	(0.012)	(0.012)	(0.015)	(0.020)

Notes: This table reports average and quantile marginal effects of price knowledge on discount utilization. The standard errors shown in parentheses are robust to heteroskedasticity (HC1) and calculated using the delta method. *** $p < 0.001$; ** $p < 0.01$; * $p < 0.05$.

These findings have important implications for policy. First, they suggest that policies aimed at improving price availability and knowledge may have progressive distributional effects. In our sample, high price consciousness is negatively correlated with income and positively correlated with being young, being a woman, and having children. To the extent that disadvantaged groups are more price conscious, population-wide gains in price knowledge (e.g., through clearer labeling or price comparison tools) may disproportionately benefit those who need the savings the most.

Second, the dependence of the effect on consciousness suggests that information provision alone has limits. For consumers with low price consciousness, simply making prices easier to learn may not lead to behavioral changes, as they lack the intrinsic motivation to act on the signal. This implies that in addition to transparency measures that lower search costs, attitudinal campaigns that encourage consumers to act on price differentials and exert their consumer power may help facilitate effective competition.

Perishable vs. Non-Perishable Products Discount utilization may naturally vary with product characteristics. Our analysis of discount utilization includes consumers' purchases across 709 products that differ substantially in storability. As emphasized by Hendel and Nevo (2013), intertemporal price discrimination in storable goods markets can arise because firms use temporary sales to price discriminate between consumers who stockpile and those who do not. Their analysis focuses on highly storable goods that can be easily stored for long periods.

To examine whether product storability matters in our setting, we divide all products into two groups: perishable items such as milk, sausages, minced meat,

[25]The marginal effect is $\frac{\partial DU_i}{\partial PK_i} = \delta + \beta_2 \cdot PC_i$, with standard error $SE = \sqrt{\text{Var}(\delta) + PC_i^2 \cdot \text{Var}(\beta_2) + 2PC_i \cdot \text{Cov}(\delta, \beta_2)}$, and t-statistic $t = \frac{\delta + \beta_2 \cdot PC_i}{SE}$. The only non-significant marginal effect occurs at the minimum value of the price consciousness index (PC = 1). However, this estimate relies on sparse data, as only *one* respondent in our sample reports a price consciousness score of one, and fewer than 40 respondents (<2%) report a score between 1 and 2.

vegetables, and fresh smoothies that typically must be consumed within a month, and non-perishable items such as soda, crispbread, jam, and frozen fish that can be stored for much longer. This classification results in 301 perishable products (42%) and 408 non-perishable products (58%).

Columns 4 and 5 of Table 3.2 present the baseline estimates for perishable and non-perishable products, respectively. We find, as anticipated, that the effect of price knowledge is somewhat stronger for non-perishable products, with an increase of about 0.8 percentage points relative to the aggregate effect reported in Column 1. For perishable products, the estimated effect is 0.8 percentage points lower but remains highly statistically significant.

We report the full specifications including control variables in Table A4 in Appendix 3.B.1, along with histograms and binned scatterplots for both groups. Neither the histograms nor the scatterplots differ qualitatively across the two categories. Furthermore, the control variables in the full specification (mirroring Column 3 in Table 3.1) retain their expected signs and significance levels. Taken together, these results suggest that while storability amplifies the effect, the relationship between price knowledge and discount utilization extends beyond storable goods.

Price Knowledge Quintiles To assess the functional form of the relationship between price knowledge and discount utilization, and to examine the empirical relevance of the binary distinction between "informed" and "uninformed" consumers often used in search-theoretic models (e.g., Varian, 1980), we re-estimate the model using price knowledge quintile indicators. The results are reported in Table A8 in Appendix 3.B.4.

The results show a clear monotonic positive relationship between price knowledge and discount utilization: discount utilization increases steadily from one price knowledge quintile to the next. This suggests that the relationship is indeed more gradual than a simple binary distinction between informed and uninformed consumers would imply, and supports the linear specification used in our main analysis as a reasonable and robust approximation. Another insight from this exercise is that we also confirm that our main results are not driven by potential large outliers.

3.3.3 The Gains Associated with Price Knowledge

We now turn to a quantification of the direct gains associated with price knowledge. We take current prices and quantities as given and thereby disregard additional equilibrium effects resulting from the fact that a greater share of informed individuals would be expected to strengthen competition.

To do this, we rely on the sales-weighted version of discount utilization (see Equation 3.8), which allows us to quantify the savings a consumer is expected to make by buying their current basket of goods at more opportune times. The simple

regression results for this specification are shown in Column 6 of Table 3.2. The estimated coefficient on price knowledge is smaller in magnitude than the coefficient in the simple-average specification by 5.5 percentage points. This difference likely arises because consumers tend to spend more on products with lower price variation, reducing the scope for finding bargains in a sales-weighted basket.[26]

We report the full results, including control variables, in Table A5 in Appendix 3.B.2. The results are qualitatively similar to Column 3 in Table 3.1, except that primarily shopping at discount stores now exhibits a detectable negative effect. The coefficients are generally smaller in magnitude across covariates.

If we consider a nuclear family consisting of a mother (31–60 years), father (31–60 years), daughter (10–13 years), and son (2–5 years), their reference budget for groceries in 2024 according to SIFO (2024) amounts to 173,640 NOK per year. With fixed quantities, a 30 percentage-point increase in price knowledge—a substantial but feasible change given its empirical distribution (see Figure 3.2)—is associated with yearly household savings of 3629 NOK (2.1%).

It is also instructive to consider the savings that a perfectly informed consumer is expected to make relative to a consumer who makes their purchases randomly throughout the quarter, akin to the uninformed consumers in Varian (1980). Assuming that a linear extrapolation of the relation between price knowledge and discount utilization to the point of $PK = 0$ is reasonable, the discount utilization of a perfectly informed consumer is given by the regression constant in Column 6 in Table 3.2, which is 4%. This suggests that a perfectly informed consumer shopping on behalf of the nuclear family pays 4% less for its groceries than someone who does not time their purchases with temporary price discounts, amounting to yearly household savings of 6925 NOK.

The potential household-level savings related to improvements in price knowledge are quantitatively noteworthy but modest. However, from a policy perspective, improvements in price knowledge add up across households and over time. Low-cost initiatives that persistently improve the price knowledge of the broader population could lead to large welfare gains. Furthermore, a core search-theoretic prediction—whose foundations we argue this chapter provides empirical support for—is that an increase in the share of informed consumers leads to lower equilibrium prices. The direct savings from better price knowledge that we calculate here should therefore be seen as a lower bound on the total savings to consumers.[27]

Finally, it is important to note that these potential savings are not uniformly

[26]Table A1 in Appendix 3.A.2 shows that the average basket coefficient of variation is lower when we weigh products according to spending as opposed to using a simple average.

[27]Most people in our sample buy from one of the leading discount chains in Norway, which has temporary sales less often than other grocery chains. The gains to price knowledge can only be expected to be larger for the general population, some of whom buy groceries from chains with more sales activity, further supporting the lower bound interpretation.

distributed; they depend heavily on the price volatility of the consumer's typical basket. When we re-estimate the interaction model using sales-weighted discount utilization (see Table A6 in Appendix 3.B.2), we find substantial heterogeneity in the economic value of being informed. For consumers with below-median basket price variation, a 30 percentage-point increase in price knowledge is associated with yearly household savings of 2396 NOK (1.4%). However, for consumers with above-median variation, the savings associated with the same increase in knowledge nearly double to 4428 NOK (2.6%). The gap is even wider when comparing a perfectly informed consumer to one who disregards sales entirely: the theoretical savings potential rises from 4167 NOK (2.4%) for low-variance baskets to 9203 NOK (5.3%) for high-variance baskets. This underscores that the monetary gains from price knowledge are highest for households that consume products subject to frequent or deep price fluctuations.

3.4 Drivers of Price Knowledge

We now turn to an examination of the drivers of price knowledge itself. The level of consumer price knowledge of groceries has been documented in many studies (see, e.g., Gabor and Granger, 1961; Conover, 1986; Vanhuele and Drèze, 2002). However, previous studies (e.g., Aalto-Setälä and Raijas, 2003; Loy, Ceynowa, and Kuhn, 2020) have focused on how price recall relates to measures that can be obtained from consumer surveys, such as demographics, reported shopping habits, and subjective price consciousness, or that are easily observed in stores by researchers, such as current product prices and their dispersion. We extend this literature by relating price knowledge to variables that are derived from transaction data. These include historical purchasing frequencies and spending on specific products, the number of store and chain visits, and other features of the respondents' shopping baskets, such as product variety.

Results from estimating Equation 3.2 with OLS are shown in Table 3.4. Standard errors are clustered at the individual level (2028 clusters) to account for within-respondent dependence across price recalls. This choice reflects the sampling design: respondents are the sampled units, while products and chains are fixed by design. Accordingly, inference is directed at the consumer population rather than a broader population of products or chains. Column 1 includes only variables derived from the survey, while Column 2 adds variables constructed from the transaction data.

The transaction-derived measures contribute substantially to explaining price knowledge ($R^2 = 8\%$ vs. $R^2 = 1\%$), underscoring the importance of revealed shopping behavior for understanding price knowledge. We therefore focus the following discussion on the results in Column 2, which represents our preferred specification.

Table 3.4: Price knowledge regressions

	Price Knowledge	
	(1)	(2)
Male	-0.015*** (0.004)	-0.015*** (0.004)
Single Person Household	-0.002 (0.007)	-0.001 (0.007)
Has Children	0.003 (0.004)	-0.001 (0.004)
Full-Time Employment	-0.005 (0.005)	-0.004 (0.004)
Higher Education	-0.013** (0.004)	-0.012** (0.004)
Rural	0.000 (0.004)	-0.002 (0.004)
Household Income > 950' NOK	0.010* (0.004)	0.008* (0.004)
Age: 40-59 Years	0.008 (0.005)	0.006 (0.004)
Age: Above 60 Years	0.004 (0.006)	0.002 (0.006)
# Frequented Grocery Chains	0.002 (0.001)	0.002 (0.001)
Mainly uses discount stores	0.006 (0.006)	0.004 (0.006)
Main Grocery Shopper	0.013** (0.004)	0.014** (0.004)
Supermarket Chain	-0.017*** (0.001)	0.033*** (0.002)
Private Label	0.044*** (0.004)	0.018*** (0.004)
3 Customer Loyalty Program Memberships	0.011** (0.004)	0.010** (0.004)
Price Sensitivity	0.001 (0.003)	0.000 (0.003)
Price Learning Effort	0.016*** (0.002)	0.015*** (0.002)
Average Weekly Chain Visits		0.006*** (0.001)
Average Weekly Chain Variety (per 10 products)		-0.010*** (0.002)
Category Size (per 10 products)		-0.029*** (0.001)
Price Coefficient of Variation		-0.120*** (0.019)
Purchased		0.068*** (0.003)
Purchased $\times$ Sales in 1000 NOK		0.043*** (0.003)
Purchased $\times$ Months Since Last Purchase		-0.001** (0.000)
Constant	-0.344*** (0.013)	-0.308*** (0.014)
Observations	69,408	69,408
Dependent variable mean	-0.27	-0.27
R^2	0.01	0.08

Notes: This table reports results from estimation of Equation 3.2. Price knowledge is calculated as the negative percentage absolute deviation between the respondents' recalled price and the true shelf price. Price sensitivity and learning effort are calculated as the factor-weighted averages of Likert-scale survey items 1–3 and 4–7, respectively. The average weekly chain variety is calculated as the average number of unique products purchased at each chain per week over the sample period. Category size is the chain-specific number of close substitutes for the product as defined by the grocery chain. The price coefficient of variation is calculated for each product $\times$ chain combination over the sample period. Purchased is an indicator of whether the respondent bought the product at a given chain during the sample period, and sales is the consumer's total product–chain spending. Column 1 includes individual characteristics and indicators for private labels and supermarket chain, and Column 2 adds variables derived from transaction data. The standard errors shown in parentheses are clustered by individual. *** $p < 0.001$; ** $p < 0.01$; * $p < 0.05$.

Demographics We see that men have poorer price knowledge than women. This replicates the findings of Loy, Ceynowa, and Kuhn (2020) and Estelami, Lehmann, and Holden (2001), as opposed to Estelami (1998); Jensen and Grunert (2014); Kenning, Hartleb, and Schneider (2011) and Wakefield and Inman (1993), who find no difference between men and women. We find no statistically precise relation between price knowledge and being single, having children, being in full-time employment, living in a rural area, or age. Consumers with higher education are less knowledgeable about grocery prices, but even though we might expect higher income to reduce price consciousness (Estelami, Lehmann, and Holden, 2001; Estelami and Lehmann, 2001; Wakefield and Inman, 1993; Loy, Ceynowa, and Kuhn, 2020), high income is, in our sample, associated with slightly more price knowledge, which is also reflected in the raw correlation. Small and sometimes counterintuitive effects of age, income, and employment on price knowledge are not new in the literature. Gaston-Breton and Raghubir (2012) point out that these sociodemographic variables are all correlated with both the ability and motivation to memorize prices in ways that can cancel out, leaving their relation to price knowledge ambiguous. For example, older consumers might have more time and incentives to think about grocery prices but also poorer memory. Similarly, high income or education can be correlated with both good memory and lower motivation to focus on grocery prices.

Consumer statements about shopping habits Price knowledge does not appear to be related, all else equal, to the number of different grocery chains respondents regularly visit or whether they primarily shop at discount stores. However, direct and indirect markers of price consciousness do independently predict price knowledge; for example, being the sole and primary grocery shopper of the household (as opposed to sharing or not having the main responsibility), or being a member of all three main customer loyalty programs (as opposed to only one or two), are both associated with higher price knowledge.[28]

Gathering information about grocery prices is strongly related to price knowledge: a one-point increase in the price learning effort index (see questions 4–7 in Table A3 in Appendix 3.A), which is less than the interquartile range, has a stronger association with price knowledge than either primary shopping responsibility or loyalty program membership. Price sensitivity, represented by questions 1–3 in Table A3 in Appendix 3.A, is almost completely unrelated to price knowledge, conditional on price learning effort.

The finding that price learning effort has an independent effect on price knowledge suggests that improving access to price information could be an effective way to raise

[28]Note that these three programs cover more than 95% of the Norwegian grocery market. The primary tangible benefit of being a member of all three main customer loyalty programs is access to personal price discounts, signaling price consciousness, consistent with the estimated positive effect.

consumer awareness. If knowledge were primarily driven by price sensitivity, shopping patterns, or demographics, such interventions would likely have limited effect. In the Norwegian grocery market, where there is currently no formal price transparency regulation or comprehensive comparison tools, collecting price information remains relatively costly for consumers. Policies that reduce these costs could both encourage more consumers to seek out prices and increase the overall level of price knowledge in the market.

Transaction data Turning now to the variables derived from the transaction data whose estimates are reported in Column 2, we find that more frequent visits to the stores of the relevant chain are related to better price recall. This is consistent with what Loy, Ceynowa, and Kuhn (2020) find using self-reported shopping frequencies. The weekly average number of unique products purchased at each chain, a variable that, to our knowledge, has not been used in the literature before, is negatively associated with price knowledge. One explanation for the negative association between basket variety and price knowledge could be that it is difficult to keep track of the prices of any given product if one tends to buy a large variety of products.

Price knowledge is decreasing in category size, defined by the number of close substitutes according to the grocery chain: Increasing category size by 10 units is associated with 3 percentage-point lower price recall accuracy in our sample. The findings are consistent with Vanhuele and Drèze (2002), who refer to the number of SKUs in a category as "clutter" and argue that increased category complexity increases the risk of confusion and poor recall.

Similarly, more price variation over time makes it harder to stay informed and predicts poorer price recall. An interquartile-range increase in the price coefficient of variation (0.09) is associated with 1.1 percentage-point lower price recall accuracy. Loy, Ceynowa, and Kuhn (2020) and Vanhuele and Drèze (2002) find a related result that more price variation within the product's category is associated with poorer recall. However, our long purchasing histories allow us to see how variation over time in the price of the product itself affects learning and recall, as opposed to the cross-sectional price variation within the product's category.

Spending 1000 NOK more on the product over the last 18 months is associated with 4.3 percentage points more accurate price recall. This is conditional on having ever bought the product and how long ago that was, so the effect of spending on price recall accuracy is not driven by familiarity with the product, but rather how important it is for the consumer to keep track of its price.[29] In other words, consumers make sure to stay more informed about products for which the potential savings are the greatest. This is in line with the predictions of Stigler (1961) concerning the incentives

[29]Replacing time since last purchase with purchasing frequency produces similar results.

for search, i.e., price knowledge acquisition.

The respondents are much more informed about products that they have experience buying: Recall accuracy is 7 percentage points higher on average for products that the consumer has purchased before. What is somewhat surprising is that this effect seems to be very persistent. Even after 12 months, the effect associated with buying the product has only diminished by approximately $12 \times (-0.001) = -1.2$ percentage points. It seems that once a consumer buys a product, they continue to stay informed about the price of that product even if they buy it infrequently.

The marketing literature has long emphasized widespread consumer inattention and limited price knowledge but has provided less evidence on how price knowledge relates to objectively measured purchasing experiences. By combining survey and transaction data, we show that price knowledge increases with the frequency of store visits, recent purchase experience, and spending on the product in question, and decreases with basket variety and infrequent shopping—suggesting that consumers acquire price knowledge through repeated exposure and focused attention. In contrast, demographics and survey-based indicators are comparatively less important in explaining price knowledge—the R^2 increases from 1% to 8% when transaction-derived variables are added. This mirrors research in marketing showing that observed purchasing behavior is substantially more predictive than individual characteristics for price knowledge (Gupta and Chintagunta, 1994; Islam et al., 2022).

3.5 Concluding Comments

This chapter provides the first direct evidence that consumers with better knowledge of grocery prices systematically pay lower prices. Using a unique dataset that links survey-based price recall measures to detailed, individual-level grocery purchase histories, we quantify the relationship between price knowledge and discount utilization. We show that consumers with more accurate knowledge of prices pay less, not by purchasing generally cheaper products or frequenting cheaper stores, but by timing their purchases to take advantage of price variation over time. Our results also show that the benefits of price knowledge are economically meaningful: a 30 percentage-point improvement in price recall is associated with an average price reduction of about 4 percent.

Our findings provide direct empirical support for a central mechanism in search-theoretic models of sales, such as Varian (1980), in which informed consumers impose a competitive constraint on firms by responding strategically to price variation. By demonstrating that price knowledge is associated with paying lower prices—also after controlling for potential confounders—we provide micro-level support for a key theoretical assumption that underlies a broad class of models in industrial organization

and marketing.

Our finding that informed consumers systematically pay lower prices implies that policies capable of increasing price knowledge can help intensify competition. Furthermore, the strong association we observe between price learning effort and recall accuracy demonstrates that price knowledge is not solely determined by fixed traits, but is responsive to information-gathering activities. Taken together, our results support the idea that price disclosure regulations and other policies that lower search costs (by making it easier for consumers to learn prices) have the potential to effectively raise price knowledge and improve market efficiency.

By linking survey-based recall data with transaction records, we provide a more detailed analysis of the determinants of price knowledge than survey data alone can offer. We find substantial heterogeneity between consumers, with greater price knowledge among those who visit stores frequently, have recently purchased the product, and spend more on it. In contrast, varied or infrequent shopping, larger product categories, and greater price variation are associated with lower knowledge. While attentive shoppers are more likely to benefit from temporary discounts, our findings suggest that targeted pricing and communication strategies aimed at less engaged consumers may offer untapped potential for firms.

Appendix

3.A Data Cleaning and Descriptive Statistics

3.A.1 Data Cleaning

A few price recalls are so extreme that they are likely erroneous entries or, at least, not based on a serious attempt at recalling the right price. We filter out all individuals who make a price recall that lies outside the 10–300 percent range of the true shelf price. Furthermore, we exclude respondents who do not make at least 18 out of the 36 possible product recalls on account of likely lack of focus and effort. This leaves us with 2028 respondents, 2 chains, 24 products, and 72,392 observations, a loss of 11% of respondents and 7% of recalled prices.[30]

3.A.2 Descriptive Statistics

The primary outcome and explanatory variables that we use in the individual-level analysis are summarized in Table A1, and the individual $\times$ chain $\times$ product-level, or recall-level, variables are summarized in Table A2. We include the mean, standard deviation, minimum, maximum, and quartile values. The variables that overlap between the tables have identical means, largely because most respondents answer all 18 price questions, but the recall-level variables have different quantiles and higher standard deviations.

[30]Fewer observations end up being used in the regression analysis due to missing values for some of the survey variables. Finally, we also checked for respondents with very few transactions in their purchasing histories, but removing these respondents does not meaningfully alter the results.

Table A1: Summary statistics: individual-level variables

	Mean	Standard Deviation	Minimum	1st Quartile	Median	3rd Quartile	Maximum
Discount Utilization (Simple)	0.04	0.04	-0.04	0.01	0.03	0.06	0.27
Discount Utilization (Sales-Weighted)	0.02	0.03	-0.07	0.00	0.01	0.03	0.22
Price Knowledge (PK)	-0.27	0.08	-0.70	-0.32	-0.27	-0.22	-0.04
Price Bias (PD)	0.02	0.13	-0.44	-0.06	0.01	0.09	0.60
Share Over-estimated	0.46	0.15	0	0.36	0.44	0.56	0.92
Share Under-estimated	0.53	0.15	0.06	0.44	0.53	0.64	1
Share Correct	0.01	0.04	0	0	0	0	0.53
Share Within 1%	0.06	0.07	0	0.03	0.06	0.08	0.56
Share Within 5%	0.17	0.11	0	0.08	0.14	0.22	0.78
Share Within 10%	0.29	0.13	0	0.19	0.28	0.36	0.89
Share Within 20%	0.50	0.15	0.06	0.40	0.50	0.58	1
Male	0.38	0.49	0	0	0	1	1
Single Person Household	0.12	0.33	0	0	0	0	1
Has Children	0.42	0.49	0	0	0	1	1
Full-Time Employment	0.61	0.49	0	0	1	1	1
Higher Education	0.71	0.45	0	0	1	1	1
Rural	0.30	0.46	0	0	0	1	1
Household Income >950′ NOK	0.54	0.50	0	0	1	1	1
Age: Below 40 Years	0.23	0.42	0	0	0	0	1
Age: 40-59 Years	0.47	0.50	0	0	0	1	1
Age: Above 60 Years	0.30	0.46	0	0	0	1	1
# Frequented Grocery Chains	2.98	1.39	1	2	3	4	9
Mainly uses discount stores	0.86	0.35	0	1	1	1	1
Average Weekly Store Visits	2.90	1.80	0.09	1.69	2.58	3.65	16.53
Main Grocery Shopper	0.74	0.44	0	0	1	1	1
Average Weekly Variety	27.98	13.23	1.39	18.80	26.15	35.09	102.72
3 Customer Loyalty Program Memberships	0.48	0.50	0	0	0	1	1
Price Consciousness	3.81	0.75	1	3.36	3.91	4.38	5
Price Sensitivity	4.12	0.79	1	3.74	4.26	4.74	5
Price Learning Effort	3.52	0.89	1	2.95	3.62	4.22	5
Basket Coefficient of Variation (Simple)	0.11	0.01	0.07	0.10	0.11	0.12	0.17
Basket Coefficient of Variation (Sales-Weighted)	0.10	0.02	0.04	0.08	0.09	0.11	0.16

Notes: This table reports summary statistics for individual-level variables for the 2,028 respondents used in the analysis.

Table A2: Summary statistics: individual × chain × product-level variables

	Mean	Standard Deviation	Minimum	1st Quartile	Median	3rd Quartile	Maximum
Price Knowledge (PK)	-0.27	0.26	-1.99	-0.38	-0.20	-0.08	0
Price Bias (PD)	0.02	0.37	-0.90	-0.22	-0.03	0.17	1.99
Share Over-estimated	0.46	0.50	0	0	0	1	1
Share Under-estimated	0.53	0.50	0	0	1	1	1
Share Correct	0.01	0.11	0	0	0	0	1
Share Within 1%	0.06	0.24	0	0	0	0	1
Share Within 5%	0.17	0.37	0	0	0	0	1
Share Within 10%	0.29	0.45	0	0	0	1	1
Share Within 20%	0.50	0.50	0	0	1	1	1
Male	0.38	0.49	0	0	0	1	1
Single Person Household	0.12	0.33	0	0	0	0	1
Has Children	0.42	0.49	0	0	0	1	1
Full-Time Employment	0.61	0.49	0	0	1	1	1
Higher Education	0.71	0.45	0	0	1	1	1
Rural	0.30	0.46	0	0	0	1	1
Household Income >950′ NOK	0.54	0.50	0	0	1	1	1
Age: Below 40 Years	0.23	0.42	0	0	0	0	1
Age: 40-59 Years	0.47	0.50	0	0	0	1	1
Age: Above 60 Years	0.30	0.46	0	0	0	1	1
# Frequented Grocery Chains	2.98	1.39	1	2	3	4	9
Mainly uses discount stores	0.86	0.35	0	1	1	1	1
Main Grocery Shopper	0.74	0.44	0	0	1	1	1
3 Customer Loyalty Program Memberships	0.48	0.50	0	0	0	1	1
Price Consciousness	3.81	0.75	1	3.36	3.91	4.38	5
Price Sensitivity	4.12	0.79	1	3.74	4.26	4.74	5
Price Learning Effort	3.52	0.89	1	2.95	3.62	4.22	5
Supermarket Chain	0.50	0.50	0	0	0.50	1	1
Private Label	0.08	0.28	0	0	0	0	1
Average Weekly Chain Visits	1.45	1.56	0	0.23	0.99	2.21	15.56
Average Weekly Chain Variety	13.99	14.37	0	1.42	9.67	23.01	89.13
Category Size	19.34	19.11	2	7	12.92	25	128
Price Coefficient of Variation	0.11	0.06	0.01	0.06	0.11	0.15	0.33
Sales	92.33	363.26	0	0	0	31.56	18,492.96
Has Purchased Product at Chain	0.31	0.46	0	0	0	1	1
Months Since Last Purchase	4.68	4.80	0	0.73	2.73	7.50	17.53

Notes: This table reports summary statistics for individual × chain × product-level variables for the 72,392 price-guess observations used in the analysis.

Table A2 shows that the mean PK of the recalled prices is -0.27, meaning that the average price recall either over- or underestimates the true shelf price by 27%. In addition to the PK defined in Equation 3.3 above, we include the share of recalled prices within different error bounds. Only 1% of the product recalls are exactly correct at the two-decimal level, i.e., approximately 724 out of the 72,392 recalled prices.

Loy, Ceynowa, and Kuhn (2020), who surveyed grocery shoppers in two stores in Northern Germany, found that the average PK in their sample is 35% and that around 14% of recalled prices lie within 5% of the true price. This is a slightly poorer performance than our sample, with an average PK of 27% and a share of recalled prices within 5% of the true price of 17%. The magnitudes are fairly similar, but of course, the results are sensitive to the products asked about (see Figure A1a), the population surveyed, and the setting in which the respondents were surveyed (see Evanschitzky, Kenning, and Vogel, 2004 for a discussion).

Equation 3.3 measures price knowledge without distinguishing between over- and underestimation. To measure the direction of the respondents' price bias, we can calculate:

$$\mathrm{PD}_{ijc} = \frac{P_{ijc}^{\mathrm{Recall}} - P_{jcmt}^{\mathrm{Shelf}}}{P_{jcmt}^{\mathrm{Shelf}}}$$

The respondent's overall price bias is then given by $\mathrm{PD}_i = \frac{1}{K_i} \sum_{k=1}^{K_i} \mathrm{PD}_{ijc}$. Finally, averaging indicators of the sign of PD_{ijc} lets us calculate the respondent's share of over- and underestimations.

The price recalls overestimate the correct prices by 2% on average. Consistent with the findings of Aalto-Setälä and Raijas (2003), who measured price knowledge among Finnish grocery shoppers, the bias of the respondents as a group is low, despite substantial errors on any given recall. However, this breaks down somewhat at the chain and product level (see Figure A1b). Although respondents only slightly underestimate the prices of the discount chain on average ($PD = -1.5\%$), they overestimate the prices of the supermarket chain by 5.2%. This likely reflects the considerable efforts that discount chains in Norway, who serve over 60% of the market, put into building and maintaining a favorable price image. Supermarkets, on the other hand, focus more on service, assortment size, and quality. Furthermore, price bias varies substantially across products. For example, the respondents are far off target, on average, for strawberry jam (60%), crispbread (49%), potato chips (33%), and frozen cod (-40%), whereas the bias is quite low for eggs (-2%), smoothies (-2%), and 10 other products with less than 10% of price bias.

The share of overestimated recalls is 46%, and the share of underestimated recalls is 53%, despite the average price bias being positive. This stems from the overestimated price recalls being larger in magnitude, which is also reflected in the median price bias (-3%) being lower than the average (2%). The shares of over- and underestimation

differ substantially from the findings of Loy, Ceynowa, and Kuhn (2020), who find that 56% of price recalls overestimate the true price, and only 36% underestimate true prices. However, they find that 8% of recalls are exact, which we suspect can be explained by differences in the number of decimal places accounted for. We do not know if some of the 8% would be counted as over- or underestimations in our study. However, if we allow for slight errors within 1%, we get a success rate of 6%, which is much closer to the results of Loy, Ceynowa, and Kuhn (2020).

The explanatory variables that we use in the main regression analyses are also included in Tables A1 and A2. To simplify the analysis, we mostly use binary indicators derived from the detailed questions included in the survey. Both tables show that the sample is quite representative in terms of demographics, although compared to the general population, more respondents have higher education (71%) and are older than 40 years old (77%), and there are more female than male respondents (62%). From the averages of price consciousness (3.81), sensitivity (4.12), and learning effort (3.52), we understand that most respondents agree or strongly agree with statements related to price sensitivity, whereas respondents are more mixed on activities that generate price knowledge (see Figure A2).

The variables that are unique to Table A1 are those related to discount utilization and shopping basket price variation, which are calculated from the full purchasing histories of the respondents, not just the products included in the survey. We see that most consumers have positive discount utilization, meaning that they achieve lower prices than they would by shopping at random. Furthermore, simple discount utilization is higher across the quartiles than sales-weighted discount utilization, which suggests that the products that consumers spend the most on are not the products they buy the most at discounted prices, despite being incentivized to look for sales on products that they can make the most savings on. One explanation is that the products people spend the most on have fewer and smaller discounts for consumers to act on. This is supported by the lower average price coefficient of variation for the respondents' sales-weighted shopping baskets, at 0.1, compared to the simple average across products, at 0.11, indicating that the products consumers spend the most on vary slightly less in price over the sample period.

Table A2 includes variables that are unique to the recall-level analysis. The share of price recalls for the supermarket and the discount chain is even, and 8% of recalls were on private label products (2 out of the 24 included products). The average weekly number of visits per chain is 1.45 and the average weekly number of unique products purchased at a given chain is 14. The average recalled price is related to a product that has 19 close substitutes, defined by the category it belongs to, and that has an average price coefficient of variation, now measured for a specific product $\times$ chain combination included in the survey, of 0.11, which happens to be identical to the

simple average across all products that the respondents purchase. This indicates that the list of products that we include in the survey is representative of all products that consumers buy in terms of price variation.

Out of all the recalled prices that are included in the analysis, 31% were given by respondents who at some point purchased the product at the relevant chain during the sample period. Among those, the average time since purchase is 4.7 months whereas the median is 2.7 months. This might suggest that most respondents buy the same products more frequently, but that a few respondents go much longer between purchases or are even one-time buyers, pulling up the average. The average spending on the product being recalled during the sample period of 18 months is 92 NOK, but this includes the 69% of recalls for products on which there is zero spending, which is reflected by the median spending of 0 NOK. The total spending distribution among those that bought the product is also skewed, with the average and 3rd quartile spending both being 300 NOK, the 95th percentile being 1200 NOK, and the observed maximum being 18,500 NOK.

Table A3: Factor loadings and index weights for price consciousness, price sensitivity, and price learning effort

Likert Scale Question	**Price Consciousness**		**Price Sensitivity**		**Price Learning Effort**	
	Loading	**Weight**	**Loading**	**Weight**	**Loading**	**Weight**
1. Price is crucial for my choice of products.	0.73	15.4%	0.883	39.7%	–	–
2. Price is crucial for my choice of store.	0.719	15.2%	0.771	34.7%	–	–
3. I buy on Sales and promotions as often as I can.	0.74	15.7%	0.571	25.7%	–	–
4. I usually compare prices between the grocery chains.	0.81	17.1%	–	–	0.86	32.9%
5. I usually check the grocery chains' flyers for prices.	0.717	15.2%	–	–	0.762	29.1%
6. I usually check the results of VGs grocery price comparison tests.	0.387	8.2%	–	–	0.411	15.7%
7. I usually check the shelf price of the product before I buy it.	0.624	13.2%	–	–	0.584	22.3%
Sum	4.727	100%	2.225	100%	2.617	100%

Notes: This table shows estimated factor loadings for each Likert scale item using diagonally weighted least squares, as implemented by the *lavaan* package in *R*. The weights put on each Likert scale question in the indices for price consciousness, sensitivity, and learning effort are calculated by dividing the factor loadings by the sum of the factor loadings.

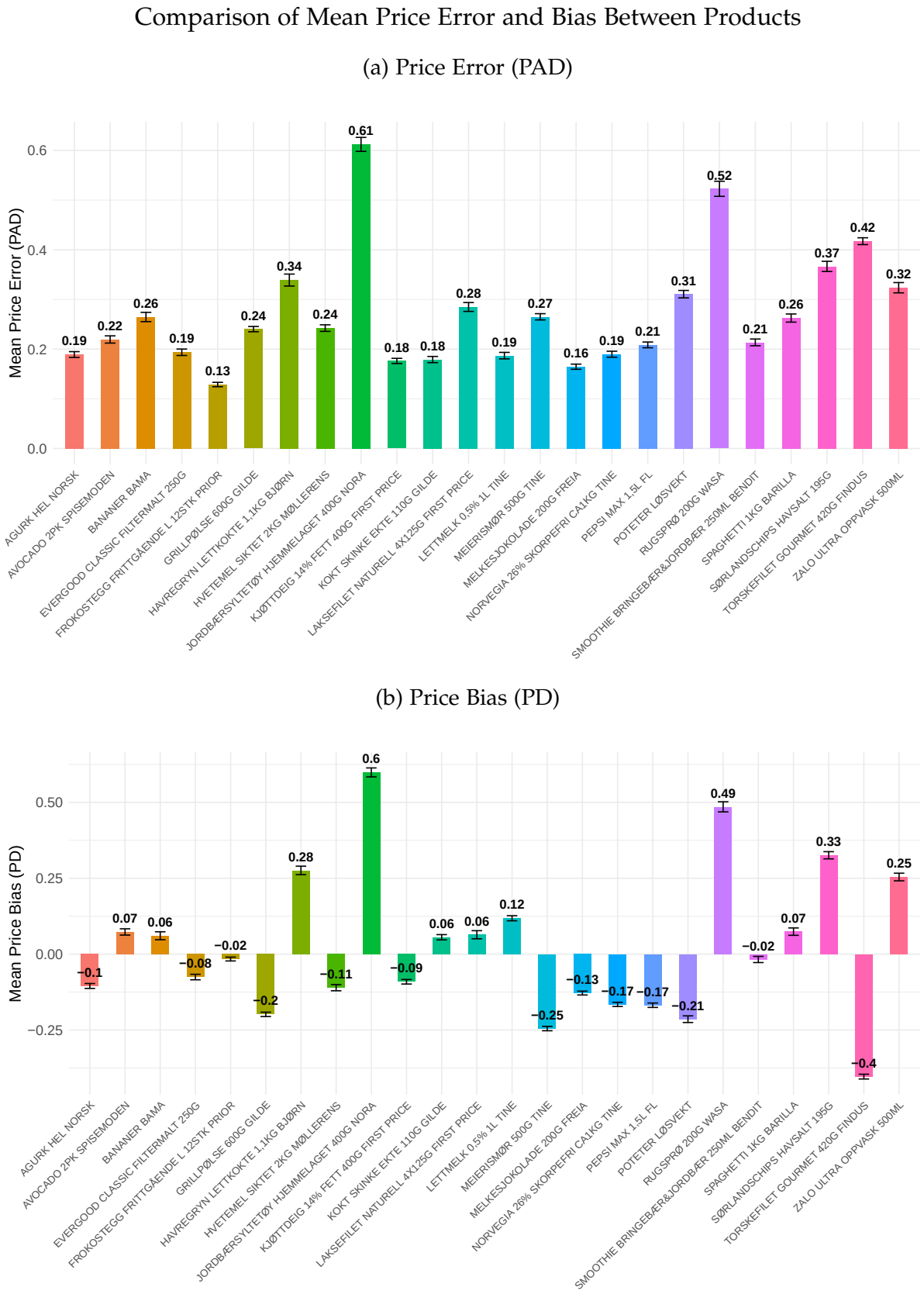


Figure A1: Price error is measured as the percentage absolute deviation between the respondent's recalled price and the true shelf price. Price bias is measured as the percentage deviation between the respondent's recalled price and the true shelf price. Confidence intervals at 95% are calculated using the Student's t-distribution.

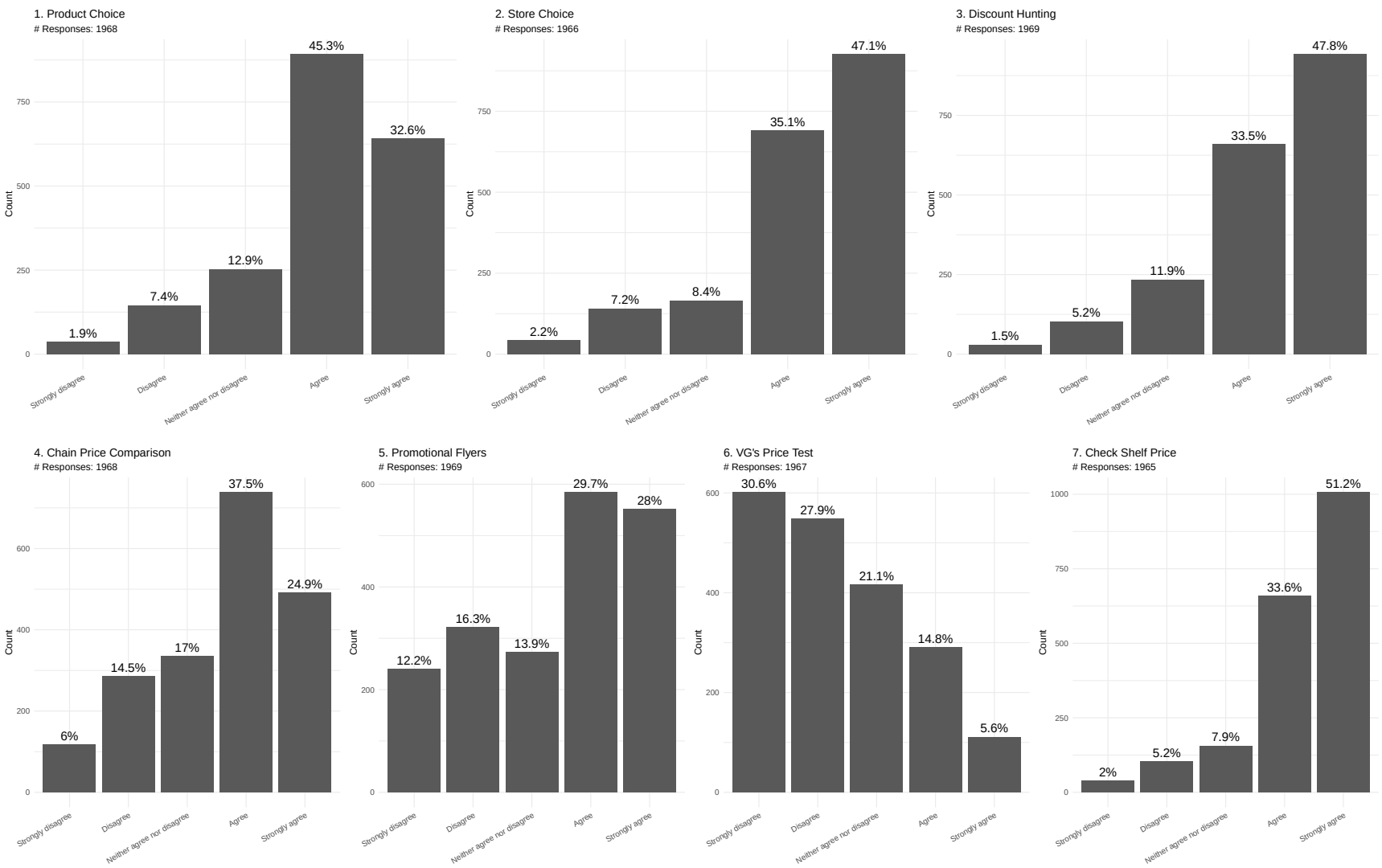


Figure A2: This figure contains bar charts displaying the distribution of answers for each Likert-scale question. The number of answers is given by the height of the bars, and the percentage share is indicated above the bars.

3.B Robustness Checks and Extensions

3.B.1 Discount Utilization for Perishable vs. Non-Perishable Products

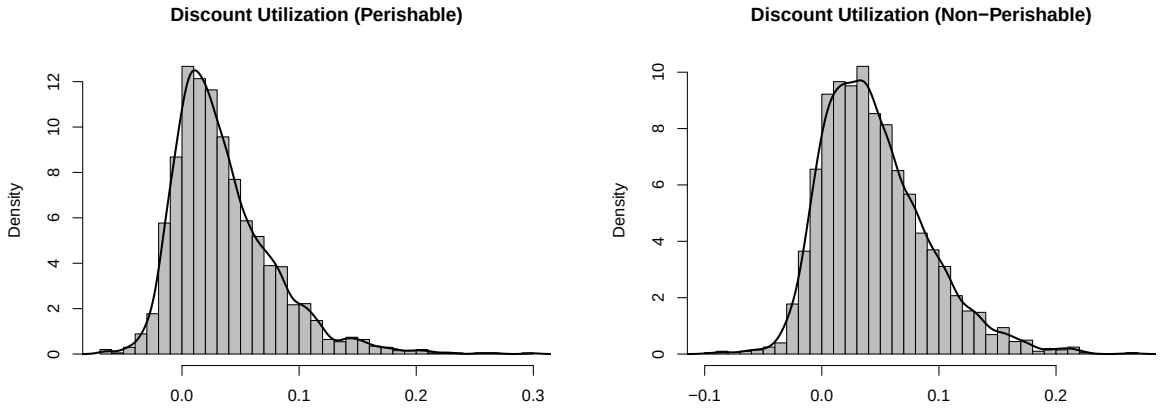


Figure A3: Histograms and density estimates of discount utilization for perishable and non-perishable products. Histogram bin sizes are chosen using Scott's rule, and density bandwidths are chosen using the Sheather–Jones plug-in estimator.

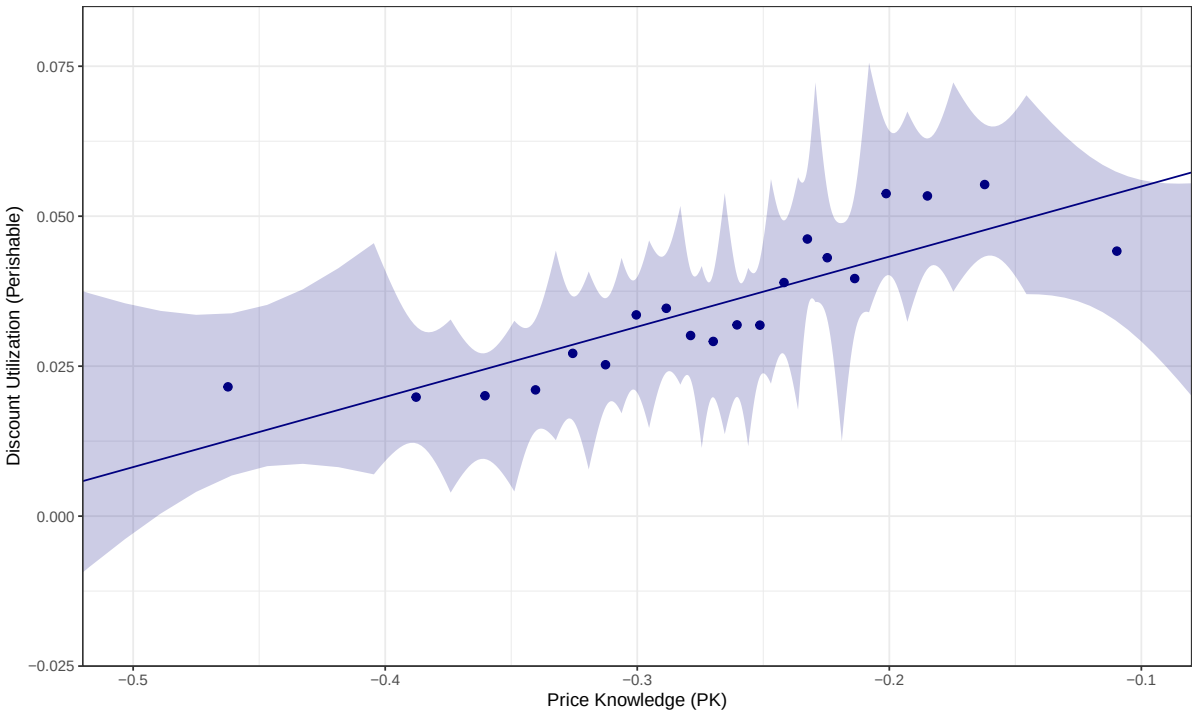


Figure A4: Estimate of the expected value of discount utilization conditional on price knowledge using a binned scatterplot with a linear fit. The number of bins is chosen to minimize the integrated mean squared error. Confidence bands at 95% are calculated using piecewise polynomials.

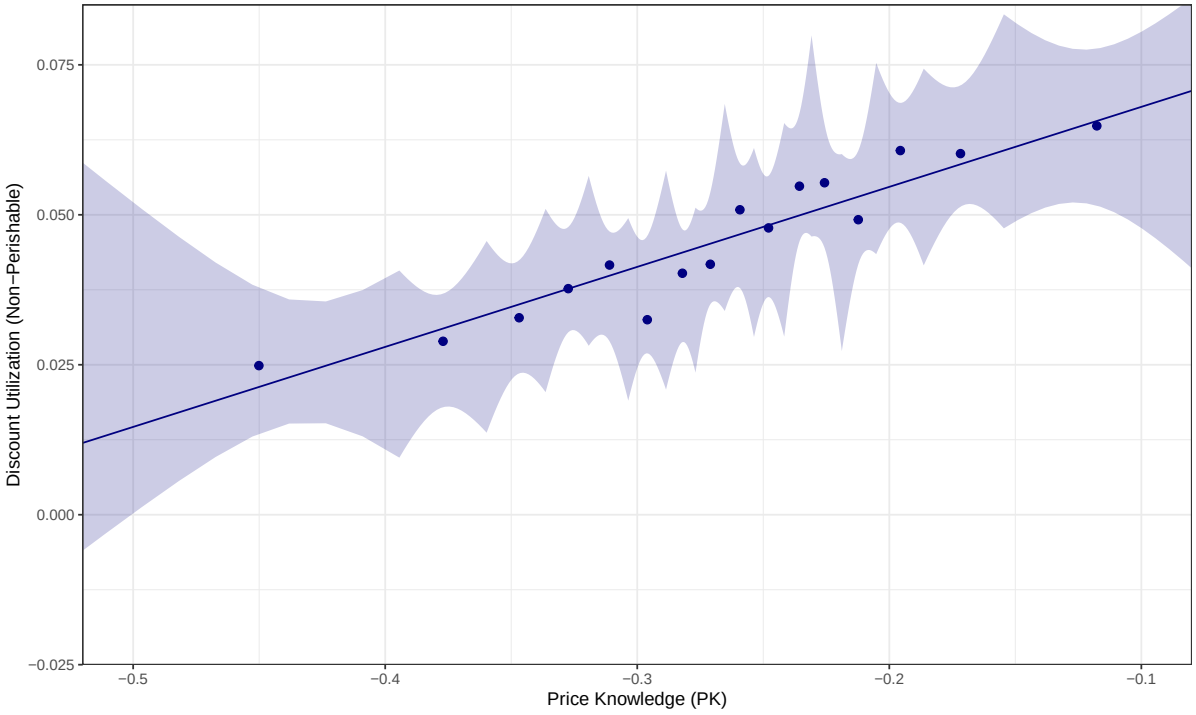


Figure A5: Estimate of the expected value of discount utilization conditional on price knowledge using a binned scatterplot with a linear fit. The number of bins is chosen to minimize the integrated mean squared error. Confidence bands at 95% are calculated using piecewise polynomials.

Table A4: Discount utilization regressions: perishable vs. non-perishable products

	DU Perishable (1)	DU Non-Perishable (2)	DU Perishable (3)	DU Non-Perishable (4)
Price Knowledge	0.117*** (0.012)	0.133*** (0.013)	0.075*** (0.011)	0.101*** (0.012)
Male			0.000 (0.002)	0.002 (0.002)
Single Person Household			0.002 (0.003)	0.006 (0.003)
Has Children			0.007*** (0.002)	0.003 (0.002)
Full-Time Employment			-0.004 (0.002)	-0.003 (0.002)
Higher Education			0.002 (0.002)	0.002 (0.002)
Rural			-0.004* (0.002)	-0.004* (0.002)
Household Income > 950' NOK			-0.002 (0.002)	0.001 (0.002)
Age: 40-59 Years			0.008*** (0.002)	0.006* (0.002)
Age: Above 60 Years			0.009** (0.003)	0.008** (0.003)
Price Sensitivity			0.003 (0.001)	0.003 (0.001)
Price Learning Effort			0.007*** (0.001)	0.008*** (0.001)
# Frequented Grocery Chains			0.000 (0.001)	0.000 (0.001)
Mainly Uses Discount Stores			-0.004 (0.003)	-0.001 (0.003)
Average Weekly Store Visits			0.004*** (0.001)	0.001 (0.001)
Main Grocery Shopper			0.004 (0.002)	0.000 (0.002)
Average Weekly Variety (per 10 products)			-0.003*** (0.001)	-0.003*** (0.001)
3 Customer Loyalty Program Memberships			0.006*** (0.002)	0.004* (0.002)
Basket Coefficient of Variation			1.122*** (0.087)	1.327*** (0.098)
Constant	0.067*** (0.003)	0.081*** (0.004)	-0.116*** (0.012)	-0.116*** (0.013)
Observations	2,028	2,028	1,928	1,928
Dependent variable mean	0.04	0.05	0.04	0.05
R^2	0.05	0.06	0.26	0.26

Notes: This table reports results from estimation of separate discount utilization regressions for perishable and non-perishable products. Discount utilization is calculated as the percentage difference between the quarterly average of daily modal (shelf) prices and the average price paid on the purchase day, averaged across purchase days and then across product $\times$ chain combinations (simple weights). Price knowledge is calculated as the negative average percentage absolute deviation between the respondent's recalled prices and true shelf prices. The average weekly variety is calculated as the average number of unique products purchased per week over the sample period. Price sensitivity and learning effort are calculated as the factor-weighted averages of Likert-scale survey items 1–3 and 4–7, respectively. Basket CV is calculated as the coefficient of variation of shelf prices over the sample period, averaged across all products in the consumer's purchase history, using simple weights. Columns 1–2 include the simple regression on price knowledge for perishable and non-perishable products, and Columns 3–4 add controls for demographics and shopping habits. The standard errors shown in parentheses are robust to heteroskedasticity (HC1). *** $p < 0.001$; ** $p < 0.01$; * $p < 0.05$.

3.B.2 Sales-Weighted Discount Utilization Regression

Table A5: Discount utilization regressions: sales weights

	Discount Utilization		
	(1)	(2)	(3)
Price Knowledge	0.070*** (0.010)	0.073*** (0.007)	0.054*** (0.007)
Male		0.000 (0.001)	0.000 (0.001)
Single Person Household		0.000 (0.002)	0.001 (0.002)
Has Children		-0.001 (0.001)	0.002 (0.001)
Full-Time Employment		-0.004* (0.002)	-0.002 (0.001)
Higher Education		-0.001 (0.001)	0.000 (0.001)
Rural		-0.006*** (0.001)	-0.003* (0.001)
Household Income > 950′ NOK		0.000 (0.001)	0.001 (0.001)
Age: 40-59 Years		0.003* (0.001)	0.004** (0.001)
Age: Above 60 Years		0.006** (0.002)	0.005** (0.002)
Price Sensitivity		0.003** (0.001)	0.001 (0.001)
Price Learning Effort			0.005*** (0.001)
# Frequented Grocery Chains			0.000 (0.000)
Mainly uses discount stores			-0.008*** (0.002)
Average Weekly Store Visits			0.003*** (0.001)
Main Grocery Shopper			0.002 (0.001)
Average Weekly Variety (per 10 products)			-0.003*** (0.001)
3 Customer Loyalty Program Memberships			0.003** (0.001)
Basket Coefficient of Variation			0.566*** (0.051)
Constant	0.040*** (0.003)	0.032*** (0.005)	-0.042*** (0.007)
Observations	2,028	1,935	1,928
Dependent variable mean	0.02	0.02	0.02
R^2	0.04	0.08	0.31

Notes: This table reports results from estimation of Equation 3.1. The dependent variable, individual-level discount utilization, is the percentage difference between the quarterly average of daily modal (shelf) prices and the average price paid on the purchase day, averaged across purchase days and then across product × chain combinations, with the latter using sales weights. Price knowledge is calculated as the negative average percentage absolute deviation between the respondent's recalled prices and true shelf prices. The average weekly variety is calculated as the average number of unique products purchased per week over the sample period. Price sensitivity and learning effort are calculated as the factor-weighted averages of Likert-scale survey items 1–3 and 4–7, respectively. Basket CV is calculated as the coefficient of variation of shelf prices over the sample period, averaged across all products in the consumer's purchase history, using sales weights. Column 1 reports the simple regression on price knowledge, Column 2 adds controls for demographics and price sensitivity, and Column 3 adds controls for shopping habits and price learning effort. The standard errors shown in parentheses are robust to heteroskedasticity (HC1). *** $p < 0.001$; ** $p < 0.01$; * $p < 0.05$.

Table A6: Discount utilization regressions: sales-weighted basket CV interactions

	Discount Utilization	
	(1)	(2)
Price Knowledge	0.070***	0.046***
	(0.010)	(0.005)
High Basket CV		0.029***
		(0.005)
Price Knowledge × High Basket CV		0.039*
		(0.018)
Constant	0.040***	0.024***
	(0.003)	(0.002)
Observations	2,028	2,028
Dependent variable mean	0.02	0.02
R^2	0.04	0.16

Notes: This table reports results from estimation of discount utilization regressions using an interaction between price knowledge and an indicator for having above-median basket price variation. Discount utilization is calculated as the percentage difference between the quarterly average of daily modal (shelf) prices and the average price paid on the purchase day, averaged across purchase days and then across product × chain combinations (sales weights). Price knowledge (PK) is calculated as the negative average percentage absolute deviation between the respondent's recalled prices and true shelf prices. Basket CV is calculated as the coefficient of variation of shelf prices over the sample period, averaged across all products in the consumer's purchase history, using sales weights. Column 1 includes the simple regression on price knowledge and Column 2 adds an interaction with Basket CV. The standard errors shown in parentheses are robust to heteroskedasticity (HC1). *** $p < 0.001$; ** $p < 0.01$; * $p < 0.05$.

3.B.3 Discount Utilization for Non-Survey Products

As discussed in Section 3.2.2, a potential concern for the causal interpretation of our main results is reverse causality. Specifically, if recently purchasing products on sale causes consumers to recall those prices more accurately, the positive association between price knowledge and discount utilization would be upward biased as an estimate of the causal effect of price knowledge, as it would partly reflect the reverse effect of discount utilization on price knowledge.

To address this, we re-estimate the relationship between price knowledge and discount utilization, but we calculate the dependent variable using only the products in the respondents' purchase history that were not included in the survey. Thus, we separate the measurement of price knowledge (based on the 18 surveyed items) from the behavior it predicts (discount utilization across the remaining ~700 items in the purchase history).

The results are presented in Table A7. We find that the association between price knowledge and discount utilization remains robust and highly statistically significant when excluding the survey products. In fact, the estimated coefficient on price knowledge is slightly larger in this specification than in the baseline results. This provides evidence against the reverse causality hypothesis; if price knowledge were partly a function of recalling prices recently paid on discount, we would expect a weaker association between price knowledge of the surveyed products and discount utilization on a different set of products.

Table A7: Discount utilization regressions: non-survey products

	Discount Utilization		
	(1)	(2)	(3)
Price Knowledge	0.137*** (0.013)	0.133*** (0.012)	0.097*** (0.011)
Male		0.002 (0.002)	0.003 (0.002)
Single Person Household		0.004 (0.003)	0.004 (0.003)
Has Children		0.005* (0.002)	0.006** (0.002)
Full-Time Employment		-0.005* (0.002)	-0.003 (0.002)
Higher Education		0.001 (0.002)	0.003 (0.002)
Rural		-0.007*** (0.002)	-0.005** (0.002)
Household Income > 950′ NOK		-0.002 (0.002)	-0.001 (0.002)
Age: 40-59 Years		0.007** (0.002)	0.007*** (0.002)
Age: Above 60 Years		0.006* (0.003)	0.007* (0.003)
Price Sensitivity		0.008*** (0.001)	0.004** (0.001)
Price Learning Effort			0.008*** (0.001)
# Frequented Grocery Chains			0.000 (0.001)
Mainly uses discount stores			-0.006 (0.003)
Average Weekly Store Visits			0.003*** (0.001)
Main Grocery Shopper			0.001 (0.002)
Average Weekly Variety (per 10 products)			-0.004*** (0.001)
3 Customer Loyalty Program Memberships			0.006*** (0.002)
Basket Coefficient of Variation			1.360*** (0.087)
Constant	0.080*** (0.004)	0.042*** (0.007)	-0.129*** (0.012)
Observations	2,028	1,935	1,928
Dependent variable mean	0.04	0.04	0.04
R^2	0.07	0.11	0.33

Notes: This table reports results from estimation of Equation 3.1. The dependent variable, individual-level discount utilization, is the percentage difference between the quarterly average of daily modal (shelf) prices and the average price paid on the purchase day, averaged across purchase days and then across non-survey product × chain combinations (simple weights). Price knowledge is calculated as the negative average percentage absolute deviation between the respondent's recalled prices and true shelf prices. The average weekly variety is calculated as the average number of unique products purchased per week over the sample period. Price sensitivity and learning effort are calculated as the factor-weighted averages of Likert-scale survey items 1–3 and 4–7, respectively. Basket CV is calculated as the coefficient of variation of shelf prices over the sample period, averaged across all products in the consumer's purchase history, using simple weights. Column 1 reports the simple regression on price knowledge (PK), Column 2 adds controls for demographics and price sensitivity, and Column 3 adds controls for shopping habits and price learning effort. The standard errors shown in parentheses are robust to heteroskedasticity (HC1). *** $p < 0.001$; ** $p < 0.01$; * $p < 0.05$.

3.B.4 Discount Utilization and Price Knowledge Quintiles

To assess the functional form of the relationship between price knowledge and discount utilization, and to evaluate the empirical validity of the binary distinction between "informed" and "uninformed" consumers often found in search-theoretic models (e.g., Varian, 1980), we re-estimate the model using quintile indicators of price knowledge. In this specification, the bottom quintile (lowest price knowledge) serves as the reference category. The results are presented in Table A8, and the corresponding quintile intervals for price knowledge are reported in Table A9.

Table A8: Discount utilization regressions: price knowledge quintiles

	Discount Utilization		
	(1)	(2)	(3)
Price Knowledge Q2 (20–40%)	0.010*** (0.002)	0.010*** (0.002)	0.010*** (0.002)
Price Knowledge Q3 (40–60%)	0.012*** (0.002)	0.011*** (0.002)	0.010*** (0.002)
Price Knowledge Q4 (60–80%)	0.023*** (0.003)	0.023*** (0.003)	0.018*** (0.002)
Price Knowledge Q5 (80–100%)	0.032*** (0.003)	0.030*** (0.003)	0.022*** (0.002)
Male		0.000 (0.002)	0.001 (0.002)
Single Person Household		0.004 (0.003)	0.004 (0.003)
Has Children		0.004* (0.002)	0.005** (0.002)
Full-Time Employment		-0.005* (0.002)	-0.003 (0.002)
Higher Education		0.001 (0.002)	0.002 (0.002)
Rural		-0.006*** (0.002)	-0.005** (0.002)
Household Income > 950′ NOK		-0.003 (0.002)	-0.001 (0.002)
Age: 40-59 Years		0.007*** (0.002)	0.007*** (0.002)
Age: Above 60 Years		0.007** (0.003)	0.008*** (0.002)
Price Sensitivity		0.007*** (0.001)	0.003** (0.001)
Price Learning Effort			0.007*** (0.001)
# Frequented Grocery Chains			0.000 (0.001)
Mainly uses discount stores			-0.003 (0.002)
Average Weekly Store Visits			0.003*** (0.001)
Main Grocery Shopper			0.002 (0.002)
Average Weekly Variety (per 10 products)			-0.003*** (0.001)
3 Customer Loyalty Program Memberships			0.005*** (0.002)
Basket Coefficient of Variation			1.205*** (0.074)
Constant	0.025*** (0.002)	-0.006 (0.005)	-0.149*** (0.010)
Observations	2,028	1,935	1,928
Dependent variable mean	0.04	0.04	0.04
R^2	0.09	0.13	0.36

Notes: This table reports results from estimation of discount utilization regressions using indicators for the quintiles of price knowledge. The dependent variable, individual-level discount utilization, is the percentage difference between the quarterly average of daily modal (shelf) prices and the average price paid on the purchase day, averaged across purchase days and then across product × chain combinations (simple weights). Price knowledge is calculated as the negative average percentage absolute deviation between the respondent's recalled prices and true shelf prices. The average weekly variety is calculated as the average number of unique products purchased per week over the sample period. Price sensitivity and learning effort are calculated as the factor-weighted averages of Likert-scale survey items 1–3 and 4–7, respectively. Basket CV is calculated as the coefficient of variation of shelf prices over the sample period, averaged across all products in the consumer's purchase history, using simple weights. Column 1 reports the simple regression on price knowledge, Column 2 adds controls for demographics and price sensitivity, and Column 3 adds controls for shopping habits and price learning effort. The standard errors shown in parentheses are robust to heteroskedasticity (HC1). *** $p < 0.001$; ** $p < 0.01$; * $p < 0.05$.

Table A9: Quintiles for Price Knowledge

Quintile	PK interval
Q1 (0–20%)	$[-0.70, -0.33]$
Q2 (20–40%)	$(-0.33, -0.28]$
Q3 (40–60%)	$(-0.28, -0.25]$
Q4 (60–80%)	$(-0.25, -0.21]$
Q5 (80–100%)	$(-0.21, -0.04]$

The results confirm a monotonic positive relationship between price knowledge and discount utilization. Relative to the reference category, discount utilization increases as we move to higher quintiles. Entering the second quintile is associated with a 1.0 percentage point increase in discount utilization. While the increase from the second to the third quintile is small (0.2 percentage points), the subsequent increases to the fourth and fifth quintiles are substantial (1.1 and 0.9 percentage points, respectively).

The quintiles around the center of mass of the distribution (PK $= -0.27$) are quite narrow, particularly the third quintile (Q3), which spans only 3 percentage points (from -0.28 to -0.25). The flattening of the effect in this region may therefore reflect that the estimated Q3 coefficient is sensitive to local idiosyncrasies rather than indicating a genuine nonlinearity in the underlying relationship between price knowledge and discount utilization. This interpretation is supported by the non-parametric evidence in the binned scatterplot (Figure 3.3), which shows a fairly linear trend through the bulk of the data despite some local variability. Consequently, we conclude that the linear assumption employed in our main specification remains a reasonable and robust approximation.

We further note that the coefficients on the demographic and behavioral control variables remain stable in magnitude and statistical significance relative to the baseline linear specification (Table 3.1).

Finally, we find no evidence of distinct thresholds of price knowledge associated with sharp changes in discount utilization. Instead, discount utilization increases gradually across the distribution. This pattern suggests that the binary classification used in theoretical models is a simplification of real-world behavior. Price knowledge is better viewed as a continuous trait, where incremental improvements lead to incremental reductions in prices paid.

3.B.5 Discount Utilization and the Share of Correct Price Recalls

Table A10 shows the OLS results of a simple regression of discount utilization on the share of recalled prices within some error bound, using the sample including outliers in PK, for comparison with the results in Table 3.1:

$$\mathrm{DU}(Q)_i^{\text{Simple}} = \alpha + \delta \cdot W_i^e + \epsilon_i \tag{3.12}$$

where W_i^e is the share of correctly recalled prices within some error bound e. Like a high PK, a high share of correctly recalled prices is predictive of more discount utilization. This is reassuring, since the share of correct recalls is robust to extreme price recalls, for example, from erroneous entry or inattention, that could significantly affect the PK.

The explained variance is larger for the share of roughly correct recalls, i.e., within 20%. This suggests that consumers do not need precise knowledge of prices in order to act on sales—a rough idea of price levels suffices. Still, the PK performs even better with an R^2 of 7% compared to the R^2 of 5% obtained by using the share of roughly correct recalls.[31] Finally, note that the larger coefficients on precise relative to rough recalls do not signify that precise recalls are more important, since a given increase in the share of precise recalls is much harder to obtain than an equal increase in the share of roughly correct recalls.

[31]This also holds when comparing like-for-like on the sample without outliers.

Table A10: Discount utilization regressions: within percentage error bounds

	Discount Utilization			
	(1)	(2)	(3)	(4)
Share Within 1%	0.064***			
	(0.012)			
Share Within 5%		0.060***		
		(0.007)		
Share Within 10%			0.051***	
			(0.006)	
Share Within 20%				0.051***
				(0.005)
Constant	0.035***	0.030***	0.025***	0.014***
	(0.001)	(0.001)	(0.002)	(0.002)
Observations	2,285	2,285	2,285	2,285
Dependent variable mean	0.04	0.04	0.04	0.04
R^2	0.02	0.03	0.04	0.05

Notes: This table reports results from estimation of discount utilization regressions using the share of correct price recalls within different error bounds and the sample including outliers in price knowledge. Discount utilization is calculated as the percentage difference between the quarterly average of daily modal (shelf) prices and the average price paid on the purchase day, averaged across purchase days and then across product $\times$ chain combinations using simple weights. Columns (1-4) uses error bounds of 1, 5, 10, and 20 percent as explanatory variables, respectively. The standard errors shown in parentheses are robust to heteroskedasticity (HC1). *** $p < 0.001$; ** $p < 0.01$; * $p < 0.05$.

3.B.6 Explaining the Likelihood of Correct Price Recalls

Table A11 reports average marginal effects from logistic regressions, estimated by maximum likelihood, in which the dependent variable is an indicator for whether a price recall falls within varying error bounds. The explanatory variables are the same as those in Column 2 of Table 3.4. Estimates are based on the full sample of price recalls, including outliers, using the following specification:

$$Pr(W^{e}_{ijc} = 1|\{\theta, \mathbf{X}_{ijc}\}) = \Lambda(\theta + \gamma^{'}\mathbf{X}_{ijc}) = \frac{1}{1 + e^{-(\theta+\gamma^{'}\mathbf{X}_{ijc})}} \tag{3.13}$$

The results are qualitatively similar to those reported in our main specification (Table 3.4) in terms of both sign and statistical significance, particularly when using the more lenient error bounds. This consistency is reassuring, as the binary indicators for correct recall are robust to outliers, confirming that our primary results are not driven by our outlier exclusion criteria.

The results in Table A11 also illustrate how the determinants of price knowledge differ between precise and approximate recall. In most cases, the average marginal effects on the probability of a correct recall share the same sign across specifications; however, the magnitudes are naturally larger for the more lenient error bounds, reflecting the higher baseline probability of achieving a rough recall.[32]

Notably, we observe a divergence regarding the time elapsed since the last purchase. While the probability of a precise recall (within 1%) diminishes significantly as time passes, we find no statistically significant decay for roughly correct recalls (within 20%). Among consumers who have purchased a product, the probability of a precise recall drops by 2.6 percentage points (from 9.1% to 6.5%) after 12 months. In contrast, the probability of a roughly correct recall decreases by only 1 percentage point (from 59% to 58%) over the same period. This suggests that while exact price information held in short-term memory fades, a general sense of the price level persists in long-term memory. This finding aligns with Vanhuele and Drèze (2002), who show that consumers can recognize deal prices even when unable to recall exact prices.

[32]This is consistent with the sigmoid shape of the logistic function, given that precise recalls are significantly rarer than rough recalls.

Table A11: Logistic price knowledge regressions: correct price recalls

	Pr(Within 1%) (1)	Pr(Within 5%) (2)	Pr(Within 10%) (3)	Pr(Within 20%) (4)
Male	-0.118* (0.055) [-0.7%]	-0.134*** (0.036) [-1.8%]	-0.131*** (0.030) [-2.6%]	-0.133*** (0.028) [-3.2%]
Single Person Household	0.094 (0.104) [0.5%]	-0.002 (0.064) [0.0%]	-0.032 (0.053) [-0.6%]	-0.037 (0.047) [-0.9%]
Has Children	0.000 (0.062) [0.0%]	0.033 (0.042) [0.4%]	0.017 (0.035) [0.3%]	0.005 (0.033) [0.1%]
Full-Time Employment	-0.040 (0.072) [-0.2%]	-0.007 (0.045) [-0.1%]	-0.039 (0.038) [-0.8%]	-0.028 (0.034) [-0.7%]
Higher Education	-0.183** (0.061) [-1.0%]	-0.172*** (0.039) [-2.4%]	-0.135*** (0.033) [-2.7%]	-0.092** (0.031) [-2.2%]
Rural	0.003 (0.059) [0.0%]	-0.025 (0.038) [-0.3%]	-0.013 (0.032) [-0.3%]	-0.010 (0.030) [-0.2%]
Household Income > 950' NOK	0.060 (0.064) [0.3%]	0.083* (0.041) [1.1%]	0.068* (0.034) [1.4%]	0.076* (0.031) [1.8%]
Age: 40-59 Years	0.061 (0.063) [0.3%]	0.030 (0.040) [0.4%]	-0.002 (0.034) [0.0%]	-0.023 (0.032) [-0.6%]
Age: Above 60 Years	-0.083 (0.100) [-0.5%]	-0.039 (0.062) [-0.5%]	-0.104* (0.052) [-2.0%]	-0.108* (0.047) [-2.6%]
# Frequented Grocery Chains	0.015 (0.019) [0.1%]	0.014 (0.012) [0.2%]	0.014 (0.010) [0.3%]	0.009 (0.010) [0.2%]
Mainly Uses Discount Stores	-0.102 (0.085) [-0.6%]	-0.005 (0.054) [-0.1%]	0.009 (0.046) [0.2%]	0.014 (0.042) [0.3%]
Main Grocery Shopper	0.067 (0.053) [0.4%]	0.069 (0.037) [0.9%]	0.074* (0.032) [1.5%]	0.098** (0.030) [2.4%]
Supermarket Chain	-0.132* (0.059) [-0.7%]	0.035 (0.028) [0.5%]	0.118*** (0.019) [2.3%]	0.200*** (0.016) [4.8%]
Private Label	-1.182*** (0.107) [-4.3%]	0.200*** (0.039) [2.8%]	0.128*** (0.033) [2.6%]	0.118*** (0.034) [2.8%]
3 Customer Loyalty Program Memberships	0.110* (0.052) [0.6%]	0.055 (0.034) [0.7%]	0.084** (0.029) [1.7%]	0.084** (0.027) [2.0%]
Price Sensitivity	-0.061 (0.045) [-0.3%]	-0.024 (0.029) [-0.3%]	-0.019 (0.023) [-0.4%]	-0.013 (0.020) [-0.3%]
Price Learning Effort	0.126*** (0.037) [0.7%]	0.122*** (0.023) [1.6%]	0.130*** (0.020) [2.6%]	0.135*** (0.018) [3.2%]
Average Weekly Chain Visits	0.075* (0.031) [0.4%]	0.055** (0.017) [0.7%]	0.051*** (0.013) [1.0%]	0.054*** (0.012) [1.3%]
Average Weekly Chain Variety (per 10 products)	-0.075* (0.031) [-0.4%]	-0.069*** (0.018) [-0.9%]	-0.077*** (0.014) [-1.5%]	-0.091*** (0.012) [-2.2%]
Category Size	-0.062*** (0.011) [-0.3%]	-0.064*** (0.007) [-0.9%]	-0.049*** (0.005) [-1.0%]	-0.093*** (0.005) [-2.2%]
Price Coefficient of Variation	-1.381*** (0.305) [-7.7%]	-1.245*** (0.197) [-16.6%]	-0.766*** (0.172) [-15.3%]	-1.251*** (0.163) [-30.0%]
Purchased	0.417*** (0.049) [2.4%]	0.465*** (0.034) [6.6%]	0.418*** (0.030) [9.2%]	0.455*** (0.030) [12.2%]
Purchased × Sales in 1000 NOK	0.303*** (0.038) [2.2%]	0.278*** (0.034) [4.5%]	0.449*** (0.037) [10.0%]	0.588*** (0.049) [13.8%]
Purchased × Months Since Last Purchase	-0.030*** (0.006) [-0.2%]	-0.020*** (0.004) [-0.3%]	-0.009** (0.003) [-0.2%]	-0.004 (0.003) [-0.1%]
Constant	-2.630*** (0.198)	-1.888*** (0.130)	-1.304*** (0.112)	-0.377*** (0.100)
Observations	73,548	73,548	73,548	73,548
Baseline Probability	6%	16%	29%	49%
Pseudo R^2	0.03	0.02	0.02	0.03

Notes: This table reports results from estimation of Equation 3.13 using maximum likelihood. Price knowledge (within) is an indicator for a recalled price being within a percentage error bound of the true shelf price. Price sensitivity and learning effort are calculated as the factor-weighted average of Likert items 1–3 and 4–7, respectively. The average weekly chain variety is calculated as the average number of unique products purchased at each chain per week over the sample period. Category size is the chain-specific number of close substitutes for the product as defined by the grocery chain. The price coefficient of variation is calculated for each product × chain combination over the sample period. Purchased is an indicator of whether the respondent bought the product at a given chain during the sample period, and sales is the consumer's total product–chain spending. Columns 1–4 parallel Column 3 in Table 3.4 using error bounds of 1, 5, 10, and 20 percent, respectively. The standard errors shown in parentheses are clustered by individual, and the average marginal effects are shown in square brackets. *** $p < 0.001$; ** $p < 0.01$; * $p < 0.05$.

3.C Full Consumer Survey

The survey used in this chapter was originally presented to respondents in Norwegian. Below, we present an English version of the complete survey, translated using Google Translate. We have not fine-tuned the translated version to match natural English usage; consequently, some formulations may appear awkward or unpolished.

The survey was distributed to Trumf members through a newsletter over the period 4–12 June 2024. The median respondent spent 8.7 minutes completing the questionnaire. Out of the 2505 respondents, 418 did not complete the survey in its entirety, yielding an attrition rate of 16.7%. Most respondents who dropped out did so before the price recall section.

There were a few inconsistencies in the Likert-scale response options. Two different Norwegian synonyms for "Strongly" ("Svært" and "Veldig") were used for the extreme response categories, and the adverb "Slightly" appeared only in the "Agree" option. It is difficult to determine whether this influenced respondents' choices, but we suspect it may have slightly shifted some answers between the two outer categories on either side.

English

Introduksjon

Price knowledge and purchasing behavior

You are hereby invited to participate in a research project led by FOOD, a collaboration between the Norwegian School of Economics (NHH) and NorgesGruppen. The study aims to explore the relationship between grocery customers' price knowledge and their purchasing behavior.

Project overview

Participation involves answering questions about demographics, shopping habits, and price awareness. No directly identifiable information about you will be shared with the FOOD project.

A selection of your purchase history from Trumf, which will be linked to the answers you provide in the survey, will be shared with the FOOD project. This includes information about selected products from product groups in the survey, sales amount, quantity/weight, chain and date of purchase, and number of purchases and average shopping cart size per week from January 1, 2023 to June 12, 2024.

Trumf will process the sales data and link it to your survey responses, but they will not have access to your responses.

Participation details

Participation in the survey is voluntary. The information is used only in the FOOD project and is treated confidentially in accordance with data protection regulations. The project will end in 2024, after which only anonymized data will be stored on a secure server at NHH, accessible to FOOD researchers.

Once you have completed the survey, you will be entered into the draw to win:

- Main prize: Trumf bonus of NOK 5,000
- Partial prizes for 10 pieces: Trumf bonus of 500 kr

The survey takes approximately **5-10 minutes** to complete.

By answering the survey, you agree to participate in the research project. If you do not wish to participate, please close the tab.

We greatly appreciate your participation!

Demografi

What is your age?

Under 30 years old
30-39 years old
40-49 years
50-59 years
60-69 years
70 years or over

What gender are you?

Woman
Man
Other:

How many members are there in your household?

1
2
3
4
5
6 or more

How many children under the age of 18 are there in your household?

0

1

2

3

4 or more

Choose the option that *best* describes your work situation:

Full-time employee

Part-time employee

Self-employed

Not at work

Student

Retired

Other:

What is your highest completed education?

Primary school

High school

Vocational school

University/college

Other:

What is your household's total annual income before taxes?

Total income includes all wage income, business income, capital income, pensions, and other taxable and tax-exempt transfers.

Under 300,000 NOK

300,000 - 600,000 kr

600,000 - 950,000 kr

950,000 - 1,350,000 kr

1,350,000 - 1,800,000 kr

Over 1,800,000 kr

How would you describe the area you live in?

City
Suburb
Rural

Handlevaner

Which of these grocery chains do you usually shop at during a typical month?

Check multiple boxes if applicable.

Kiwi
Rema 1000
Coop Extra
Menu
Spar/Eurospar
Coop Note
Coop Mega
Coop Price
Joker
Bottom price
Coop Market
The convenience store
The food corner
Oda

You did not check these chains. Which of them would it have been practical for you to shop at in your everyday life?

Check multiple boxes if applicable.

» Kiwi
» Rema 1000
» Coop Extra
» Menu
» Spar/Eurospar
» Coop Note
» Coop Mega

- Coop Price
- Joker
- Bottom price
- Coop Market
- The convenience store
- The food corner
- Oda

Which of these chains do you shop at the most?

- Kiwi
- Rema 1000
- Coop Extra
- Menu
- Spar/Eurospar
- Coop Note
- Coop Mega
- Coop Price
- Joker
- Bottom price
- Coop Market
- The convenience store
- The food corner
- Oda

How many times do you shop for groceries (on average) per week?

0-1

2-3

4-5

6 or more

How many loyalty programs are you a member of (Æ, Trumf, Coop)?

1

2

3 or more

Are you primarily responsible for purchasing groceries for the household?

Yes

No

Shared responsibility

Priskunnskap - Intro

You will now be asked what you think the current shelf price is for 18 different grocery products.

Any offers or promotions should be taken into account, but you can disregard personal offers and coupons.

We ask that you answer as best you can based on your own memory.

Frokostprodukter

What is the current shelf price of **200g Wasa Rye Crackers** in the following stores?



	Number of kroner:
Kiwi	
Menu	

What is the current shelf price of **400g Nora homemade strawberry jam** in the following stores?



	Number of kroner:
Kiwi	
Menu	

What is the current shelf price of **110g sliced cooked ham from Gilde** in the following stores?



	Number of kroner:
Kiwi	
Menu	

What is the current shelf price of **1.1kg Bjørn easy-cooked oatmeal from AXA** in the following stores?



	Number of kroner:
Kiwi	
Menu	

Kjøtt og fisk

What is the current shelf price of **400g of minced meat from First Price** in the following stores?



	Number of kroner:
Kiwi	
Menu	

What is the current shelf price of **600g Gilde grill sausages** in the following stores?



	Number of kroner:
Kiwi	
Menu	

What is the current shelf price of **4x125g salmon fillets from First Price** in the following stores?



	Number of kroner:
Kiwi	
Menu	

What is the current shelf price of **420g cod fillet from Findus** in the following stores?



	Number of kroner:
Kiwi	
Menu	

Meieriprodukter

What is the current shelf price of **1l Tine low-fat milk 0.5% fat** in the following stores?



	Number of kroner:
Kiwi	
Menu	

What is the current shelf price of **12 breakfast eggs from Prior** in the following stores?



	Number of kroner:
Kiwi	
Menu	

What is the current shelf price of **500g Tine dairy butter** in the following stores?



	Number of kroner:
Kiwi	
Menu	

What is the current shelf price of **1kg of Norvegia yellow cheese** in the following stores?



	Number of kroner:
Kiwi	
Menu	

Husholdningsprodukter og tørrvarer

What is the current shelf price of **2kg of Møllerens sifted wheat flour** in the following stores?



	Number of kroner:
Kiwi	
Menu	

What is the current shelf price of **500ml of Zalo Ultra dishwashing detergent** in the following stores?



	Number of kroner:
Kiwi	
Menu	

What is the current shelf price of **1kg of spaghetti from Barilla** in the following stores?



	Number of kroner:
Kiwi	
Menu	

What is the current shelf price of **250g Evergood Classic filter ground coffee** in the following stores?



	Number of kroner:
Kiwi	
Menu	

Brus, snacks og søtsaker

What is the current shelf price of **200g Freia milk chocolate** in the following stores?



	Number of kroner:
Kiwi	
Menu	

What is the current shelf price of **1.5l Pepsi Max** in the following stores (without deposit)?



	Number of kroner:
Kiwi	
Menu	

What is the current shelf price of **195g Sørlandschips sea salt** in the following stores?



	Number of kroner:
Kiwi	
Menu	

What is the current shelf price of **250ml Bendit smoothie with raspberries and strawberries** in the following stores?



	Number of kroner:
Kiwi	
Menu	

Frukt og grønt

What is the current shelf price of **a Norwegian cucumber from Gartner** in the following stores?



	Number of kroner:
Kiwi	
Menu	

What is the current shelf price for **2 packs of ripe avocados from Bama** in the following stores?



	Number of kroner:
Kiwi	
Menu	

What is the current shelf price of **1kg of bananas from Bama** in the following stores?



	Number of kroner:
Kiwi	
Menu	

What is the current shelf price of **1kg of potatoes in bulk** in the following stores?



	Number of kroner:
Kiwi	
Menu	

Prisbevissthet - 1-by-1

Finally, we ask you to indicate the extent to which you agree with 10 different statements.

Price is crucial for my choice of *products* .

Strongly disagree	Disagree	Neither agree nor disagree	Slightly agree	Strongly agree

Price is crucial for my choice of *store* .

Strongly disagree	Disagree	Neither agree nor disagree	Slightly agree	Strongly agree

I buy on offers and promotions as often as I can.

Strongly disagree | Disagree | Neither agree nor disagree | Slightly agree | Strongly agree

I usually compare prices between chains.

Strongly disagree | Disagree | Neither agree nor disagree | Slightly agree | Strongly agree

I usually check the grocery chains' customer flyers for prices.

Strongly disagree | Disagree | Neither agree nor disagree | Slightly agree | Strongly agree

I usually check the results in VG's price comparison test.

Strongly disagree | Disagree | Neither agree nor disagree | Slightly agree | Strongly agree

I usually check the shelf price of the product before I buy it.

Strongly disagree | Disagree | Neither agree nor disagree | Slightly agree | Strongly agree

I have good knowledge of grocery prices.

Strongly disagree | Disagree | Neither agree nor disagree | Slightly agree | Strongly agree

I have better knowledge about the prices of groceries I buy frequently.

Strongly disagree | Disagree | Neither agree nor disagree | Slightly agree | Strongly agree

I have better knowledge about the prices of groceries that cost a lot.

Strongly disagree | Disagree | Neither agree nor disagree | Slightly agree | Strongly agree

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